MATHEMATICS

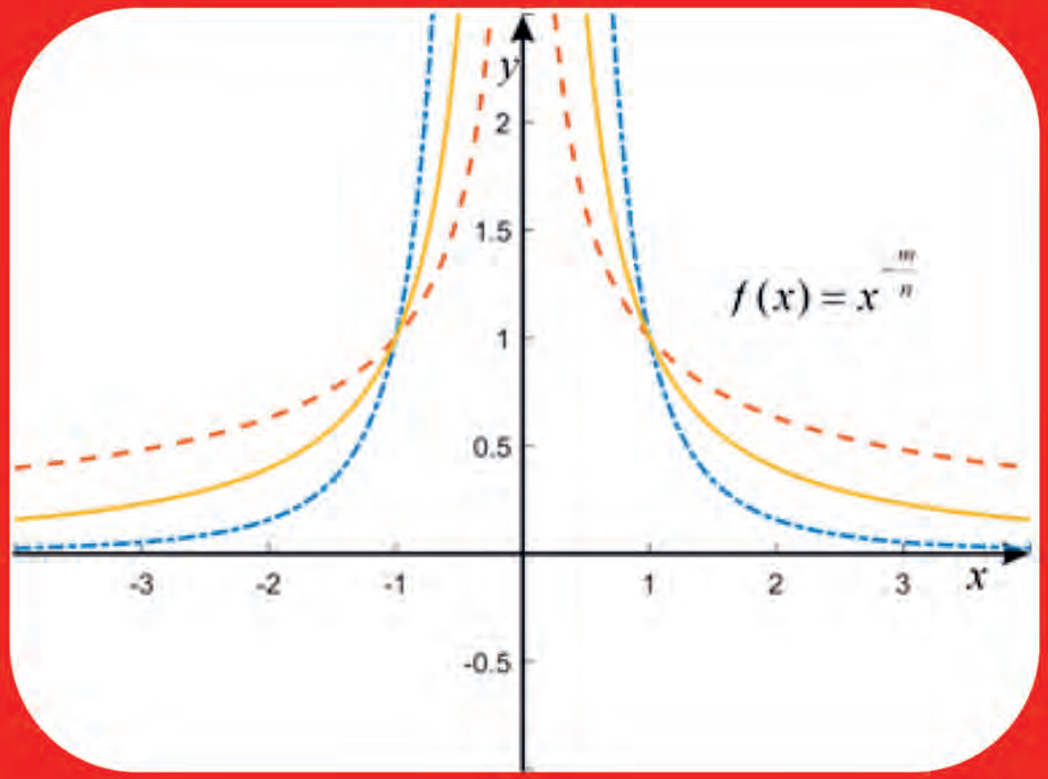
STUDENT'S TEXTBOOK GRADE 11

ISBN 978-99990-0-026-0

MATHEMATICS STUDENT'S TEXTBOOK GRADE 11

MATHEMATICS

STUDENT'S TEXTBOOK GRADE 11

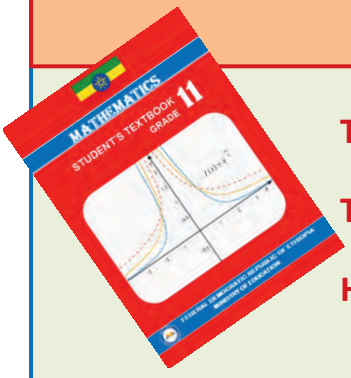


FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA
MINISTRY OF EDUCATION

FDRE
MoE

FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA
MINISTRY OF EDUCATION

Take Good Care of This Textbook



This Textbook is the property of your school.

Take good care not to damage or lose it.

Here are 10 ideas to help take care of the book:

1. Cover the book with protective materials, such as plastic, old newspapers or magazines.
2. Always keep the textbook in a clear dry place.
3. Be sure your hands are clean when you use the textbook.
4. Do not write on the cover or inside pages.
5. Use a piece of paper or cardboard as a textbook mark.
6. Never tear or cut out any pictures or pages.
7. Repair any torn pages with paste or tape.
8. Pack the textbook carefully when you place it in your school bag.
9. Handle the textbook with care when you passing it to another person.
10. When using a new textbook for the first time, lay it on its back. Open only a few pages at a time. Press lightly along the bound edge as you turn the pages. This will keep the cover in good condition.

MATHEMATICS

STUDENT'S TEXTBOOK GRADE 11

Writers:

Tilahun Abebaw (PhD)

Abera Abate (PhD)

Editors:

Zewdu Desalegn (MSc) (Content Editor)

Solomon Melesse (Prof., PhD) (Curriculum Editor)

Melaku Wakuma (PhD) (Language Editor)

Illustrator:

Zerihun Kinfe (PhD)

Designer:

Berie Getie (MSc)

Evaluators:

Dawit Ayalneh Tebekew (MSc)

Matebie Alemayehu Wasihun (MA)

Mustefa Kedir Edao (BEd)

Tesfaye Sileshi (MA)

FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA
MINISTRY OF EDUCATION

First Published August 2023 by the Federal Democratic Republic of Ethiopia, Ministry of Education, under the General Education Quality Improvement Program for Equity (GEQIP-E) supported by the World Bank, UK's Department for International Development/DFID-now merged with the Foreign, Common wealth and Development Office/FCDO, Finland Ministry for Foreign Affairs, the Royal Norwegian Embassy, United Nations Children's Fund/UNICEF), the Global Partnership for Education (GPE), and Danish Ministry of Foreign Affairs, through a Multi Donor Trust Fund.

© 2023 by the Federal Democratic Republic of Ethiopia, Ministry of Education. All rights reserved. The moral rights of the author have been asserted. No part of this textbook reproduced, copied in a retrieval system or transmitted in any form or by any means including electronic, mechanical, magnetic, photocopying, recording or otherwise, without the prior written permission of the Ministry of Education or licensing in accordance with the Federal Democratic Republic of Ethiopia as expressed in the *Federal Negarit Gazeta*, Proclamation No. 410/2004 - Copyright and Neighboring Rights Protection.

The Ministry of Education wishes to thank the many individuals, groups and other bodies involved – directly or indirectly – in publishing this Textbook. Special thanks are due to Hawassa University for their huge contribution in the development of this textbook in collaboration with Addis Ababa University, Bahir Dar University and Jimma University.

Copyrighted materials used by permission of their owners. If you are the owner of copyrighted material not cited or improperly cited, please contact the Ministry of Education, Head Office, Arat Kilo, (P.O.Box 1367), Addis Ababa Ethiopia.

Printed by:

GRAVITY GROUP IND LLC

13th Industrial Area,

Sharjah, UNITED ARAB EMIRATES

Under Ministry of Education Contract no. MOE/GEQIP-E/LICB/G-01/23

ISBN: 978-99990-0-026-0

Table of Contents

Unit 1 RELATIONS AND FUNCTIONS 1

1.1 Relations 3
1.2 Inverse of Relations and Their Graphs 12
1.3 Types of Functions 22
1.4 Composition of Functions 64
1.5 Inverse Functions and their Graphs 68
1.6 Applications of Relations and Functions 76
Summary 80
Review Exercise 81

Unit 2 RATIONAL EXPRESSIONS AND RATIONAL FUNCTIONS 83

2.1 Rational Expressions 84
2.2 Rational Equations and Rational Inequalities 102
2.3 Rational Functions and Their Graphs 109
2.4 Applications 124
Summary 128
Review Exercise 129

Unit 3 MATRICES 131

3.1 The Concepts of a Matrix 133
3.2 Operations on Matrices 137
3.3 Special Types of Matrices 157
3.4 Elementary Row Operations of Matrices 163
3.5 Systems of Linear Equations with Two or Three Variables 174
3.6 Solutions of Systems of Linear Equations 188
3.7 Inverse of a Square Matrix 192
3.8 Applications 199
Summary 204
Review Exercise 205

Unit 4 DETERMINANTS AND THEIR PROPERTIES 207

4.1 Determinants of Matrices of Order 2 209
4.2 Minors and Cofactors of Elements of Matrices 211
4.3 Determinants of Matrices of Order 3 214
4.4 Properties of Determinants 220
4.5 Inverse of a Square Matrix of Order 2 and 3 230
4.6 Solutions of Systems of Linear Equations Using Cramer's Rule 235
4.7 Applications 242
Summary 248
Review Exercise 249

Unit 5 VECTORS 251

5.1 Revision on Vectors and Scalars 252
5.2 Representation of vectors 254
5.3 Vector Product 274
5.4 Application of Scalar and Cross Product 292
5.5 Application of Vectors 296
5.6 Applications 306
Summary 309
Review Exercise 311

Unit 6 TRANSFORMATIONS OF THE PLANE 313

6.1 Introduction 314
6.2 Translation 315
6.3 Reflection 322
6.4 Rotation 337
6.5 Applications 349
Summary 352
Review Exercise 354

Unit 7 STATISTICS 355

7.1 Types of Data 357
7.2 Introduction to Grouped Data 360
7.3 Graphical Representation of Grouped Data 368
7.4 Measures of Central Tendency and Their Interpretation 371
7.5 Real-life Application of Statistics 405
Summary 409
Review Exercise 411

Unit 8 PROBABILITY 413

8.1 Introduction 414
8.2 Fundamental Principle of Counting 417
8.3 Permutations and Combinations 423
8.4 Binomial Theorem 437
8.5 Random Experiments and Their Outcomes 440
8.6 Events 444
8.7 Probability of an Event 453
8.8 Real-life Application of Probability 473
Summary 477
Review Exercise 479

Welcoming Message to Grade 11 Students

Dear students, you are welcome to Grade 11 mathematics educations. This is a golden opportunity in your academic career. Joining this grade level is a new experience and a transition to Grade 12 mathematics educations. At this grade level, you are expected to receive new and advanced opportunities that can help you learn and grow in the field of Mathematics, in life and work. Enjoy it!

Introduction about the Students' Textbook

Dear students, this textbook is organized in such a way that there is an introductory remark about how to use the textbook, some suggestions about how to care the textbook so that it will serve for many beneficiaries for a long period of time, and basic units are addressed after the introduction section.

The Grade 11 textbook is organized having 8 units. The units include: Relations and Functions, Rational Expressions and Rational Functions, Matrices, Determinants and their Properties, Vectors, Transformation of the Plane, Statistics, and probability**.** Each unit is composed of an introduction, learning outcomes, key words, lessons, and summary and review exercise.

In a unit, a brief introduction about the unit is provided. Following a brief introduction of a unit, the expected learning outcomes are presented. Once the objectives are communicated, the different lessons of the unit are presented. In the end, summary and review exercise are packed, respectively.

As mentioned earlier, a unit in the textbook is divided into different lessons. In every lesson of each unit, you learn about mathematics frequently through five components. That is, structurally, the lessons in this textbook usually

have five components; namely, Activity, Definitions/Theorems/Notes, Examples and Solutions, and Exercises.

The most important part in this process is to practice problems by yourself based on what your teacher shows and explains. Your teacher will also give you feedback and assistance, and facilitate further learning. In such a way, you will be able not only to acquire new knowledge and skills but also to develop them further. Herewith, a brief explanation of each sub-component of a lesson is forwarded.

Activity

This part of the lesson requires you to revise what you have learnt at different stages of your mathematics education and reflect on the topic under discussion by using this background knowledge. The activity also introduces you to what you are going to learn in the respective new lesson or topic.

Definition/Theorem/Note

This part of the textbook presents and explains new concepts/ definitions/theorems.

Example and Solution

Here, the textbook provides you specific examples and thereby helps you to improve your understanding of the new content. In this part, your teacher will give you explanations and/or solutions and you are, therefore, advised to listen to your teacher's explanations very carefully and participate actively in the process. You are also required to refer to the solution part of the textbook for your review and self-learning. In this sub-section, a student is

further expected to study on the remaining examples that a classroom teacher will not explain during his/her classroom instruction.

Exercise

Under this part of the textbook, you will be required to solve the problems or questions that are given as exercise individually, in pairs or groups so as to further practice what you have learnt in the examples. That is, a teacher may provide selected exercise questions for you as class work, homework or project work. When you are doing the exercises either individually, in pairs or groups, you are expected to share your opinions with your friends, listen to others' ideas carefully and compare your ideas with others. However, it is always advisable to make some efforts individually before working in pair or groups in order to have a better understanding and input in the process.

In some of the cases, some exercises may not be covered by a classroom teacher. In that case, a student is expected to study the remaining items of the exercise a teacher will not cover during classroom instruction. A teacher will provide the possible answers for those items in the exercises after completing your study, and you are expected to check what you will do against the respective teacher's answers.

Checking in every step

Throughout a lesson, students shall check their progress and the respective teacher should do the same. It is also true that students are required to check the correct answers and solutions for the exercises at the end of each unit. If you want to practice more, therefore, you need to go to the review exercises. Doing the Review Exercises is always recommended since they help you develop a concrete view of the lesson contents.

Generally, students are advised to practice, drill and exercise each activity, example, and exercise on a daily basis as mastering mathematical skills take a long period of time and frequent practice to be second nature. You are also requested to read the notes with deep understanding.

UNIT 1

RELATIONS AND FUNCTIONS

Unit Outcomes

By the end of this unit, you will be able to:

- Generalize patterns using explicitly defined and recursively defined functions.
- Know the inverse of a given relation.
- Know types of functions.
- Recognize real valued functions.
- Know how to find compositions of functions.
- Recognize inverse of a function.
- Sketch the graph of the inverse function.
- Understand how to apply relation and function in real life situations.

Unit Contents

1.1 Relations

1.2 Inverses of Relations and Their Graphs

1.3 Types of Functions

1.4 Composition of Functions

1.5 Inverse Functions and Their Graphs

1.6 Applications

Summary

Review Exercise

- composition of functions
- domain
- function
- greatest integer (floor) function
- identity function
- inverse function
- modulus
- power function
- range
- relation
- signum (sgn) function
- vertical line test

Introduction

In your daily life, you come across many patterns that characterize relationships between individuals such as brother and sister, father and daughter or son, teacher and student, etc. In mathematics also you come across many relationships such as a line l is perpendicular to line m; a number a is less than a number b; a number c is a factor of a number d, etc.

In this unit, the concepts about relations and functions that you have learned in Unit 1 of Grade 10 will be revised and some special types of functions such as power functions, absolute value functions, signum functions and the greatest integer functions and their properties will be discussed. Furthermore, some applications of relations and functions will also be considered.

1.1 Relations

Relation: Domain and Range

In this section, we will revise about the concept of a relation and determine domain and range of relations. Graphs of relations that are defined using numbers in the coordinate plane will also be considered.

Activity 1.1

1. Mention at least five different relationships between family members of a certain family.
2. Let $A = \{1, 2\}$ and $B = \{a, b\}$.
   a. Find all subsets of $A \times B$.
   b. Find the set of the first and the set of the second coordinates of all members of the sets in (a).

From your responses in Activity 1.1, observe that relationships between individuals form a set of ordered pairs of individuals and all the nonempty subsets of $A \times B$ are sets of ordered pairs.

Note

Any set of ordered pairs is a relation.

Definition 1.1

Given two sets A and B, any set R of ordered pairs (x, y), where $x \in A$ and $y \in B$, is called a **relation from A to B** and for $(x, y) \in R$, x is called the first component or coordinate and y is called the second component or coordinate of (x, y).

Given a relation R from A to B;

a. the set of all the first components or coordinates of the elements of R is called the **domain of R and it is denoted by Dom(R).**

That is, $\text{Dom}(\text{R}) = \{x \in A \mid (x, y) \in \text{R} \text{ for some } y \in \text{B}\}$.

b. the set of all the second components or coordinates of the elements of R is called the **range of R, denoted by Ran(R).**

That is, $\text{Ran}(\text{R}) = \{y \in B \mid (x, y) \in \text{R} \text{ for some } x \in \text{A}\}$.

Example 1

Find the domain and range of the relation R = {(45, 65), (48, 68), (41, 62), (46, 66), (50, 70)}.

Solution

Dom(R) = {41, 45, 46, 48, 50}, the set of all the first coordinates.

Ran(R) = {62, 65, 66, 68, 70}, the set of all the second coordinates.

Note

i. Given an ordered pair (x, y), x is called the first component or the first coordinate of the given ordered pair and y is called the second component or the second coordinate of the ordered pair (x, y).

ii. For an ordered pair (x, y), order is important; that is, $(x, y) = (y, x)$ only when $x = y$.

iii. For two ordered pairs (a,b) and (c,d); $(a,b) = (c,d)$ if and only if $a = c$ and $b = d$.

Example 2

In a certain school, there are four sections of Grade 11 and each section is related with the number of students in the given section as in Table 1 below.

Table 1

Sections	Number of Students
Section 1	49
Section 2	51
Section 3	48
Section 4	50

a. Find the relation R defined by Table 1.

b. Find the domain and range of the relation defined by Table 1.

Solution

a. The information given in Table 1 can be represented as a set of ordered pairs. In this case, the first component represents the section and the second component represents the number of students in the given section. The ordered pairs (Section 1, 49), (Section 2, 51), (Section 3, 48) and (Section 4, 50) define a relation between a section and the number of students in the given section. This information is given as a set of ordered pairs by:

R={(Section 1, 49), (Section 2, 51), (Section 3, 48), (Section 4, 50)}.

b. Domain of R is given by Dom(R)={Section 1, Section 2, Section 3, Section 4}, which is the set of all the first components or coordinates of the elements of R and the range of R is given by Ran(R) = {49,51,48,50} , which is the set of all second components or coordinates of the elements of R.

Example 3

Given a relation $R = \{(x, y) : y = x^2 \text{ and } x \in \{-2, -1, 0, 1, 2\}\}$.

a. list all the elements of R;

b. find the domain and range of R.

Solution

a. When $x = -2, y = (-2)^2 = 4$, when $x = -1, y = (-1)^2 = 1$, when $x = 0, y = 0^2 = 0$, when $x = 1, y = 1^2 = 1$ and when $x = 2, y = 2^2 = 4$.

Thus, $R = \{(-2,4), (-1,1), (0,0), (1,1), (2,4)\}$.

b. Dom(R) $= \{-2, -1, 0, 1, 2\}$, the set of all the first coordinates of the elements of R and Ran{R}$=\{0,1,4\}$, the set of all the second coordinates of the elements of R.

Exercise 1.1

1. Let $R = \{(1,1), (2,-2), (10, 20), (5,25), (10,100)\}$. Find the domain and range of R.

2. Table 2 gives the marks of a certain student in four subjects out of 100.

Table 2

Subjects	Marks (out of 100)
English	89
Mathematics	91
Physics	88
Chemistry	90

   a. List all the elements of a relation R defined by Table 2.

   b. Find the domain and range of R in (a).

3. Let R be a relation defined by $R = \{(x, y) : y = x^2 - 2 \text{ and } x \in \{0, 1, 2, 3, 4, 5\}\}$.

   a. list all the elements of R;

   b. find the domain and range of R.

Representations of Relations (1)

A relation may consist of a finite set of ordered pairs or an infinite set of ordered pairs. Furthermore, a relation may be defined by several different methods, such as:

a. a set of ordered pairs;
b. a correspondence between domain and range;
c. a graph;
d. an equation;
e. an inequality or a combination of these , and so on.

Example 4

Given R= {(1, 2), (3, 4), (4, 4), (5, 6)}. Find the domain and range of R.

Solution

R is defined by a finite set of ordered pairs.

- Dom(R) = {1,3,4,5}, the set of all the first coordinates of the elements of R and
- Ran(R) = {2,4,6}, the set of all the second coordinates of the elements of R.

Example 5

A relation R is defined by a diagram in **Figure 1.1** below. Find the domain and range of R.

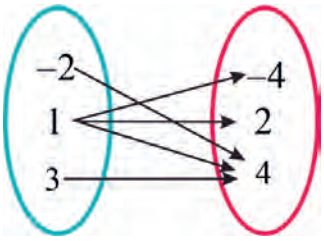


Figure 1.1

Solution

R, as a set of ordered pairs, is given by $R=\{(1,2),(1,4),(1,-4),(-2,4),(3,4)\}$.

Then

a. $Dom(R)=\{-2,1,3\}$, the set of all the first coordinates of elements of R;

b. $Ran(R)=\{-4,2,4\}$, the set of all the second coordinates of elements of R.

Example 6

Find the domain and range of the relation R defined by the graph given in **Figure 1.2**.

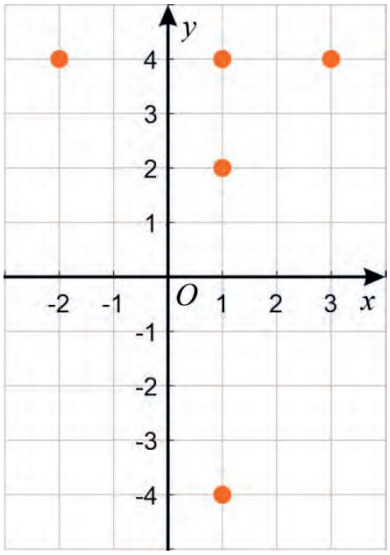


Figure 1.2

Solution

R as a set of ordered pairs is given by $R=\{(-2,4),(1,-4),(1,2),(1,4),(3,4)\}$. Then

a. $Dom(R)=\{-2,1,3\}$, the set of all the first coordinates of elements of R and

b. $Ran(R)=\{-4,2,4\}$, the set of all the second coordinates of elements of R.

Exercise 1.2

1. Find the domain and range of the relation

   $R = \{(-2,-5),(-1,-3),(0,-1),(1,1),(2,3)\}$.

2. List all the elements of the relation R defined by **Figure 1.3** and find its domain and range:

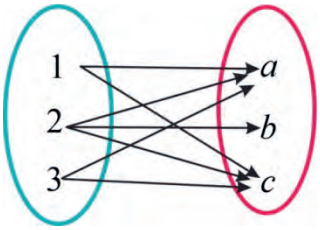


Figure 1.3

3. Find the domain and range of the relation R given in **Figure 1.4**.

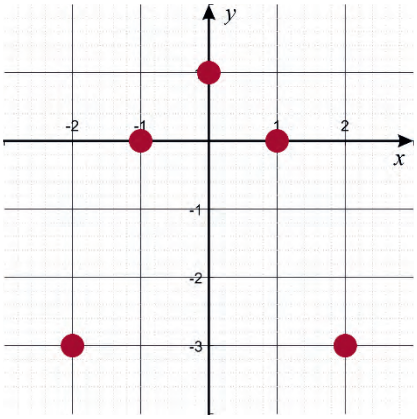


Figure 1.4

4. Find the domain and range of the relation, $R = \{(x,y): x,y \in \mathbb{R} \text{ and } y = x^2\}$.

Representations of Relations (2)

Example 7

A relation R is defined by $R = \{(x,y): x \in \mathbb{R}, y \in \mathbb{R} \text{ and } y = 2x\}$.

Find the domain and range of R.

Solution

R is defined as an infinite set of ordered pairs. From the definition of R, the first coordinate which is in the domain of R can be any real number and also the second coordinate which is in the range of R can be any real number, because for any $y \in \mathbb{R}$, you have $x = \frac{y}{2} \in \mathbb{R}$ and $(x, y) \in \mathrm{R}$.

Thus,

a. the domain of R is the set of all real numbers, $\mathbb{R}$ and
b. the range of R is the set of all real numbers, $\mathbb{R}$.

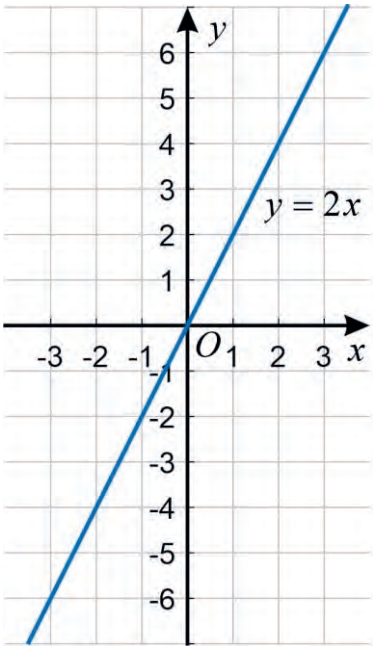


Figure 1.5

Example 8

Draw the graph of the relation R given by

$\mathrm{R} = \{(x, y) \mid x, y \in \mathbb{R}, y \le x + 1 \text{ and } y + x \le 2\}$. and find the domain and range of R.

Solution

First draw the lines $y = x + 1$ and $y + x = 2$ in the coordinated plane. Here, you can use a table of values. Then, the two lines divide the coordinate plane into four regions and by taking a point on each of the four regions, we can determine the region(s) that represents the graph of R as given in Figure 1.6 and from the graph;

a. $\text{Dom}(R) = \mathbb{R}$;

b. $\text{Ran}(R) = \left(-\infty, \frac{3}{2}\right]$.

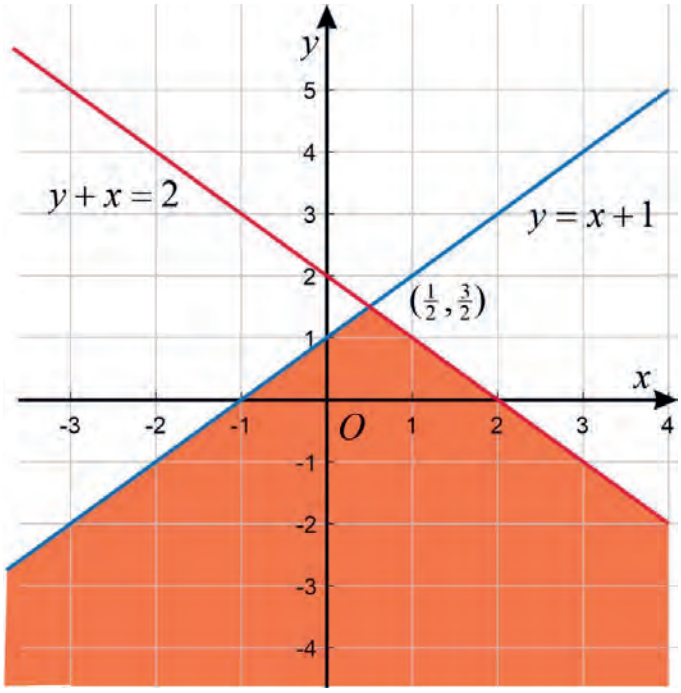


Figure 1.6

Exercise 1.3

1. Find domain and range of the relation $R = \{(x, y) : x, y \in \mathbb{R} \text{ and } y = x^2\}$.
2. Draw the graph of the relation R given by
   $R = \{(x, y) : x, y \in \mathbb{R}, y \leq x - 1 \text{ and } y + x \geq 3\}$ and find the domain and range of R.

1.2 Inverse of Relations and Their Graphs

Inverse of Relations

Activity 1.2

Consider the relations $R = \{(a,1),(b,2),(c,3),(d,4)\}$ and $S = \{(1,a),(2,b),(3,c),(4,d)\}$.

a. Interchange the order of the first and the second components of the ordered pairs in R and obtain a new relation.

b. Interchange the order of the first and the second components of the ordered pairs in S and obtain a new relation.

From Activity 1.2, observe that from a given relation you can form another relation by interchanging the order of the first and the second components of the ordered pairs in the given relation and these two relations are called **inverse relations of each other.**

Definition 1.2

The inverse of a relation is a relation formed by interchanging the order components of each of the ordered pairs in the given relation.

That is, if $R = \{(x,y) : (x,y) \in R\}$ is a relation, then the inverse of R, denoted by R^{-1}, is the relation defined by $R^{-1} = \{(y,x) : (x,y) \in R\}$.

Example 1

Let $R = \{(1,3),(2,5)(3,7),(4,11),(5,23)\}$. Then, find R^{-1}.

Solution

The elements of R^{-1} are given by interchanging the order of the components of the elements of R. That is, $R^{-1} = \{(3,1),(5,2)(7,3),(11,4),(23,5)\}$.

Example 2

Table 3 below gives a relation R between a person's age and the person's maximum recommended heart rate. Find R^{-1}.

Table 3

Age (years) x	Maximum Recommended Heart Rate (Beats per Minute) y
20	200
30	190
40	180
50	170
60	160

Solution

The relation R is given as a set of ordered pairs by:

$R = \{(20,200),(30,190),(40,180),(50,170),(60,160)\}$.

Then, the inverse of R is given by interchanging the order of the first and the second coordinates of ordered pairs of R; that is,

$R^{-1} = \{(200,20),(190,30),(180,40),(170,50),(160,60)\}$.

Example 3

For the relation R defined by the diagram in **Figure 1.7**, find R^{-1}.

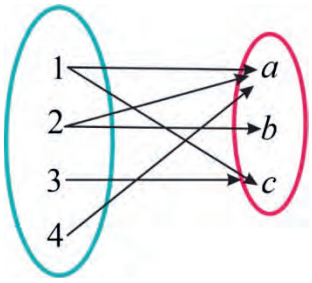


Figure 1.7

Solution

The relation R is defined by a diagram and from the diagram in **Figure 1.7** R is given by $R = \{(1,a),(1,c),(2,a),(2,b),(3,c),(4,a)\}$.

Then, $R^{-1} = \{(a,1),(c,1),(a,2),(b,2),(c,3),(a,4)\}$. It is obtained by interchanging the order of the first and the second components of all the ordered pairs in R.

Example 4

Let $R = \{(x,y) : x, y \in \mathbb{R} \text{ and } y = 3x - 5\}$.. Find R^{-1}.

Solution

$$R^{-1} = \{(x,y) : x, y \in \mathbb{R} \text{ and } x = 3y - 5\} = \{(x,y) : x, y \in \mathbb{R} \text{ and } y = \frac{x+5}{3}\}.$$

Exercise 1.4

Find the inverse of each of the following relations.

1. $R = \{(1,1),(2,4),(3,9),(4,16),(5,25)\}$.
2. R is defined by the diagram given in Figure 1.8.

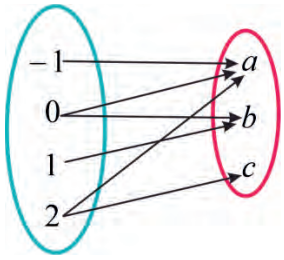


Figure 1.8

3. $R = \{(x,y) : x, y \in \mathbb{R} \text{and } y = 4x + 2\}$

Domain and Range of Inverse Relations

Activity 1.3

Consider $R = \{(2,20),(3,19),(4,18),(5,17),(6,16)\}$.

a. Find R^{-1}

b. Find the domain and range of R.

c. Find the domain and range of R^{-1}

From Activity 1.3, observe the following results for the given relation R.

a. Domain of R^{-1} = Range of R;

b. Range of R^{-1} = Domain of R.

As relations are often specified by expressions involving first component, say x, and a second component, say y, it is natural for us that we want to work with the concept of the inverse relation in that setting. The inverse relation can be formed by interchanging the roles of x and y in the defining expression.

That is, given a relation $R = \{(x,y) \mid (x,y) \in R\}$, we have two ways of defining R^{-1};

a. interchanging x and y in the ordered pair (x,y) and keep the relation; that is,

$$R^{-1} = \{(y,x) : (x,y) \in R\}.$$

b. keep the ordered pair (x,y) as it is and interchange the roles of x and y in the relation; that is, $R^{-1} = \{(x,y) : (y,x) \in R\}$.

Example 5

Let $R = \{(-3,2),(-2,3),(-1,4),(0,5),(1,6),(2,7),(3,8)\}$. Then find the domain and the range of R^{-1}.

Solution

R^{-1} is obtained by interchanging the order of the components of each of the ordered points in R.

That is, $R^{-1}=\{(2,-3),(3,-2),(4,-1),(5,0),(6,1),(7,2),(8,3)\}$ and then,

i. the domain of R^{-1} is $\text{Dom}(R^{-1})=\{2,3,4,5,6,7,8\}$ and

ii. the range of R^{-1} is $\text{Ran}(R^{-1})=\{-3,-2,-1,0,1,2,3\}$.

Example 6

Given $R=\{(x,y)\mid x,y\in\mathbb{R} \text{ and } y=3x+6\}$.

a. Find the inverse of R. b. Find the domain and range of R^{-1}.

Solution

a. First interchange variables x and y in the equation $y=3x+6$ to obtain $x=3y+6$. Then solving for y in terms of x gives you $y=\frac{1}{3}x-2$.

Thus, $R^{-1}=\left\{(x,y)\mid x,y\in\mathbb{R} \text{ and } y=\frac{1}{3}x-2\right\}$.

R^{-1} can also be written as $R^{-1}=\{(y,x)\mid x,y\in\mathbb{R} \text{ and } y=3x+6\}$.

Observe that the two expressions of R^{-1} define the same relation.

b. $\text{Dom}(R)=\mathbb{R}=\text{Ran}(R^{-1})$ and $\text{Ran}(R)=\mathbb{R}=\text{Dom}(R^{-1})$.

Example 7

Find the inverse of $R=\{(x,y)\mid x,y\in\mathbb{R}, y\geq x-3 \text{ and } y\leq x+5\}$.

Solution

$R^{-1} = \{(x,y) \mid x,y \in \mathbb{R},\ x \geq y-3 \text{ and } x \leq y+5\}$

$= \{(x,y) \mid x,y \in \mathbb{R}, -y \geq -x-3 \text{ and } -y \leq -x+5\}$ (Solving for y in terms of x).

$= \{(x,y) \mid x,y \in \mathbb{R},\ y \leq x+3 \text{ and } y \geq x-5\}$.

That is, $R^{-1} = \{(x,y) \mid x,y \in \mathbb{R},\ y \leq x+3 \text{ and } y \geq x-5\}$

Exercise 1.5

Find the inverse and also the domain and range of the inverse of each of the following relations.

a. $R = \{(1,3), (2,4), (3,9)\}$

b. $S = \{(x,y) \mid x,y \in \mathbb{R} \text{ and } y = 2x-4\}$

c. $T = \{(x,\ y) \mid x,y \in \mathbb{R}, y \geq 2x+1 \text{ and } y \leq x-2\}$

Graphs of Inverse Relations

Activity 1.4

Consider the ordered pairs $(1,2), (-3,1), (-2,-4)$ and $(2,-3)$.

1. Plot the given points on the coordinate plane;
2. Find the image of each of the ordered pairs given with reflection of the coordinate plane on the line $y = x$.

From your responses in Activity 1.4, observe that the image of any given ordered pair (a, b) with respect to the reflection of the coordinate plane on the line $y = x$ is (b, a).

The graphs of relations and their inverses are related in an interesting way. First, note that, in the coordinate system ordered pairs (a, b) and (b, a) are the

images of each other with respect to the reflection of the coordinate plane on the line $y = x$.

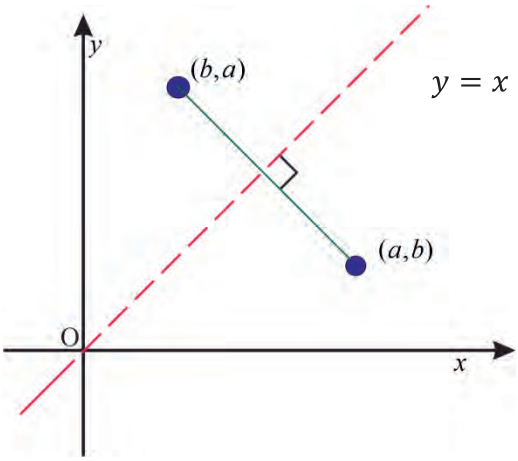


Figure 1.9: Image of (a,b) with respect to the reflection on the line $y = x$.

Therefore, the graph of the inverse of a given relation is the image of the graph of R with respect to the reflection of the coordinate plane on the line $y = x$.

Example 8

Given $R = \{(0,0),(1,2),(2,4),(3,6)\}$, plot the graphs of R and R^{-1} on the same coordinate plane.

Solution

First let us find R^{-1} by interchanging the order of the components of the ordered pairs of R and obtain $R^{-1} = \{(0,0),(2,1),(4,2),(6,3)\}$.

Then, plotting each of the ordered pairs in both R and R^{-1} gives us the following figure.

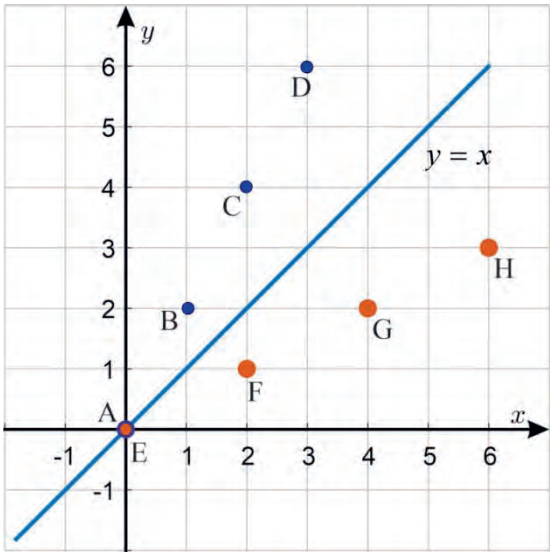


Figure 1.10: Graphs of R and R^{-1}

In Figure 1.10, the points E, F G and H are images of the points A, B, C and D after they are reflected with respect to the line $y = x$.

Example 9

Draw the graph of the inverse of the relation, $R = \{(x, y) | x, y \in \mathbb{R} \text{ and } y = 3x + 6\}$ in the coordinate plane.

Solution

First $R^{-1} = \{(x, y) | x, y \in \mathbb{R} \text{ and } x = 3y + 6\}$ and solving for y in terms of x gives you

$$R^{-1} = \left\{(x, y) | x, y \in \mathbb{R} \text{ and } y = \frac{1}{3}x - 2\right\}.$$

First, draw the graph of R and reflect it with respect to the line $y = x$ and obtain the graph of R^{-1} as in **Figure 1.11.**

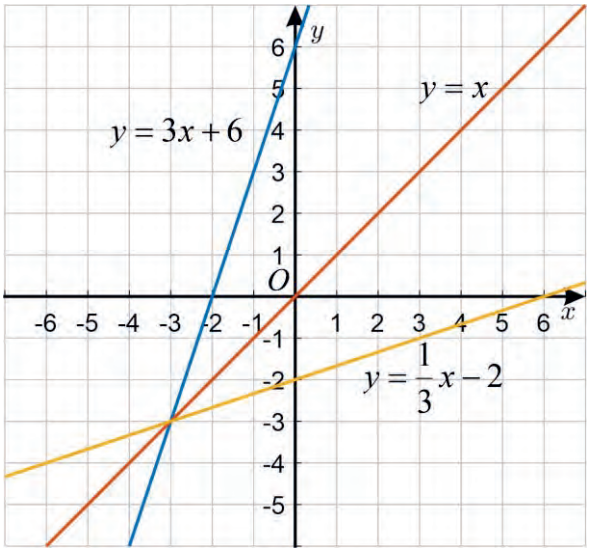


Figure 1.11: Graphs of $y = 3x + 6$ and $y = \frac{1}{3}x - 2$.

Example 10

Graph $R = \{(x, y) \mid x, y \in \mathbb{R} \text{ and } y = x - 4\}$ and its inverse in the same coordinate plane.

Solution

First, let us find R^{-1}.

Thus, $R^{-1} = \{(x, y) \mid x, y \in \mathbb{R} \text{ and } x = y - 4\} = \{(x, y) \mid x, y \in \mathbb{R} \text{ and } y = x + 4\}$. Then, we draw the graph of R in the coordinated plane and reflect it with respect to the line $y = x$ to obtain the graph of R^{-1} as in **Figure 1.12**.

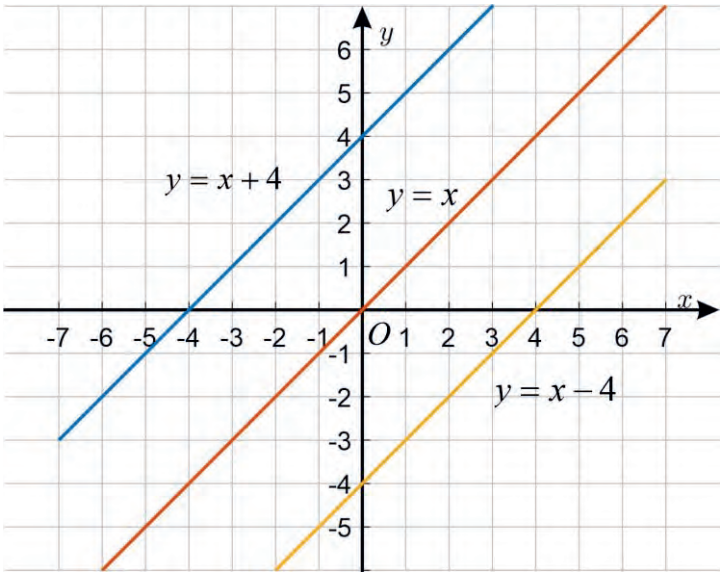


Figure 1.12: Graphs of $y = x + 4$ and $y = x - 4$,

Exercise 1.6

Draw the graph of the given relation and its inverse on the same coordinate plane for each of the following relations.

a. $R = \{(2,4), (3,9), (4,16)\}$.

b. $T = \{(x, y) | x, y \in \mathbb{R} \text{ and } y = x + 5\}$.

c. $S = \{(x, y) \mid x, y \in \mathbb{R} \text{ and } y = 2x - 2\}$.

1.3 Types of Functions

In this section, you will be introduced with special types of relations called functions. Some special types of functions: power functions, absolute value functions, signum functions and the greatest integer function will be considered.

Functions

Activity 1.5

What differences do you observe between the relations R_1 and R_2 given in **Figure 1.13**?

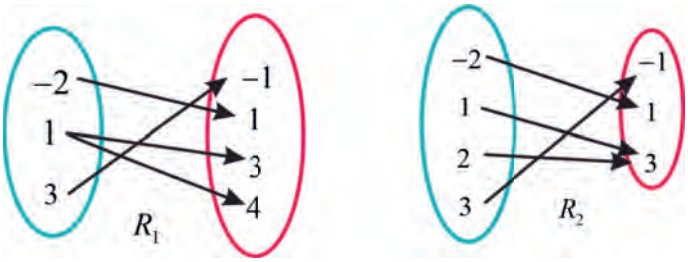


Figure 1.13

From your responses in Activity 1.5, observe that:

i. there are relations in which one first component is related to two or more different second components.

ii. there are relations in which every first component is related to only one corresponding second coordinate.

Relations of the second type are called functions of which its formal definition is given as follows.

Definition 1.3

A relation in which each object from the set of the first components of ordered pairs of the relation is related with exactly one object from the set of second components of the ordered pairs of the relation is called a **function**.

Example 1

Determine if $R = \{(-1,0),(0,-3),(2,-3),(3,0),(4,5)\}$ is a function.

Solution

From the ordered pairs in R, you can see that every first component is related to exactly one second component; that is, -1 is related with 0 only, 0 is related with -3 only, 2 is related with only -3, 3 is related with 0 only and 4 is related with only 5.

Therefore, R is a function.

Example 2

Determine if $R = \{(6,1),(7,3),(0,2),(6,4)\}$ is a function.

Solution

$\text{Dom}(R) = \{6,7,0\}$ and $\text{Ran}(R) = \{1, 2, 3, 4\}$.

From the set of first components, you can see that, the relation has two ordered pairs with 6 as a first component: (6,1) and (6,4), but $1 \neq 4$.

Therefore, the relation R is not a function.

Example 3

Show that the relation $R = \{(x, y) : y = x + 1\}$ is a function.

Solution

For any given variable x, there is only one variable $y = x + 1$. That is, if (x, y) and (x, z) are in R, then $y = x + 1$ and $z = x + 1$. This implies, $y = z$.

Example 4

Let $R = \{(x, y) : x, y \in \mathbb{R} \text{ and } x^2 - y^2 = 0\}$. Show that R is not a function.

Solution

$(1,-1)$ and $(1,1)$ are both in R, because $1^2 - (-1)^2 = 1 - 1 = 0$ and $1^2 - 1^2 = 1 - 1 = 0$, but $-1 \neq 1$. Thus, R is not a function.

Exercise 1.7

Which of the following relations are functions?

a. $R_1 = \{(1,1),(2,3),(3,4),(5,5)\}$

b. $R_2 = \{(1,0),(1,3),(4,5),(6,6)\}$

c. $R_3 = \{(x, y) : x, y \in \mathbb{R} \text{ and } y = x + 3\}$

d. $R_4 = \{(x, y) : x, y \in \mathbb{R} \text{ and } x^2 + y^2 = 1\}$

Vertical Line Test

Notation

If f is a function from A to B, then we denote it by $f : A \to B$ or $A \xrightarrow{f} B$ and

a. Dom(f) = A

b. Range of f is a subset of B.

c. If $(x, y) \in f$, then we write $y = f(x)$ and $f(x)$ is read as f of x.

Note

For any given relation R, to determine whether it is a function or not, check whether the relation gives two different second coordinates for the same first coordinate. In other words, given a relation R, if $(x, y) \in R$ and $(x, z) \in R$ implies $y = z$ for any x, y and z, then R is a function. Otherwise, R is not a function.

Example 5

Let $R=\{(x,y): x,y\in\mathbb{R} \text{ and } y=2x+1\}$. Then, show that R is a function.

Solution

Suppose $(x,y)\in$ R and $(x,z)\in$ R. Then $y=2x+1$ and $z=2x+1$.

This implies $y=2x+1=z$ and hence R is a function.

Note

The Vertical Line Test

A set of points in the coordinate plane represents a function if and only if no two points of the given set are on the same vertical line.

Example 6

Use the Vertical Line Test to determine if each one of the relations in the following graphs is function.

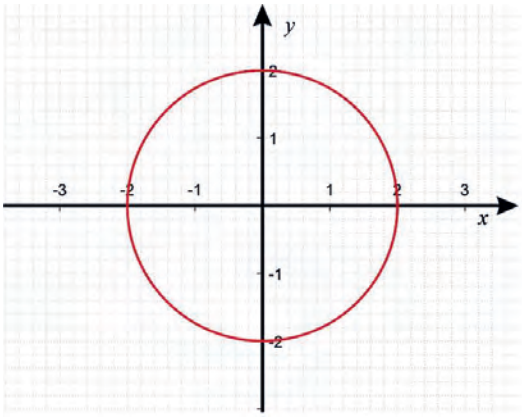


Figure 1.14: Graph of a relation R

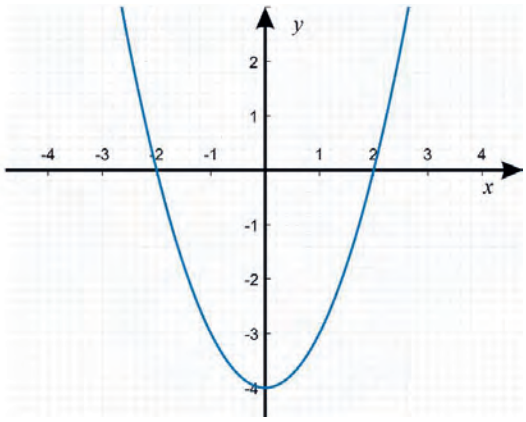


Figure 1.15: Graph of a relation S

Solution

a. You can see from the graph of R that it is possible to find a vertical line intersecting the graph more than once as it can be seen in **Figure 1.16**.

Hence, R does not represent a function.

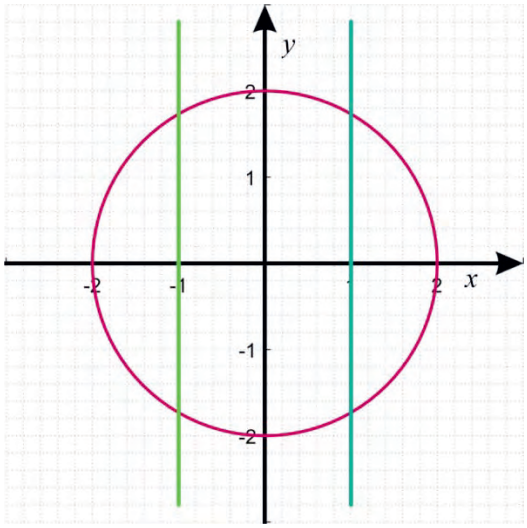


Figure 1.16

b. Any vertical line intersects the graph of S exactly once as shown in **Figure 1.17**.

Hence, S is a function.

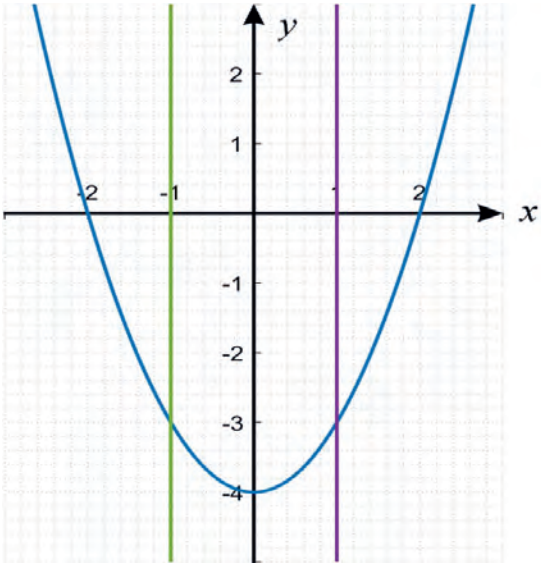


Figure 1.17

Exercise 1.8

1. Determine if the following relations are functions.
   a. $R_1 = \{(x, y) | x, y \in \mathbb{R} \text{ and } y = 3x + 1\}$
   b. $R_2 = \{(x, y): x, y \in \mathbb{R} \text{ and } x^2 + y^2 = 9\}$
2. Which of the following graphs represent graph(s) of a function?

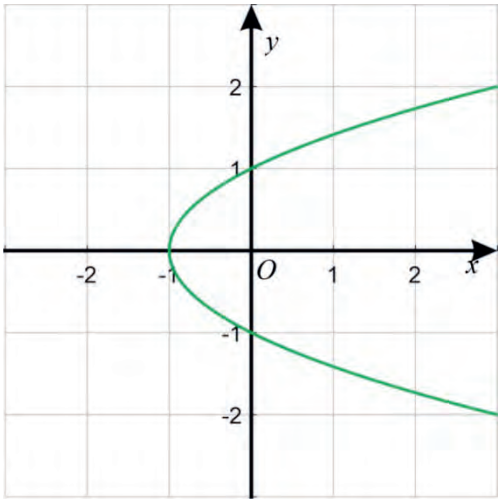


Figure 1.18

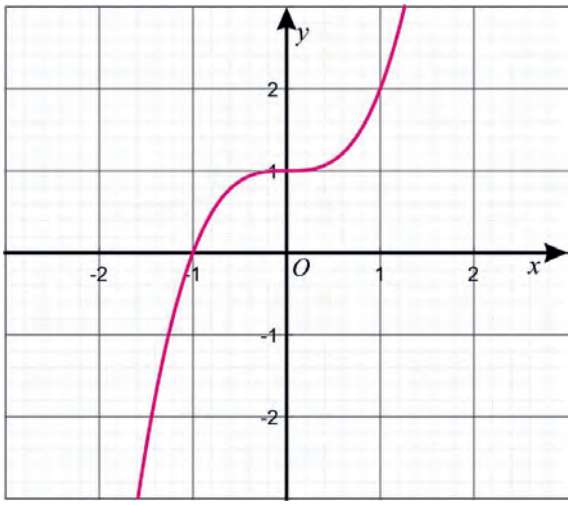


Figure 1.19

1.3.1 Power Functions with Their Graphs

Basic Properties of Power Functions (1)

Activity 1.6

a. A square is cut out of a cardboard, with each side having length x. Find the area of the square in terms of its side length x.
b. Find the volume of a cube with each edge having length t.
c. Find the area of a circle of radius r.

From your responses in Activity 1.6, observe that:

a. the area of the square is given as a function of x, which is a power of x;
b. the volume of the cube is given as a function of t, which is a power of t;

c. the area of the circle is given as a function of r, which is a constant multiple of power of r.

These three functions are examples of power functions; that means, they are functions that are some constant times a power of the given variable.

Definition 1.4

A function of the form $f(x) = ax^r$ for some nonzero real number a and a real number r is called a power function.

Example 1

Which of the following functions are power functions?

a. $f(x) = 3x^2$ b. $g(x) = 5x^{\frac{2}{3}}$ c. $h(x) = 2^x$

d. $l(x) = x^{\frac{4}{3}}$ e. $m(x) = \log_3 x$ f. $n(x) = -4x^{-3}$

Solution

The functions $f(x)$, $g(x)$, $l(x)$ and $n(x)$ are power functions whereas $h(x)$ and $m(x)$ are not a power functions; $h(x)$ is an exponential function and $m(x)$ is a logarithmic function.

Next, you will learn the behaviors of power functions and these will be done by considering different examples of power functions and generalize the behavior of these power functions.

Example 2

Find the domain and range of each of the following functions and draw their graphs.

a. $f(x) = x^2$ b. $g(x) = x^4$

Solution

a. $\text{Dom}(f) = \mathbb{R}$ and $\text{Ran}(f) = [0, \infty)$.

b. $\text{Dom}(g) = \mathbb{R}$ and $\text{Ran}(g) = [0, \infty)$.

To draw the graphs of both f and g, use the table of values given below.

x	-2	-1	0	1	2
$f(x)$	4	1	0	1	4
$g(x)$	16	1	0	1	16

Then, using the values in the table, we draw the graphs of f and g as in **Figure 1.20**.

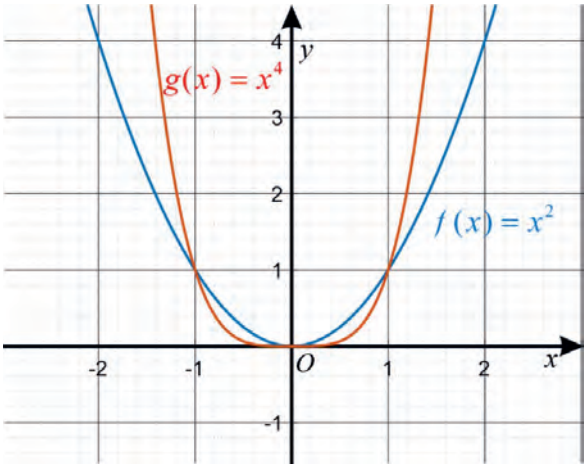


Figure 1.20: Graphs of $f(x) = x^2$ and $g(x) = x^4$

Using f and g in Example 2, we can generalize the properties of power functions of the form $f(x) = x^n$ for an even natural number n as follows.

I. Functions of the form $f(x) = x^n$ for an even natural number n have similar properties and some of these common properties are given as follows:

a. Domain of the function is $\mathbb{R}$; b. Range of the function is $[0, \infty)$;

b. The graph of the function is a curve that is similar to the curves in **Figure 1.21**.

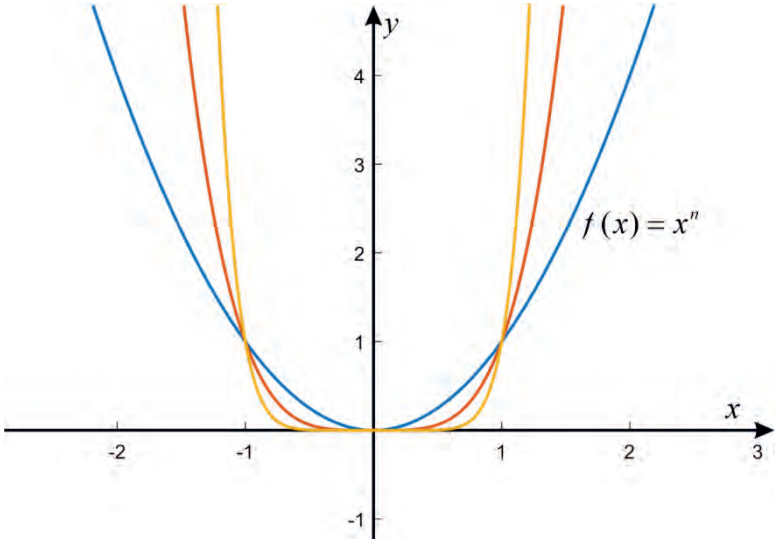


Figure 1.21: Graph of $f(x) = x^n$ for a positive even integer n.

Example 3

Find the domain and range of each of the following functions and draw their graphs.

a. $f(x) = x^3$ b. $g(x) = x^5$

Solution

a. $\text{Dom}(f) = \mathbb{R}$ and $\text{Ran}(f) = \mathbb{R}$.

b. $\text{Dom}(g) = \mathbb{R}$ and $\text{Ran}(g) = \mathbb{R}$.

To draw the graphs of both f and g, use the table of values given below.

x	-2	-1	0	1	2
$f(x)$	-8	-1	0	1	8
$g(x)$	-32	-1	0	1	32

Then, using the values in the table, we draw the graphs of f and g as in **Figure 1.22.**

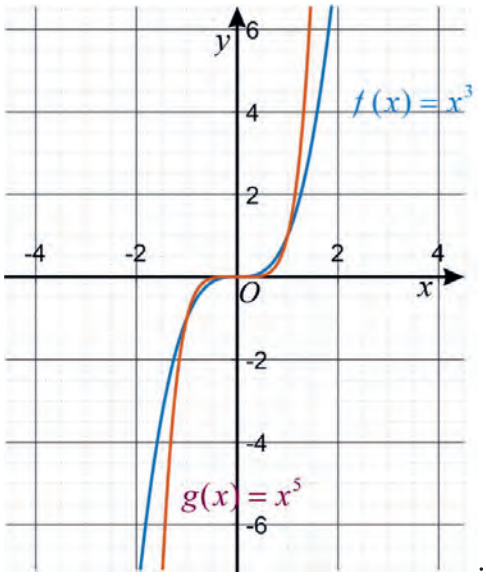


Figure 1.22: Graphs of $f(x) = x^3$ and $g(x) = x^5$

II. Functions of the form $f(x) = x^n$ for an odd natural number n have similar properties and some of these common properties are given as follows:

a. Domain of the function is $\mathbb{R}$;

b. Range of the function is $\mathbb{R}$;

c. The graph of the function is a curve that has the same shape as the curves in **Figure 1.23.**

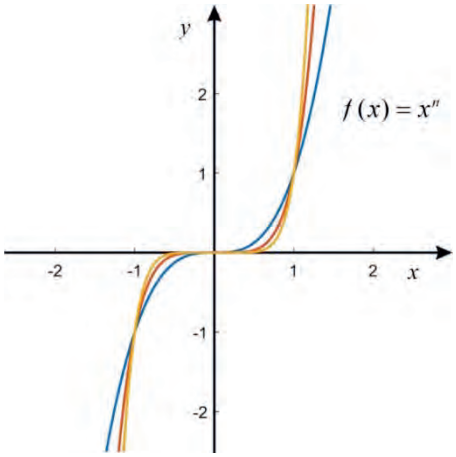


Figure 1.23: Graph of $f(x) = x^n$ for a positive odd integer n.

Exercise 1.9

For each of the following functions, find the domain, range and sketch the graph.

a. $f(x) = x^6$ b. $g(x) = x^8$ c. $h(x) = x^7$ d. $l(x) = x^9$

Basic Properties of Power Functions (2)

Example 4

Find the domain and range of each of the following functions and draw their graphs.

a. $f(x) = x^{\frac{1}{2}}$ b. $g(x) = x^{\frac{1}{4}}$

Solution

a. $\text{Dom}(f) = [0, \infty)$ and $\text{Ran}(f) = [0, \infty)$.

b. $\text{Dom}(g) = [0, \infty)$ and $\text{Ran}(g) = [0, \infty)$.

To draw the graphs of both functions f and g, use the table given below.

x	0	1	⋯	4	⋯	16
$f(x)$	0	1	⋯	2	⋯	4
$g(x)$	0	1	⋯	$\sqrt[4]{4}$	⋯	2

Then, using the values in the table, the graphs of f and g are given in **Figure 1.24**.

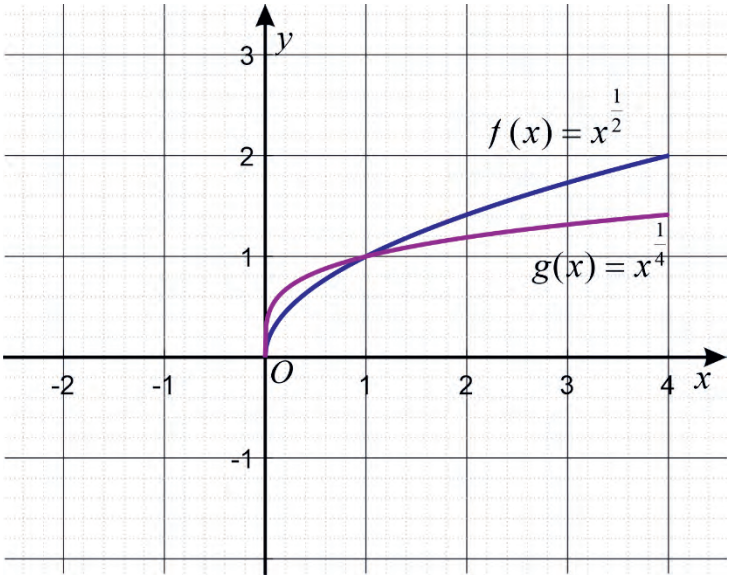


Figure 1.24: Graphs of $f(x) = x^{\frac{1}{2}}$ and $g(x) = x^{\frac{1}{4}}$

III. Functions of the form $f(x) = x^{\frac{1}{n}}$ for an even natural number n have similar properties and some of these common properties are:

a. Domain of the function is $[0, \infty)$;

b. Range of the function is $[0, \infty)$;

c. The graph of the function is given in **Figure 1.25**.

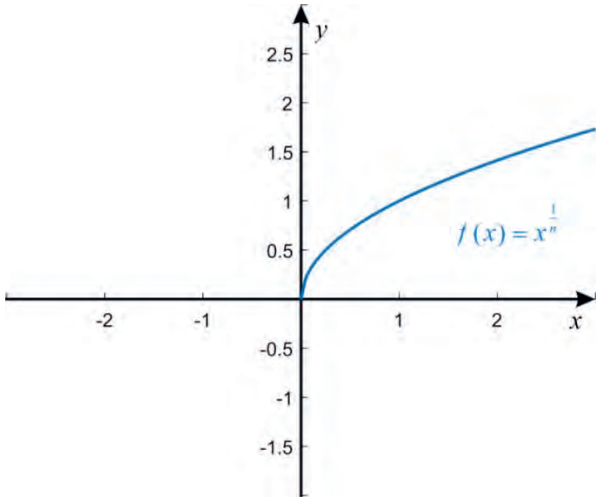


Figure 1.25: Graph of $f(x) = x^{\frac{1}{n}}$, where n is an even natural number.

Example 5

Find the domain and range of each of the following functions and draw their graphs.

a. $f(x) = x^{\frac{1}{3}}$ b. $g(x) = x^{\frac{1}{5}}$

Solution

a. $\text{Dom}(f) = \mathbb{R}$ and $\text{Ran}(f) = \mathbb{R}$.

b. $\text{Dom}(g) = \mathbb{R}$ and $\text{Ran}(g) = \mathbb{R}$.

To draw the graphs of both functions f and g, use the tables given below.

x	-32	$\cdots$	-8	$\cdots$	-1	0	1	$\cdots$	8	$\cdots$	32
$f(x)$	$\sqrt[3]{-32}$	$\cdots$	-2	$\cdots$	-1	0	1	$\cdots$	2	$\cdots$	$\sqrt[3]{32}$
$g(x)$	-2	$\cdots$	$\sqrt[5]{-8}$	$\cdots$	-1	0	1	$\cdots$	$\sqrt[5]{8}$	$\cdots$	2

Then, using the values in the table, the graphs of f and g are given in **Figure 1.26.**

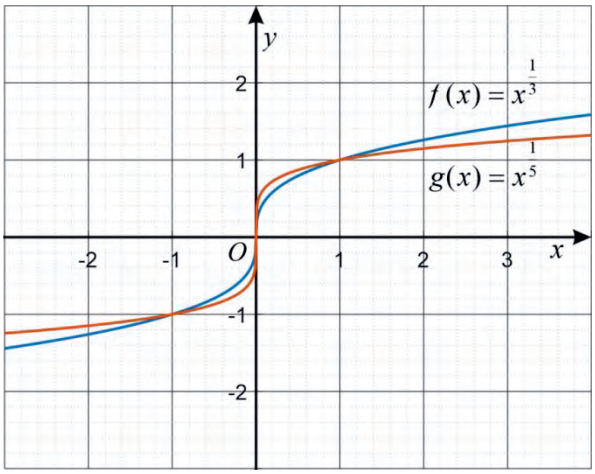


Figure 1.26: Graphs of $f(x)=x^{\frac{1}{3}}$ and $g(x)=x^{\frac{1}{5}}$

IV. Functions of the form $f(x)=x^{\frac{1}{n}}$ for an odd natural number n have similar properties and some of these common properties are:

a. Domain of the function is $\mathbb{R}$;

b. Range of the function is $\mathbb{R}$;

c. The graph of the function is a curve that has the same shape as the curve in **Figure 1.27**.

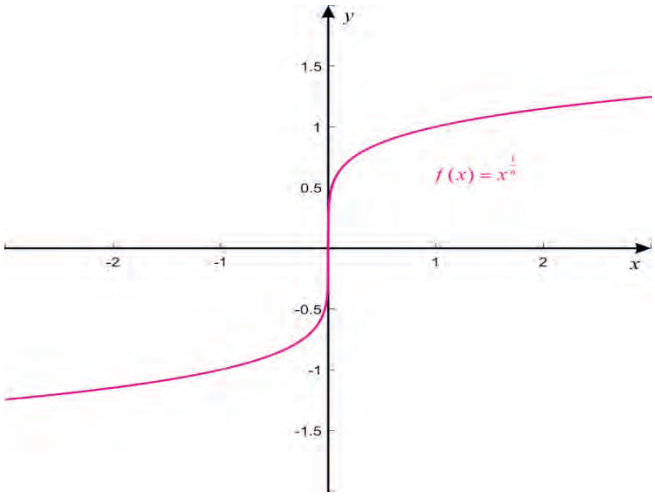


Figure 1.27: Graph of $f(x)=x^{\frac{1}{n}}$, where n is an odd natural number.

Exercise 1.10

For each of the following functions, find the domain, range and sketch the graph.

a. $f(x) = x^{\frac{1}{6}}$ b. $g(x) = x^{\frac{1}{8}}$ c. $h(x) = x^{\frac{1}{7}}$ d. $l(x) = x^{\frac{1}{9}}$

Basic Properties of Power Functions (3)

Example 6

Find the domain and range of each of the following functions and sketch their graphs.

a. $f(x) = x^{\frac{3}{2}}$ b. $g(x) = x^{\frac{5}{4}}$

Solution

a. $\text{Dom}(f) = [0, \infty)$ and $\text{Ran}(f) = [0, \infty)$.

b. $\text{Dom}(g) = [0, \infty)$ and $\text{Ran}(g) = [0, \infty)$.

To draw the graphs of both functions f and g, use the table given below,

x	0	1	⋯	4	⋯	16
$f(x)$	0	1	⋯	8	⋯	64
$g(x)$	0	1	⋯	$4\sqrt{2}$	⋯	32

Then, using the values in the table, the graphs of f and g are given in **Figure 1.28.**

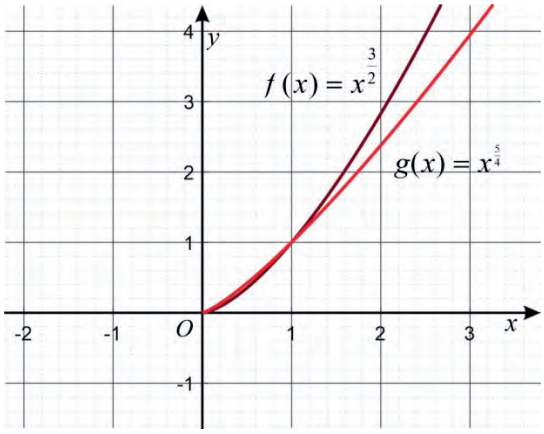


Figure 1.28: Graphs of $f(x) = x^{\frac{3}{2}}$ and $g(x) = x^{\frac{5}{4}}$

V. Functions of the form $f(x) = x^{\frac{m}{n}}$ for an odd natural number m and an even natural number n have similar properties and some of these common properties are:

a. Domain of the function is $[0, \infty)$.

b. Range of the function is $[0, \infty)$.

c. The graph of the function is a curve that has the same shape as the curves in **Figure 1.29.**

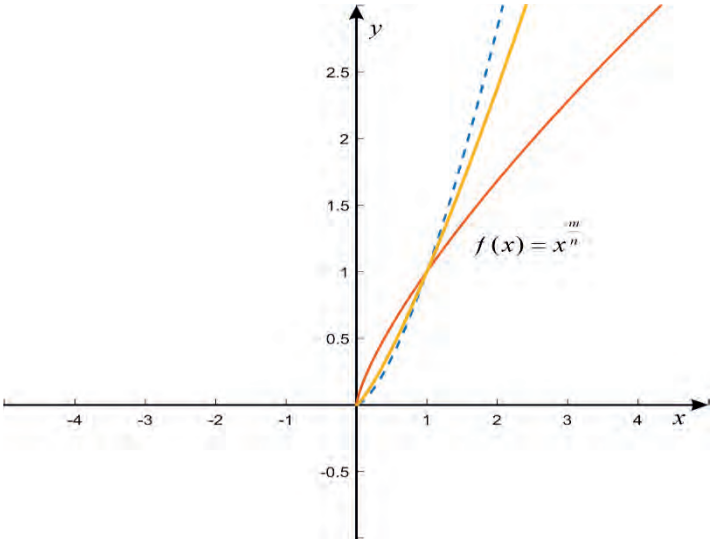


Figure 1.29: Graph of $f(x) = x^{\frac{m}{n}}$, where m and n are odd and even natural numbers respectively.

Example 7

Find the domain and range of each of the following functions and sketch their graphs.

a. $f(x) = x^{\frac{2}{3}}$ b. $g(x) = x^{\frac{4}{5}}$

Solution

a. $\text{Dom}(f) = \mathbb{R}$ and $\text{Ran}(f) = [0, \infty)$.

b. $\text{Dom}(g) = \mathbb{R}$ and $\text{Ran}(g) = [0, \infty)$.

To draw the graphs of both functions f and g, use the table given below,

x	-32	$\cdots$	-8	$\cdots$	-1	0	1	$\cdots$	8	$\cdots$	32
$f(x)$	$4\sqrt{4}$	$\cdots$	4	$\cdots$	1	0	1	$\cdots$	4	$\cdots$	$4\sqrt{4}$
$g(x)$	16	$\cdots$	$\sqrt[5]{8^4}$	$\cdots$	1	0	1	$\cdots$	$\sqrt[5]{8^4}$	$\cdots$	16

Then, using the values in the table, the graphs of f and g are given in **Figure 1.30.**

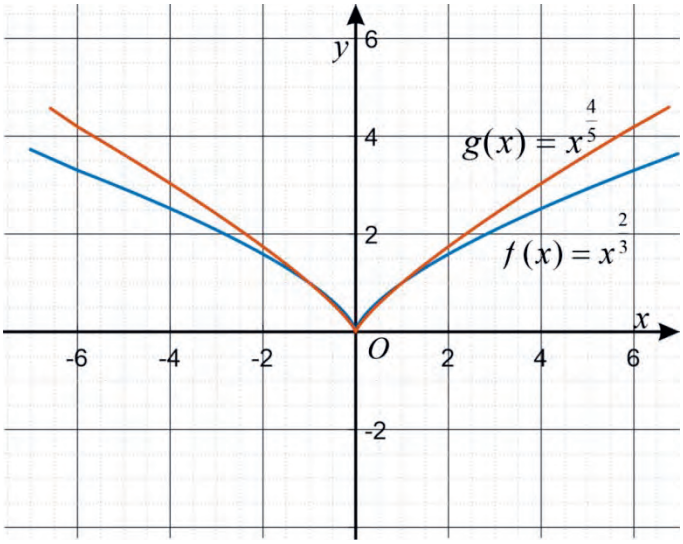


Figure 1.30: Graphs of $f(x) = x^{\frac{2}{3}}$ and $g(x) = x^{\frac{4}{5}}$

VI. Functions of the form $f(x) = x^{\frac{m}{n}}$ for an even natural number m and an odd natural number n have similar properties and some of these common properties are:

a. Domain of the function is $\mathbb{R}$.

b. Range of the function is $[0, \infty)$.

c. The graph of the function is a curve that has the same shape as the curves in **Figure 1.31.**

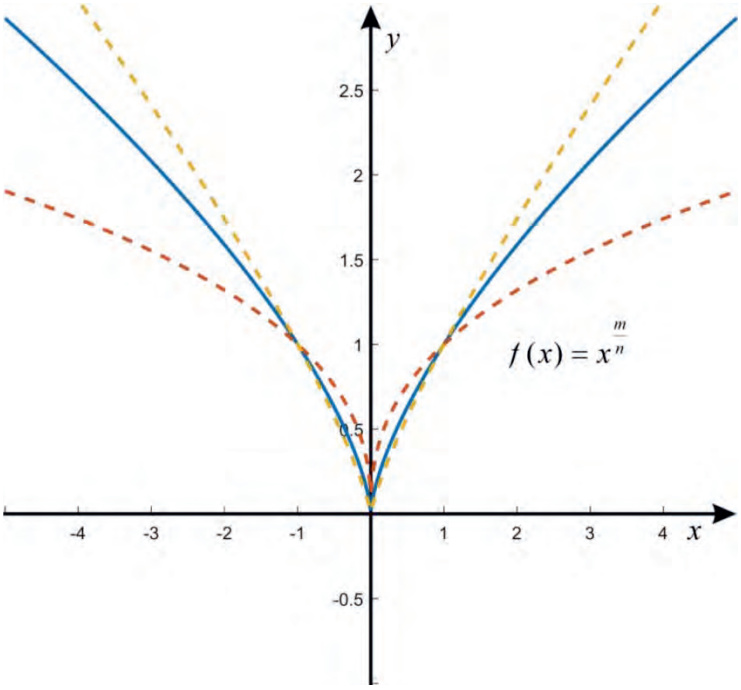


Figure 1.31: Graph of $f(x) = x^{\frac{m}{n}}$ for an even natural number m and an odd natural number n

Exercise 1.11

For each of the following functions, find the domain, range and sketch the graph.

a. $f(x) = x^{\frac{5}{6}}$

b. $g(x) = x^{\frac{7}{8}}$

c. $h(x) = x^{\frac{6}{5}}$

d. $l(x) = x^{\frac{4}{7}}$

Basic Properties of Power Functions (4)

Example 8

Find the domain and range of each of the following functions and sketch their graphs.

a. $f(x) = x^{-\frac{1}{2}}$

b. $g(x) = x^{-\frac{1}{4}}$

Solution

a. $\text{Dom}(f) = (0, \infty)$ and $\text{Ran}(f) = (0, \infty)$.

b. $\text{Dom}(g) = (0, \infty)$ and $\text{Ran}(g) = (0, \infty)$.

To draw the graphs of both functions f and g, use the table given below.

x	1	⋯	4	⋯	16
$f(x)$	1	⋯	0.5	...	0.25
$g(x)$	1	⋯	$\frac{1}{\sqrt{2}}$	⋯	0.5

Then, using the values in the table, the graphs of f and g are given in **Figure 1.32.**

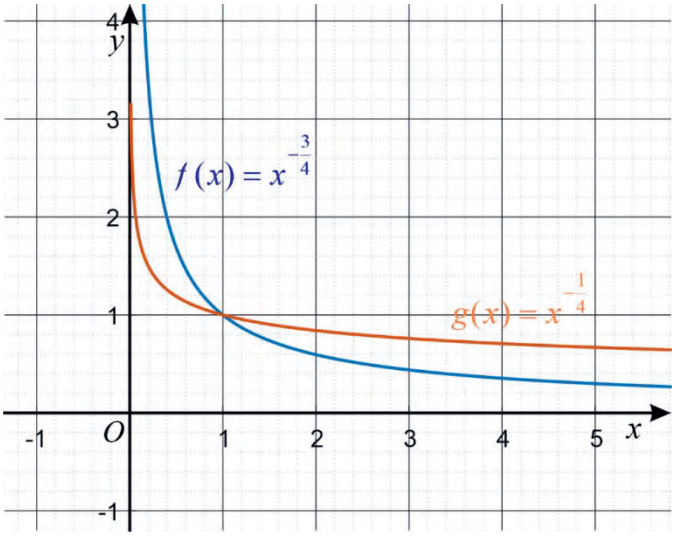


Figure 1.32: Graphs of $f(x) = x^{-\frac{1}{2}}$ and $g(x) = x^{-\frac{1}{4}}$

VII. Functions of the form $f(x) = x^{-\frac{1}{n}}$ for an even natural number n have similar properties and some of these common properties are:

a. Domain of the function is $(0, \infty)$.

b. Range of the function is $(0, \infty)$.

c. The graph of the function is a curve that has the same shape as the curves in **Figure 1.33.**

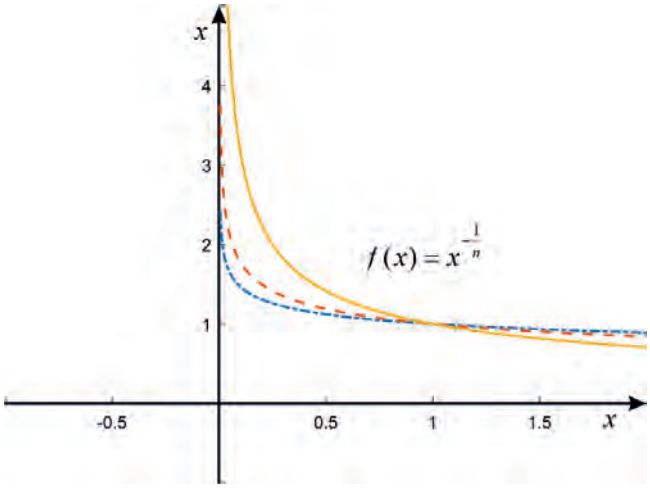


Figure 1.33: Graph of $f(x)=x^{-\frac{1}{n}}$ for an even natural number n.

Example 9

Find the domain and range of each of the following functions and sketch their graphs.

a. $f(x)=x^{-\frac{1}{3}}$ b. $g(x)=x^{-\frac{1}{5}}$

Solution

a. Domain of f is $\mathbb{R}\backslash\{0\}$ and Range of f is $\mathbb{R}\backslash\{0\}$.

b. Domain of g is $\mathbb{R}\backslash\{0\}$ and Range of g is $\mathbb{R}\backslash\{0\}$.

To draw the graphs of both functions f and g, use the table given below.

x	-32	$\cdots$	-8	$\cdots$	-1	1	$\cdots$	8	$\cdots$	32
$f(x)$	$-\frac{1}{2\sqrt[3]{4}}$	$\cdots$	-0.5	...	-1	1	$\cdots$	0.5	$\cdots$	$\frac{1}{2\sqrt[3]{4}}$
$g(x)$	-0.5	$\cdots$	$-\frac{1}{\sqrt[5]{8}}$	$\cdots$	-1	1	$\cdots$	$\frac{1}{\sqrt[5]{8}}$	$\cdots$	0.5

Then, using the values in the table, the graphs of f and g are given in **Figure 1.34.**

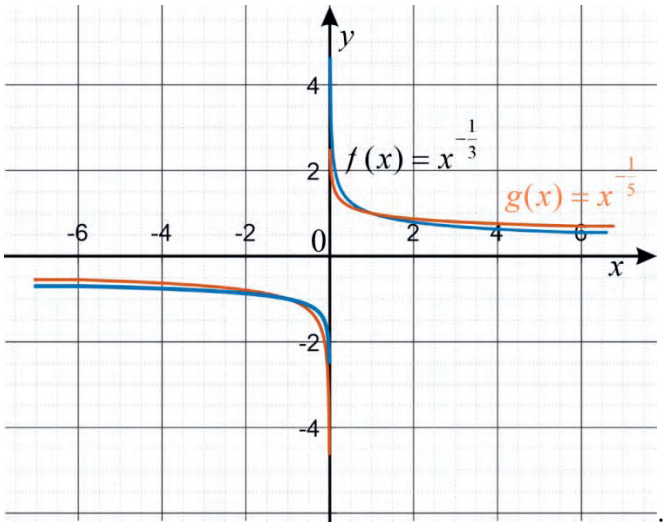


Figure 1.34: Graphs of $f(x)=x^{-\frac{1}{3}}$ and $g(x)=x^{-\frac{1}{5}}$

VIII. Functions of the form $f(x)=x^{-\frac{1}{n}}$ for an odd natural number n have similar properties and some of these common properties are:

a. Domain of the function is $\mathbb{R}\setminus\{0\}$.

b. Range of the function is $\mathbb{R}\setminus\{0\}$.

c. The graph of the function is a curve that has the same shape as the curves in **Figure 1.35.**

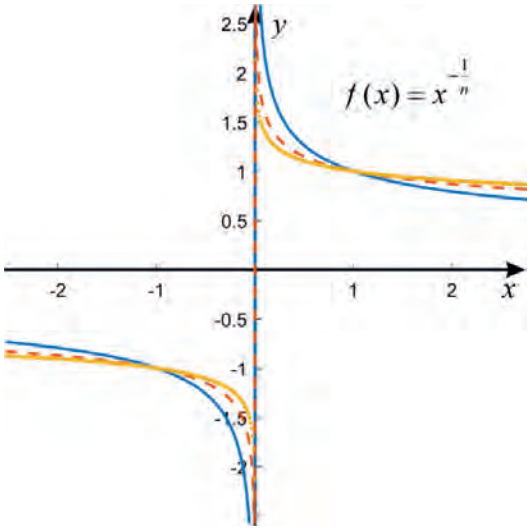


Figure 1.35: Graph of $f(x)=x^{-\frac{1}{n}}$ for an odd natural number n.

Exercise 1.12

For each of the following functions, find the domain, range and sketch the graph.

a. $f(x) = x^{-\frac{1}{6}}$ b. $g(x) = x^{-\frac{1}{8}}$ c. $h(x) = x^{-\frac{1}{7}}$ d. $l(x) = x^{-\frac{1}{9}}$

Basic Properties of Power Functions (5)

Example 10

Find the domain and range of each of the following functions and sketch their graphs.

a. $f(x) = x^{-\frac{3}{2}}$ b. $g(x) = x^{-\frac{2}{3}}$

Solution

a. $\text{Dom}(f) = (0, \infty)$ and $\text{Ran}(f) = (0, \infty)$.

b. $\text{Dom}(g) = \mathbb{R}\backslash\{0\}$ and $\text{Ran}(g) = (0, \infty)$.

To draw the graphs of both functions f and g, use the tables given below,

x	$\frac{1}{2}$	1	4	9
$f(x)$	$\sqrt{8}$	1	$\frac{1}{8}$	$\frac{1}{27}$

x	-8	-1	1	8
$g(x)$	$\frac{1}{4}$	1	1	$\frac{1}{4}$

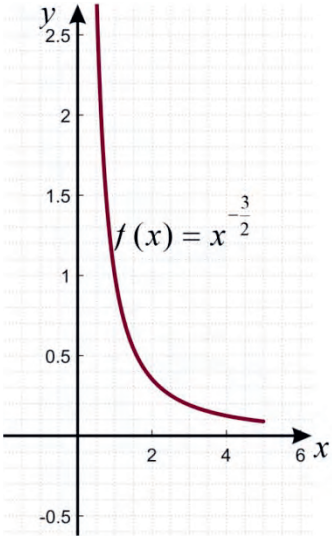


Figure 1.36: Graph of $f(x) = x^{-\frac{3}{2}}$

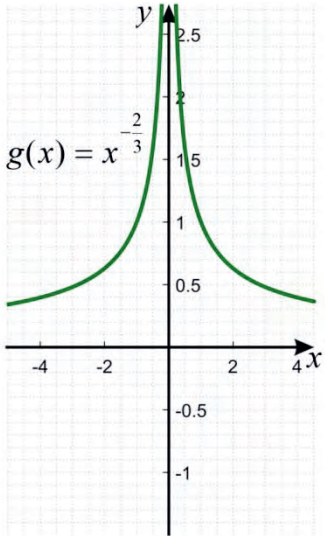


Figure 1.37: Graph of $g(x) = x^{-\frac{2}{3}}$

IX. Functions of the form $f(x) = x^{-\frac{m}{n}}$ for an odd natural number m and an even natural number n have similar properties and some of these common properties are:

a. Domain of the function is $(0, \infty)$.

b. Range of the function is $(0, \infty)$.

c. The graph of the function is a curve that has the same shape as the curves in **Figure 1.38.**

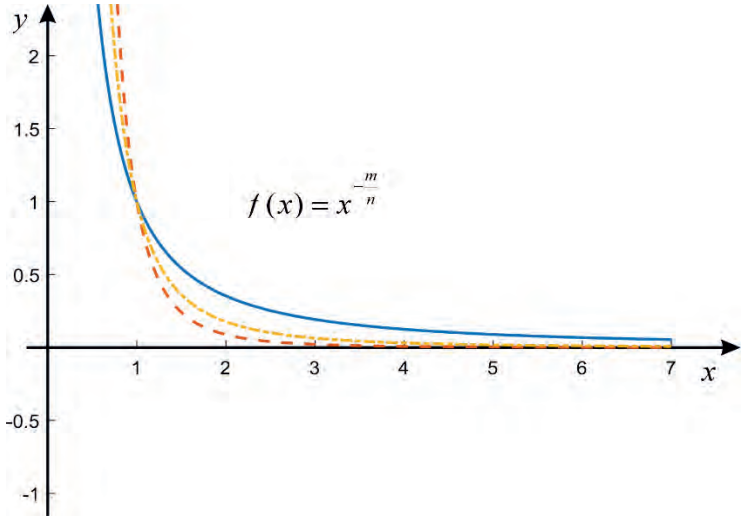


Figure 1.38: Graph of $f(x) = x^{-\frac{m}{n}}$, m is odd and n is even natural numbers.

X. Functions of the form $f(x) = x^{-\frac{m}{n}}$ for an even natural number m and an odd natural number n have similar properties and some of these common properties are:

a. Domain of the function is $\{x \in \mathbb{R} : x \neq 0\}$.

b. Range of the function is $(0, \infty)$.

c. The graph of the function is given in **Figure 1.39**.

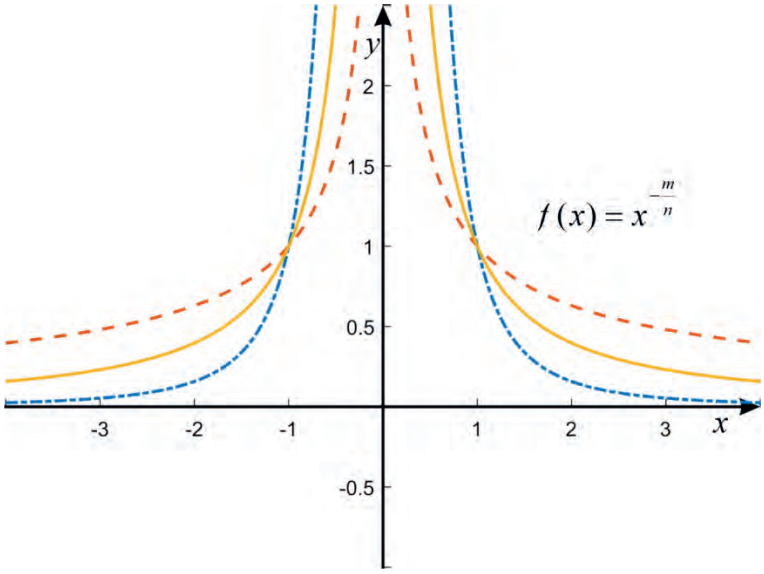


Figure 1.39: Graph of $f(x) = x^{-\frac{m}{n}}$, m is even and n is odd.

Exercise 1.13

For each of the following functions, find the domain, range and sketch the graph.

a. $f(x) = x^{-\frac{5}{4}}$

b. $g(x) = x^{-\frac{6}{5}}$

c. $h(x) = x^{-\frac{7}{6}}$

d. $l(x) = x^{-\frac{6}{7}}$

1.3.2 Modulus (Absolute Value) Function

Revision of Absolute Values

Activity 1.7

a. Find the distance from the origin to the point represented by -3 on the number line.

b. Find the distance from the origin to the point represented by 3 on the number line.

c. Find the distance from origin to the point represented by 0 on the number line.

From your response in Activity 1.7, observe that points represented by -3 and 3 on a number line are both 3 units from the origin, as in Figure 42 and the distance from the origin to itself is 0.

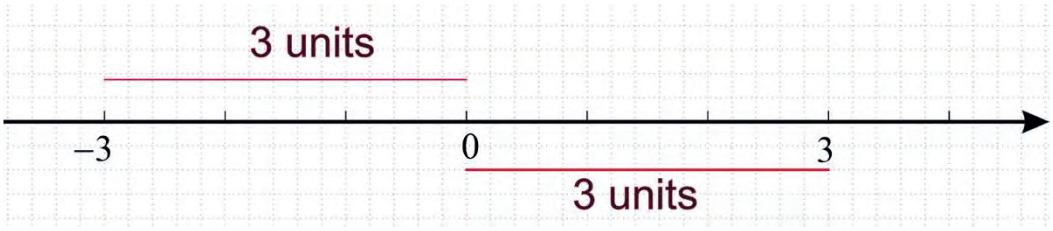


Figure 1.40

Distance on a number line is defined using the concept of absolute value.

Definition 1.5

Given a real number a, the absolute value (modulus) of a, denoted by $|a|$, is the nonnegative real number defined by

$$|a| = \begin{cases} a & \text{if } a > 0 \\ 0 & \text{if } a = 0 \\ -a & \text{if } a < 0 \end{cases}$$

Example 1

Find the absolute value of each of the following numbers.

a. 6 b. 0 c. -5

Solution

a. $|6| = 6$, because $6 > 0$.

b. $|0| = 0$, by the definition of absolute value.

c. $|-5| = -(-5) = 5$, because $-5 < 0$.

Properties of Absolute Value

Activity 1.8

a. Find the absolute value of each of the numbers 4 and -4.

b. Compare the two results in (a).

From your responses in Activity 1.8, observe that $|4|$ and $|-4|$ are equal.

Thus we have the following general property of absolute value.

Property I

For any real number a, $|a| = |-a|$.

Activity 1.9

1. If $a = 3$ and $b = 4$, then
   - a. find both $|ab|$and $|a||b|$;
   - b. compare $|ab|$ and $|a||b|$;
   - c. find both $\left|\frac{a}{b}\right|$ and $\frac{|a|}{|b|}$;
   - d. compare $\left|\frac{a}{b}\right|$ and $\frac{|a|}{|b|}$
   - e. find both $|a+b|$ and $|a|+|b|$
   - f. Compare $|a+b|$ and $|a|+|b|$
2. If $a = -3$ and $b = -4$, then
   - a. find both $|ab|$and $|a||b|$;
   - b. compare $|ab|$ and $|a||b|$;
   - c. find both $\left|\frac{a}{b}\right|$ and $\frac{|a|}{|b|}$;
   - d. compare $\left|\frac{a}{b}\right|$ and $\frac{|a|}{|b|}$
   - e. find both $|a+b|$ and $|a|+|b|$
   - f. Compare $|a+b|$ and $|a|+|b|$
3. If $a = 3$ and $b = -4$, then
   - a. find both $|ab|$and $|a||b|$;
   - b. compare $|ab|$ and $|a||b|$;
   - c. find both $\left|\frac{a}{b}\right|$ and $\frac{|a|}{|b|}$;
   - d. compare $\left|\frac{a}{b}\right|$ and $\frac{|a|}{|b|}$
   - e. find both $|a+b|$ and $|a|+|b|$
   - f. Compare $|a+b|$ and $|a|+|b|$

From Activity 1.9, you have observed the following properties of absolute values.

Property II

For any two real numbers a and b,

i. $|ab| = |a||b|$ ii. $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$ for $b \neq 0$.

iii. $|a + b| \leq |a| + |b|$ (Triangular Inequality).

Activity 1.10

Let $a = \frac{2}{5}$ and $b = -3$. Then

a. Find both $|a|$ and $|b|$.

b. Compare $|a|$ with both $\frac{2}{5}$ and $-\frac{2}{5}$.

c. Compare $|b|$ with 3 and -3.

d. Find both $|-a|$ and $|-b|$.

From your responses in Activity 1.10, observe that $\left|\frac{2}{5}\right| \geq \frac{2}{5}$ and $\left|\frac{2}{5}\right| \geq -\frac{2}{5}$ and also $|-3| \geq 3$ and $|-3| \geq -3$,

Thus, we have the following properties of absolute value.

Property III

a. For any real number a, $|a| \geq a$ and $|a| \geq -a$.

b. For any given positive real number a, $|a| = a$ and $|-a| = -(-a) = a$.
Thus, $|x| = a$ if and only if $x = a$ or $x = -a$.

c. For any real number x, $|x|$ can never be negative, as it is the distance from the origin to the point represented by x on the number line. Thus, $|x| \geq 0$.

Example 2

Solve each of the following equations.

a. $|x| = 7$ b. $|x| = 0$ c. $|x| = -3$ d. $|2x + 6| = 10$

Solution

a. If $|x| = 7$, then $x = 7$ or $x = -7$

b. If $|x| = 0$, then $x = 0$.

c. If $|x| = -3$, then there is no number for x that is a solution for this equation.

d. If $|2x + 6| = 10$, then $2x + 6 = 10$ or $2x + 6 = -10$.

This implies $2x = 10 - 6 \Rightarrow 2x = 4 \Rightarrow x = 2$ or

$2x + 6 = -10 \Rightarrow 2x = -10 - 6 \Rightarrow 2x = -16 \Rightarrow x = -8$.

Thus, the values of x that make the statement $|2x + 6| = 10$ true are 2 and -8.

Exercise 1.14

1. Find the absolute value of each of the following numbers.

   a. $-\frac{2}{5}$ b. $\frac{1}{7}$ c. $-\frac{5}{4}$

2. Compare the absolute values of each of the following pairs of numbers.

   a. -4.5 and 3.75 b. $\frac{2}{9}$ and $-\frac{5}{12}$

3. Solve each of the following equations.

   a. $|x| = 5$ b. $|x - 1| = 0$ c. $|3x + 6| = 9$ d. $|x + 1| = -2$

Modulus Functions

Definition 1.6

The function $f: \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = |x|$ is called the absolute value (or modulus) function.

Example 3

Given $f(x) = |x|$, determine the values of each of the following.

$f(-1)$, $f(0)$ and $f(1)$

Solution

a. $f(-1) = |-1| = -(-1) = 1$;

b. $f(0) = |0| = 0$;

c. $f(1) = |1| = 1$.

Note

For any real number x, $|x|$ is defined and $|x| \geq 0$. Thus,

a. the domain of $f(x) = |x|$ is $\mathbb{R}$.

b. the range of $f(x) = |x|$ is $[0, \infty)$.

Example 4

Draw the graph of the function $f(x) = |x|$.

Solution

Consider the following table of values of f.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x)$	4	3	2	1	0	1	2	3	4

Then using the table, you can draw the graph of the absolute value function $f(x) = |x|$ on the coordinate plane, as in **Figure 1.41.**

Observe that, $f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$.

So the graph of $f(x) = |x|$ is the graph of $y = x$ for $x \geq 0$ and the graph of $y = -x$ for $x < 0$.

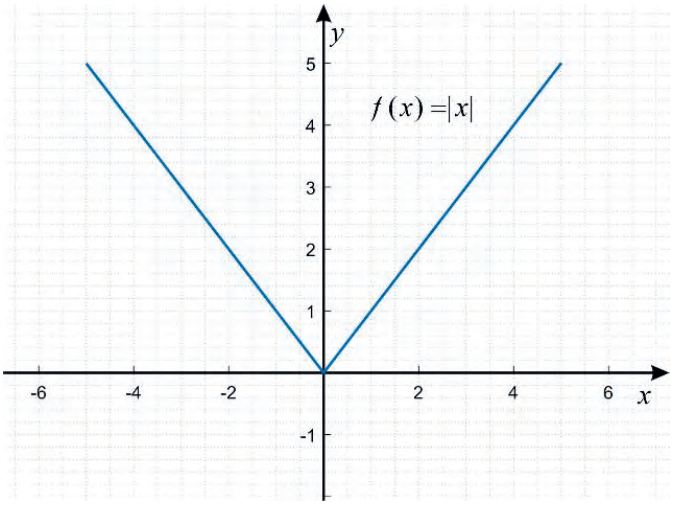


Figure 1.41: Graph of $f(x) = |x|$.

Observe that the graph of the absolute function $f(x) = |x|$

a. is a continuous curve; that is, it has no break;

b. passes through $(0, 0)$ and has a sharp corner at$(0, 0)$;

c. is symmetric with respect to y-axis; that is, the part of the graph of the function for $x > 0$ is the image of the reflection of the graph of the function for $x < 0$ with respect to the y −axis.

Example 5

Find the domain, range and sketch the graph of the function $f(x) = -|x|$

Solution

As $f(x) = -|x|$ defined for all x in $\mathbb{R}$ and $-|x| \leq 0$ for all $x \in \mathbb{R}$,

a. $\text{Dom}(f) = \mathbb{R}$ b. $\text{Ran}(f) = (-\infty, 0]$.

To draw the graph of f consider the following table of values of the function:

x	-3	-2	-1	0	1	2	3
$f(x)$	-3	-2	-1	0	-1	-2	-3

Then, plotting the points given in the table and joining these points with proper curves gives us the graph of f, **Figure 1.42.**

Observe that $f(x) = -|x| = \begin{cases} -x \text{ if } x \geq 0 \\ -(-x) = x \text{ if } x < 0 \end{cases}$.

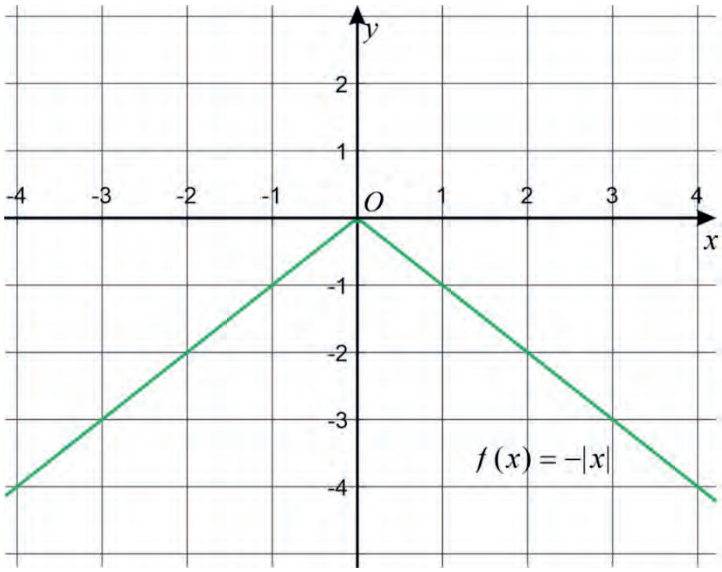


Figure 1.42: Graph of $f(x) = -|x|$

Example 6

Find the domain, range and sketch the graph of the function $f(x) = |x-1|$.

Solution

As $f(x) = |x - 1|$ defined for all x in $\mathbb{R}$ and $|x - 1| \geq 0$ for all $x \in \mathbb{R}$,

a. $\text{Dom}(f) = \mathbb{R}$ b. $\text{Ran}(f) = [0, \infty)$.

To draw the graph of f consider the following table of values of the function.

x	-3	-2	-1	0	1	2	3
$f(x)$	4	3	2	1	0	1	2

Then, plotting the points given in the table and joining these points with proper curves gives us the graph of f, **Figure 1.43.**

Observe that $f(x) = \begin{cases} x-1, \text{when } x \geq 1; \\ -x+1, \text{when } x < 1. \end{cases}$

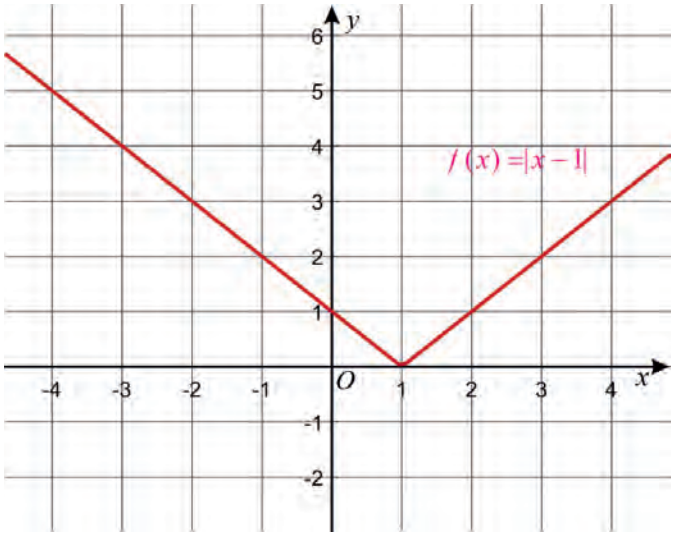


Figure 1.43: Graph of $f(x) = |x - 1|$

Example 7

Find the domain, range and then sketch the graph of the function $f(x) = |x| + 1$.

Solution

As $f(x) = |x| + 1$ defined for all x in $\mathbb{R}$ and $|x| + 1 \geq 1$ for all $x \in \mathbb{R}$,

a. $\text{Dom}(f) = \mathbb{R}$ b. $\text{Ran}(f) = [1, \infty)$.

To draw the graph of f consider the following table of values of the function:

x	-3	-2	-1	0	1	2	3
$f(x)$	4	3	2	1	2	3	4

Then, plotting the points given in the table and joining these points with proper curves gives you the graph of f, **Figure 1.44.**

Observe that when $x \geq 0, f(x) = x + 1$ and when $x < 0, f(x) = -x + 1$.

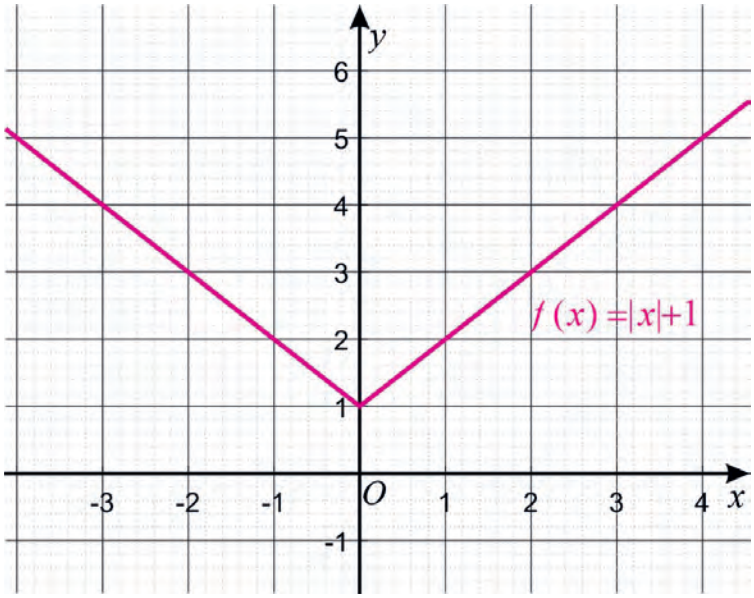


Figure 1.44: Graph of $f(x) = |x| + 1$

Example 8

Find the domain, range and sketch the graph of the function $f(x) = |2x|$.

Solution

As $f(x) = |2x|$ defined for all x in $\mathbb{R}$ and $|2x| \geq 0$ for all $x \in \mathbb{R}$,

a. $\text{Dom}(f) = \mathbb{R}$ b. $\text{Ran}(f) = [0, \infty)$.

To draw the graph of f consider the following table of values of the function:

x	-3	-2	-1	0	1	2	3
$f(x)$	6	4	2	0	2	4	6

Then, plotting the points given in the table and joining these points with proper curves gives you the graph of f, **Figure 1.45.**

Observe that when $x \geq 0, f(x) = 2x$ and when $x < 0, f(x) = -2x$.

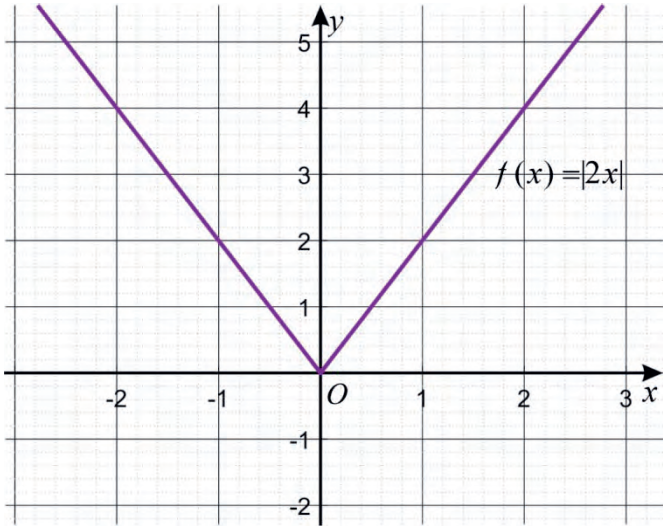


Figure 1.45: Graph of $f(x) = |2x|$

Exercise 1.15

Find the domain, range and sketch the graph of each of the following functions.

a. $f(x) = |x+1|$ b. $g(x) = |x| - 1$ c. $h(x) = 2 - |x|$ d. $h(x) = |5x|$

1.3.3 Signum Function

Activity 1.11

Consider the function $f(x) = 2$ for all $x \in (-\infty, 0]$ and $g(x) = 3$ for all $x \in (0, \infty)$.

a. Find the domains and ranges of both f and g.

b. Let $h(x) = f(x)$ for all $x \in (-\infty, 0]$ and $h(x) = g(x)$ for all $x \in (0, \infty)$. Find the domain and range of h.

c. Sketch the graphs of f, g and h.

From your responses in Activity 1.11, you have observed that the functions f and g are constant functions and the function h is defined piecewise, one value for $x \leq 0$ and another value for $x > 0$.

One of the common piecewise functions is the signum function and its definition is given below.

Definition 1.7

The function defined by $f(x)=\begin{cases}1 & \text{for } x>0;\\ 0 & \text{for } x=0;\\ -1 & \text{for } x<0.\end{cases}$ and is called the signum function denoted by $sgn\ x$ and read as signum x.

That is, $sgn\,x=\begin{cases}1 & \text{for } x>0;\\ 0 & \text{for } x=0;\\ -1 & \text{for } x<0.\end{cases}$ and

Note

The function $sgn\ x=\begin{cases}1 & \text{for } x>0\\ 0 & \text{for } x=0\\ -1 & \text{for } x<0\end{cases}$ has the following properties.

a. Its domain is $\mathbb{R}$

b. Its range is the set $\{-1,0,1\}$

For any real number x, $\frac{|x|}{x}=\begin{cases}\frac{x}{x}=1 & \text{for } x>0\\ \text{undefined} & \text{for } x=0\\ \frac{-x}{x}=-1 & \text{for } x<0\end{cases}$

Therefore, we can define the signum function using absolute value as:

$$sgn\ x=\begin{cases}\frac{|x|}{x} & \text{for } x\neq 0\\ 0 & \text{for } x=0\end{cases}$$

Now we draw the graph of the signum function using the table given below.

x	-4	-3	-2	-1	0	1	2	3	4
$sgn\ x$	-1	-1	-1	-1	0	1	1	1	1

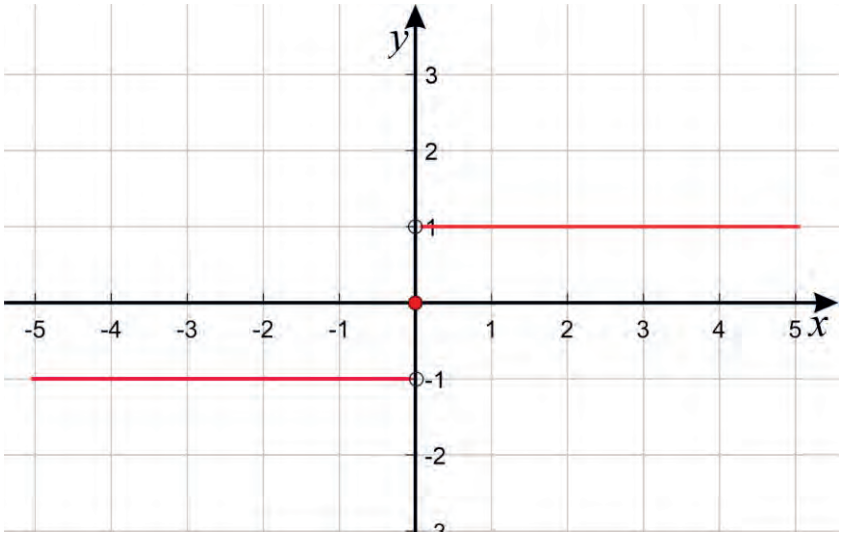


Figure 1.46: Graph of $f(x) = \operatorname{sgn} x$

Example 1

Determine the domain and range of the function $f(x) = sgn\, x + 2$ and sketch its graph.

Solution

From the definition of the signum function:

$$f(x) = \operatorname{sgn} x + 2 = \begin{Bmatrix} 1 & \text{for } x > 0 \\ 0 & \text{for } x = 0 \\ -1 & \text{for } x < 0 \end{Bmatrix} + 2 = \begin{Bmatrix} 3 & \text{for } x > 0 \\ 2 & \text{for } x = 0 \\ 1 & \text{for } x < 0 \end{Bmatrix}$$

Then the domain of f is $\mathbb{R}$ and its range is the set $\{1,2,3\}$.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x)$	1	1	1	1	2	3	3	3	3

Then, using the table, the graph of f is sketched in **Figure 1.47.**

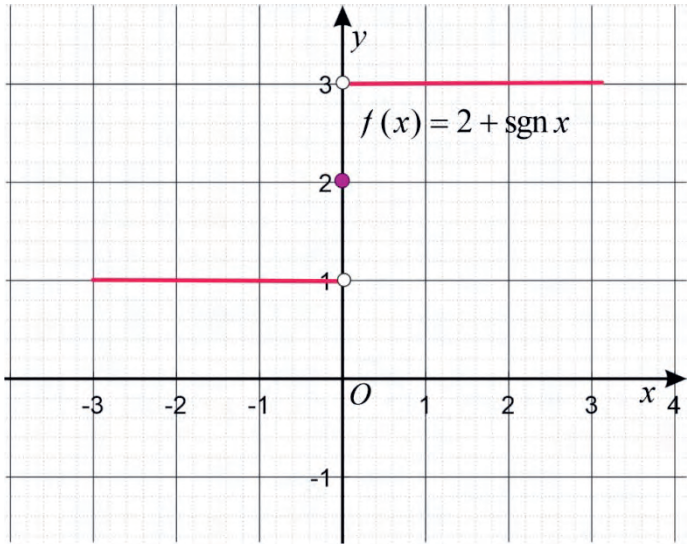


Figure 1.47: Graph of $f(x) = 2 + sgn\ x$

Example 2

Determine the domain and range of the function $f(x) = 4sgn\ x$ and sketch its graph.

Solution

From the definition of the signum function:

$$f(x) = 4\text{sgn}\,x = 4 \times \begin{cases} 1 & \text{for } x > 0 \\ 0 & \text{for } x = 0 \\ -1 & \text{for } x < 0 \end{cases} = \begin{cases} 4 & \text{for } x > 0 \\ 0 & \text{for } x = 0 \\ -4 & \text{for } x < 0 \end{cases}$$

Then, the domain of f is $\mathbb{R}$ and its range is the set $\{-4, 0, 4\}$.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x)$	-4	-4	-4	-4	0	4	4	4	4

Then, using the above table, the graph of f is sketched in **Figure 1.48.**

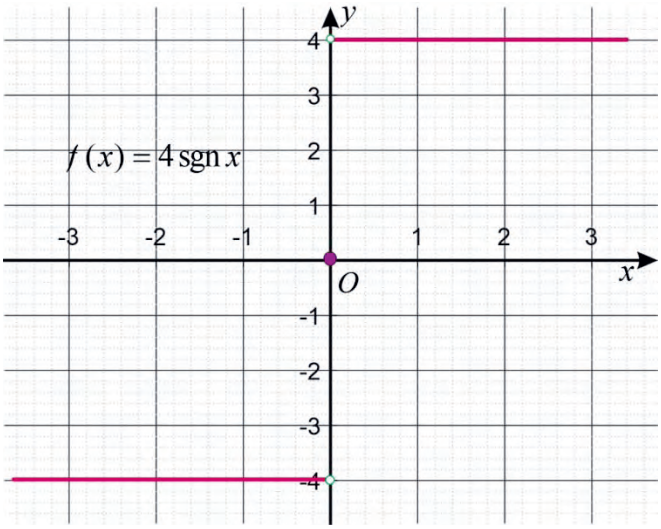


Figure 1.48: Graph of $f(x) = 4sgnx$

Exercise 1.16

Find the domain and range of each of the following functions and sketch its graph.

a. $f(x) = 2\text{sgn}\, x$

b. $g(x) = -3\text{sgn}\, x$

c. $h(x) = 1 + \text{sgn}\, x$

d. $l(x) = 3 - \text{sgn}\, x$

1.3.4 The Greatest Integer (Floor or Step) Function

Greatest Integer Function

Activity 1.12

For each of the following numbers, find the greatest integer that is less than or equal to the given number.

a. 3.7 b. −1.9 c. 6 d. −4

From Activity 1.12, you have observed that you can find the greatest integer that is less than or equal to a given number, for example, the greatest integer less than or equal to -1.9 is -2.

Definition 1.8

The function defined by $f(x)$ =the greatest integer that is less than or equal to x, denoted by $f(x) = \lfloor x \rfloor$ is called the Greatest Integer (Floor or Step) Function.

Example 1

Evaluate the following.

a. $\lfloor 10.7 \rfloor$ b. $\lfloor -12.4 \rfloor$ c. $\lfloor -18 \rfloor$

Solution

a. The greatest integer that is less than or equal to 10.7 is 10, that is $\lfloor 10.7 \rfloor = 10$.

b. The greatest integer that is less than or equal to -12.4 is -13, that is $\lfloor -12.4 \rfloor = -13$.

c. The greatest integer that is less than or equal to -18 is -18, that is $\lfloor -18 \rfloor = -18$.

Exercise 1.17

Evaluate each of the following.

a. $\lfloor \pi \rfloor$ c. $\lfloor \sqrt{2} \rfloor$ e. $\left\lfloor -\frac{23}{25} \right\rfloor$

b. $\left\lfloor \frac{23}{25} \right\rfloor$ d. $\lfloor -\pi \rfloor$ f. $\lfloor -\sqrt{2} \rfloor$

Graph of the Greatest Integer Function

Given the function $f(x) = \lfloor x \rfloor$:

a. Dom(f) is the set of all real numbers.

b. Ran(f) is the set of all integers.

To understand the behavior of this function in terms of a graph, let us construct a table of values as in the following table:

x	$-3 \leq x < -2$	$-2 \leq x < -1$	$-1 \leq x < 0$	$0 \leq x < 1$	$1 \leq x < 2$	$2 \leq x < 3$
$y = [x]$	-3	-2	-1	0	1	2

The table shows us that the function increases to the next higher integer any time the x-value becomes an integer and the graph of the function $y = \lfloor x \rfloor$ is given in **Figure 1.49**.

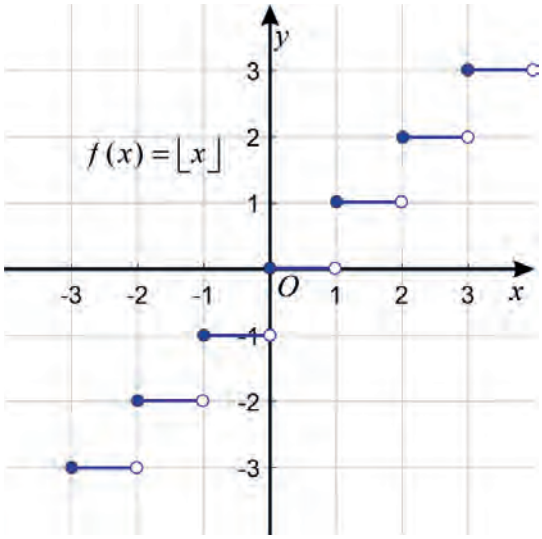


Figure 1.49: Graph of $f(x) = [x]$

Example 2

Find the domain, range and sketch the graph of $f(x) = [2x]$.

Solution

The function $f(x) = [2x]$ is defined for all real numbers and the set of its functional values is the set of integers. Therefore,

a. Dom(f) is the set of all real numbers.

b. Ran(f) is the set of all integers.

x	$-1 \leq x < -0.5$	$-0.5 \leq x < 0$	$0 \leq x < 0.5$	$0.5 \leq x < 1$	$1 \leq x < 1.5$
$y = [2x]$	-2	-1	0	1	2

Using the points in the table, the graph of the function $f(x) = [2x]$ is sketched in **Figure 1.50**.

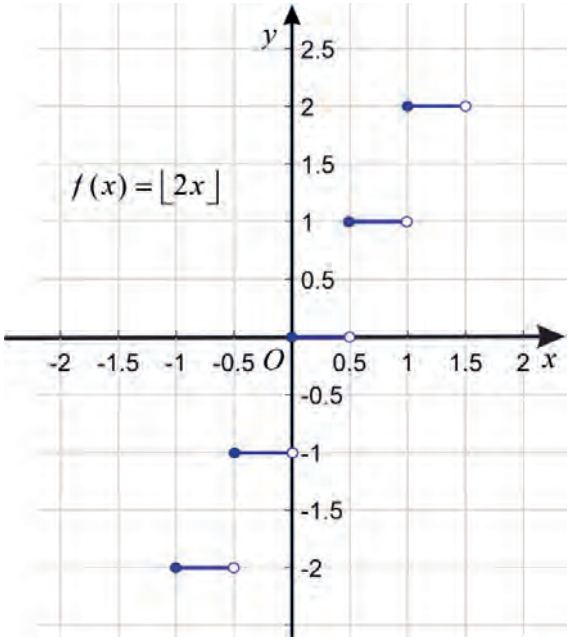


Figure 1.50: Graph of $f(x)=\lfloor 2x \rfloor$.

Example 3

Find the domain, range and then sketch the graph of $f(x)=\lfloor x+1 \rfloor$.

Solution

The function $f(x)=\lfloor x+1 \rfloor$ is defined for all real numbers and the set of its functional values is the set of integers. Therefore,

a. Dom(f) is the set of all real numbers.

b. Ran(f) is the set of all integers.

x	$-2 \leq x < -1$	$-1 \leq x < 0$	$0 \leq x < 1$	$1 \leq x < 2$	$2 \leq x < 3$
$y = [x+1]$	-1	0	1	2	3

Using the points in the above table, the graph of the function $f(x) = \lfloor x + 1 \rfloor$ is sketched in **Figure 1.51**.

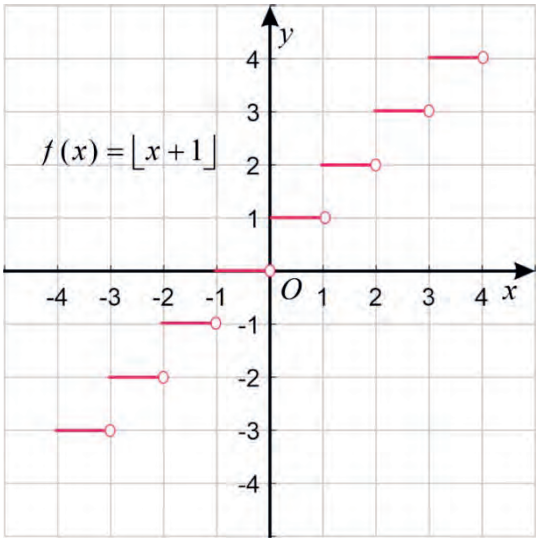


Figure 1.51: Graph of $f(x) = \lfloor x + 1 \rfloor$.

Example 4

Find the domain, range and sketch the graph of $f(x) = 3\lfloor x \rfloor$.

Solution

The function $f(x) = 3\lfloor x \rfloor$ is defined for all real numbers and the set of its functional values is the set of all integer multiples of 3. Therefore,

a. Dom(f) is the set of all real numbers.

b. Ran(f) is the set of all integer multiples of 3.

x	$-2 \leq x < -1$	$-1 \leq x < 0$	$0 \leq x < 1$	$1 \leq x < 2$	$2 \leq x < 3$
$y = 3\lfloor x \rfloor$	-6	-3	0	3	6

Using the points in the table, the graph of the function $f(x) = 3\lfloor x \rfloor$ is sketched in **Figure 1.52**.

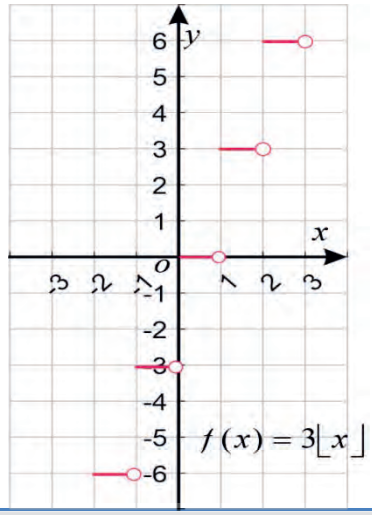


Figure 1.52: Graph of $f(x) = 3\lfloor x \rfloor$

Exercise 1.18

Sketch the graph of each of the following functions.

a. $f(x) = \lfloor 4x \rfloor$ b. $g(x) = 2\lfloor x \rfloor$ c. $h(x) = \lfloor x + 3 \rfloor$

Equations involving the Greatest Integer Function:

When working with equations that involve the greatest integer function, we use the following formula: $\lfloor x \rfloor = m$, where m is an integer if and only if $m \le x < m+1$.

For example, $\lfloor x \rfloor = 6$ if and only if $6 \le x < 7$.

Example 5

Solve the equation $\lfloor x + 3 \rfloor = 10$.

Solution

First, rewrite the equation using the inequalities $10 \le x + 3 < 11$.

Then, solve the inequalities: $10 \le x + 3 < 11 \Rightarrow 10 - 3 \le x < 11 - 3$

$$\Rightarrow 7 \le x < 8$$

Therefore, the solution set is the interval $[7,8)$.

Example 6

Solve the equation $\lfloor 1 + \lfloor x \rfloor \rfloor = 5$.

Solution

First let us replace $\lfloor x \rfloor$ with z. This is called a "change of variable" and it will make the equation easier to work with.

$$\lfloor 1 + \lfloor x \rfloor \rfloor = 5 \Longrightarrow \lfloor 1 + z \rfloor = 5$$

Then, let us solve the equation, $\lfloor 1 + z \rfloor = 5$.

Replace the given equation with the inequalities, $5 \leq 1 + z < 6$

Solve the inequality

$$5 \leq 1 + z < 6 \Longrightarrow 4 \leq z < 5 \Longrightarrow 4 \leq \lfloor x \rfloor < 5$$

Since $\lfloor x \rfloor$ is an integer, the only way to satisfy the above inequalities is for $\lfloor x \rfloor = 4$.

Let us determine the value of x.

Again, using the inequalities, we know $4 \leq x < 5$

Therefore, the solution set of the given equation is the interval $[4,5)$.

Exercise 1.19

Solve each of the following equations.

a. $\lfloor x + 4 \rfloor = 6$

b. $\lfloor 3 + \lfloor x \rfloor \rfloor = 5$

1.4 Composition of Functions

Activity 1.13

Consider the functions $f(x) = x + 3$ and $g(x) = x^2$. What result will you have if

a. you substitute $x + 3$ to x in $g(x)$?

b. you substitute x^2 to x in $f(x)$?

From your responses in Activity 1.13, observe that:

a. you get $g(x+3)=(x+3)^2$ and

b. you get $f(x^2) = x^2 + 3$.

$g(x+3)=(x+3)^2$ is the result of substituting $f(x) = x + 3$ into g(x).

That is, $g(x+3)=g(f(x))$, which is called the composition g by f.

Similarly $f(x^2) = x^2 + 3$ is the result of substituting $g(x)=x^2$ into $f(x)$.

That is, $f(x^2)=f(g(x))$, which is called the composition f by g.

Note that, $g(f(x))$ is denoted by $(g\circ f)(x)$ and $f(g(x))$ is denoted by $(f\circ g)(x)$.

Definition 1.9

Let $f: A \to B$ and $g: B \to C$ be functions, Then the function h defined by $h(x)=g(f(x))$ for all $x \in A$ and $f(x)$ is in the domain of g, is called the composition of g by f, denoted by $g\circ f$. That is, $(g\circ f)(x)=g(f(x))$, for all $x \in$ A and $f(x)$ is in the domain of g.

Example 1

Let $f=\{(-1,1),(0,2),(1,-1),(2,5)\}$ and

$g=\{(-1,2),(0,0),(1,3),(2,1),(3,4),(4,2),(5,5)\}$.

Then, find $g\circ f$ and $f\circ g$.

Solution

$(g\circ f)(-1)=g(f(-1))=g(1)=3$,

$(g\circ f)(0)=g(f(0))=g(2)=1$,

$(g\circ f)(1)=g(f(1))=g(-1)=2$,

$(g\circ f)(2)=g(f(2))=g(5)=5$.

Therefore, $g\circ f=\{(-1,3),(0,1),(1,2),(2,5)\}$.

$(f \circ g)(-1) = f(g(-1)) = f(2) = 5,$

$(f \circ g)(0) = f(g(0)) = f(0) = 2$

$(f \circ g)(2) = f(g(2)) = f(1) = -1$

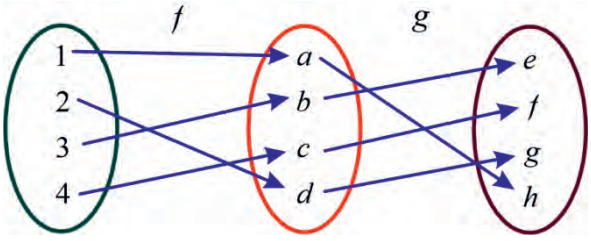
$(f \circ g)(4) = f(g(4)) = f(2) = 5.$

Therefore, $f \circ g = \{(-1,5),(0,2),(2,-1),(4,5)\}$

Observe that, $(f \circ g)(1)$ is not defined, because $g(1) = 3$ and 3 is not in the domain of f.

Example 2

Consider functions f and g given in the following diagram.



Then, determine $g \circ f$.

Solution

First observe that $f = \{(1,a),(2,d),(3,b),(4,c)\}$ and $g = \{(a,h),(b,e),(c,f),(d,g)\}$.

Thus,

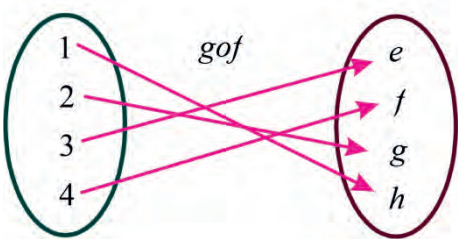
i. $(g \circ f)(1) = g(f(1)) = g(a) = h$

ii. $(g \circ f)(2) = g(f(2)) = g(d) = g$

iii. $(g \circ f)(3) = g(f(3)) = g(b) = e$

iv. $(g \circ f)(4) = g(f(4)) = g(c) = f$

Therefore, $g \circ f = \{(1,h),(2,g),(3,e),(4,f)\}$.



Example 3

Let $f(x)=x+1$ and $g(x)=x^2-2$. Then find $f\circ g$ and $g\circ f$.

Solution

a. $(f\circ g)(x)=f(g(x))=f(x^2-2)=(x^2-2)+1=x^2-1$

b. $(g\circ f)(x)=g(f(x))=g(x+1)=(x+1)^2-2=x^2+2x-1$

Example 4

Let $f(x)=x^2+1$ and $g(x)=\sqrt{x}$. Then find

a. $(g\circ f)(2)$ b. $(g\circ f)(x)$

Solution

a. $(g\circ f)(2)=g(f(2))=g(5)=\sqrt{5}$

b. $(g\circ f)(x)=g(f(x))=g(x^2+1)=\sqrt{x^2+1}$.

Note

For two functions f and g, the domain of $g\circ f$ is the set of all x in the domain of f, such that $(g\circ f)(x)=g(f(x))$ is defined.

Exercise 1.20

1. Let $f(x)=x^2+2x$ and $g(x)=3x-5$. Then find

   a. $(f\circ g)(3)$ b. $(g\circ f)(3)$

   c. $(g\circ f)(0)$ d. $(f\circ g)(0)$

2. Let $f(x)=\sqrt{x}$ and $g(x)=2x-5$. Then determine

   a. $(g\circ f)(x)$ b. $(f\circ g)(x)$

   c. Domain of $g\circ f$ d. Domain of $f\circ g$

1.5 Inverse Functions and Their Graphs

In Section 1.1, you have learned about inverses of relations. In this section, you will learn about the inverses of functions. As functions are special types of relations, the idea of inverse of a function is a continuation of the idea of inverse of a relation, but the only problem here is that the inverse of a function may not be a function.

Inverse Function

Activity 1.14

1. Find the inverse of each of the following functions.

   a. $f=\{(1,2),(2,3),(3,4),(4,5)\}$ c. $h=\{(x,y)\mid x,y\in\mathbb{R}, y=x^2+1\}$

   b. $g=\{(-1,a),(0,b),(3,4),(4,5)\}$ d. $l=\{(x,y)\mid x,y\in\mathbb{R}, y=2x+1\}$

2. From (1) above, which inverses are functions and which are not? (Give your reasons).

From your responses in Activity 1.14, you have observed that the inverse of a function may or may not be a function.

Note

a. Given a function f, if its inverse is also a function, then we say that f is invertible and we denote its inverse by f^{-1}.

b. Given a function $y = f(x)$, if f is invertible then to find its inverse;

I. interchange x and y in the expression $y = f(x)$, that is., $x = f(y)$;

II. solve y in terms of x, (if possible);

III. write f^{-1} as $y = f^{-1}(x)$.

Example 1

Find the inverse of each of the following functions.

a. $f(x) = 3x + 2$ b. $g(x) = 6 - 2x$ c. $h(x) = \frac{x+3}{2x-4}, x \neq 2$

Solution

a. $y = f(x) = 3x + 2$.

- Interchange y and x and obtain $x = 3y + 2$
- Solve y for x and obtain $y = \frac{1}{3}x - \frac{2}{3}$.
- Therefore, $f^{-1}(x) = \frac{1}{3}x - \frac{2}{3}$

b. $y = g(x) = 6 - 2x$

- Interchange y and x and obtain $x = 6 - 2y$
- Solve for y in terms of x and obtain $y = -\frac{1}{2}x + 3$.
- Therefore, $g^{-1}(x) = -\frac{1}{2}x + 3$

c. $y = h(x) = \frac{x+3}{2x-4}, x \neq 2$

- Interchange y and x in the formula and obtain $x = \frac{y+3}{2y-4}$

- Solve for y in terms of x and obtain $y=\dfrac{4x+3}{2x-1}$.
- Therefore, $h^{-1}(x)=\dfrac{4x+3}{2x-1}$ for $x\neq\dfrac{1}{2}$.

Note

There are functions whose inverses are not functions.

Example 2

Show that the inverse of the function $f(x)=x^3-x+1$ is not a function.

Solution

$f(-1)=(-1)^3-(-1)+1=1$ and $f(1)=1^3-1+1=1$.

This implies that $f(-1)=f(1)$.

Thus $(1,-1)$ and $(1,1)$ are elements of the inverse of f, but $-1\neq 1$.

Thus, the inverse of f is not a function.

Exercise 1.21

Find the inverses of each of the following functions.

a. $f(x)=2x+3$ b. $f(x)=-3x+2$ c. $f(x)=\dfrac{x+5}{3x-4}$ for $x\neq\dfrac{4}{3}$.

Identity Function

Definition 1.10

Given a set A, if $f:A\to A$ is a function given by $f(x)=x$ for all $x\in A$, then f is called the identity function on A and denoted by I.

Activity 1.15

Let $f=\{(1,2),(2,3),(3,4),(4,1)\}$ a function defined on a set $A=\{1,2,3,4\}$ and I be the identity on A as, $I=\{(1,1),(2,2),(3,3),(4,4)\}$.

Then determine

a. $I\circ f$ b. $f\circ I$ c. Compare $I\circ f$ and $f\circ I$.

In Activity 1.15, the two compositions, $f\circ I$ and $I\circ f$, are equal and both are equal to f.

Example 3

Let $f(x)=2x-3$ and $I(x)=x$. Then

$(I\circ f)(x)=I(f(x))=I(2x-3)=2x-3=f(x)$ and

$(f\circ I)(x)=f(I(x))=f(x)=2x-3$.

This implies $I\circ f=f=f\circ I$.

Note

In general, if f is a function and I is the identity function, then $I\circ f=f=f\circ I$

Activity 1.16

Consider the function $f=\{(1,2),(2,3),(3,4),(4,5)\}$ from $A=\{1,2,3,4\}$ to $B=\{2,3,4,5\}$. Then find,

a. f^{-1} b. $f^{-1}\circ f$ c. $f\circ f^{-1}$

In Activity 1.16, you can observe that $f^{-1}\circ f$ is the identity on A and $f\circ f^{-1}$ is the identity on B.

Note

Given a function f, a function g is the inverse of f if and only if $g \circ f = I$ and $f \circ g = I$.

Example 4

Let $f(x) = 3x - 1$ then $f^{-1}(x) = \frac{1}{3}x + \frac{1}{3}$.

- $(f \circ f^{-1})(x) = f(f^{-1}(x)) = f\left(\frac{1}{3}x + \frac{1}{3}\right) = 3\left(\frac{1}{3}x + \frac{1}{3}\right) - 1 = x = I(x)$ and
- $(f^{-1} \circ f)(x) = f^{-1}(f(x)) = f^{-1}(3x - 1) = \frac{1}{3}(3x - 1) + \frac{1}{3} = x = I(x)$.

This implies, $f \circ f^{-1} = I = f^{-1} \circ f$.

Example 5

Determine if each of the following pairs of functions are inverse of each other.

a. $f(x) = 2x - 5$ and $g(x) = -2x + 5$.

b. $f(x) = \frac{x+2}{x-2}$ and $g(x) = \frac{2x+2}{x-1}$.

Solution

Let us use the idea of composition to determine if f and g are inverses of each other.

a. $(f \circ g)(x) = f(g(x)) = f(-2x + 5) = 2(-2x + 5) - 5 = -4x + 5$ and

$(g \circ f)(x) = g(f(x)) = g(2x - 5) = -2(2x - 5) + 5 = -4x + 15$.

This implies $g \circ f \neq I$ and $f \circ g \neq I$. Thus g is not the inverse of f.

From this example, observe that $g(x) = -f(x)$ and hence $f^{-1} \neq -f$ .The inverse of f is $f^{-1}(x) = \frac{1}{2}x + \frac{5}{2}$

Using your knowledge in finding the inverses of functions, you can find that,

b. $(g\circ f)(x)=g(f(x))=g\left(\frac{x+2}{x-2}\right)=\frac{2\left(\frac{x+2}{x-2}\right)+2}{\left(\frac{x+2}{x-2}\right)-1}=\left(\frac{4x}{x-2}\right)\left(\frac{x-2}{4}\right)=x=I(x)$ and

similarly you can show that $(f\circ g)(x)=x=I(x)$.

Hence, g is an inverse for f, that is, $f^{-1}(x)=g(x)$.

Exercise 1.22

1. Let $f(x)=3x-1$ and $\mathrm{I}(x)=x$. Determine $\mathrm{I}\circ f$ and $f\circ \mathrm{I}$.
2. Let $f(x)=2x+4$ and $\mathrm{I}(x)=x$.
   a. Determine $f^{-1}(x)$.
   b. Show that $(f^{-1}\circ f)(x)=\mathrm{I}(x)$ and
   c. Show that $(f\circ f^{-1})(x)=\mathrm{I}(x)$.

Graphs of Inverse Functions

As functions are relations, from our discussions on graphs of inverse relations, recall that the graph of the inverse of a given function can be obtained by reflecting the graph of the function with respect to the line $y=x$.

Example 6

Draw the graph of the inverse of each of the following functions.

a. $f=\{(1,3),(2,1),(3,2),(4,4)\}$ b. $g(x)=2x+1$

Solution

a. $f^{-1}=\{(3,1),(1,2),(2,3),(4,4)\}$ and the graph of f^{-1} is obtained by reflecting the graph of f with respect to the line $y=x$, as given in **Figure 1.53.**

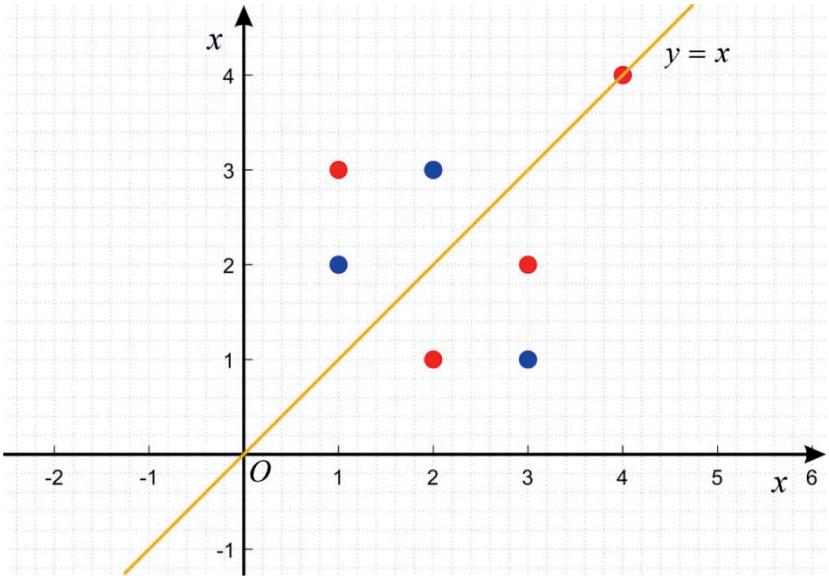


Figure 1.53: Graph of f and f^{-1}

b. Let $y=2x+1$. Then, interchanging x and y in the given equation gives us $x=2y+1$. Solving for y in terms of x gives you $y=\frac{1}{2}x-\frac{1}{2}$; that is, $g^{-1}(x)=\frac{1}{2}x-\frac{1}{2}$.

Therefore, the graph of g^{-1} is obtained by reflecting the graph of g with respect to the line $y=x$ as given in **Figure 1.54.**

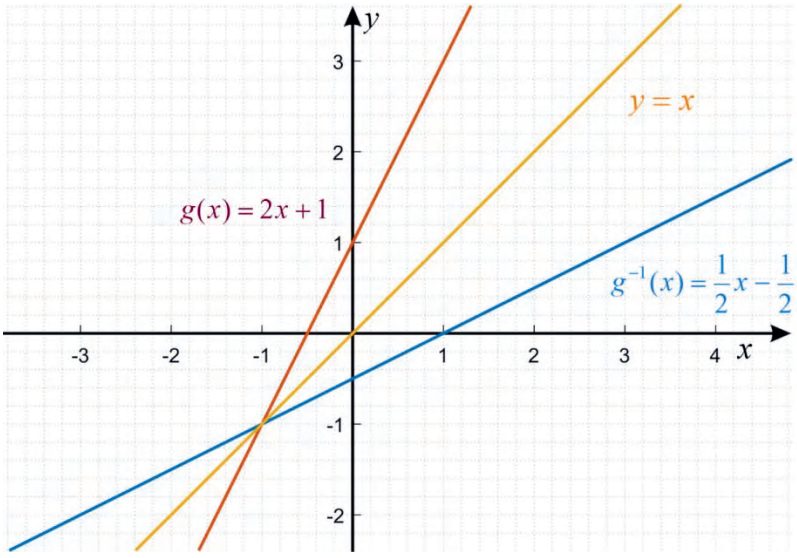


Figure 1.54 : Graphs of $g(x)=2x+1$ and $g^{-1}(x)=\frac{1}{2}x-\frac{1}{2}$

Exercise 1.23

1. Use compositions of functions to determine if each pairs of functions are inverses of each other.

   a. $f(x)=8x$ and $g(x)=\frac{x}{8}$

   b. $f(x)=\frac{2}{3}x+2$ and $g(x)=\frac{3}{2}x+3$

   c. $f(x)=5x-7$ and $g(x)=\frac{x+5}{7}$

   d. $f(x)=\frac{x}{2+x}$ and $g(x)=\frac{2x}{1-x}$

2. Sketch the graphs of the functions and its inverse on the same coordinate plane for each of the following functions.

   a. $f(x)=x-2$ b. $g(x)=5x-6$

1.6 Applications of Relations and Functions

In this section, different applications of relations and functions are given.

Example 1

Let $p+3q=30$ be an equation involving two variables p (price) and q (quantity). Indicate the meaningful domain and range of this function when

a. the price is considered as an independent variable;

b. the quantity is considered as an independent variable.

Solution

a. When price (p) is taken as an independent variable, you have $q(p)=10-\frac{1}{3}p$.

Domain: $0\le p\le 30$ Range: $0\le q\le 10$

b. When quantity (q) is taken as an independent variable, you have $p(q)=30-3q$.

Domain: $0\le q\le 10$ Range: $0\le p\le 30$

Example 2

A car rental company charges an initial fee, which is also called a flat fee of Birr 300 and an additional Birr 15 per kilometer to rent a van.

a. Write a function that approximates the cost y (in Birr) in terms of x, the number of kilometers driven.

b. How much would an 80 kilometer trip cost?

Solution

a. The total cost of renting a van is equal to the rate per kilometer times the number of kilometers driven plus the cost for the flat fee.

That is, $y=15x+300$

b. To calculate the cost of a 80 kilometers trip, substitute 80 for x in the equation:

$$y = 15x + 300$$

Thus, $y(80) = (15 \times 80) + 300 = 1500$.

Therefore, the cost of 80 kilometers trip is 1500 Birr.

Example 3

A man that buys and sells newspapers buys newspapers for price of Birr 6 per newspaper and sells newspapers at price of Birr 10 per newspaper. The unsold newspapers at the end of the day can be sold at 4 Birr per newspaper to a wastepaper dealer. If the man bought 300 newspapers, sold 200 newspapers and another 50 newspapers to a wastepaper dealer, what is the profit of the man in that day?

Solution

The overall profit to the man depends on the number of newspapers that he sells in relation to the number of newspapers he bought at the beginning of the day.

Purchase: 300×6 Birr $= 1800$ Birr

Sale: $(200 \times 10 \text{ Birr}) + (50 \times 4 \text{ Birr}) = 2000 \text{ Birr} + 200 \text{ Birr} = 2200 \text{ Birr}$.

Profit: $\text{Sale} - \text{Purchase} = 2200\text{Birr} - 1800\text{Birr} = 400\text{Birr}$.

Therefore, the profit of the man on that particular day is 400 Birr.

Example 4

Assume that for a closed economy, $E = C + I + G$, where E is total expenditure, C is expenditure on consumption of goods, I is expenditure on investment on goods and G is Government spending.

For equilibrium, we must have E = Y, where Y is the total income received.

For a certain economy, it is given that $C = 15 + 0.90\,Y$, $I = 20 + 0.05\,Y$ and $G = 25$.

a. Find the equilibrium values of Y, C and I.

b. How will these change if there is no Government spending?

Solution

Given that $E = C + I + G$ and $E = Y$. Thus, we have

a. $Y = C + I + G = (15 + 0.90\,Y) + (20 + 0.05\,Y) + 25 = 60 + 0.95\,Y$

This implies $Y(1 - 0.95) = 60 \Rightarrow Y = \dfrac{60}{0.05} = 1200$

For this value of Y, we have $C = 15 + 0.90\,Y = 15 + 0.90 \times 1200 = 1095$

and $I = 20 + 0.05\,Y = 20 + 0.05 \times 1200 = 80$

b. If there is no government spending, that is, $G = 0$, then closed economy equation becomes $Y = C + I = (15 + 0.90\,Y) + (20 + 0.05\,Y) = 35 + 0.95\,Y$

or $Y(1 - 0.95) = 35$, that is, $Y = \dfrac{35}{0.05} = 700$

For this value of Y, we have, $C = 15 + 0.90\,Y = 15 + 0.90 \times 700 = 645$

and $I = 20 + 0.05\,Y = 20 + 0.05 \times 700 = 55$.

Exercise 1.24

1. An electrician charges a base fee of Birr70 plus Birr 150 for each hour of work.
   a. Create a table that shows the amount that the electrician charges for 1,2, 3, and 4 hours of work.
   b. Let x represent the number of hours and y represent the amount charged for x hours. Is this relation a function?

2. A firm produces an item whose production cost function is $C(x)=200+60x$, where x is the number of items produced.
   a. If entire stock is sold at a price of each item which is Birr 800, then determine the revenue function.
   b. If the total number of items produced was 1000 and entire stock is sold at a price of each item which is Birr 800, find the profit of the firm.

Problem Solving

1. A company producing dry cells introduces production bonus for its employees which increases the cost of production. The daily cost of production $C(x)$ for x number of cells is Birr $(3.5x + 12{,}000)$.
   a. If each cell is sold for Birr 6, determine the number of cells that should be produced to ensure no loss.
   b. If the selling price is increased by 50 percent, what would be the break-even point?
   c. If at least 6000 cells can be sold daily, what price the company should charge per cell to guarantee no loss?
2. The force applied to a spring varies directly with the distance that the spring is stretched. When 30 N of force is applied, the spring stretches 3 m.
   a. Write a variation model using k as the constant of variation.
   b. Find k.
   c. How many meters will the spring stretch when 5N force pressure is applied?

Summary

1. A relation is any set of ordered pairs.
2. Given two sets A and B, a relation from A to B is any subset of $A \times B$.
3. Given a relation R, the inverse of R is given by $R^{-1} = \{(y,x): (x,y) \in R\}$
4. If R is a relation, then Domain of R^{-1} = Range of R and Range of R^{-1} = Domain of R.
5. The graph of the inverse of a given relation can be obtained by reflecting the graph of the relation with respect to the line $y = x$.
6. A function is a special type of a relation in which no two of the ordered pairs in it have the same first element.
7. A function of the form $f(x)=ax^r$, where a is a nonzero real number and r is a real number is called power function.
8. The function defined by $f(x)=|x|$, where $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$ is called the absolute value function.
9. The function defined by $f(x)=\operatorname{sgn} x$, where $\operatorname{sgn} x = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$ is called the signum function.
10. The function defined by $f(x)=\lfloor x \rfloor$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x is called the floor function or the greatest integer function.
11. If f is a function from A into B and g is a function from B into C, then the composition of f by g is the function defined by $(f \circ g)(x)=f(g(x))$.
12. For two functions f and g, g and f are inverse functions of each other if and only if $g(f(x))=x$ and $f(g(x))=x$.

Review Exercise

1. Find the domain and range of each of the following relations.
   a. $R_1 = \{(1,0),(2,2),(3,2),(4,5)\}$.
   b. $R_2 = \{(-1,-2),(0,1),(1,2),(2,5)\}$.
   c. $R_3 = \{(x,y) \mid x,y \in \mathbb{R} \text{ and } y = x-1\}$.
   d. $R_4 = \{(x,y) \mid x,y \in \mathbb{R} \text{ and } y = x^2+1\}$.
2. Find inverse, domain and range of the inverses of each of the relations in Question Number 1 above.
3. Find the inverse of each of the following functions.
   a. $f(x) = 3x-2$ b. $g(x) = x^3+5$ c. $h(x) = \dfrac{2x+3}{4x-5}$
4. Find the domain and range of each of the following functions.
   a. $f(x) = 3x^{\frac{4}{5}}$ b. $g(x) = -2x^{\frac{5}{4}}$ c. $h(x) = x^{\frac{3}{5}}$
5. Given $f(x) = 3 + 2|x|$:
   a. evaluate f at each of the following numbers.
      i. -5 ii. 0 iii. $-\frac{5}{2}$ iv. 3
   b. find the domain and range of f.
   c. sketch the graph of f.
6. Given $f(x) = 2 - 3sgn\ x$:
   a. evaluate f at each of the following numbers.
      i. -3 ii. 0 iii. $\frac{1}{2}$ iv. 4
   b. find the domain and range of f.
   c. sketch the graph of f.

7. Let $f(x) = \lfloor x - 3 \rfloor$. Then
   a. evaluate f at each of the following numbers.
      i. -2 ii. 0 iii. $-\frac{5}{4}$ iv. 5
   b. find the domain and rang of f.
   c. sketch the graph of f.
8. Determine if each pair of the following functions are inverses of each other.
   a. $f(x) = x - 1$ and $g(x) = x + 1$
   b. $f(x) = \frac{2x+3}{x-1}$ and $g(x) = \frac{x+3}{2x+1}$
   c. $f(x) = 2x - 3$ and $g(x) = 3 - 2x$
9. The area A of a square is directly proportional to the square of the length s of its sides.
   a. Write a general model of the proportionality with k as the constant of proportion.
   b. If the length of the sides is doubled, what effect will that have on the area?
   c. If the length of the sides is tripled, what effect will that have on the area?

UNIT 2

RATIONAL EXPRESSIONS AND RATIONAL FUNCTIONS

Unit Outcomes

By the end of this unit, you will be able to:

- Recognize a rational expression.
- Define a rational expression and a rational function.
- Know methods and procedures in simplifying rational expressions.
- Understand efficient methods in solving rational equations and inequalities.
- Develop efficient methods in solving rational equations and inequalities.
- Know basic concept and specific facts about rational functions.

Unit Contents

2.1 Rational Expressions
2.2 Rational Equations and Rational Inequalities
2.3 Rational Functions and Their Graphs
2.4 Applications
Summary
Review Exercise

- graphs of rational functions
- oblique asymptote
- horizontal asymptote
- operations on rational expressions
- partial fractions
- rational equations
- rational expression
- rational inequality
- vertical asymptote
- zeros of a rational function

Introduction

You can add, subtract or multiply polynomial expressions using the basic rules defined on the set of real numbers. Then, in these computational processes the result gives another polynomial expression. On the other hand, if you divide a polynomial expression by another polynomial expression, the result may not be a polynomial expression. In this unit, you will learn about quotients of polynomial expressions which are called rational expressions.

2.1 Rational Expressions

2.1.1 Rational Expressions

Activity 2.1

1. Which of the following numbers are rational numbers?

   a. $1.\overline{12}$ b. $\frac{2}{3}$ c. π d. $\sqrt{2}$

2. Which of the following are polynomial expressions?

   a. x^3+2x-1 b. $3\sqrt{x}+5x+3$ c. 3^x+x-9

From your responses in Activity 2.1, observe that:

a. a rational number is a number that can be written as a quotient of two integers.

b. An expression that can be written in the form $a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$, where n is a non-negative integer and $a_0, \ldots, a_n$ are real numbers, is called a polynomial expression.

Rational expressions are defined as quotients of polynomials.

Definition 2.1

Given two polynomials $p(x)$ and $q(x)$ with $q(x) \neq 0$, an expression that can be expressed in the form $\frac{p(x)}{q(x)}$ is called a rational expression. In a rational expression $\frac{p(x)}{q(x)}$, the polynomial $p(x)$ is called numerator and $q(x)$ is called denominator.

Example

Which of the following are rational expressions?

a. $\frac{2x+3}{5x^2+9x-4}$

b. $\frac{6x^3+4x-7}{x^2+1}$

c. $\frac{2}{x^3-1}$

d. x^3+3x-1

e. $\frac{2x^2+x}{\sqrt{x^2+1}}$

Solution

a. $\frac{2x+3}{5x^2+9x-4}$ is a rational expression, because both $2x+3$ and $5x^2+9x-4$ are polynomial expressions.

b. $\frac{6x^3+4x-7}{x^2+1}$ is a rational expression, because both $6x^3+4x-7$ and x^2+1 are polynomial expressions.

c. $\frac{2}{x^3-1}$ is a rational expression, because 2 is a constant polynomial and x^3-1 is a polynomial expression.

d. x^3+3x-1 is a rational expression, because the expression x^3+3x-1 can be expressed as $\dfrac{x^3+3x-1}{1}$ and both x^3+3x-1 and 1 are polynomial expressions.

e. $\dfrac{2x^2+x}{\sqrt{x^2+1}}$ is not a rational expression, because the denominator $\sqrt{x^2+1}$ is not a polynomial expression.

Exercise 2.1

Which of the following expressions are rational expressions?

a. $\dfrac{x+3}{x^3+x-9}$ b. $\dfrac{1}{x^3+\sqrt{x}+1}$ c. $\dfrac{x^2-x}{\sqrt{x^4+4x^2+4}}$

d. $\dfrac{x^{\frac{3}{2}}+2x-1}{6x^3+4x-7}$ e. $2x^4-3x^3+2x^2+3x-1$

2.1.2 Domains of Rational Expressions

Activity 2.2

Determine the value of the rational expression $\dfrac{2x^2+x}{x^2-1}$ for each of the following values of x if it is defined.

a. $x=2$ b. $x=-1$ c. $x=1$ d. $x=0$

From your responses in Activity 2.2, observe that a given rational expression is not defined for those numbers that are the zeros of the expression x^2-1, which is the denominator of the given rational expression.

Definition 2.2

Given a rational expression $\dfrac{p(x)}{q(x)}$, the set of all real numbers x such that $q(x)\neq 0$ is called the domain of $\dfrac{p(x)}{q(x)}$. That is, the domain of $\dfrac{p(x)}{q(x)}$ is $\{x\in\mathbb{R}: q(x)\neq 0\}$.

Example

Determine the domain of each of the following rational expressions:

a. $\frac{2x^2+x}{6x+3}$ b. $\frac{3}{x^2+5x+6}$ c. $\frac{5x+9}{x^2+1}$

Solution

a. The denominator of the given rational expression is $6x + 3$ and $6x + 3 = 0 \Longrightarrow x = -\frac{1}{2}$.

Therefore, the domain of $\frac{2x^2+x}{6x+3}$ is $\left\{x \in \mathbb{R} \mid x \neq -\frac{1}{2}\right\}$.

b. The denominator of the given rational expression is $x^2 + 5x + 6$ and $x^2 + 5x + 6 = 0$

implies $(x+2)(x+3) = 0 \Rightarrow x = -2$ or $x = -3$.

Thus, the domain of $\frac{3}{x^2+5x+6}$ is $\{x \in \mathbb{R} \mid x \neq -2 \text{ and } x \neq -3\}$.

c. The denominator of the given rational expression $x^2 + 1$ and $x^2 + 1 \neq 0$ for all $x \in \mathbb{R}$. Therefore, domain of $\frac{5x+9}{x^2+1}$ is $\mathbb{R}$, the set of all real numbers.

2.1.3 Simplifications of Rational Expressions

Activity 2.3

Write each one of the following rational numbers in its lowest term.

a. $\frac{4}{6}$ b. $\frac{10}{100}$ c. $\frac{15}{28}$ d. $\frac{18}{3}$

From your responses in Activity 2.3, observe the following two important points to write a given rational number in its lowest form:

i. first factorize both the numerator and the denominator; and

ii. cancel out the common factors of both the numerator and the denominator.

Recall from your knowledge about rational numbers that:

i. a rational number $\frac{a}{b}$, where a and b are integers and $b \neq 0$, is in its lowest term if the greatest common factor of a and b is 1;

ii. given three integers a, b, c, with $b \neq 0$ and $c \neq 0$, you have:

$$\frac{ac}{bc} = \frac{a}{b}.$$

Definition 2.3

A rational expression $\frac{p(x)}{q(x)}$ is said to be in lowest term or in its simplest form, if only common factor of the numerator $p(x)$ and the denominator $q(x)$ is 1.

Example 1

Simplify the rational expression $\frac{x+1}{x^2-1}$.

Solution

$\frac{x+1}{x^2-1} = \frac{(x+1)\times 1}{(x+1)(x-1)}$ (Factorize both the numerator and the denominator.)

$= \frac{1}{x-1}$, for $x \neq -1$. (Cancel out the common factor, $x+1$, of both the numerator and denominator.)

Observe that for $x = -1$, the expression $\frac{x+1}{x^2-1}$ is not defined, but the expression $\frac{1}{x-1}$ is defined at $x = -1$, and its value is $\frac{1}{-1-1} = -\frac{1}{2}$.

Therefore, $\frac{x+1}{x^2-1} = \frac{1}{x-1}$, for $x \neq -1$.

To simplify a given rational expression, first you have to determine the domain of the given expression.

Steps to simplify a given rational expression $\frac{p(x)}{q(x)}$:

Step 1: Determine the domain of the expression $\frac{p(x)}{q(x)}$;

Step 2: Factorize both $p(x)$and $q(x)$ (if possible); and

Step 3: Cancel out the common factors of both $p(x)$ and $q(x)$ to obtain the simplified form of the given rational expression.

Example 2

Simplify each of the following rational expressions.

a. $\frac{3x+6}{x^2+4x+4}$ b. $\frac{x^3+2x^2+x}{x^2-1}$ c. $\frac{x-1}{x^2-4x+3}$

Solution

a. $x^2+4x+4=0$ implies $x=-2$. Then, the domain of $\frac{3x+6}{x^2+4x+4}$ is $\{x \in \mathbb{R} | x \neq -2\}$.

Then $\frac{3x+6}{x^2+4x+4}=\frac{3(x+2)}{(x+2)^2}$ (Factorize both $3x+6$ and x^2+4x+4.)

$=\frac{3(x+2)}{(x+2)(x+2)}=\frac{3}{x+2}$, for $x \neq -2$ (Cancel out $x+2$ from both the numerator and the denominator.)

Therefore, $\frac{3x+6}{x^2+4x+4}=\frac{3}{x+2}$, for $x \neq -2$ in a simplified form as 1 is the only common factor of 3 and $x+2$.

b. $x^2-1=0$ implies $x=1$ or $x=-1$.

Then domain of $\frac{x^3+2x^2+x}{x^2-1}$ is $\{x \in \mathbb{R} | x \neq -1, 1\}$.

$\frac{x^3+2x^2+x}{x^2-1} = \frac{x(x+1)(x+1)}{(x+1)(x-1)}$ (Factorize both $x^3 + 2x^2 + x$ and $x^2 - 1$.)

$= \frac{x(x+1)}{x-1} = \frac{x^2+x}{x-1}$, for $x \neq -1$ (Cancel out $x + 1$ from the numerator and the denominator.)

Therefore, $\frac{x^3+2x^2+x}{x^2-1} = \frac{x^2+x}{x-1}$ for $x \neq -1$.

c. $x^2 - 4x + 3 = 0$ implies $x = 1$ and $x = 3$.

Then the domain of $\frac{x-1}{x^2-4x+3}$ is $\{x \in \mathbb{R} | x \neq 1, 3\}$.

$\frac{x-1}{x^2-4x+3} = \frac{(x-1)}{(x-1)(x-3)}$ (Factorize $x^2 - 4x + 3$.)

$= \frac{(x-1)\times 1}{(x-1)(x-3)} = \frac{1}{x-3}$, for $x \neq -1$ (Cancel out $x - 1$ from both the numerator and the denominator.)

Therefore, $\frac{x-1}{x^2-4x+3} = \frac{1}{x-3}$ for $x \neq 1$.

Exercise 2.2

1. Find the domain of each of the following rational expressions.

a. $\frac{2}{x^2+3}$ b. $\frac{x+1}{x^2-4}$ c. $\frac{x^4-5x+2}{x^3+6x^2+5x}$

2. Simplify each of the following rational expressions.

a. $\frac{2x-4}{x^2-x-2}$ b. $\frac{x^2+5x+6}{x^2-9}$ c. $\frac{x+2}{x^2+2x+2}$

2.1.4 Operations with Rational Expressions

Revision of Operations on Rational Numbers

Activity 2.4

Compute each of the following.

a. $\frac{1}{2}+\frac{2}{3}$ b. $\frac{1}{2}-\frac{2}{3}$ c. $\frac{1}{2}\times\frac{2}{3}$ d. $\frac{1}{2}\div\frac{2}{3}$

From your knowledge on rational numbers, observe that the results in Activity 2.4 are obtained by using the rules of addition, subtraction, multiplication and division of rational numbers.

Let $\frac{a}{b}$ and $\frac{c}{d}$ be rational numbers with $b \neq 0$ and $d \neq 0$. The basic four operations, addition, subtraction, multiplication and division on rational numbers are given as follows:

i. $\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}$

ii. $\frac{a}{b}-\frac{c}{d}=\frac{ad-bc}{bd}$

iii. $\frac{a}{b}\times\frac{c}{d}=\frac{ac}{bd}$

iv. $\frac{a}{b}\div\frac{c}{d}=\frac{a}{b}\times\frac{d}{c}=\frac{ad}{bc}, (c\neq 0)$

Addition and Subtraction of Rational Expressions

Addition and subtraction of rational expressions are extensions of the same operations on the set of rational numbers and they are defined as follows:

consider two rational expressions $\frac{p(x)}{q(x)}$ and $\frac{r(x)}{s(x)}$, with $q(x) \neq 0$ and $s(x) \neq 0$. Then,

a. $\frac{p(x)}{q(x)}+\frac{r(x)}{s(x)}=\frac{p(x)s(x)+r(x)q(x)}{q(x)s(x)}$

b. $\frac{p(x)}{q(x)}-\frac{r(x)}{s(x)}=\frac{p(x)s(x)-r(x)q(x)}{q(x)s(x)}$

Example 1

Perform each of the following operations:

a. $\frac{x+1}{x^2+2x}+\frac{3x}{x+5}$

b. $\frac{x+1}{x^2+2x}-\frac{3x}{x+5}$

Solution

a. $\frac{x+1}{x^2+2x}+\frac{3x}{x+5}=\frac{(x+1)(x+5)+3x(x^2+2x)}{(x^2+2x)(x+5)}=\frac{x^2+6x+5+3x^3+6x^2}{x^3+7x^2+10x}$

Therefore, $\frac{x+1}{x^2+2x}+\frac{3x}{x+5}=\frac{3x^3+7x^2+6x+5}{x^3+7x^2+10x}$.

b. $\frac{x+1}{x^2+2x}-\frac{3x}{x+5}=\frac{(x+1)(x+5)-3x(x^2+2x)}{(x^2+2x)(x+5)}=\frac{x^2+6x+5-3x^3-6x^2}{x^3+7x^2+10x}$

Therefore, $\frac{x+1}{x^2+2x}-\frac{3x}{x+5}=\frac{-3x^3-5x^2+6x+5}{x^3+7x^2+10x}$.

Multiplication and Division of Rational Expressions

Multiplication and division of rational expressions are extensions of the same operations on the set of rational numbers and they are defined as follows:

consider two rational expressions $\frac{p(x)}{q(x)}$ and $\frac{r(x)}{s(x)}$, with $q(x)\neq 0$ and $s(x)\neq 0$. Then,

i. $\frac{p(x)}{q(x)}\times\frac{r(x)}{s(x)}=\frac{p(x)r(x)}{q(x)s(x)}$;

ii. $\frac{p(x)}{q(x)}\div\frac{r(x)}{s(x)}=\frac{p(x)}{q(x)}\times\frac{s(x)}{r(x)}=\frac{p(x)s(x)}{q(x)r(x)}$, $(r(x)\neq 0)$.

Example 2

Perform each of the following operations:

a. $\frac{x+1}{x-1}\times\frac{x+2}{x^2+3x}$ b. $\frac{x+1}{x-1}\div\frac{x+2}{x^2+3x}$

Solution

a. $\frac{x+1}{x-1}\times\frac{x+2}{x^2+3x}=\frac{(x+1)(x+2)}{(x-1)(x^2+3x)}=\frac{x^2+3x+2}{x^3+2x^2-3x}$

b. $\frac{x+1}{x-1}\div\frac{x+2}{x^2+3x}=\frac{x+1}{x-1}\times\frac{x^2+3x}{x+2}=\frac{(x+1)(x^2+3x)}{(x-1)(x+2)}=\frac{x^3+4x^2+3x}{x^2+x-2}$

Exercise 2.3

Perform each of the following operations and simplify the given expression.

a. $\frac{x-1}{x+2}+\frac{x+3}{x-4}$

b. $\frac{x-1}{x+2}-\frac{x+3}{x-4}$

c. $\frac{x-1}{x+2}\times\frac{x+3}{x-4}$

d. $\frac{x-1}{x+2}\div\frac{x+3}{x-4}$

2.1.5 Decomposition of Rational Expressions into Partial Fractions

Activity 2.5

Consider the rational expression $\frac{x+1}{x^2-3x+2}$. Then,

a. factorize x^2-3x+2;

b. show that $\frac{x+1}{x^2-3x+2}=\frac{-2}{x-1}+\frac{3}{x-2}$.

In Activity 2.5, the fractional expressions $\frac{-2}{x-1}$ and $\frac{3}{x-2}$ are called partial fractions and writing $\frac{x+1}{x^2-3x+2}$ as $\frac{-2}{x-1}+\frac{3}{x-2}$ is decomposing the given expression as a sum of partial fractions.

The main objective in this discussion is to determine how to write a given rational expression as a sum of partial fractions. The idea of writing a rational expression as a sum of partial fractions is important, especially in a branch of Mathematics called Calculus and you will be introduced to the topic calculus in Grade 12.

Definition 2.4

Two polynomials $p(x)$ and $q(x)$ are equal if both polynomials have the same degree and the terms of the same degree in both polynomials have equal coefficients.

Example 1

Let $p(x)=3x^3+5x^2-7x+c$ and $q(x)=ax^4+bx^3+5x^2-7x+4$.

Find the values of the constants a, b and c if $p(x) = q(x)$ for all $x \in \mathbb{R}$.

Solution

$p(x) = q(x)$ for all $x \in \mathbb{R}$ if and only if terms of the same degree in both polynomials have equal coefficients.

i. The coefficient of the term with degree 4 in $p(x)$ is 0 and the coefficient of the term with degree 4 in $q(x)$ is a. This implies, $a = 0$.

ii. The coefficient of the term with degree 3 in $p(x)$ is 3 and the coefficient of the term with degree 3 in $q(x)$ is b. This implies, $b = 3$.

iii. The constant term in $p(x)$ is c and the constant term in $q(x)$ is 4. This implies, $c = 4$.

Definition 2.5

A rational expression $\frac{p(x)}{q(x)}$ is called proper rational expression if the degree of $p(x)$ is less than degree of $q(x)$, otherwise it is called an improper rational expression.

Note

Note that, given an improper rational expression $\frac{p(x)}{q(x)}$, using long division of polynomial expressions, we can write it as $\frac{p(x)}{q(x)} = f(x) + \frac{r(x)}{q(x)}$, where $f(x)$ and $r(x)$ are polynomial expressions and the degree of $r(x)$ is less than the degree of $q(x)$.

Example 2

Write the rational expression $\frac{5x^3+7x^2+5x+9}{x^2+1}$ in the form of $f(x) + \frac{r(x)}{q(x)}$, where $f(x)$ and $r(x)$ are polynomial expressions and the degree of $r(x)$ is less than the degree of $q(x)$.

Solution

Using long division on polynomial expressions, the polynomial expression $5x^3 + 7x^2 + 5x + 9$ can be written as $5x^3 + 7x^2 + 5x + 9 = (x^2 + 1)(5x + 7) + 2$.

Then

$$\frac{5x^3 + 7x^2 + 5x + 9}{x^2 + 1} = (5x + 7) + \frac{2}{x^2 + 1}$$

In writing a given rational expression as a sum of partial fractions, it is always important to factorize a polynomial expression into a product of polynomial expressions of smaller degree whenever possible.

The following Theorem is important in factorizing a polynomial into a product of polynomials of smaller degrees.

Theorem 2.1

Any non-constant polynomial expression with real coefficients can be factorized as a product of linear and/or irreducible quadratic factors with the possibility of some powers.

Example 3

Factorize each of the following polynomials, if possible.

a. $x^2 + 5x + 6$ b. $x^3 - 4x^2 + 7x - 6$

Solution

a. $x^2 + 5x + 6 = (x + 2)(x + 3)$

b. $x^3 - 4x^2 + 7x - 6 = (x - 2)(x^2 - 2x + 3)$ and the quadratic polynomial $x^2 - 2x + 3$ is irreducible as its discriminant $(-2)^2 - 4(1)(3) = -8$ and $-8 < 0$.

Exercise 2.4

1. Determine the constants a, b, c and d, if the polynomials $p(x) = ax^5 + 5x^4 + 3x^2 + d$ and $q(x) = 2x^5 + bx^4 + cx^3 + 3x^2 - 4$ are equal for all $x \in \mathbb{R}$.
2. Write each of the following rational expressions as $f(x) + \frac{r(x)}{q(x)}$, where $f(x), r(x)$ and $q(x)$ are polynomials and the degree of $r(x)$ is less than the degree of $q(x)$.
   a. $\frac{x^3+1}{x^2+2x+5}$ b. $\frac{x^2-5}{x^2+1}$ c. $\frac{x+1}{x^3+4x+5}$
3. Factorize each of the following polynomials, if possible.
   a. $x^3 + x^2 - 2x$ b. $x^4 + 4x^2 + 4$

Steps in decomposing a given rational expression as a sum of partial fractions:

I. If $ax + b$, for $a \neq 0$, is a factor in the denominator of the given rational expression and $(ax + b)^2$ is not a factor, then the partial fraction that corresponds to this factor is $\frac{A}{ax+b}$, where A is a constant that we have to determine.

Example 4

Decompose $\frac{6x+2}{x^2+6x+5}$ as a sum of partial fractions.

Solution

The given rational expression is a proper rational expression because the degree of $6x + 2$ is 1, which is less than the degree of $x^2 + 6x + 5$, which is 2.

First let us factorize $x^2 + 6x + 5$.

That is, $x^2 + 6x + 5 = (x + 5)(x + 1)$.

Then, $\frac{6x+2}{x^2+6x+5} = \frac{A}{x+5} + \frac{B}{x+1}$ for some constants A and B.

Now, let us determine A and B and to do so write:

$$\frac{6x+2}{x^2+6x+5} = \frac{A}{x+1} + \frac{B}{x+5} = \frac{A(x+5)+B(x+1)}{(x+1)(x+5)} = \frac{(A+B)x+(5A+B)}{x^2+6x+5}$$

Then, for $x \neq -1$ and $x \neq -5$, the polynomial expressions, $6x + 2$ and $(A + B)x + (5A + B)$ are equal.

That is, $A + B = 6$ and $5A + B = 2$, the coefficients of terms of the same degree are equal.

Then we can solve the two equations simultaneously as follows:

$$-\begin{cases} A + B = 6 \\ 5A + B = 2 \end{cases}$$
$$-4A = 4$$

This implies $A = -1$ and from the equation $A + B = 6$, we have $B = 6 - (-1) = 7$.

Hence, $\frac{6x+2}{x^2+6x+5} = \frac{-1}{x+1} + \frac{7}{x+5}$ (written as a sum of partial fractions).

II. If $(ax+b)^k$, for $a \neq 0$, is a factor in the denominator, for some integer $k > 1$ and $(ax+b)^{k+1}$ is not a factor, then the corresponding partial fractions that correspond to this factor are $\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_k}{(ax+b)^k}$, where $A_1, \cdots, A_k$ are constants that we have to determine.

Example 5

Decompose $\frac{x+5}{x^2+4x+4}$ as a sum of partial fractions.

Solution

First let us factorize x^2+4x+4.

That is, $x^2+4x+4=(x+2)^2$.

Then, $\dfrac{x+5}{x^2+4x+4}=\dfrac{A_1}{x+2}+\dfrac{A_2}{(x+2)^2}=\dfrac{A_1(x+2)+A_2}{(x+2)^2}=\dfrac{A_1x+(2A_1+A_2)}{(x+2)^2}$.

For all $x \in \mathbb{R}$ and $x \neq -2$, we have $x+5=A_1x+(2A_1+A_2)$.

This implies $A_1=1$ and $2A_1+A_2=5$ (Coefficients of terms of the same degree are equal). Then, solving for A_2 gives you $A_2=5-2A_1=5-2(1)=3$.

Therefore, $\dfrac{x+5}{x^2+4x+4}=\dfrac{1}{x+2}+\dfrac{3}{(x+2)^2}$ (written as a sum of partial fractions).

Exercise 2.5

Decompose each of the following rational expressions as a sum of partial fractions.

a. $\dfrac{x+1}{x^2+4x+3}$

b. $\dfrac{2x-3}{x^2+x-2}$

c. $\dfrac{x+2}{x^2+6x+9}$

d. $\dfrac{x-1}{x^2-4x+4}$

III. If ax^2+bx+c, for $a \neq 0$, is a factor in the denominator with $b^2-4ac<0$ and $(ax^2+bx+c)^2$ is not a factor, then the partial fraction that correspond to this factor is $\dfrac{Ax+B}{ax^2+bx+c}$, where A and B are constants that we have to determine.

Example 6

Write $\dfrac{x^2+3x+5}{(x+2)(x^2+x+1)}$ as a sum of partial fractions.

Solution

The quadratic expression x^2+x+1 is irreducible, because the discriminant:

$$1^2-4(1)(1)=-3 \text{ and } -3<0.$$

Then, $\dfrac{x^2+3x+5}{(x+2)(x^2+x+1)}=\dfrac{A}{x+2}+\dfrac{A_1x+B_1}{x^2+x+1}$.

This implies,

$$\frac{x^2+3x+5}{(x+2)(x^2+x+1)} = \frac{A\left(x^2+x+1\right)+(A_1x+B_1)(x+2)}{(x+2)(x^2+x+1)}$$

$$= \frac{\left(A+A_1\right)x^2+\left(A+2A_1+B_1\right)x+(A+2B_1)}{(x+2)(x^2+x+1)}.$$

Thus, for $x\neq -2$, $x^2+3x+5=(A+A_1)x^2+\left(A+2A_1+B_1\right)x+(A+2B_1)$.

Thus, from the equality of the two polynomial expressions, we have

$$\begin{cases} A+A_1=1 \\ A+2A_1+B_1=3 \\ A+2B_1=5 \end{cases}$$

Solving the three equations simultaneously gives us $A=1$, $A_1=0$ and $B_1=2$.

Therefore, $\dfrac{x^2+3x+5}{(x+2)(x^2+x+1)}=\dfrac{1}{x+2}+\dfrac{2}{x^2+x+1}$ (written as a sum of partial fractions).

IV. If $(ax^2+bx+c)^k$, for $a \neq 0$ and $b^2-4ac<0$, is a factor in the denominator for $k>1$ and $\left(ax^2+bx+c\right)^{k+1}$ is not a factor, then the corresponding sum of partial fractions to this factor is:

$$\frac{A_1x+B_1}{ax^2+bx+c}+\frac{A_2x+B_2}{(ax^2+bx+c)^2}+\cdots+\frac{A_kx+B_k}{\left(ax^2+bx+c\right)^k},$$

where $A_1,\ldots,A_k$ and $B_1,\ldots,B_k$ are constants that we have to determine.

Example 7

Write $\frac{6x^2+5x+2}{(x^2+2x+3)^2}$ as a sum of partial fractions.

Solution

The quadratic expression x^2+2x+3 is irreducible, because

$2^2-4(1)(3)=4-12=-8$ and $-8<0$

Then, $\frac{6x^2+5x+2}{(x^2+2x+3)^2}=\frac{A_1x+B_1}{x^2+2x+3}+\frac{A_2x+B_2}{(x^2+2x+3)^2}$.

Thus

$6x^2+5x+2=A_1x^3+(2A_1+B_1)x^2+(3A_1+2B_1+A_2)x+(3B_1+B_2)$ for all $x \in \mathbb{R}$.

This implies, $A_1=0,\ 2A_1+B_1=6, 3A_1+2B_1+A_2=5, 3B_1+B_2=2$ (the coefficients of terms of the same degree are equal).

Then, solving four equations simultaneously gives you,

$A_1=0,\ B_1=6,\ A_2=-7,\ B_2=-16$.

Therefore, $\frac{6x^2+5x+2}{(x^2+2x+3)^2}=\frac{6}{x^2+2x+3}+\frac{-7x-16}{(x^2+2x+3)^2}$ (written as a sum of partial fractions).

Note

To decompose an improper fraction of the form $\frac{p(x)}{q(x)}$, first write $\frac{p(x)}{q(x)} = f(x) + \frac{r(x)}{q(x)}$, where the degree of *r(x)* is less than the degree of *q(x)* and then write $\frac{r(x)}{q(x)}$ as a sum of partial fractions using any one of the methods discussed above.

Example 8

Write $\frac{x^3+7x+1}{x^2+3x+2}$ as a sum of partial fraction.

Solution

Since the degree of the numerator is not less than degree of the denominator, perform polynomial long division.

$$\frac{x^3+7x+1}{x^2+3x+2} = x-3+\frac{14x+7}{x^2+3x+2}$$

Then write $\frac{14x+7}{x^2+3x+2}$ as a sum of partial fractions.

That is, $\frac{14x+7}{x^2+3x+2} = \frac{14x+7}{(x+1)(x+2)} = \frac{A}{x+1}+\frac{B}{x+2}$.

This implies $\frac{14x+7}{(x+1)(x+2)} = \frac{(x+1)B+(x+2)A}{(x+1)(x+2)} = \frac{(A+B)x+(2A+B)}{(x+1)(x+2)}$.

The denominators of the two expressions are equal, so we require the equality of the numerators:

$14x+7 = (A+B)x+(2A+B)$.

Thus, solve the following system of two linear equations:

$$\begin{cases} A+B=14 \\ 2A+B=7 \end{cases}$$

Solving these two equations gives you: $A=-7$ and $B=21$.

Therefore, $\frac{x^3+7x+1}{x^2+3x+2}=x-3+\frac{14x+7}{x^2+3x+2}=x-3+\frac{-7}{x+1}+\frac{21}{x+2}$.

Exercise 2.6

Write each of the following rational expressions as a sum of partial fractions.

a. $\frac{2x^2+6x-6}{x^3-1}$ b. $\frac{x^2+7x}{x^2+7x-8}$ c. $\frac{x^2-3}{x^3+x^2+x+1}$

d. $\frac{42-14x}{x^3-4x^2+x-4}$ e. $\frac{x^3+2x+1}{x^2+x-2}$

2.2 Rational Equations and Rational Inequalities

2.2.1 Rational Equations

Activity 2.6

1. Solve each of the following equations.
   a. $\frac{x+1}{5}=\frac{3}{2}$ b. $\frac{2x}{3}=\frac{1}{3}$
2. Solve each of the following equations;
   a. $x^2-4x+3=0$ b. $x^2-3x=0$

In your responses of Activity 2.6, you need to revise how to solve linear equations for question number 1 and quadratic equations for question number 2 from your earlier grade mathematics.

In this section, you will learn how to solve equations involving rational expressions.

Definition 2.6

Any equation that involves only rational expressions is called a rational equation. That is, a rational equation is an equation that can be reduced to the form $\frac{p(x)}{q(x)}=0$, where $p(x)$ and $q(x)$ are polynomial expressions and $q(x)\neq 0$.

Example 1

Write each of the following equations as $\frac{p(x)}{q(x)}=0$, where $p(x)$ and $q(x)$ are polynomial expressions and $q(x)\neq 0$.

a. $\frac{1}{x+1}=\frac{2x}{x-1}$

b. $3x=\frac{2x+5}{x^2-2}$

Solution

a. $\frac{1}{x+1}=\frac{2x}{x-1}$ implies $\frac{1}{x+1}-\frac{2x}{x-1}=0.$

$\Rightarrow \frac{1(x-1)-2x(x+1)}{(x+1)(x-1)}=0.$ (Simplifying the rational expression.)

That is, $\frac{-2x^2-x-1}{x^2-1}=0$ is the required rational equation.

b. $3x=\frac{2x+5}{x^2-2}$ implies $3x-\frac{2x+5}{x^2-2}=0$

$\Rightarrow \frac{3x\left(x^2-2\right)-1(2x+5)}{x^2-2}=0$ (Simplifying the rational expression)

Thus, $\frac{3x^3-8x-5}{x^2-2}=0$ is the required rational equation.

Steps in Solving Rational Equations

Given a rational equation, solving the rational equation is finding all the possible numbers in the domain of the given rational expression that satisfy the given rational equation.

The following are steps that you can follow to solve rational equations.

Step 1: Reduce the given rational equation to the form $\frac{p(x)}{q(x)}=0$. (without simplifying the given expression).

Step 2: Solve the equation $q(x)=0$.

Step 3: Solve the equation $p(x)=0$.

Therefore, the solution set is the set of all real numbers in Step 3 without the set of numbers obtained in Step 2.

Example 2

Solve each of the following rational equations:

a. $x-\frac{1}{2}=\frac{1}{3}x$ b. $2-\frac{x}{x+1}=\frac{3}{x+1}$ c. $\frac{2}{x^2-9}=\frac{x}{x+3}$

Solution

a. $x-\frac{1}{2}=\frac{1}{3}x$ implies $x-\frac{1}{3}x-\frac{1}{2}=0$. (Subtracting $\frac{1}{3}x$ from both sides.)

Then, $\frac{6x-2x-3}{3\times 2}=0$ (Simplifying the rational expression.).

This implies, $\frac{4x-3}{6}=0\Rightarrow 4x-3=0$ (a linear equation)

Then, solving the linear equation $4x-3=0$ gives us $x=\frac{3}{4}$.

Therefore, the solution set of the given equation is $\left\{\frac{3}{4}\right\}$.

b. $2-\frac{x}{x+1}=\frac{3}{x+1}$ implies $2-\frac{x}{x+1}-\frac{3}{x+1}=0$. (Subtracting $\frac{3}{x+1}$ from both sides.)

This implies, $\frac{2(x+1)-(x+3)}{x+1}=0 \Rightarrow \frac{x-1}{x+1}=0.$

Then, determine the value(s) of x that make the denominator zero.

That is, $x+1=0 \Rightarrow x=-1.$

Thus, $x=-1$ will be excluded from the solution set of the given rational equation.

Then, solving the equation $x-1=0$ gives us $x=1.$

Therefore, the solution set of the given equation is $\{1\}$.

c. $\frac{2}{x^2-9}=\frac{x}{x+3}$ implies $\frac{2}{x^2-9}-\frac{x}{x+3}=0$. (Subtracting $\frac{x}{x+3}$ from both sides.)

Then, $\frac{2}{(x-3)(x+3)}-\frac{x}{x+3}=0.$

This implies $\frac{2-x(x-3)}{(x-3)(x+3)}=0 \Rightarrow \frac{2-x^2+3x}{(x-3)(x+3)}=0.$

Find the value(s) of x that make the denominator zero.

That is, $(x-3)(x+3)=0 \Rightarrow x-3=0$ or $x+3=0$

That is, $x=3$ or $x=-3$ and both $x=3$ and $x=-3$ are excluded from the solution set of the given rational equation.

Then, $2-x^2+3x=0$ and this is a quadratic equation. Thus, using the quadratic formula:

$$x=\frac{-3\pm\sqrt{9-4(-1)(2)}}{2(-1)}$$

That is, $x=\frac{-3\pm\sqrt{17}}{-2}.$

Therefore, the solution set of the given equation is $\left\{\frac{3-\sqrt{17}}{2},\frac{3+\sqrt{17}}{2}\right\}.$

Exercise 2.7

Solve each of the following equations.

a. $\frac{x^2-5x+6}{x^2-2}=0$

b. $\frac{x}{x-1}=\frac{3}{x+1}$

2.2.2 Rational Inequalities

Definition 2.7

Any inequality that involves only rational expressions is called a rational inequality.

That is, a rational inequality is an inequality that can be reduced to the form $\frac{p(x)}{q(x)}\leq 0$, $\frac{p(x)}{q(x)}<0$, $\frac{p(x)}{q(x)}\geq 0$ or $\frac{p(x)}{q(x)}>0$, where $p(x)$ and $q(x)$ are polynomial expressions and $q(x)\neq 0$.

Example 1

Which of the following are rational inequalities?

a. $1>\frac{2}{3x}$

b. $2^x<\frac{2}{x+5}$

c. $\frac{1}{x}-\frac{2}{x^2}\leq\frac{5}{x^2+1}$

Solution

a. Since both 1 and $\frac{2}{3x}$ are rational expressions, $1>\frac{2}{3x}$ is a rational inequality.

b. 2^x is not a rational expression. Thus, $2^x<\frac{2}{x+5}$ is not a rational inequality

c. $\frac{1}{x}-\frac{2}{x^2}\leq\frac{5}{x^2+1}\Leftrightarrow\frac{x-2}{x^2}\leq\frac{5}{x^2+1}$ and both $\frac{x-2}{x^2}$ and $\frac{5}{x^2+1}$ are rational expressions.

Thus, $\frac{1}{x}-\frac{2}{x^2}\leq\frac{5}{x^2+1}$ is a rational inequality.

The solution set of a rational inequality is the set of all real numbers in the domain of the given rational expression that satisfy the given inequality.

Steps in Solving Rational Inequalities

Step 1: Determine the domain of the rational inequality.

Step 2: Write the given inequality to the form $\frac{p(x)}{q(x)} > 0$, $\frac{p(x)}{q(x)} \geq 0$, $\frac{p(x)}{q(x)} < 0$ or $\frac{p(x)}{q(x)} \leq 0$.

Step 3: Find the zeros of both $p(x)$ and $q(x)$.

Step 4: Divide the number line into different intervals using the numbers that are not in the domain of the inequality in Step 1 and numbers in the Step 3.

The method is called the sign chart method.

Step 5: Find the interval that satisfies the given inequality using the product rule.

Example 2

Solve the inequality $\frac{x+1}{x-1} > 0$.

Solution

To determine the domain of the given inequality, solve the equation $x - 1 = 0$.

This implies, $x = 1$ and then the domain of the given inequality is $\{x \in \mathbb{R} | x \neq 1\}$.

Solve the equation $x + 1 = 0$. This implies $x = -1$.

Using the numbers -1 and 1, divide the number line into three intervals as in the table below.

	$x < -1$	-1	$-1 < x < 1$	1	$x > 1$
$x + 1$	$- - -$	0	$+ + +$	$+$	$+ + +$
$x - 1$	$- - -$	$-$	$- - -$	0	$+ + +$
$\frac{x+1}{x-1}$	$+ + +$	0	$- - -$	Undefined	$+ + +$

Then, from the given sign chart, observe that:

i. $\frac{x+1}{x-1}$ is positive on $(-\infty,-1)\cup(1,\infty)$;

ii. $\frac{x+1}{x-1}$ is negative on $(-1,1)$;

iii. $\frac{x+1}{x-1}$ is zero at $x=-1$ and

iv. $\frac{x+1}{x-1}$ is undefined at $x=1$.

Therefore, the solution set of the given inequality is $(-\infty,-1)\cup(1,\infty)$.

Example 3

Solve the inequality $\frac{1}{3}-\frac{2}{x^2}\geq\frac{5}{3x}$.

Solution

The given rational inequality is not defined when $x=0$.

Then, the domain of the given inequality is $\{x\in\mathbb{R}|x\neq 0\}$.

$\frac{1}{3}-\frac{2}{x^2}\geq\frac{5}{3x}\Rightarrow\frac{1}{3}-\frac{2}{x^2}-\frac{5}{3x}\geq 0$, which implies, $\frac{x^2-6-5x}{3x^2}\geq 0$.

Hence, the given inequality $\frac{1}{3}-\frac{2}{x^2}\geq\frac{5}{3x}$ is equivalent to the inequality $\frac{x^2-6-5x}{3x^2}\geq 0$, that is, both inequalities have the same solution set.

$$x^2-6-5x=0\Rightarrow x^2-5x-6=0\Rightarrow(x-6)(x+1)=0$$

This implies $x-6=0$ or $x+1=0\Rightarrow x=6$ or $x=-1$.

Then, divide the number line into four different intervals using the numbers -1, 0 and 6 as in the table below.

	$x<-1$	-1	$-1<x<0$	0	$0<x<6$	6	$x>6$
$x-6$	− − −	−	− − −	−	− − −	0	+ + +
$x+1$	− − −	0	+ + +	+	+ + +	+	+ + +
$3x^2$	+ + +	+	+ + +	0	+ + +	+	+ + +
$\frac{(x-6)(x+1)}{3x^2}$	+ + +	0	− − −	Undefined	− − −	0	+ + +

Then, from the given sign chart, observe that:

i. $\frac{(x-6)(x+1)}{3x^2}$ is positive on $(-\infty,-1) \cup (6,\infty)$;

ii. $\frac{(x-6)(x+1)}{3x^2}$ is negative on $(-1,0) \cup (0,6)$;

iii. $\frac{(x-6)(x+1)}{3x^2}$ is zero at $x = -1$ and $x = 6$;

iv. $\frac{(x-6)(x+1)}{3x^2}$ is undefined at $x = 0$.

Therefore, the solution set of the inequality $\frac{x^2-6-5x}{3x^2} \geq 0$ is $(-\infty,-1] \cup [6,\infty)$, which is also the solution set of the inequality $\frac{1}{3} - \frac{2}{x^2} \geq \frac{5}{3x}$.

Exercise 2.8

Solve each of the following inequalities.

a. $\frac{x-1}{x+1} \geq 0$

b. $\frac{1}{3} + \frac{2}{x^2} < \frac{5}{3x}$

2.3 Rational Functions and Their Graphs

2.3.1 Rational Functions

Activity 2.7

Which of the following expressions are rational expressions?

a. $\frac{x+1}{x-1}$

b. $\frac{2}{x^2+3x-1}$

c. $\frac{\sqrt{x+1}}{x+2}$

From your responses in Activity 2.7, observe that rational expressions are expressions that can be written as a quotient of two polynomial expressions and that are used to define rational functions.

Definition 2.8

A function f that can be expressed in the form $f(x) = \frac{p(x)}{q(x)}$, where both $p(x)$ and $q(x)$ are polynomial expressions and $q(x) \neq 0$, is called a rational function.

Example 1

Which of the following are rational functions?

a. $f(x) = \frac{x^3+6x+9}{x^2+4x+1}$ b. $g(x) = \frac{3x-5}{(x+1)^{\frac{3}{2}}}$ c. $p(x) = 3x^2 + 4x - 5$

Solution

a. $f(x)=\frac{x^3+6x+9}{x^2+4x+1}$ is a rational function, because both $x^3 + 6x + 9$ and $x^2 + 4x + 1$ are polynomial expressions.

b. $g(x)=\frac{3x-5}{(x+1)^{\frac{3}{2}}}$ is not a rational function, because $(x + 1)^{\frac{3}{2}}$ is not a polynomial expression.

c. $p(x)=3x^2+4x-5$ is a polynomial function and any polynomial function, $p(x)$, is a rational function, because $p(x)$ can be expressed as $p(x)=\frac{p(x)}{1}$ and the denominator 1 is constant, which is a polynomial expression.

Domains of Rational Functions

Activity 2.8

For each of the following rational functions, find the number(s) that make the denominator zero.

a. $f(x)=\frac{x+1}{x-1}$ b. $g(x)=\frac{3}{x+1}$

c. $h(x)=\frac{x^2+1}{x^2-1}$ d. $l(x)=\frac{x^2-x-12}{x^2+x-6}$

For each of the rational functions in Activity 2.8, at the value of x that makes the denominator zero, the given rational function is not defined, as division by zero is not possible and such number(s) must not be in the domain of the given rational function. Thus, the definition of domain of a rational function is given as follows.

Definition 2.9

The domain of a rational function $f(x)=\frac{p(x)}{q(x)}$ is the set of all real numbers such that $q(x)$ is not zero. That is, $\text{Dom}(f)=\{x\in\mathbb{R}\mid q(x)\neq 0\}$.

Example 2

Find the domain of each of the following rational functions:

a. $f(x)=\frac{3x+5}{x^2+3x+2}$

b. $g(x)=\frac{2x^2+x-1}{x^2+x+4}$

Solution

a. Consider the denominator, x^2+3x+2 of $f(x)$..

Then, $x^2+3x+2=0\Rightarrow x=-1$ or $x=-2$.

Therefore, Dom$(f)=\{x\in\mathbb{R}|x\neq -1$ and $x\neq -2\}$.

b. Consider the denominator, x^2+x+4, of $g(x)$, which is a quadratic expression.

Since the discriminant of x^2+x+4 : $1^2-4(1)(4)=-15$ and $-15<0$, $x^2+x+4\neq 0$ for all $x\in\mathbb{R}$.

Therefore, Dom$(g)=\mathbb{R}$.

Exercise 2.9

1. Which of the following are rational functions?

   a. $f(x)=\dfrac{2x^2+3x-5}{x^3-2x^2+5x-1}$ b. $g(x)=\dfrac{2x+9}{\sqrt{x^2+x-2}}$

   c. $h(x)=\dfrac{1}{3}x^3+4x^2-x+5$

2. Find the domain of each of the following rational functions.

   a. $f(x)=\dfrac{x^2+x+5}{x^2+4x+3}$ b. $g(x)=\dfrac{x+6}{x^2+2x+2}$

2.3.2 Graphs of Rational Functions

Asymptotes to the graph of a Rational Function

Activity 2.9

Consider the function $f(x)=\dfrac{1}{x}$.

1. Complete the following tables.

x	1	0.5	0.1	0.01	0.001	0.0001	0.00001
$f(x)$							

x	-1	-0.5	-0.1	-0.01	-0.001	-0.0001	-0.00001
$f(x)$							

2. Complete the following tables.

x	1	10	100	1000	10000	100000
$f(x)$						

x	-1	-10	-100	-1000	-10000	-100000
$f(x)$						

The domain of $f(x) = \frac{1}{x}$ is $\{x \in \mathbb{R} | x \neq 0\}$ and

1. from the two tables of Activity 2.9(1), observe that;
   i. as x approaches 0 from the right, the value of the function $f(x)$ increases without bound and
   ii. as x approaches 0 from the left, the value of the function $f(x)$ decreases without bound.

Note that, x approaches to 0 from the right means, x takes numbers very close to 0, but numbers that are greater than 0 (in this case we mean positive numbers), and x approaches to 0 from the left means, x takes numbers very close to 0, but numbers less than 0 (in this case we mean negative numbers).

These two behaviors of the function $f(x)$ near $x = 0$ are denoted as follows:

a. if x approaches 0 from the right (this concept is denoted by $x \longrightarrow 0^+$), then $f(x)$ increases without bound (this concept is denoted by $f(x) \longrightarrow \infty$);
b. if x approaches 0 from the left (this concept is denoted by $x \longrightarrow 0^-$), then $f(x)$ decreases without bound (this concept is denoted by $f(x) \longrightarrow -\infty$).

The line $x = 0$ is called a vertical asymptote to the graph of $f(x)$.

2. From the two tables of Activity 2.9(2), observe that:
   a. as x increases without bound, $f(x)$ approaches 0 and it is expressed by: $f(x) \to 0$ as $x \to \infty$.
   b. as x decreases without bound, $f(x)$ approaches 0 and it is expressed by $f(x) \to 0$ as $x \to -\infty$.

   These two information imply that, as $x \to \infty$ or $x \to -\infty$, the graph of $f(x)$ approaches the horizontal line $y = 0$.Thus, the horizontal line $y = 0$ is called a horizontal asymptote to the graph of $f(x)$. With these given information, we can sketch the graph of $f(x) = \frac{1}{x}$ as in Figure 2.1

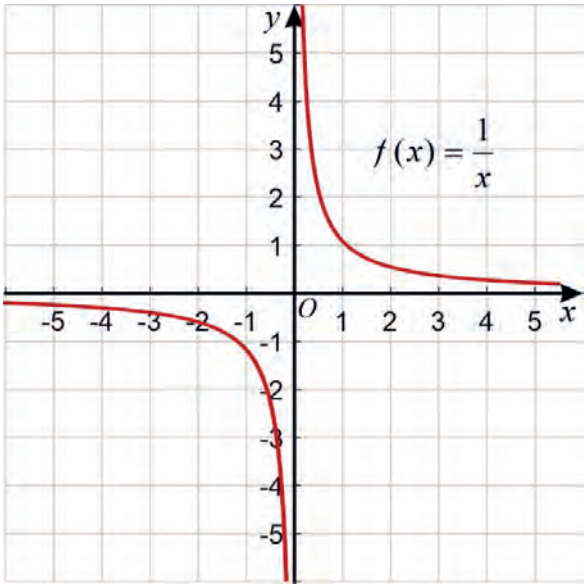


Figure 2.1: Graph of $f(x)=\frac{1}{x}$

To sketch the graph of a rational function $f(x)=\frac{p(x)}{q(x)}$, the following points are important:

a. the domain of the given function;

b. the behavior of the function near the points that make the denominator zero; and

c. the behavior of the function for very large positive values and for very small negative values of the variable.

Definition 2.10

Given a rational function $f(x)=\frac{p(x)}{q(x)}$;

a. If $f(x)\to\infty$ or $f(x)\to-\infty$ as $x\to a$ from the left or $x\to a$ from the right, then the line $x=a$ is called a vertical asymptote to the graph of $f(x)$.

b. If $f(x)\to b$ as $x\to\infty$ or as $x\to-\infty$, then the line $y=b$ is called a horizontal asymptote to the graph of $f(x)$.

Consider a rational function $f(x)=\frac{p(x)}{q(x)}$, where, $p(x)=a_nx^n+a_{n-1}x^{n-1}+\cdots+a_0$, with $a_n\neq 0$ and $q(x)=b_mx^m+b_{m-1}x^{m-1}+\cdots+b_0$ with $b_m\neq 0$.

1. If $p(a) \neq 0$ and $q(a) = 0$, then the line $x = a$ is a vertical asymptote to the graph of $f(x)$.
2. If both $p(a) = 0$ and $q(a) = 0$, then the graph of $f(x)$ has a "hole" at $x = a$ or we have to make further simplification to decide.
3. If $n < m$, then the line $y = 0$ is a horizontal asymptote to the graph of $f(x)$.
4. If $n = m$, then the line $y = \frac{a_n}{b_n}$ is a horizontal asymptote to the graph of $f(x)$.
5. If $n = m + 1$, then the graph of $f(x)$ has an asymptote $y = ax + b, a \neq 0$, and because the line is an oblique line, $y = ax + b$ is called an oblique asymptote to the graph of $f(x)$.
6. If $n > m + 1$, the graph of $f(x)$ has no horizontal asymptote and it has no oblique asymptote.

Example 1

Find the vertical, horizontal or oblique asymptotes to the graph of each of the following functions (if any):

a. $f(x) = \dfrac{1}{x-2}$ b. $g(x) = \dfrac{x+1}{x^2+3x+2}$ c. $h(x) = \dfrac{3x^2+1}{x^2+4x+4}$

d. $l(x) = \dfrac{x^3+3x+1}{x^2+1}$ e. $m(x) = \dfrac{x^3+x^2+x+1}{x+2}$

Solution

a. $x - 2 = 0$ implies $x = 2$. Then, $\text{Dom}(f) = \{x \in \mathbb{R} | x \neq 2\}$.

Let $p(x) = 1$ and $q(x) = x - 2$.

Then $f(x) = \frac{1}{x-2} = \frac{p(x)}{q(x)}$.

$p(2) = 1 \neq 0$ and $q(2) = 0$.

Therefore, the line $x = 2$ is a vertical asymptote to the graph of $f(x)$.

The degree of $p(x)$ is 0 and the degree of $q(x)$ is 1. That is, the degree of $p(x)$ is less than the degree of $q(x)$. Here, the line $y = 0$ is a horizontal asymptote to the graph of $f(x)$.

b. $x^2 + 3x + 2 = 0$ implies $x = -1$ or $x = -2$.

Thus, $\text{Dom}(g) = \{x \in \mathbb{R} | x \neq -1, -2\}$.

$$g(x) = \frac{x+1}{x^2+3x+2} = \frac{x+1}{(x+2)(x+1)} = \frac{1}{x+2} \text{ for } x \neq -1.$$

Therefore, $x = -2$ is the only vertical asymptote to the graph of $g(x)$ and the line $y = 0$ is a horizontal asymptote to the graph of $g(x)$, because the degree of $x + 1$ is 1 and the degree of $x^2 + 3x + 2$ is 2. That is, the degree of $x + 1$ is less than the degree of $x^2 + 3x + 2$.

c. $x^2 + 4x + 4 = 0$ implies $x = -2$. Then the domain of $h(x)$ is $\{x \in \mathbb{R} | x \neq -2\}$.

$h(x) = \frac{3x^2+1}{x^2+4x+4} = \frac{3x^2+1}{(x+2)^2}$ and $3(-2)^2 + 1 = 7 \neq 0$.

This implies the line $x = -2$ is the vertical asymptote to the graph of $h(x)$.

Since the degree of $3x^2 + 1$ is 2 and the degree of $x^2 + 4x + 4$ is also 2, then the line $y = \frac{3}{1} = 3$ is the horizontal asymptote to the graph of $h(x)$.

d. Since $x^2 + 1 \neq 0$ for any real number x, the graph of $l(x)$ has no vertical asymptote.

Since the degree of $x^3 + 3x + 1$ is 3 and the degree of $x^2 + 1$ is 2. That is, the degree of $x^3 + 3x + 1$ is one more than the degree of $x^2 + 1$, the using long division, you can write

$$l(x) = \frac{x^3+3x+1}{x^2+1} = x + \frac{2x+1}{x^2+1}.$$

Therefore, the line $y = x$ is an oblique asymptote to the graph of $l(x)$.

e. $x+2=0$ implies $x=-2$. Thus the domain of $m(x)$ is $\{x \in \mathbb{R} \mid x \neq -2\}$.

$(-2)^3 + (-2)^2 + (-2) + 1 = -5 \neq 0$.

This implies $x=-2$ is a vertical asymptote to the graph of $m(x)$.

The degree of $x^3 + x^2 + x + 1$ is 3, the degree of $x+2$ is 1 and $3 > 1+1$. Thus, the graph of $m(x) = \frac{x^3 + x^2 + x + 1}{x+2}$ has no horizontal asymptote and it also has no oblique asymptote.

Exercise 2.10

Find the vertical, horizontal and oblique asymptote(s) (if any) to the graph of each of the following functions.

a. $f(x)=\dfrac{1}{x+2}$ b. $g(x)=\dfrac{x+2}{x^2+5x+6}$ c. $h(x)=\dfrac{x^2+3}{x^2+6x+9}$

d. $l(x)=\dfrac{x^3+2x+1}{x^2+4}$ e. $m(x)=\dfrac{x^4+2x^3-3x+1}{x-2}$

Graphs of Rational Functions

To sketch the graph of a given rational function $f(x)=\frac{p(x)}{q(x)}$ the following points are important.

1. Determine the domain of the given function.
2. Determine the point(s) at which the graph of $f(x)$ intersects x-axis, if any. These points are called x-intercepts and they can be obtained by solving the equation $p(x)=0$,
3. Determine the point at which the graph of $f(x)$ intersects y-axis, if any. This point is called y-intercept and it can be obtained by solving for y in the expression $y=f(x)$ for $x=0$ provided that, $x=0$ is in the domain of $f(x)$. If $x=0$ is not in the domain of $f(x)$, then the graph of $f(x)$ has no y-intercept.
4. Determine the vertical asymptote(s) if any. That is, find a so that $p(a)\neq 0$ and $q(a)=0$. Then the vertical line $x=a$ is a vertical asymptote to the graph of $f(x)$.
5. Determine the horizontal or oblique asymptotes (if there is any).

Example 2

Sketch the graph of the rational function $f(x) = \frac{1}{x^2}$.

Solution

Domain

$x^2 = 0 \implies x = 0$. Thus, $\text{Dom}(f) = \{x \in \mathbb{R} | x \neq 0\}$.

Intercepts

x-intercept(s): As $1 \neq 0$, the graph of $f(x)$ has no x-intercept.

y-intercept: Since $x = 0$ is not on the domain of $f(x)$, the graph of $f(x)$ has no y-intercept.

Asymptote(s)

Let $p(x) = 1$ and $q(x) = x^2$. Then, $f(x) = \frac{p(x)}{q(x)}$:

i. $p(0) = 1$ and $q(0) = 0$ and this implies $x = 0$ is a vertical asymptote to the graph of $f(x)$.

ii. The degree of $p(x)$ is 0 and the degree of $q(x)$ is 2, that is, the degree of $p(x)$ is less than the degree of $q(x)$, the line $y = 0$ is a horizontal asymptote to the graph of $f(x)$.

You can use the following table to sketch the graph of $f(x)$.

x	-10	-1	-0.1	0.1	1	10
$f(x)$	0.01	1	100	100	1	0.01

The graph of $f(x)$ is given in **Figure 2.2**.

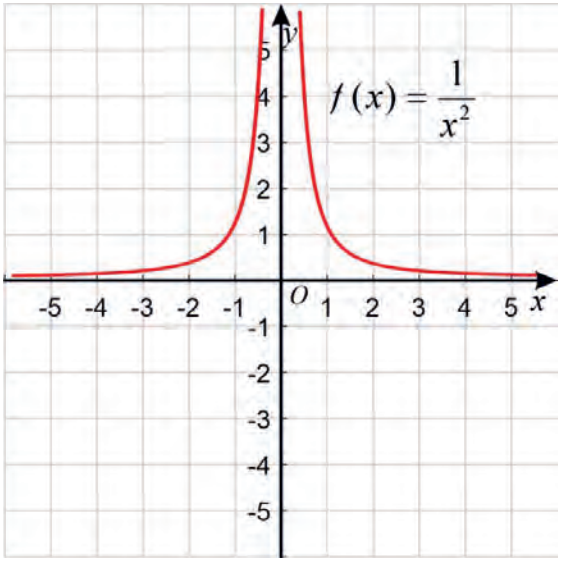


Figure 2.2: Graph of $f(x) = \frac{1}{x^2}$

Example 3

Sketch the graph of the rational function $g(x) = \frac{x+1}{x^2+5x+6}$.

Solution

Domain:

$x^2 + 5x + 6 = 0$ implies $x = -2$ or $x = -3$.

Thus, Dom$(g) = \{x \in \mathbb{R} | x \neq -2, -3\}$.

Intercepts:

x-intercept(s): $x+1=0$ implies $x = -1$.

This implies, $(-1,0)$ is the x-intercept of the graph of $g(x)$.

y-intercept: $g(0) = \frac{0+1}{0^2+5(0)+6} = \frac{1}{6}$.

This implies, $\left(0, \frac{1}{6}\right)$ is the y-intercept of the graph of $g(x)$.

Asymptote(s):

Vertical asymptote(s):

Observe that $-2+1=-1\neq 0$, $(-2)^2+5(-2)+6=0$, $-3+1=-2\neq 0$ and $(-3)^2+5(-3)+6=0$.

This implies, both $x=-2$ and $x=-3$ are vertical asymptotes to the graph of $g(x)$.

Horizontal/Oblique Asymptote:

Since the degree of $x+1$ is 1, the degree of the degree of x^2+5x+6 is 2 and $1<2$, the line $y=0$ is a horizontal asymptote to the graph of $g(x)$.

You can use the following table to sketch the graph of $g(x)$.

x	-4	0	1	2
$g(x)$	$-\frac{3}{2}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{3}{20}$

The graph of $g(x)$ is given in **Figure 2.3**.

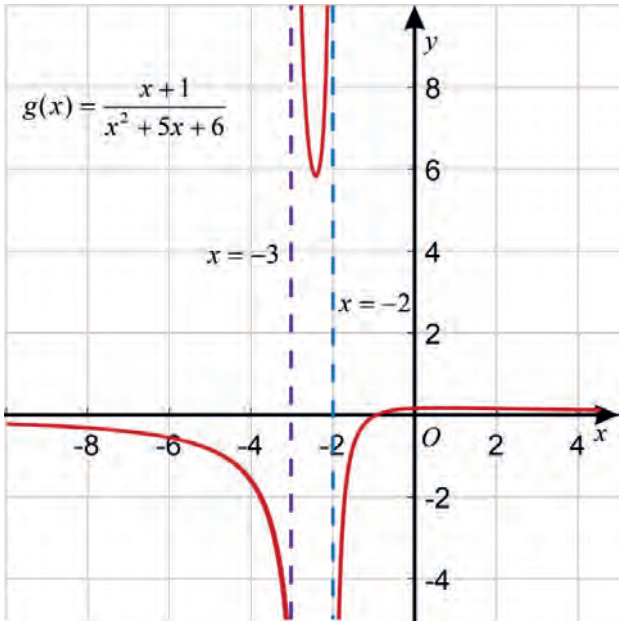


Figure 2.3: Graph of $g(x)=\dfrac{x+1}{x^2+5x+6}$

Example 4

Sketch the graph of the rational function $h(x)=\dfrac{x^2}{x^2+3x+2}$.

Solution

Domain:

$x^2+3x+2=0$ implies $x=-1$ or $x=-2$.

Then, Dom$(h)=\{x\in\mathbb{R}|x\neq -1,-2\}$.

Intercepts:

x-intercept(s): $x^2=0$ implies $x=0$.

This implies, $(0,0)$ is the x-intercept of the graph of $h(x)$.

y-intercept: $\dfrac{0^2}{0^2+3(0)+2}=0$ implies $(0,0)$ is the y-intercept of the graph of $h(x)$.

Asymptotes:

Vertical Asymptote(s):

- $(-1)^2=1\neq 0$ and $(-1)^2+3(-1)+2=0$.

This implies the line $x=-1$ is a vertical asymptote to the graph of $h(x)$.

- $(-2)^2=4\neq 0$ and $(-2)^2+3(-2)+2=0$.

This implies the line $x=-2$ is vertical asymptote to the graph of $h(x)$.

Horizontal/Oblique Asymptote:

Since the degree of x^2 is 2, the degree of x^2+3x+2 is 2 and $2=2$, the line $y=\frac{1}{1}=1$ is the horizontal asymptote to the graph of $h(x)$.

We can use the following table to sketch the graph of $h(x)$.

x	0	1	2	3	4	5
$h(x)$	0	$0.1\dot{6}$	$0.\dot{3}$	0.45	$0.5\dot{3}$	≈ 0.595

Then, the graph of $h(x)$ is given in **Figure 2.4.**

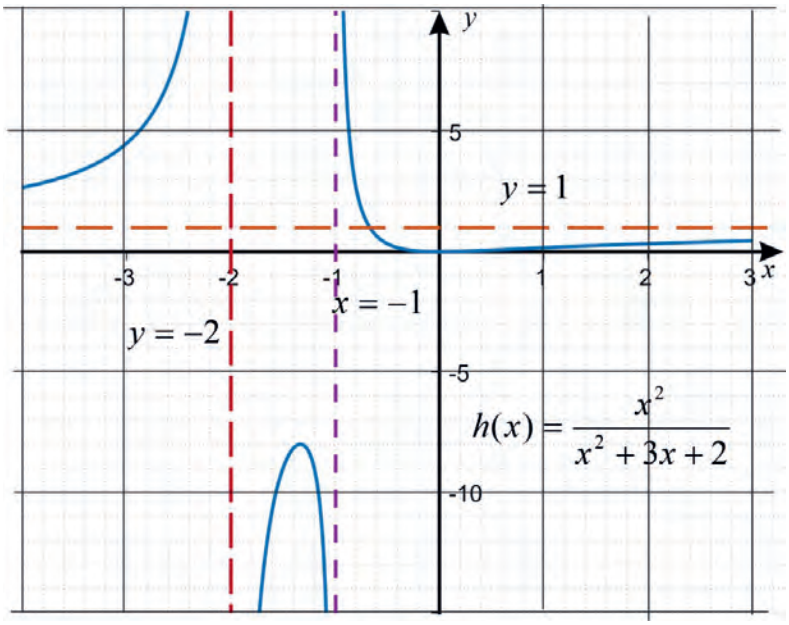


Figure 2.4: Graph of $h(x) = \frac{x^2}{x^2+3x+2}$

Example 5

Sketch the graph of the rational function $l(x) = \dfrac{x^3+1}{x^2+4x+4}$.

Solution

Domain:

$x^2+4x+4=0$ implies $(x+2)^2=0$.

This implies, $x=-2$

Thus, Dom$(l) = \{x \in \mathbb{R} \mid x \neq -2\}$.

Intercepts:

***x*-intercept(s):** $x^3 + 1 = 0$ *i*mplies $x = -1$.

This implies $(-1,0)$ is the x-intercept of the graph of $l(x)$.

***y*-intercept:** For $x=0$, $h(0) = \frac{1}{4}$ and hence $\left(0, \frac{1}{4}\right)$ is the y-intercept of the graph of $l(x)$.

Asymptotes

Vertical Asymptote(s)

$(-2)^3+1=-8+1=-7\neq 0$ and $(-2)^2+4(-2)+4=0$.

This implies, the line $x=-2$ is a vertical asymptote to the graph of $l(x)$.

Horizontal/Oblique Asymptote:

The degree of x^3+1 is 3, the degree of x^2+4x+4 is 2 and $3>2$.

Then, using long division we can write:

$$\frac{x^3+1}{x^2+4x+4}=(x-4)+\frac{12x+17}{x^2+4x+4}$$

Therefore, the line $y=x-4$ is an oblique asymptote to the graph of $l(x)$.

We can use the following table to sketch the graph of $l(x)$.

x	-3	-1	0	2
$g(x)$	-24	0	$\frac{1}{4}$	$\frac{9}{16}$

The graph of l is given in **Figure 2.5.**

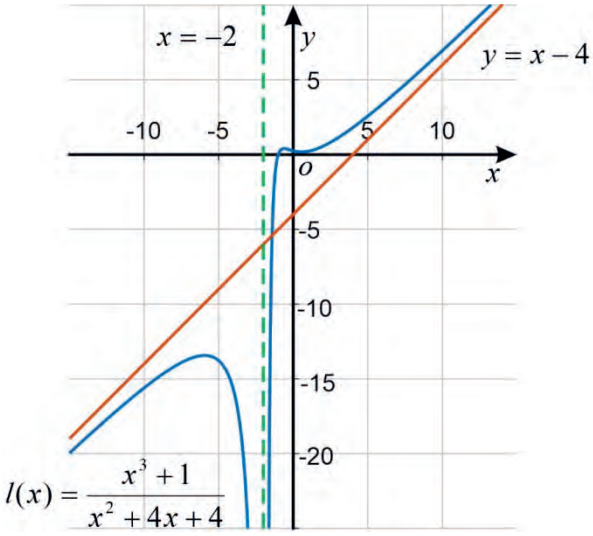


Figure 2.5: Graph of $l(x)=\frac{x^3+1}{x^2+4x+4}$

Exercise 2.11

Sketch the graphs of each of the following rational functions.

a. $f(x)=\dfrac{4}{x+2}$ b. $g(x)=\dfrac{5x}{6x-2}$ c. $h(x)=\dfrac{1}{x^2+x-12}$

d. $l(x)=\dfrac{2x-1}{-2x^2-5x+3}$ e. $m(x)=\dfrac{x^2+1}{x+3}$

2.4 Applications

Rational formulas can be useful tools for representing real-life situations and for finding answers to real problems. Equations representing direct, inverse, and joint variation are examples of rational formulas that can model many real-life situations.

The following steps are used in solving word problems.

Step 1: Understanding the problem

Step 2: Setting up the equation(s)/or expressions using variables.

Step 3: Solving the given equation(s) for the given variables.

Step 4: Interpreting the results.

Example 1

Person A takes 2 hours to plant 50 flower bulbs and person B takes 3 hours to plant 45 flower bulbs. If they are working together, how long should it take them to plant 160 bulbs?

Solution

To solve this problem, consider the planting rate of the two persons; that is, how many bulbs each person can plant in one hour.

A: $\dfrac{50\text{ Bulbs}}{2\text{ Hours}}=\dfrac{25\text{ Bulbs}}{1\text{ Hour}}$ and B: $\dfrac{45\text{ Bulbs}}{3\text{ Hours}}=\dfrac{15\text{ Bulbs}}{1\text{ Hour}}$

Now, combine the hourly rates of the two persons to determine the rate of the work when they work together.

A and B together: $\frac{25 \text{ Bulbs}}{1\text{Hour}} + \frac{15 \text{ Bulbs}}{1 \text{ Hour}} = \frac{40 \text{ Bulbs}}{1 \text{ Hour}}$.

That is, when A and B work together, they plant 40 flower bulbs in an hour.

From the work formula, $W = rt$, we have $r = \frac{W}{t}$, where r is the combined work rate, W is the amount of work that must be done and t is the amount of time it will take to complete the work.

The required information is, how much time it will take to do the required work at the given rate. To do so we have to solve the rational equation.

$$\frac{40}{1} = \frac{160}{t}.$$

Solving this equation for t gives us that $t = 4$ hours.

Thus, if A and B work together, it will take 4 hours for them to plant 160 bulbs together.

Example 2

A sample of 80 g of a particular ice cream contains 15 gram of fat and 20 g of carbohydrate.

a. How much fat does 350 g of the same ice cream contain?

b. How much carbohydrate does 350 g of the same ice cream contain?

Solution

a. If the amount of fat in 350 g is x g, then $\frac{x}{350} = \frac{15}{80}$..

Solving for x gives us, $x = \frac{15}{80} \times 350$ g$= 65.625$ g

b. If the amount of carbohydrate in 350 g is y g, then $\frac{y}{350} = \frac{20}{80}$.

Solving for y gives us, $y = \frac{20}{80} \times 350\text{g} = 87.5\text{g}$.

Example 3

A large mixing tank contains 100 gallons of water into which 5 pounds of sugar have been mixed. A tap will open pouring 10 gallons per minute of water into the tank. At the same time sugar is poured into the tank at a rate of 1 pound per minute. Find the concentration (pounds per gallon) of sugar in the tank after 10 minutes. Compare the concentration after 10 minutes with the concentration at the beginning.

Solution

Let t be the number of minutes since the tap opened. Since the water increases at 10 gallons per minute, and the sugar increases at 1 pound per minute, these are constant rates of change, that is,

$W(t) = 100 + 10t$ in gallons (the amount of water in the tank after t minutes) and

$S(t) = 5 + t$ in pounds (the amount of sugar in the tank after t minutes).

The concentration C will be the ratio of pounds of sugar to gallons of water.

That is, $\mathrm{C}(t) = \dfrac{5+t}{100+10t}$.

Thus, the concentration after 10 minutes is given by

$$\mathrm{C}(10) = \frac{5+10}{100+10(10)} = \frac{15}{200} = 0.075.$$

The concentration at the beginning was $\mathrm{C}(0) = \dfrac{5}{100} = 0.05$.

Since $0.075 > 0.05$, there is high concentrations after 10 minutes than at the beginning.

Exercise 2.12

1. Two pipes can fill a tank in 6 hr. The larger pipe works twice as fast as the smaller pipe. How long would it take each pipe to fill the tank if they worked separately?
2. Legesse bought USD 150 with Birr 7500. At this rate, how many USD can he buy with Birr 30000?
3. Sofia drove her car 45 kilometers and she ran out of gas and had to walk 2 kilometers to a gas station. Her speed of driving is 15 times her speed of walking. If the total time for the drive and walk was $1\frac{1}{2}$ hours, what was Sofia's driving speed?

Problem Solving

1. Zeru can complete a certain job in 5 days and Tulu can complete the same job in 3 days. How long will it take them to complete the job if they work together?
2. In one morning, if the shadow of a building of 20 meters long is 15 meters long, how long is the shadow of a tree of 10 meters long?

Summary

1. Let $p(x)$ and $q(x)$ be two polynomial expressions. An expression of the form $\frac{p(x)}{q(x)}$, where $q(x) \neq 0$, is a rational expression

2. Given two rational expressions $\frac{p(x)}{q(x)}$ and $\frac{r(x)}{s(x)}$ with $q(x) \neq 0$ and $s(x) \neq 0$, we define

   a. $\frac{p(x)}{q(x)} + \frac{r(x)}{s(x)} = \frac{p(x)s(x)+r(x)q(x)}{q(x)s(x)}$

   b. $\frac{p(x)}{q(x)} - \frac{r(x)}{s(x)} = \frac{p(x)s(x)-r(x)q(x)}{q(x)s(x)}$

   c. $\frac{p(x)}{q(x)} \times \frac{r(x)}{s(x)} = \frac{p(x)r(x)}{q(x)s(x)}$

   d. $\frac{p(x)}{q(x)} \div \frac{r(x)}{s(x)} = \frac{p(x)}{q(x)} \times \frac{s(x)}{r(x)} = \frac{p(x)s(x)}{q(x)r(x)}, (r(x) \neq 0)$

3. A function f that can be expressed in the form $f(x) = \frac{p(x)}{q(x)}$, where both $p(x)$ and $q(x)$ are polynomials and $q(x) \neq 0$, is called a rational function.

4. The domain of a rational function $f(x) = \frac{p(x)}{q(x)}$ is the set of all real numbers such that $q(x)$ is not zero.

5. Given a rational function $f(x) = \frac{p(x)}{q(x)}$;

   a. If $f(x) \to \infty$ or $f(x) \to -\infty$ as $x \to a$ from the left or $x \to a$ from the right, then the line $x = a$ is called a vertical asymptote to the graph of $f(x)$.

   b. If $f(x) \to b$ as $x \to \infty$ or as $x \to -\infty$, then the line $y = b$ is called a horizontal asymptote to the graph of $f(x)$.

Review Exercise

1. Find the domain and simplify each of the following rational expressions.

   a. $\dfrac{x-1}{x^2-1}$ b. $\dfrac{x^2-5x+6}{x^2-4}$

   c. $\dfrac{4x^2-36}{x^2-6x+9}$ d. $\dfrac{x^2+1}{x^3+2x^2+x}$

2. Find the domain and perform each of the following indicated operations.

   a. $\dfrac{x-1}{x+2}+\dfrac{x+3}{x^2+6x+5}$ b. $\dfrac{x-1}{x+2}-\dfrac{x+3}{x^2+6x+5}$

   c. $\dfrac{(x-1)}{(x+2)}\times\dfrac{(x+3)}{(x^2+6x+5)}$ d. $\dfrac{x-1}{x+2}\div\dfrac{x+3}{x^2+6x+5}$

3. Write each of the following rational expressions as sum of partial fractions.

   a. $\dfrac{5x+6}{x^2-4}$ b. $\dfrac{x+6}{x^2-4x+4}$

   c. $\dfrac{6x+5}{x^4+4x^2}$ d. $\dfrac{x^2+x+2}{x^2+6x+8}$ e. $\dfrac{x^3+2x^2+2}{x^2-x-6}$

4. Solve each of the following rational equations.

   a. $x+\dfrac{4}{x+3}=2$ b. $1-\dfrac{5}{x}-\dfrac{6}{x^2}=0$ c. $\dfrac{3x}{x+4}+\dfrac{2}{x^2+6x+8}=\dfrac{1}{x+2}$

5. Solve each of the following rational inequalities.

   a. $\dfrac{x+3}{x-1}>0$ b. $\dfrac{2x}{x-4}\le 1$ c. $\dfrac{1}{3}-\dfrac{2}{x^2}\ge\dfrac{5}{3x}$ d. $\dfrac{5}{x-1}\ge\dfrac{4}{x+2}$

6. For each of the following rational functions:

   i. determine the domain;

   ii. find the x-intercept(s) and y-intercept of the graph of the function, if any;

   iii. find the vertical asymptote(s) and horizontal/oblique asymptote to the graph of the function, if any;

iv. sketch the graph of each of the following functions.

a. $f(x)=\dfrac{1}{x-4}$ b. $g(x)=\dfrac{x+2}{x^2+5x+6}$

c. $h(x)=\dfrac{x^2+1}{x^2-4x+4}$ d. $k(x)=\dfrac{x+6}{x^2+4}$ e. $l(x)=\dfrac{x^3+6x^2+5x}{x^2-4x+4}$

7. Let The ratio of boys to girls in a certain high school is 4: 5. If the total number of students in the school is 1260, find the number of boys and the number of girls in the school.

UNIT 3

MATRICES

Unit Outcomes

By the end of this unit, you will be able to:

- Know basic concepts about matrices.
- Perform operation on matrices.
- Differentiate types of a matrix.
- Know specific ideas, methods and principles concerning matrices.
- Formulate elementary row/column operations.
- Define inverse of an invertible matrix.
- Use elementary row operations to find inverse of square matrices of order 2 and 3.
- Define system of linear equations.
- Solve linear system of equations by using elementary row operations.
- Apply matrix concepts to model, solve and analyze real-world situations.

Unit Contents

3.1 The Concepts of a Matrix

3.2 Operations on Matrices

3.3 Special Types of Matrices

3.4 Elementary Row Operations on Matrices

3.5 Systems of Linear Equations with Two or Three Variables

3.6 Solutions of Systems of Linear Equations

3.7 Inverse of a Square Matrix

3.8 Applications

Summary

Review Exercise

- augmented matrix
- column matrix
- consistent system
- diagonal matrix
- elementary row operations
- inconsistent system
- inverse matrix
- non-singular matrix
- reduced-echelon form
- row echelon form
- row matrix
- scalar matrix
- singular matrix
- skew-symmetric matrix
- symmetric matrix
- transpose of a matrix
- triangular matrix
- zero matrix

Introduction

Systems of linear equations occur in many areas such as in Geometry, Engineering, Business, Economics, Biology, etc. Solving such systems of linear equations is one of the most important applications of mathematics to other fields.

In Unit 3 of Grade 9, you have learned how to solve systems of linear equations in two variables. In this unit, you will learn how to solve systems of linear equations in a more systematic way.

Consider the following two linear equations in the variables x and y, $x - y = 1$ and $2x + y = 7$. A common solution for these two equations can be found by working only on the numbers involved in the two equations. These numbers form a rectangular array of numbers as follows:

$$\begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & 7 \end{pmatrix}$$

Such array of numbers is called a matrix. Matrices (plural form of a matrix) are involved in different applications of mathematics.

3.1 The Concepts of a Matrix

Activity 3.1

1. Consider the seats in your classroom.
   a. How many rows of seats are there in your classroom?
   b. How many columns of seats are there is your classroom?
   c. For each of the students in your class, identify the row and column that the student is found.
2. Consider the following table, which gives the sections and the numbers of male and female students in each section of Grade 11 students of a certain school.

	Male	Female
Section 1	25	26
Section 2	28	24
Section 3	30	20
Section 4	15	35

   a. Find the number of female students in Sections 1, 2, 3 and 4.
   b. Find the number of male students in Sections 1, 2, 3 and 4.

In most classrooms, seats are arranged in rows and columns. In such arrangements, one can assign an ordered pair of numbers to each seat which indicates the position of the row and the position of the column that the seat is found.

The information in Activity 3.1 (2), can be displayed as an array of numbers as follows:

$$\begin{pmatrix} 25 & 26 \\ 28 & 24 \\ 30 & 20 \\ 15 & 35 \end{pmatrix}.$$

This array of numbers is a matrix and a formal definition of matrices is given below.

Definition 3.1

Let m and n be positive integers. A rectangular array of numbers A in $\mathbb{R}$ that is written as:

$$\mathrm{A} = \begin{pmatrix} a_{11} & a_{12} \cdots & a_{1n} \\ a_{21} & a_{22} \cdots & a_{2n} \\ \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} \cdots & a_{mn} \end{pmatrix}$$

is called an m by n matrix (or $m \times n$) matrix in $\mathbb{R}$.

In Definition 3.1:

a. the number m is the number of rows, n is the number of columns and $m \times n$ is called the size of the matrix and it is read as "m by n".

b. The number a_{ij} is called the ij-entry or ij -element of A. That is, numbers in the given array are called elements or entries of A. The ij -entry is an element appearing in the i^{th} row and the j^{th}-column of the matrix.

c. Matrix A can be abbreviated by writing as $\mathrm{A} = (a_{ij})_{m\times n}$.

d. Matrices are usually denoted by capital letters like A, B, C etc and entries of matrices by small letters like a, b, c etc.

Example 1

Find the size and determine its elements/or entries for each of the following matrices.

a. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

b. $B = \begin{pmatrix} -2 & 10 \\ 3 & -1 \\ 0 & 5 \end{pmatrix}$

Solution

a. A has two rows and two columns. Therefore, A is a 2×2 matrix.

The entries of A are:

i. $a_{11} = 1$ is the entry found in the first row and in the first column of A;

ii. $a_{12} = 2$ is the entry found in the first row and in the second column of A;

iii. $a_{21} = 3$ is the entry found in the second row and in the first column of A; and

iv. $a_{22} = 4$ is the entry found in the second row and in the second column of A.

b. B has three rows and two columns. Therefore, B is a 3×2 matrix.

The entries of B are:

i. $b_{11} = -2$ is the entry found in the first row and in the first column of B;

ii. $b_{12} = 10$ is the entry found in the first row and in the second column of B;

iii. $b_{21} = 3$ is the entry found in the second row and in the first column of B;

iv. $b_{22} = -1$ is the entry found in the second row and in the second column of B;

v. $b_{31} = 0$ is the entry found in the third row and in the first column of B; and

vi. $b_{32} = 5$ is the entry found in the third row and in the second column of B.

Example 2

Find the size of the matrix and determine its elements/or entries for each of the following matrices:

a. $C = \begin{pmatrix} 1 \\ 0 \\ -2 \\ \frac{1}{4} \end{pmatrix}$. b. $D = \left(2, 1, 3, 0, -1, \frac{1}{4}\right)$ c. $E = (-4)$.

Solution

a. Matrix C has four rows and one column. Therefore, C is a 4×1 matrix.

The entries of C are:

i. $c_{11} = 1$ is the entry found in the first row and in the first column of C;

ii. $c_{21} = 0$ is the entry found in the second row and in the first column of C;

iii. $c_{31} = -2$ is the entry found in the third row and in the first column of C; and

iv. $c_{41} = \frac{1}{4}$ is the entry found in the fourth row and in the first column of C.

b. Matrix D has one row and six columns. Therefore, D is a 1×6 matrix.

The entries of D are:

i. $d_{11} = 2$ is the entry found in the first row and in the first column of D;

ii. $d_{12} = 1$ is the entry found in the first row and in the second column of D;

iii. $d_{13} = 3$ is the entry found in the first row and in the third column of D;

iv. $d_{14} = 0$ is the entry found in the first row and in the fourth column of D;

v. $d_{15} = -1$ is the entry found in the first row and in the fifth column of D; and

vi. $d_{16} = \frac{1}{4}$ is the entry found in the first row and in the sixth column of D.

c. Matrix E has one row and one column. Therefore, E is a 1×1 matrix.

The entry of E is $e_{11} = -4$, the entry found in the first row and in the first column of E.

Exercise 3.1

1. Let $A=\begin{pmatrix}1 & 2 & 1 & 0\\ 3 & 4 & 5 & -1\\ 2 & 2 & 7 & 10\end{pmatrix}$. Then determine each of the following entries (if it exists).

   a. a_{11} b. a_{23} c. a_{32} d. a_{41} e. a_{14}

2. Determine the size and the elements/or entries of each of the following matrices

   a. $A=\begin{pmatrix}2 & 3\\ 3 & 1\\ 1 & 1\end{pmatrix}$ b. $B=\begin{pmatrix}1 & 2 & 1\\ -1 & 4 & 0\end{pmatrix}$ c. $C=\begin{pmatrix}1 & 2 & 1 & 3\end{pmatrix}$

   d. $D=\begin{pmatrix}1\\ 1\\ 2\\ 2\end{pmatrix}$ e. $E=(5)$

3. Let $A=(a_{ij})_{2\times 3}$ and $a_{ij}=j-i$. Then determine A.

3.2 Operations on Matrices

Applications of matrices involve different operations on matrices and some of these operations are addition, subtraction and multiplication of matrices by scalars and matrix multiplications. These operations will be considered in this section.

3.2.1 Equality of Matrices

Activity 3.2

Consider the matrices $A=\begin{pmatrix}1 & 1\\ 2 & 2\end{pmatrix}$, $B=\begin{pmatrix}3 & 2\\ 1 & 4\end{pmatrix}$ and $C=(c_{ij})_{2\times 2}$, where $c_{ij}=i$ for $i=1,2$ and $j=1,2$.

a. Compare each of the corresponding entries of A and B.

b. Compare each of the corresponding entries of A and C.

From your responses in Activity 3.2, you have observed that matrices A and C have the same size and the corresponding entries of A and C are equal and such matrices are called equal matrices.

For most operations on matrices, equality of two matrices is required and it is defined as follows.

Definition 3.2

Two matrices $A=(a_{ij})_{m\times n}$ and $B=(b_{ij})_{m\times n}$ with the same size are said to be equal, written as A=B, if their corresponding entries are equal. That is, A = B, if $a_{ij}=b_{ij}$ for $i=1,2,\ldots,m$ and $j=1,2,\ldots,n$.

Let $A=\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ and $B=\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$. If A = B, then $a_{11}=b_{11}$, $a_{12}=b_{12}$, $a_{21}=b_{21}$ and $a_{22}=b_{22}$.

Example

Find the values of the unknowns in each of the following, if A = B.

1. $A=\begin{pmatrix} 1 & 2 \\ -3 & a \end{pmatrix}$ and $B=\begin{pmatrix} 1 & b \\ -3 & 4 \end{pmatrix}$
2. $A=\begin{pmatrix} a+b & -2 \\ 5 & 1 \end{pmatrix}$ and $B=\begin{pmatrix} 3 & -2 \\ 5 & a-b \end{pmatrix}$
3. $A=\begin{pmatrix} a+b & 5 & e \\ 3 & -1 & 4 \\ 0 & 1 & 9 \end{pmatrix}$ and $B=\begin{pmatrix} 1 & 5 & 7 \\ a-b & -1 & 4 \\ d & 1 & c \end{pmatrix}$

Solution

1. If A = B, then the corresponding entries of the two matrices are equal.
   That is, $a = 4$ and $b = 2$.
2. If A = B, then the corresponding entries of the two matrices are equal.

That is, $a + b = 3$ and $a - b = 1$. Adding the two equations gives you $2a = 4 \Longrightarrow a = 2$.

Then solving for b gives you $b = 3 - 2 = 1$.

3. If A = B, then the corresponding entries of the two matrices are equal.

This implies $a+b=1$, $a-b=3$, $c = 9$, $d = 0$ and $e = 7$.

Then let us solve the two equations

$$\begin{aligned} a+b&=1\\ a-b&=3 \end{aligned}.$$

Adding the two equations gives $2a = 4$ which implies $a = 2$. Then substituting $a = 2$ in either of the equations $a + b = 1$ or $a - b = 3$ and solving for b gives us $b = -1$.

Therefore, if A = B, then $a = 2, b = -1, c = 9, d = 0$ and $e = 7$.

Note that equality of matrices are defined only when the two matrices have the same size.

Exercise 3.2

1. Let $\mathrm{A} = \begin{pmatrix} 3 & a & 2 \\ b & 1 & 5 \\ 2 & 4 & c \end{pmatrix}$ and $\mathrm{B} = \begin{pmatrix} d & 5 & 2 \\ 4 & e & 5 \\ f & 4 & 6 \end{pmatrix}$. If A = B, then determine the values of a, b, c, d, e and f.

2. Let $\mathrm{A} = \begin{pmatrix} 8 & a+b \\ 5 & -1 \end{pmatrix}$ and $\mathrm{B} = \begin{pmatrix} 8 & 3 \\ 5 & a-b \end{pmatrix}$. If $\mathrm{A} = \mathrm{B}$, then determine the values of a and b.

3. Let $A=\begin{pmatrix} a+b & 2 & 3 \\ a-b & 0 & -1 \\ 4 & 5 & 6 \end{pmatrix}$ and $B=\begin{pmatrix} 2 & 2 & c \\ 4 & 0 & -1 \\ 4 & d & 6 \end{pmatrix}$. If $A=B$, then determine the values of a, b, c and d.

3.2.2 Additions of Matrices

Activity 3.3

The following two tables give the numbers of child and adult shoes of both sexes, male and female, of a certain company in two different shops: Shop A and Shop B.

Number of Shoes in Shop A

	Male	Female
Child	65	42
Adult	111	154

Number of Shoes in Shop B

	Male	Female
Child	15	21
Adult	19	28

a. Find the number of child and adult shoes of both sexes that the company has in both shops and present your solutions in a table form.

b. The following table gives the number of shoes of each type the company sells in a particular week from Shop A. Find the number of shoes of each type remaining in Shop A at the end of the week.

Shop A

	Male	Female
Child	33	20
Adult	90	120

In Activity 3.3, observe that,

a. to obtain the number of shoes of each type in both shops, you have added the corresponding entries in the two tables;

b. to obtain the number of shoes of each type in Shop A after a sell in the given week, you have subtracted the corresponding entries in the two tables in the proper order.

This leads us to define the first operation on matrices, addition of matrices.

Addition of two matrices is defined provided that the two matrices have the same size. Thus, the sum of the two matrices is a matrix of the same size as the two given matrices and its entries are obtained by adding the corresponding entries of the two matrices.

Definition 3.3

Let m and n be positive integers and $A=(a_{ij})_{m\times n}$ and $B=(b_{ij})_{m\times n}$ be matrices in $\mathbb{R}$. Then

a. the sum of A and B, denoted by A + B, is an $m\times n$ matrix obtained by adding the corresponding entries of A and B.

That is, $A+B=(a_{ij}+b_{ij})_{m\times n}$

b. the subtraction of B from A, denoted by $A-B$, is an $m \times n$ matrix obtained by subtracting the entries of B from the corresponding entries of A.

That is, $A-B=(a_{ij}-b_{ij})_{m\times n}$.

Let $A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B=\begin{pmatrix} e & f \\ g & h \end{pmatrix}$. Then,

i. $A+B=\begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}$ ii. $A-B=\begin{pmatrix} a-e & b-f \\ c-g & d-h \end{pmatrix}$

Example 1

Let $A=\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B=\begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}$. Then, compute A+B and A − B.

Solution

Both A and B are 2×2 matrices and hence both $A + B$ and $A - B$ are defined.

Thus,

a. $A + B = \begin{pmatrix} 1+0 & 2-1 \\ 3+1 & 4+2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 4 & 6 \end{pmatrix}$.

b. $A - B = \begin{pmatrix} 1-0 & 2-(-1) \\ 3-1 & 4-2 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix}$

Example 2

Let $A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 2 & 3 \\ 4 & 1 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 2 & 5 \\ 1 & -4 & 3 \\ 0 & 0 & -7 \end{pmatrix}$. Then compute A+B and $A - B$.

Solution

Both A and B are 3×3 matrices and hence both $A + B$ and $A - B$ are defined.

Thus,

a. $A+B = \begin{pmatrix} 1-1 & -1+2 & 2+5 \\ 0+1 & 2-4 & 3+3 \\ 4+0 & 1+0 & 5-7 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 7 \\ 1 & -2 & 6 \\ 4 & 1 & -2 \end{pmatrix}$

b. $A-B = \begin{pmatrix} 1-(-1) & -1-2 & 2-5 \\ 0-1 & 2-(-4) & 3-3 \\ 4-0 & 1-0 & 5-(-7) \end{pmatrix} = \begin{pmatrix} 2 & -3 & -3 \\ -1 & 6 & 0 \\ 4 & 1 & 12 \end{pmatrix}$

Activity 3.4

Let $A = \begin{pmatrix} -1 & 2 \\ 3 & -4 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 3 & -2 \\ 4 & 5 & 2 \end{pmatrix}$, $0_{2\times2} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $0_{2\times3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Then find

a. $A + 0_{2\times2}$
b. $0_{2\times2} + A$
c. $B + 0_{2\times3}$
d. $0_{2\times3} + B$

From Activity 3.4, observe that,

(i) $A + 0_{2\times2} = A = 0_{2\times2} + A$;

(ii) $B + 0_{2\times3} = B = 0_{2\times3} + B$.

Thus, matrices $0_{2\times2}$ and $0_{2\times3}$ are called the zero matrices of their respective order and do not change matrices on matrix addition.

Definition 3.4

Let m and n be positive integers. Consider the $m \times n$ matrix whose all entries are zero and we denote this matrix by $0_{m\times n}$ or simply by 0 if the order is clear.

That is, $0_{m\times n} = \begin{pmatrix} 0 & 0 \ldots & 0 \\ 0 & 0 \ldots & 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 \ldots & 0 \end{pmatrix}_{m\times n}$ and it is called the zero matrix.

Observe that, for any an $m \times n$ matrix $A = (a_{ij})_{m\times n}$, if 0 denotes the $m \times n$ zero matrix, then $A + 0 = (a_{ij} + 0)_{m\times n} = (a_{ij})_{m\times n} = (0 + a_{ij})_{m\times n} = 0 + A = A$.

Therefore, the zero matrix is the identity matrix for matrix addition.

Exercise 3.3

1. Let $A = \begin{pmatrix} 1 & 3 \\ -2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. Compute each of the following matrices.

   a. $A + B$ b. $A + C$ c. $A - B$ d. $A - C$

2. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -2 & 0 & 1 \\ 4 & 5 & 6 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 & 4 \\ 4 & 0 & -1 \\ 5 & 2 & -2 \end{pmatrix}$. Compute each of the following matrices.

   a. $A + B$ b. $A - B$ c. $B - A$ d. $A + 0$ e. $0 + B$

3.2.3 Properties of Matrix Addition

Activity 3.5

Let $A = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ 1 & 4 \end{pmatrix}$ and $C = \begin{pmatrix} 0 & -2 \\ 1 & -5 \end{pmatrix}$. Then find

a. $A + B$ b. $B+A$ c. $A+(B+C)$ d. $(A+B)+C$

From Activity 3.5, you observed that:

(i) $A+B = B+A$; that is, addition of the two given matrices is commutative.

(ii) $A+(B+C)=(A+B)+C$; that is, addition of the given three matrices is associative.

In general, addition of matrices is both commutative and associative.

Commutative and Associative Properties of Matrix Addition

Let m and n be positive integers and A, B and C be $m \times n$ matrices in $\mathbb{R}$. Then

a. $A + B = B + A$ (Addition of Matrices is Commutative);

b. $A + (B + C) = (A + B) + C$ (Addition of Matrices is Associative).

Example

Let $A = \begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -2 \\ 2 & -5 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & -1 \\ -2 & 5 \end{pmatrix}$. Then determine

a. $A + B$ b. $B+A$ c. $A+(B+C)$ d. $(A+B)+C$

Solution

a. $A+B = \begin{pmatrix} 1+1 & 1+(-2) \\ 3+2 & 1+(-5) \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 5 & -4 \end{pmatrix}$

b. $B+A = \begin{pmatrix} 1+1 & -2+1 \\ 2+3 & -5+1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 5 & -4 \end{pmatrix}$.

c. $B+C=\begin{pmatrix}1+1 & -2+(-1)\\ 2+(-2) & -5+5\end{pmatrix}=\begin{pmatrix}2 & -3\\ 0 & 0\end{pmatrix}$ and

$A+(B+C)=\begin{pmatrix}1+2 & 1+(-3)\\ 3+0 & 1+0\end{pmatrix}=\begin{pmatrix}3 & -2\\ 3 & 1\end{pmatrix}$.

d. $A+B=\begin{pmatrix}2 & -1\\ 5 & -4\end{pmatrix}$ and $(A+B)+C=\begin{pmatrix}2+1 & -1+(-1)\\ 5+(-2) & -4+5\end{pmatrix}=\begin{pmatrix}3 & -2\\ 3 & 1\end{pmatrix}$

From (a) and (b), observe that $A+B=B+A$ and from (c) and (d) observe also that $A+(B+C)=(A+B)+C$.

Exercise 3.4

Let $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$, $B=\begin{pmatrix}2 & 3\\ -1 & -2\end{pmatrix}$ and $C=\begin{pmatrix}1 & 1\\ 2 & 2\end{pmatrix}$. Determine each of the following matrices.

a. $A+B$ b. $B+A$ c. $(A+B)+C$ d. $A+(B+C)$

3.2.4 Scalar Multiplication of Matrices

Activity 3.6

The following table gives the marks of Nigisti and Tadesse out of 50 in three subjects: English, Mathematics and Biology.

	Nigisti	Tadesse
English	36	34
Mathematics	40	42
Biology	38	39

a. Convert the mark of each subject of the two students out of 100.

b. Convert the mark of each subject of the two students in the three subjects out of 25.

In Activity 3.6, to obtain the marks of both students in the three subjects, you have to multiply

a. each mark of both students by 2 for the first question;

b. each mark of both students by $\frac{1}{2}$ for the second question.

In the following definition, multiplying a matrix by a number or scalar is defined.

Definition 3.5

Let $A = (a_{ij})_{m\times n}$ be a matrix in $\mathbb{R}$ and $k \in \mathbb{R}$. The scalar multiple of A by k, denoted by kA, is the $m \times n$ matrix defined by:

$$kA = k\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = \begin{pmatrix} ka_{11} & ka_{12} & \cdots & ka_{1n} \\ ka_{21} & ka_{22} & \cdots & ka_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ k\,a_{m1} & ka_{m2} & \cdots & ka_{mn} \end{pmatrix}.$$

That is, the entries of kA are obtained by multiplying each entry of A by k.

If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and k is a scalar, then $kA = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}$.

Example

Let $A = \begin{pmatrix} 1 & 0 \\ 2 & -3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 & 1 \\ -2 & -1 & 3 \end{pmatrix}$. Then find 3A and $-\frac{2}{3}$B.

Solution

a. $3A = \begin{pmatrix} 3\times 1 & 3\times 0 \\ 3\times 2 & 3\times(-3) \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 6 & -9 \end{pmatrix}$.

b. $-\frac{2}{3}B = \begin{pmatrix} \left(-\frac{2}{3}\right)\times 2 & \left(-\frac{2}{3}\right)\times 1 & \left(-\frac{2}{3}\right)\times 1 \\ \left(-\frac{2}{3}\right)\times(-2) & \left(-\frac{2}{3}\right)\times(-1) & \left(-\frac{2}{3}\right)\times 3 \end{pmatrix} = \begin{pmatrix} -\frac{4}{3} & -\frac{2}{3} & -\frac{2}{3} \\ \frac{4}{3} & \frac{2}{3} & -2 \end{pmatrix}$

Activity 3.7

Let $A=\begin{pmatrix}0 & 1\\ -1 & 2\end{pmatrix}$ and $B=\begin{pmatrix}1 & 1\\ 2 & 0\end{pmatrix}$. Then find,

a. $3(A+B)$ b. $3A+3B$ c. $2(3A)$

d. $(2\times 3)A$ e. $(2+3)A$ f. $2A+3A$ g. $1A$

From your solutions in Activity 3.7, observe that,

a. $3(A+B)=3A+3B$ b. $2(3A)=(2\times 3)A$

c. $(2+3)A=2A+3A$ d. $1A=A$

These properties are true for any two matrices A and B of the same size and for any two scalars r and s.

Properties of Scalar Multiplication of Matrices

Let m and n be positive integers, A and B be $m\times n$ matrices and $r,s\in\mathbb{R}$. Then:

a. $r(A+B)=rA+rB$. (Scalar multiplication is distributive over additions of matrices.)

b. $(r+s)A=rA+sA$. (Scalar multiplications are distributive over additions of numbers.)

c. $(rs)A=r(sA)$

d. $1A=A$

Scalar Multiplication of Matrices and Subtraction of Matrices

If A and B are $m\times n$ matrices, then

i. $(-1)A=-A$ and $A+(-A)=0_{m\times n}=(-A)+A$. This implies $-A$ is additive inverse of A;

ii. the subtraction of two matrices A and B, $A-B$, is given by $A-B=A+(-1)B$.

Exercise 3.5

1. Let $A=\begin{pmatrix}1 & 2\\ 4 & 3\end{pmatrix}$ and $B=\begin{pmatrix}3 & -1\\ 2 & -4\end{pmatrix}$. Then determine each of the following.

   a. $4A$ b. $-\frac{1}{3}B$ c. $2A+3B$

   d. matrix C such that $A+2C=B$ e. $A-\frac{1}{2}B$ f. $3A-2B$

2. Let $A=\begin{pmatrix}0 & 1 & 2\\ 1 & -1 & 4\\ 5 & 1 & 6\end{pmatrix}$ and $B=\begin{pmatrix}1 & 0 & 1\\ 1 & 0 & 0\\ 0 & 0 & 2\end{pmatrix}$. Then find

   a. $3A$ b. $-\frac{1}{2}B$ c. $3A+2B$

   d. matrix C such that $A-2C=B$ e. $A-5B$ f. $3A-2B$

3. Let $A=\begin{pmatrix}1 & -2 & 3\\ 0 & -5 & 2\end{pmatrix}$, $B=\begin{pmatrix}-3 & 2 & 1\\ 4 & 0 & -2\end{pmatrix}$ and $C=\begin{pmatrix}2 & 2 & 0\\ 8 & 0 & -6\end{pmatrix}$.

   Then, determine $2A+(-3)B+\frac{1}{2}C$.

3.2.5 Multiplication of Matrices

Activity 3.8

Consider the following two matrices A and B, where A shows the type and quantity of certain types of food items that Paulos (P) and Meti (M) want to buy and matrix B shows the cost of the items in two supermarkets X and Y.

$$\begin{array}{c} \\ \\ P\\ M \end{array}\begin{array}{c} \begin{array}{ccc} C & L & Po \end{array}\\ \begin{pmatrix} 2 & 4 & 5\\ 1 & 7 & 3 \end{pmatrix} \end{array}$$

$$\begin{array}{c} \\ \\ C\\ L\\ Po \end{array}\begin{array}{c} \begin{array}{cc} X & Y \end{array}\\ \begin{pmatrix} 120 & 110\\ 55 & 60\\ 35 & 30 \end{pmatrix} \end{array}$$

where, C = Cereal Packets, L = Loaves of Bread and Po = Potatoes (Kg)

a. Calculate the shopping bill for items in each supermarket.

b. Where should they buy, in terms of shopping bill, at X or Y?

In Activity 3.8, to determine the shopping bill for items in each supermarket, you have to add the products of the respective entries of rows of matrix A with the columns of matrix B.

Using this idea, multiplications of matrices are defined as follows.

Definition 3.6

Let $A = (a_{ij})_{m\times p}$ and $B = (b_{kl})_{p\times n}$ be matrices in $\mathbb{R}$, where m, n and p are positive integers. The product of A and B, denoted by AB, is the $m \times n$ matrix $AB = (c_{ij})_{m\times n}$, where, $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots .. + a_{ip}b_{pj}$, for $i = 1, \ldots, m$ and $j = 1, \ldots, n$.

That is, the ij^{th} entry of the product AB is the sum of the product of the corresponding entries of the i^{th} row of A with the j^{th} column of B.

1. If $A = (a \quad b)$ and $B = \begin{pmatrix} c \\ d \end{pmatrix}$, then $AB = ac + bd$, a real number.
2. If $A = \begin{pmatrix} a \\ b \end{pmatrix}$ and $B = (c \quad d)$, then AB $= \begin{pmatrix} ac & ad \\ bc & bd \end{pmatrix}$.
3. If A $= \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and B $= \begin{pmatrix} d & e \\ f & g \end{pmatrix}$, then AB $= \begin{pmatrix} ad+bf & ae+bg \\ cd+df & ce+dg \end{pmatrix}$

Observe that, if A is an $m \times n$ matrix and B is an $s \times t$ matrix, then AB is defined only when $n = s$ and the product will be an $m \times t$ matrix.

That is, the product AB is defined only when the number of columns of the first matrix,A, is equal to the number of rows of the second matrix, B.

In this case, we say that the two matrices are compatible for matrix multiplication.

Example 1

Determine AB for each of the following:

a. A $= (1 \quad 2)$ and B $= \begin{pmatrix} 3 \\ 4 \end{pmatrix}$.

b. A $= \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and B $= (3 \quad 4)$.

c. A $= \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and B $= \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}$

d. A $= \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix}$ and B $= \begin{pmatrix} 0 & 1 & -2 \\ 4 & 9 & 3 \end{pmatrix}$.

Solution

a. A is a 1×2 matrix and B is a 2×1 matrix, then the product AB is a 1×1 matrix or a real number.

That is, $AB = \begin{pmatrix}1 & 2\end{pmatrix}\begin{pmatrix}3\\4\end{pmatrix} = (1 \times 3) + (2 \times 4) = 3 + 8 = 11.$

b. A is a 2×1 matrix and B is a 1×2 matrix, then the product AB is a 2×2. That is $AB = \begin{pmatrix}1\\2\end{pmatrix}\begin{pmatrix}3 & 4\end{pmatrix} = \begin{pmatrix}1 \times 3 & 1 \times 4\\2 \times 3 & 2 \times 4\end{pmatrix} = \begin{pmatrix}3 & 4\\6 & 8\end{pmatrix}.$

c. A is a 2×2 matrix and B is a 2×2 matrix, then the product AB is a 2×2.

That is $AB = \begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}\begin{pmatrix}5 & 7\\6 & 8\end{pmatrix} = \begin{pmatrix}1(5)+2(6) & 1(7)+2(8)\\3(5)+4(6) & 3(7)+4(8)\end{pmatrix} = \begin{pmatrix}17 & 23\\39 & 53\end{pmatrix}.$

d. A is a 2×2 matrix and B is a 2×3 matrix, then the product AB is defined and it is a 2×3 matrix, given by

$$AB = \begin{pmatrix}1(0)+3(4) & 1(1)+3(9) & 1(-2)+3(3)\\2(0)+7(4) & 2(1)+7(9) & 2(-2)+7(3)\end{pmatrix} = \begin{pmatrix}12 & 28 & 7\\28 & 65 & 17\end{pmatrix}.$$

Note that, for matrices A and B in (d) of Example 1, the product BA is not defined, because B is a 2×3 matrix and A is 2×2 matrix.

That is, the number of columns of B is not equal to the number of rows of A.

Example 2

Consider the following two matrices $A = \begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}$ and $B = \begin{pmatrix}1 & 1\\2 & 2\end{pmatrix}$.

Then, $AB = \begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}\begin{pmatrix}1 & 1\\2 & 2\end{pmatrix} = \begin{pmatrix}3 & 3\\3 & 3\end{pmatrix}$ and $BA = \begin{pmatrix}1 & 1\\2 & 2\end{pmatrix}\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix} = \begin{pmatrix}2 & 2\\4 & 4\end{pmatrix}$.

This implies both AB and BA are defined, but $AB \neq BA$.

The example illustrates that, in general, multiplication of matrices is not commutative.

Example 3

Let $A = (a_{ij})_{3\times 2}$ and $B = (b_{ij})_{m\times n}$. Find numbers m and n such that

a. AB is defined;

b. BA is defined;

c. both AB and BA are defined.

Solution

It is given that A is a 3 by 2 matrix.

a. the product AB is defined, if $m = 2$ and n can be any positive integers;

b. the product BA is defined, if $n = 3$ and m can be any positive integers;

c. both products AB and BA are defined only when $m = 2$ and $n = 3$.

Exercise 3.6

Compute the product AB for each of the following cases.

a. $A = \begin{pmatrix}1 & 2\end{pmatrix}$ and $B = \begin{pmatrix}4\\3\end{pmatrix}$.

b. $A = \begin{pmatrix}1\\2\end{pmatrix}$ and $B = \begin{pmatrix}1 & 3\end{pmatrix}$

c. $A = \begin{pmatrix}3 & 1\\2 & 4\end{pmatrix}$ and $B = \begin{pmatrix}2 & 1\\3 & 7\end{pmatrix}$

d. $A = \begin{pmatrix}1 & 2\\3 & 1\end{pmatrix}$ and $B = \begin{pmatrix}1 & 0 & -2\\2 & 3 & 1\end{pmatrix}$

Properties of Multiplication of Matrices

Definition 3.7

Let $A = (a_{ij})_{m\times n}$ be a matrix. Then A is said to be a square matrix, if the number of rows and the number columns of A are equal, $m = n$. In this case, A is called a square matrix of order n.

Example 4

a. $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is a square matrix of order 2.

b. $B = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ is a square matrix of order 3.

Activity 3.9

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $B = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix}$, $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

Then, compute

a. AI_2 and I_2A **b.** BI_3 and I_3B

In Activity 3.9, you have observed that $AI_2 = A = I_2A$ and $BI_3 = B = I_3B$. That is,

a. I_2 is the identity matrix for multiplication of matrices of order 2;

b. I_3 is the identity matrix for multiplication of matrices of order 3.

Next, we define the identity matrix for matrix multiplication.

Definition 3.8

The square matrix order n of defined by, $I_n = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}_{n\times n}$, where 1 on the main diagonal and 0 other wise is called identity matrix for multiplication.

That is, for any square matrix A of order n, $AI_n = A = I_nA$.

The identity matrix of order 2 is given by $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, the identity matrix of order 3 is $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and so on .

Activity 3.10

Let $A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}$ and $C = \begin{pmatrix} 2 & 2 \\ 1 & 3 \end{pmatrix}$. Then compute

a. $A(B + C)$ and $AB+AC$.
b. $(B+C)A$ and $BA+CA$
c. $A(BC)$ and $(AB)C$
d. $2(AB)$, $A(2B)$ and $(2A)B$.

In your responses in Activity 3.10, you have observed the following properties of matrix multiplication:

a. $A(B + C) = AB+AC$ and $(B+C)A=BA+CA$, (multiplication of matrices is distributive over addition of matrices);
b. $A(BC) = (AB)C$, (matrix multiplication of matrices is associative);
c. $2(AB) = A(2B)=(2A)B$.

These properties are summarized as follows.

Properties of Matrix Multiplication

Let A, B and C be matrices in $\mathbb{R}$ and $k \in \mathbb{R}$. Assume that the indicated operations on matrices are defined. Then

a. $A(B + C) = AB + BC$ (Multiplication of matrices is left distributive over addition).
b. $(A + B)C = AC + BC$ (Multiplication of matrices is right distributive over addition).
c. $A(BC) = (AB)C$ (Multiplication of matrices is associative).
d. $k(AB) = (kA)B = A(kB)$
e. If 0 is the zero matrix of appropriate size, then $0A = 0$ and $A0 = 0$.

From your knowledge in the set of real numbers, recall that:

(i) if a and b are real numbers such that $ab = 0$, then either $a = 0$ or $b = 0$ (zero product rule);

(ii) if a, b and c are real numbers such that $ab = ac$ and $a \neq 0$, then $b = c$.

That is, cancellation of multiplication works on the set of real numbers.

However, these properties are not true for multiplications of matrices in general as it can be seen in the following examples.

Example 5

If $A = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix}$, then find AB.

Solution

$AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and observe that neither A nor B are the zero matrix.

Example 6

If $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$ and $C = \begin{pmatrix} -1 & -1 \\ 3 & 4 \end{pmatrix}$, then find both AB and AC.

Solution

$AB = \begin{pmatrix} 5 & 7 \\ 10 & 14 \end{pmatrix}$ and $AC = \begin{pmatrix} 5 & 7 \\ 10 & 14 \end{pmatrix}$.

Then observe that, $AB = AC$, but $B \neq C$.

Exercise 3.7

1. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$ and $C = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. Then find:
   a. AB b. BA c. (AB)C
   d. A(BC) e. I_2A f. BI_2
2. Let A be a 2×3 matrix and B be an $m \times n$ matrix. Find m and n so that:
   a. AB is defined b. BA is defined c. both AB and BA are defined
3. Let $A = \begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix}$. Then determine AB.
4. Let $A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -1 \\ 2 & 2 \end{pmatrix}$ and $C = \begin{pmatrix} 4 & 0 \\ -1 & 1 \end{pmatrix}$. Determine AB and AC.

3.2.6 Transpose of a Matrix and Its Properties

Activity 3.11

Let $A=\begin{pmatrix}1 & 2 & 3\\ -4 & 3 & 0\end{pmatrix}$ and $B=\begin{pmatrix}1 & -4\\ 2 & 3\\ 3 & 0\end{pmatrix}$. Can you mention how B is obtained from A?

From your responses in Activity 3.11, you have observed that the rows of A are columns of B and the columns of A are rows of B.

That is, B is obtained from A by changing rows of A to columns of B and columns of A to rows of B and this process is called transposing a matrix.

Definition 3.9

Let $A=(a_{ij})_{m\times n}$ be an $m\times n$ matrix. Then the matrix $B=(b_{ji})_{n\times m}$ where $b_{ji}=a_{ij}$ for $1\le j\le n$ and $1\le i\le m$ is called the transpose of A and denoted by A^t.

That is, if $A=\begin{pmatrix}a_{11} & a_{12} & \dots & a_{1n}\\ a_{21} & a_{22} & \dots & a_{2n}\\ \vdots & \vdots & \ddots & \vdots\\ a_{m1} & a_{m2} & \dots & a_{mn}\end{pmatrix}$, then $A^t=\begin{pmatrix}a_{11} & a_{21} & \dots & a_{m1}\\ a_{12} & a_{22} & \dots & a_{m2}\\ \vdots & \vdots & \ddots & \vdots\\ a_{1n} & a_{2n} & \dots & a_{mn}\end{pmatrix}$.

Example 1

Find the transpose of each of the following matrices.

a. $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$

b. $B=\begin{pmatrix}1 & -4\\ 3 & 0\\ 2 & 1\end{pmatrix}$

Solution

a. $A^t=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}^t=\begin{pmatrix}1 & 3\\ 2 & 4\end{pmatrix}$

b. $B^t=\begin{pmatrix}1 & -4\\ 3 & 0\\ 2 & 1\end{pmatrix}^t=\begin{pmatrix}1 & 3 & 2\\ -4 & 0 & 1\end{pmatrix}$.

Activity 3.12

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & -1 \\ 2 & -2 \end{pmatrix}$. Then find,

a. $(A + B)^t$ and $A^t + B^t$ **b.** $(3A)^t$ and $3A^t$

c. $(AB)^t$ and B^tA^t **d.** $(A^t)^t$

From your response in Activity 3.12, observe that;

a. $(A + B)^t = A^t + B^t$ b. $(3A)^t = 3A^t$

c. $(AB)^t = B^tA^t$ d. $(A^t)^t = A$.

These properties of transposes of matrices are true in general and they are given as follows.

Properties of Transposes of Matrices

Let A and B be matrices such that the given operations are defined and $k \in \mathbb{R}$. Then

a. $(A + B)^t = A^t + B^t$ b. $(kA)^t = kA^t$ c. $(AB)^t = B^tA^t$ d. $(A^t)^t = A$

Example 2

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$. Then, compute $(A + B)^t$, $5A^t$, $(AB)^t$ and $(B^t)^t$

Solution

$A^t = \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$ and $B^t = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$. Then,

a. $(A+B)^t = A^t + B^t = \begin{pmatrix} 1+0 & 3+1 \\ 2+2 & 1+0 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}$.

b. $5A^t = \begin{pmatrix} 5 \times 1 & 5 \times 3 \\ 5 \times 2 & 5 \times 1 \end{pmatrix} = \begin{pmatrix} 5 & 15 \\ 10 & 5 \end{pmatrix}$.

c. $(AB)^t = B^tA^t = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} (0 \times 1) + (1 \times 2) & (0 \times 3) + (1 \times 1) \\ (2 \times 1) + (0 \times 2) & (2 \times 3) + (0 \times 1) \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 6 \end{pmatrix}$

d. $(B^t)^t = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}^t = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} = B$.

Exercise 3.8

1. Let $A=\begin{pmatrix}1 & 3\\ 4 & 5\end{pmatrix}$ and $B=\begin{pmatrix}0 & 1\\ 2 & 3\end{pmatrix}$ Then find

   a. A^t b. B^t c. $(A+B)^t$

   d. $(AB)^t$ e. $(B^t)^t$ f. $(4AB)^t$

2. Let $A=\begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{pmatrix}$ and $B=\begin{pmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 2\end{pmatrix}$. Then find

   a. A^t b. $(A+B)^t$ c. B^t

   d. $(AB)^t$ e. $(B^t)^t$ f. $(2AB)^t$

3.3 Special Types of Matrices

3.3.1 Symmetric and Skew Symmetric Matrices

Activity 3.13

Let $A=\begin{pmatrix}1 & 2\\ 2 & 4\end{pmatrix}$ and $B=\begin{pmatrix}0 & 2\\ -2 & 0\end{pmatrix}$. Find

a. A^t b. B^t c. $B+B^t$

From your responses in Activity 3.13, observe that

a. $A^t=A$ b. $B^t=-B$ c. $B+B^t=0$

Definition 3.10

Let $A=(a_{ij})_{n\times n}$ be a square matrix of order n. Then A is said to be

a. a symmetric matrix, if $A=A^t$.

   That is, A is a symmetric matrix, if $a_{ij}=a_{ji}$ for $1\le i,j\le n$.

b. a skew-symmetric matrix, if $A=-A^t$.

   That is, A is a skew symmetric matrix if $A+A^t=0_{n\times n}$.

   Equivalently A is skew symmetric matrix if $a_{ij}=-a_{ji}$ for $1\le i,j\le n$.

Note that, if matrix $A = (a_{ij})_{n\times n}$ is skew symmetric matrix, then $a_{ii} = 0$ for $i = 1, 2, \ldots, n$.

Example 1

Identify the following matrices as symmetric, skew-symmetric or neither.

I_n, for a positive integer n, $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 9 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -2 & 3 \\ 2 & 0 & -4 \\ -3 & 4 & 0 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 0 & 3 \\ 5 & 0 & 7 \\ -3 & 2 & 4 \end{pmatrix}$.

Solution

a. $I_n^t = I_n$ and hence I_n is a symmetric matrix.

b. $A^t = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 9 \end{pmatrix} = A$, hence A is a symmetric matrix.

c. $B^t = \begin{pmatrix} 0 & 2 & -3 \\ -2 & 0 & 4 \\ 3 & -4 & 0 \end{pmatrix}$ and $B^t = -B$. This implies B is a skew-symmetric matrix.

d. $C^t = \begin{pmatrix} 1 & 5 & -3 \\ 0 & 0 & 2 \\ 3 & 7 & 4 \end{pmatrix}$. This implies $C \neq C^t$ and $C \neq -C^t$, hence C is neither a symmetric matrix nor a skew-symmetric matrix.

Example 2

Consider the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ -4 & 3 & 8 \\ 6 & -10 & 5 \end{pmatrix}$. Then find

a. $B = \frac{1}{2}(A + A^t)$ b. $C = \frac{1}{2}(A - A^t)$ c. $B + C$

Solution

The transpose of A is $A^t = \begin{pmatrix} 1 & -4 & 6 \\ 2 & 3 & -10 \\ 3 & 8 & 5 \end{pmatrix}$

Thus, $A + A^t = \begin{pmatrix} 2 & -2 & 9 \\ -2 & 6 & -2 \\ 9 & -2 & 10 \end{pmatrix}$ and $A - A^t = \begin{pmatrix} 0 & 6 & -3 \\ -6 & 0 & 18 \\ 3 & -18 & 0 \end{pmatrix}$

Then,

a. matrix $B = \frac{1}{2}(A + A^t) = \frac{1}{2}\begin{pmatrix} 2 & -2 & 9 \\ -2 & 6 & -2 \\ 9 & -2 & 10 \end{pmatrix} = \begin{pmatrix} 1 & -1 & \frac{9}{2} \\ -1 & 3 & -1 \\ \frac{9}{2} & -1 & 5 \end{pmatrix}$ is a symmetric matrix;

b. matrix $C = \frac{1}{2}(A - A^t) = \frac{1}{2}\begin{pmatrix} 0 & 6 & -3 \\ -6 & 0 & 18 \\ 3 & -18 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 & -\frac{3}{2} \\ -3 & 0 & 9 \\ \frac{3}{2} & -9 & 0 \end{pmatrix}$ is a skew-symmetric matrix.

Observe that $B + C = \frac{1}{2}(A + A^t) + \frac{1}{2}(A - A^t) = A$ hence A is written as sum of a symmetric matrix and a skew-symmetric matrix.

Exercise 3.9

1. Identify each of the following matrices as symmetric, skew-symmetric or neither

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 3 & 0 & 5 \end{pmatrix},\ B = \begin{pmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{pmatrix},\ C = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 5 \end{pmatrix},\ D = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}.$$

2. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 1 & 5 \end{pmatrix}$. Find a symmetric matrix B and a skew-symmetric matrix C so that $A = B + C$

3. Let $A = \begin{pmatrix} 1 & a & 2 \\ 0 & 1 & 0 \\ b & c & 5 \end{pmatrix}$. Find the values a, b, c so that A is a symmetric

3.3.2 Diagonal Matrices and Triangular Matrices

Definition 3.11

Let $A=(a_{ij})_{n\times n}$ be a square matrix of order n.

a. The elements $a_{11}, a_{22}, \ldots, a_{nn}$ are called diagonal elements of A.

b. The sum of the diagonal elements of A is called the **trace** of A and denoted by tr(A). That is, $\text{tr}(A)=a_{11}+a_{22}+\cdots+a_{nn}$

c. A is called a diagonal matrix if $a_{ij}=0$ for all $i\neq j$. That is, A is a diagonal matrix, if

$$A=\begin{pmatrix} a_{11} & 0 & \ldots & 0 \\ 0 & a_{22} & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & a_{nn} \end{pmatrix}$$

d. A is called a scalar matrix if A is a diagonal matrix and $a_{ii}=\alpha$ for some $\alpha\in\mathbb{R}$ and for $i=1,2,\ldots,n$. That is, A is scalar matrix, if

$$A=\begin{pmatrix} \alpha & 0 & \ldots & 0 \\ 0 & \alpha & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \alpha \end{pmatrix}_{n\times n} \quad \text{for some } \alpha\in\mathbb{R}.$$

Definition 3.12

Let $A = (a_{ij})_{n\times n}$ be a square matrix of order n.

a. If $a_{ij} = 0$ for all $i > j$, then A is called an upper triangular matrix. That is,

A is said to be an upper triangular matrix if $A = \begin{pmatrix} a_{11} & a_{12} & \dots a_{1n} \\ 0 & a_{22} & \dots a_{2n} \\ \vdots & \vdots & \ddots \vdots \\ 0 & 0 & \dots a_{nn} \end{pmatrix}$;

b. A is called a strictly upper triangular matrix if $a_{ij} = 0$ for all $i \geq j$. That is,

A is said to be a strictly upper triangular matrix if

$A = \begin{pmatrix} 0 & a_{12} & \dots a_{1(n-1)} & a_{1n} \\ 0 & 0 & \dots a_{2(n-1)} & a_{2n} \\ \vdots & \vdots & \ddots \vdots & \vdots \\ 0 & 0 & \dots 0 & 0 \end{pmatrix}$;

c. A is called a lower triangular matrix if $a_{ij} = 0$ for all $i < j$. That is A is said to be a lower triangular matrix if

$A = \begin{pmatrix} a_{11} & 0 & \dots & 0 & 0 \\ a_{21} & a_{22} & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{n(n-1)} & a_{nn} \end{pmatrix}$;

d. A is called a strictly lower triangular matrix if $a_{ij} = 0$ for all $i \leq j$ That is A is said to be a strictly lower triangular matrix if

$A = \begin{pmatrix} 0 & 0 & \dots & 0 & 0 \\ a_{21} & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{n(n-1)} & 0 \end{pmatrix}$.

Example

Identify each of the following matrices as diagonal, scalar or triangular matrices.

a. $A=\begin{pmatrix}1&0&3\\0&-2&-6\\0&0&5\end{pmatrix}$ b. $B=\begin{pmatrix}0&0&0\\0&0&0\\-1&2&0\end{pmatrix}$ c. $C=\begin{pmatrix}-2&0&0\\0&-2&0\\0&0&-2\end{pmatrix}$

Solution

a. $A=\begin{pmatrix}1&0&3\\0&-2&-6\\0&0&5\end{pmatrix}$ is an upper triangular matrix, because all the entries below the main diagonal are 0.

b. $B=\begin{pmatrix}0&0&0\\0&0&0\\-1&2&0\end{pmatrix}$ is a strictly lower triangular matrix, because all the entries below the main diagonal and on the main diagonal are 0.

c. $C=\begin{pmatrix}-2&0&0\\0&-2&0\\0&0&-2\end{pmatrix}$ is both a scalar and a diagonal matrix, because all the entries below and above the main diagonals are 0 and the entries in the main diagonal are all -2.

Exercise 3.10

1. Identify each of the following matrices as diagonal, scalar or triangular matrices.

$A=\begin{pmatrix}1&2&3\\0&1&0\\0&0&5\end{pmatrix}$, $B=\begin{pmatrix}0&0&0\\-1&0&0\\2&-3&-3\end{pmatrix}$, $C=\begin{pmatrix}5&0&0\\0&5&0\\0&0&5\end{pmatrix}$, $D=\begin{pmatrix}0&0\\0&2\end{pmatrix}$.

2. Let $A=\begin{pmatrix}1&a&2\\0&1&0\\b&c&5\end{pmatrix}$. Find a, b, c such that A is an upper triangular matrix.

3.4 Elementary Row Operations of Matrices

Activity 3.14

Consider the system

$$x + y = 1$$
$$x - 2y = -11$$

a. Multiply the first equation by 2 and add the result to the second equation.

b. Multiply the result in (a) by $\frac{1}{3}$ and solve for x.

The operations you have performed in Activity 3.14 are operations on the rows of the matrix $\begin{pmatrix} 1 & 1 & 1 \\ 1 & -2 & -11 \end{pmatrix}$ of the given system.

A familiar method of solving a set of linear equations is by elimination of the unknowns. We shall systematize this method by working only on the numbers involved in the given set of linear equations.

Definition 3.13

Let m and n be positive integers and A be an $m \times n$ matrix with entries in $\mathbb{R}$. An elementary row operations on A is any one of the following operation.

Swapping: Interchanging two rows of matrix A:

Swapping of the i^{th} row and the j^{th} row of A is denoted by $R_i \leftrightarrow R_j$.

Scaling: Multiplying a row of A by a non-zero constant:

Multiplying the i^{th} row by a nonzero scalar k is denoted by $R_i \to kR_i$.

Pivoting: Adding a nonzero constant multiple of one row of A onto another row of A.

Adding k times the i^{th} row of A onto the j^{th} row is denoted by $R_j \to R_j + kR_i$

Example 1

Let $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$. Apply each of the following elementary row operations one after the other on A and obtain a new matrix.

a. $R_1 \leftrightarrow R_2$ b. $R_2 \to R_2 - 4R_1$ c. $R_2 \to -\frac{1}{10}R_2$

Solution

a. $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}$ (Interchanging R_1 by R_2).

b. $\begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix} \xrightarrow{R_2 \to R_2 - 4R_1} \begin{pmatrix} 1 & 3 \\ 0 & -10 \end{pmatrix}$ (Replacing R_2 by $R_2 - 4R_1$)

c. $\begin{pmatrix} 1 & 3 \\ 0 & -10 \end{pmatrix} \xrightarrow{R_2 \to -\frac{1}{10}R_2} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ (Multiplying R_2 by $-\frac{1}{10}$)

Thus, $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ is an upper triangular matrix.

Example 2

Let $A = \begin{pmatrix} 2 & 4 & 6 \\ 4 & 5 & 6 \\ 2 & 7 & 12 \end{pmatrix}$. Apply appropriate elementary row operations to change A into an upper triangular matrix.

Solution

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 4 & 5 & 6 \\ 2 & 7 & 12 \end{pmatrix} \xrightarrow{R_1 \to \frac{1}{2}R_1} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 7 & 12 \end{pmatrix}$$ (Multiplying R_1 by $\frac{1}{2}$)

$$\xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - 4R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 3 & 6 \end{pmatrix}$$ (Replacing R_2 by $R_2 - 4R_1$ and R_3 by $R_3 - 2R_1$)

$$\xrightarrow{R_3 \to R_3 + R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix} \text{(Replacing } R_3 \text{by } R_3 + R_2\text{)}$$

The matrix $\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix}$ is an upper triangular matrix obtained from A.

Definition 3.14

The inverse operation of an elementary row operation is an elementary row operation of the same type. That is,

a. The inverse of the elementary row operation $R_i \leftrightarrow R_j$ is $R_j \leftrightarrow R_i$, which is again an elementary row operation.

b. For a nonzero real number k, the inverse of the elementary row operation $R_j \longrightarrow kR_j$ is $R_j \longrightarrow \frac{1}{k}R_j$ which is again an elementary row operation.

c. For a scalar k the inverse of the elementary row operation $R_i \longrightarrow R_i + kR_j$ is $R_i \longrightarrow R_i - kR_j$ which is also an elementary row operation.

Example 3

Let $A = \begin{pmatrix} 1 & 2 \\ 1 & 5 \end{pmatrix}$. Then, find the inverse of the following elementary row operations:

$$A = \begin{pmatrix} 1 & 2 \\ 1 & 5 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 5 \\ 1 & 2 \end{pmatrix} \text{ (Interchanging } R_1 \text{ and } R_2\text{)}$$

$$\xrightarrow{R_2 \to R_2 - R_1} \begin{pmatrix} 1 & 5 \\ 0 & -3 \end{pmatrix} \text{ (Replacing } R_2 \text{ by } R_2 - R_1\text{)}$$

$$\xrightarrow{R_2 \to -\frac{1}{3}R_2} \begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix} \text{(Replacing } R_2 \text{by} -\frac{1}{3}R_2\text{)}$$

Solution

Let us apply the inverse elementary row operations of the above three elementary row operations on the matrix $\begin{pmatrix}1 & 5\\0 & 1\end{pmatrix}$ as follows.

$\begin{pmatrix}1 & 5\\0 & 1\end{pmatrix} \xrightarrow{R_2 \longrightarrow -3R_2} \begin{pmatrix}1 & 5\\0 & -3\end{pmatrix}$ (Multiplying R_2 by -3, which is the inverse of the elementary row operation , multiplying $R_2$by $-\frac{1}{3}$).

$\xrightarrow{R_2 \longrightarrow R_2+R_1} \begin{pmatrix}1 & 5\\1 & 2\end{pmatrix}$ (Replacing $R_2$by $R_2 + R_1$, which is the inverse of the elementary row operation, replacing $R_2$by $R_2 - R_1$).

$\xrightarrow{R_2 \leftrightarrow R_1} \begin{pmatrix}1 & 2\\1 & 5\end{pmatrix}$ (Interchanging $R_2$and R_1)

In Example 3, observe that

a. $R_2 \longrightarrow -3R_2$ is the inverse of $R_2 \longrightarrow -\frac{1}{3}R_2$

b. $R_2 \longrightarrow R_2 + R_1$ is the inverse of $R_2 \longrightarrow R_2 - R_1$ and

c. $R_2 \leftrightarrow R_1$ is the inverse of $R_1 \leftrightarrow R_2$

Elementary Column Operations

There are also column operations on matrices.

Swapping: Interchanging two columns of a matrix.

Scaling: Multiplying a column of a matrix by a non-zero constant.

Pivoting: Adding a constant multiple of one column of a matrix onto another column.

However, in this textbook, we use only elementary row operations in solving system of linear equations.

Exercise 3.11

1. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3 & 6 & -5 \end{pmatrix}$. Find the matrix obtained from A by applying the following elementary row operation.

   a. $R_1 \leftrightarrow R_3$ b. $R_2 \rightarrow R_2 - 4R_1$ c. $R_3 \rightarrow \frac{1}{3}R_3$

2. Let $B = \begin{pmatrix} 1 & 1 & 2 \\ -2 & 1 & -5 \\ 2 & 3 & -1 \end{pmatrix}$. Determine the inverse of each of the following elementary row operations on B.

   a. $R_2 \leftrightarrow R_3$ b. $R_3 \rightarrow R_3 + 2R_1$ c. $R_2 \rightarrow 4R_2$

Row Echelon Form (REF) of a Matrix

Definition 3.15

Given two matrices A and B of the same size, A and B are said to be row-equivalent if one of the matrices is obtained from the other by successive applications of a finite number of elementary row operations and this relation is denoted by $A \cong B$.

Example 4

Let $A = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}$. Find three matrices that are row equivalent to A.

Solution

$$A=\begin{pmatrix}1 & -2\\ 1 & 1\end{pmatrix}\xrightarrow{R_1\leftrightarrow R_2}\begin{pmatrix}1 & 1\\ 1 & -2\end{pmatrix}\ (\text{Interchanging } R_1 \text{and } R_2)$$

$$\xrightarrow{R_2\to R_2-R_1}\begin{pmatrix}1 & 1\\ 0 & -3\end{pmatrix}\ (\text{Replacing } R_2 \text{ by } R_2-R_1)$$

$$\xrightarrow{R_2\to -\frac{1}{3}R_2}\begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}\ (\text{Replacing } R_2 \text{ by } -\frac{1}{3}R_2)$$

Then the matrices $\begin{pmatrix}1 & 1\\ 1 & -2\end{pmatrix}$, $\begin{pmatrix}1 & 1\\ 0 & -3\end{pmatrix}$ and $\begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}$ are all row equivalent to matrix A.

Note

Row equivalence of matrices helps us to transform a given matrix to a matrix with greater number of zeros than the original matrix and this simplifies operations on matrices.

Definition 3.16 (Row Echelon Form (REF))

An $m \times n$ matrix A is said to be in it **Row Echelon Form (REF)** if it satisfies the following.

(a) All rows consisting entirely zeros, if any, are at the bottom of the matrix.

(b) If R_i and $R_{(i+1)}$ are any two successive nonzero rows of A, the number of zeros preceding the leading element of the $(i+1)^{th}$ row is greater than the number of such zeros in i^{th} row.

Note

By a zero row we mean that a row in which all the entries of the row are zero.

The matrix $A=\begin{pmatrix}1 & a & b & c & d\\ 0 & 0 & 2 & e & f\\ 0 & 0 & 0 & 0 & g\end{pmatrix}$ is in Row Echelon Form.

Example 5

Determine if each of the following matrices are in Row Echelon Form.

$$A=\begin{pmatrix}1&2&2\\0&1&1\\0&0&1\end{pmatrix}, B=\begin{pmatrix}1&-1&2\\0&1&0\\0&0&1\\0&0&0\end{pmatrix}, C=\begin{pmatrix}1&1&-1&0\\0&0&0&0\\0&0&1&3\\0&0&0&1\end{pmatrix},$$

$$D=\begin{pmatrix}1&1&0&4\\0&1&0&0\\0&0&1&3\\0&0&1&1\end{pmatrix} \text{ and } E=\begin{pmatrix}1&2&1&-2\\0&2&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix}.$$

Solution

a. Matrices A, B and E are in Row Echelon Form.

b. Matrix C is not in Row Echelon Form, because C has a zero row, the second row, but this zero row is not at the bottom as the third row is not a zero row.

c. Matrix D is not in row echelon form, because the number of zeros preceding the leading element of the third and the fourth row are both two.

Exercise 3.12

Determine if each of the following matrices in **Row Echelon Form**.

a. $\begin{pmatrix}1&2&3\\0&0&6\\1&1&0\end{pmatrix}$ b. $\begin{pmatrix}1&2&1&3\\0&4&5&0\\0&0&0&1\end{pmatrix}$ c. $\begin{pmatrix}1&2&1&2\\0&0&0&3\\0&1&1&4\end{pmatrix}$

Reduced Row Echelon Form (RREF) of a Matrix

Activity 3.15

Consider the matrices $A=\begin{pmatrix}1&2&3\\0&0&0\\0&1&2\end{pmatrix}$ and $B=\begin{pmatrix}1&1&1\\1&2&3\\0&2&1\end{pmatrix}$.

a. Give your reason why matrices A and B are not in Row Echelon Form

b. Apply the elementary row operation $R_2 \leftrightarrow R_3$ on A and obtain matrix C.

c. Apply the elementary row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - 2R_2$ respectively on matrix B and obtain matrix D.

d. What is the relationship between matrices C and D in (b) and (c) above respectively?

The matrices you have obtained in your responses in *Activity 3.15* are matrices in Row Echelon Form. Thus, any matrix is row equivalent to a matrix in Row Echelon Form.

Example 6

Reduce the matrix $A=\begin{pmatrix}1&3&0&2\\2&1&1&2\\-1&0&-2&-4\end{pmatrix}$ into a matrix in Row Echelon Form.

Solution

A is not in REF, because the number of zeros preceding the first nonzero element in the second row is zero which is the same as the number of zeros preceding the first nonzero element in the first row.

Therefore, to reduce A into a matrix in REF, choose appropriate elementary row operations.

$$A=\begin{pmatrix}1&3&0&2\\2&1&1&2\\-1&0&-2&-4\end{pmatrix}\xrightarrow[R_3\to R_3+R_1]{R_2\to R_2-2R_2}\begin{pmatrix}1&3&0&2\\0&-5&1&-2\\0&3&-2&-2\end{pmatrix}$$

$$\xrightarrow{R_3\to R_3+\frac{3}{5}R_2}\begin{pmatrix}1&3&0&2\\0&-5&1&-2\\0&0&-2&-\frac{4}{5}\end{pmatrix}=B.$$

The last matrix, B is in Row Echelon Form (REF), obtained from A by applying the three successive elementary row operations, $R_2\to R_2-2R_2$, $R_3\to R_3+R_1$ and $R_3\to R_3+\frac{3}{5}R_2$.

Definition 3.17

Given positive integers m and n, an $m\times n$ matrix A is said to be in **Reduced Row Echelon Form (RREF)** if A is in **Row Echelon Form (REF)** and the leading element in each nonzero row is 1 and it is the only nonzero number in its column.

Example 7

Determine which of the following matrices are in Reduced Row Echelon Form.

$$I_3=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix},\quad A=\begin{pmatrix}1&0&0\\0&1&2\\0&0&1\end{pmatrix},\quad B=\begin{pmatrix}1&0&2&5\\0&1&2&6\\0&0&0&0\end{pmatrix},$$

$$C=\begin{pmatrix}1&0&0\\0&0&1\\0&1&0\end{pmatrix},\quad D=\begin{pmatrix}1&0&0\\0&0&1\\0&0&0\\0&0&0\end{pmatrix}.$$

Solution

The matrices I_3, B and D are in RREF, because all the three matrices satisfy the requirements of the definition of a matrix in RREF.

(a) Matrix A is not in RREF, because the leading element of the third row of A, which is 1, is not the only nonzero element in its column.

(b) Matrix C is not in RREF, because the number of zeros preceding the leading element of the second row is two and the number of zeros preceding the leading element of the third row is one.

However, you can reduce both A and C in the above example to RREF by applying appropriate elementary operations.

Activity 3.16

Consider the matrices $A=\begin{pmatrix}1&0&0\\0&1&2\\0&0&1\end{pmatrix}$ and $C=\begin{pmatrix}1&0&0\\0&0&1\\0&1&0\end{pmatrix}$.

a. Apply the elementary row operation $R_2 \to R_2 - 2R_3$ on matrix A.

b. Apply the elementary row operation $R_2 \leftrightarrow R_3$ on matrix C.

In your responses for (a) and (b) of Activity 3.16, observe that the matrices obtained after the proposed elementary operations are in Reduced Row Echelon Form (RREF).

Activity 3.16 also illustrates that you can use proper elementary operations to reduce a matrix into a matrix in Reduced Row Echelon Form (RREF).

Remark

Every $m \times n$ matrix is row equivalent to an $m \times n$ matrix in Reduced Row Echelon Form (RREF).

Example 8

Reduce the matrix $A=\begin{pmatrix}1&0&0&2\\0&1&1&2\\0&0&-2&-4\\0&0&0&0\end{pmatrix}$ into a matrix in Reduced Row Echelon Form.

Solution

A is not in RREF, because

a. the first nonzero element in the third row is -2, not 1;

b. the first nonzero element of the third row is not the only nonzero element in its column.

Therefore, to reduce A into a matrix in RREF, choose appropriate elementary row operations.

$$A=\begin{pmatrix}1&0&0&2\\0&1&1&2\\0&0&-2&-4\\0&0&0&0\end{pmatrix}\xrightarrow{R_3\to-\frac{1}{2}R_3}\begin{pmatrix}1&0&0&2\\0&1&1&2\\0&0&1&2\\0&0&0&0\end{pmatrix}$$

$$\xrightarrow{R_2\to R_2-R_3}\begin{pmatrix}1&0&0&2\\0&1&0&0\\0&0&1&2\\0&0&0&0\end{pmatrix}=B.$$

The last matrix, B, obtained from A by applying the two successive elementary row operations, $R_3\to-\frac{1}{2}R_3$ and $R_2\to R_2-R_3$ is in RREF.

Exercise 3.13

1. Reduce each of the following matrices into **Row Echelon Form**.

a. $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 1 & 7 \end{pmatrix}$ b. $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 5 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$ c. $\begin{pmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 1 & 3 \\ 1 & 4 & 1 & 4 \end{pmatrix}$

2. Reduce each of the following matrices into Reduced Row Echelon Form.

a. $\begin{pmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \end{pmatrix}$ b. $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 1 \end{pmatrix}$ c. $\begin{pmatrix} 1 & 2 & 3 & 1 \\ 4 & 5 & 6 & 2 \\ 7 & 8 & 9 & 3 \end{pmatrix}$

3.5 Systems of Linear Equations with Two or Three Variables

One of the main reasons of studying matrices is to solve systems of linear equations. In this section, you will learn techniques or methods of solving systems of linear equations.

Activity 3.17

The sum of two natural numbers is 30 and the second number is 4 more than the first number. Find the numbers.

In your responses in Activity 3.17, if you let the first number by x and these second number by y, then you need to find a common solution to the two solutions $x + y = 30$ and $y = x + 4$.

The two equations $x + y = 30$ and $y = x + 4$ are called linear equations and the two equations together form a set of linear equations called a system of linear equations.

Definition 3.18

An equation of the form $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$, where $a_1, a_2, \cdots, a_n, b$ are real numbers, is called a linear equation in then variables $x_1, x_2, \cdots, x_n$. The variables $x_1, x_2, \cdots, x_n$ are also called unknowns.

Example 1

a. The equation $2x - y = 5$ is a linear equation in two variables, x and y .

b. The equation $2x + 3y + 4z = 5$ is a linear equation in three variables, x, y and z.

c. The equation $x_1 + \pi x_2 + 4x_3 + x_4 + \frac{1}{3}x_5 = 0$ is a linear equation in five variables, x_1, x_2, x_3, $x_4$and x_5.

Remark

A linear equation does not involve product of variables, quotient of variables or roots of variables and all the variables that occur in the equation are only to power one.

Example 2

Give your reasons why each of the following equations is not a linear equation. $3x^2 + 5y = 7$, $xy - 2z + w = 1$ and $x - \sqrt{y} + 4z = 0$.

Solution

a. The equation $3x^2 + 5y = 7$ is not a linear equation, because it involves x^2.

b. The equation $xy - 2z + w = 1$ in not a linear equation, because it involves xy.

c. The equation $x - \sqrt{y} + 4z = 0$ is not a linear equation, because it involves $\sqrt{y}$.

Activity 3.18

Consider the equation $x - y = 5$.

a. Substitute $x = 2$ and $y = 3$ and determine the result.

b. Substitute $x = 6$ and $y = 1$ and determine the result.

In your responses in Activity 3.18,

a. the ordered pair $(2, 3)$ gives you a false statement;

b. the ordered pair $(6, 1)$ gives you a true statement.

Thus, the ordered pair $(6, 1)$ is a solution to the given equation, whereas the ordered pair $(2, 3)$ is not a solution.

Definition 3.19

An $n-$tuple of numbers $(c_1, c_2, \cdots, c_n)$ is said to be a solution to a linear equation

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

if the equation is satisfied when we substitute $x_1 = c_1, x_2 = c_2, \cdots, x_n = c_n$ in the equation, that is, if $a_1c_1 + a_2c_2 + \cdots + a_nc_n = b$.

The set of all the possible solutions of a given linear equation is called the **solution set** or the **general solution set** of the given linear equation.

Example 3

Consider the linear equation $2x + y = 4$.

a. Which of the ordered pairs $(2,0), (0,2)$ and $(1,2)$ are solutions of the given linear equation?

b. Find the solution set of the given linear equation.

Solution

a. For $x = 2$ and $y = 0$, we have $(2 \times 2) + 0 = 4 + 0 = 4$.

This implies, the ordered pair $(2,\ 0)$ is a solution to the given linear equation.

For $x = 0$ and $y = 2$, we have $(2 \times 0) + 2 = 0 + 2 \neq 4$.

This implies, the ordered pair $(0,\ 2)$ is not a solution to the given linear equation.

For $x = 1$ and $y = 2$, you have$(2 \times 1) + 2 = 2 + 2 = 4$.

This implies, the ordered pair $(1,\ 2)$ is a solution to the given linear equation.

b. From the given equation if you solve for y in terms of x gives you

$2x + y = 4 \Rightarrow y = -2x + 4$.

This implies, the solution set of the given linear equation is the set of all ordered pairs of numbers (x, y) such that $2x + y = 4$ and $y = 4 - 2x$.

Therefore, the solution set of the given equation is $\{(x, 4 - 2x): x \in \mathbb{R}\}$

Exercise 3.14

1. Give your reasons why each of the following equations is not a linear equation.

   $x + 5y^2 = 7$, $x + yz + w = 0$ and $x - y + 4\sqrt{z} = 0$.

2. Consider the linear equation $x - 3y = 5$.

   a. Which of the ordered pairs $(1,1)$, $(2, -1)$and $(5,0)$ are solutions of the given linear equation?

   b. Find the solution set of the given linear equation.

Augmented Matrix of a System of Linear Equations

The main objective of this section is to find the set of all common solutions of a finite number of linear equations having the same variables.

Definition 3.20

Let $a_{ij}, b_i \in \mathbb{R}$, for $i = 1,2,\ldots,m$ and $j = 1,2,\ldots,n$. A finite set of linear equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

is called a system of m linear equations in n variables; $x_1, x_2, \ldots, x_n$

Activity 3.19

Let $\mathrm{A} = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$, $\mathrm{X} = \begin{pmatrix} x \\ y \end{pmatrix}$ and $\mathrm{B} = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$.

a. Find AX

b. Find the values of x and y such that $\mathrm{A}X = \mathrm{B}$.

In Activity 3.19 part (b), to find the values of x and y, you have to find a solution for the system $\begin{matrix} x + 2y = 1 \\ x + 3y = -4 \end{matrix}$, which is a system of two linear equations in two variables.

Then, the expression $\mathrm{AX} = \mathrm{B}$ is called a matrix form of the system.

Remark

Consider the following system of linear equations.

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

The system can be written in matrix form as $\mathrm{AX} = \mathrm{B}$,where

$\mathrm{A} = (a_{ij})_{m\times n}$, $\mathrm{X} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ and $\mathrm{B} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$

The matrix $A = (a_{ij})_{m\times n}$ is called the coefficient matrix of the system and the matrix

$$(A|B) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right)$$

obtained by adjoining B to A is called the **augmented matrix** of the system.

The coeeficient matrix and the augmented matrix of a given system play important roles in determining the solution set of the given system.

Example 4

Find the coefficient and the augmented matrix of the system.

$$\begin{aligned} x + y + z &= 8 \\ x + 2y &= 5 \\ y + 3z &= 9 \end{aligned}$$

Solution

The system can be rewritten in matrix form as:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ 5 \\ 9 \end{pmatrix}.$$

The matrix $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix}$ is the coefficient matrix of the system and the matrix

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 8 \\ 1 & 2 & 0 & 5 \\ 0 & 1 & 3 & 9 \end{array}\right)$$

is the augmented matrix of the given system.

Exercise 3.15

Find the coefficient matrix and the augmented matrix of each of the following systems of linear equations.

a. $\begin{aligned} 2x+y&=5 \\ x-3y&=7 \end{aligned}$

b. $\begin{aligned} x+y+z&=1 \\ 2x-y-z&=0 \\ 3x+5y-9z&=-1 \end{aligned}$

Solution Set of a System of Linear Equations (1)

Definition 3.21

Consider a system AX=B of m linear equations in n variables.

An order n −tuple of numbers $C=(c_1,c_2,\cdots,c_n)^t$ is a solution to the system $AX=B$ if $AC=B$ and the set of all solutions of $AX=B$ is called the solution set of the given system.

Example 5

Solve the following system of linear equations

$$\begin{aligned} x+2y&=1 \\ y+3z&=0 \end{aligned}.$$

Solution

Solving for y in terms of z from the second equation gives us $y=-3z$.

Substituting $y=-3z$ in the first equation, $x+2y=1$, and solving for x in terms of z gives us $x=1-2y=1-2(-3z)=1+6z$.

Therefore, the solution set of the given system is $\{(1+6z,-3z,z):z\in\mathbb{R}\}$.

If we let $z=0$, then $x=1+6(0)=1$ and $y=-3(0)=0$.

Therefore, (1, 0, 0) is a particular solution to the given system.

Activity 3.20

Consider the system

$$x - y = 1$$
$$-2x + 2y = 0$$

Multiply the first equation by -2, add the two equations and obtain the result.

In your responses to the question in Activity 3.20, you have obtained the expression $0 = 1$, which is false and hence the given system does not have a solution.

Definition 3.22

A system of linear equations that has at least one solution is called a **consistent system** and a system that does not have a solution is called an **inconsistent system**.

Example 6

Determine if the following system of linear equation is consistent.

$$x - 2y = 1$$
$$-x + 3y = 0$$

Solution

Adding the two equations gives us $y = 1$.

Then, from the first equation, $x - 2(1) = 1 \Longrightarrow x = 3$.

This implies $(3,1)$ is a solution to the given system, hence the given system is a consistent system.

Example 7

Determine if the following system of linear equations is consistent.

$$x - y = 1$$
$$-2x + 2y = 0$$

Solution

Multiplying the first equation by 2 and adding the two equations give us $0 = 2$, which is always false.

Therefore, the given system has no solution and hence the system is an inconsistent system.

Exercise 3.16

Determine if each of the following system of linear equations is consistent or inconsistent.

a. $\begin{aligned} 2x - y &= 2 \\ x + y &= 1 \end{aligned}$

b. $\begin{aligned} -x + 2y &= 5 \\ 2x - 4y &= -4 \end{aligned}$

Solution Set of a System of Linear Equations (2)

Activity 3.21

Find the point(s) of intersections of each of the following pairs of lines (if there is any).

a. $l_1: x + y = 2$ and $l_2: x - y = 0$;

b. $s_1: x - y = 2$ and $s_2: -2x + 2y = 6$;

c. $t_1: x - y = 1$ and $t_2: -2x + 2y = -2$.

From your responses in Activity 3.21, observe that,

a. l_1 and l_2 intersect exactly at one point;

b. s_1 and s_2 do not intersect and they are two parallel lines;

c. t_1 and t_2 are identical lines.

To find out all the possibilities regarding the solutions of systems of linear equations, consider the case when you have a system of two linear equations in two variables.

Consider a general system of two linear equations in two variables

$$a_1x + b_1y = c_1 \quad (a_1, b_1 \text{not both zero})$$
$$a_2x + b_2y = c_2 \quad (a_2, b_2 \text{not both zero})$$

These are equations of two lines, say $l_1: a_1x + b_1y = c_1$ and

$l_2: a_2x + b_2y = c_2$.

There are only three possibilities about the intersection points of these two lines.

a. The lines l_1 and l_2 intersect exactly at one point and this point of intersection is the only solution of the given system of linear equations, see Figure 3.1.

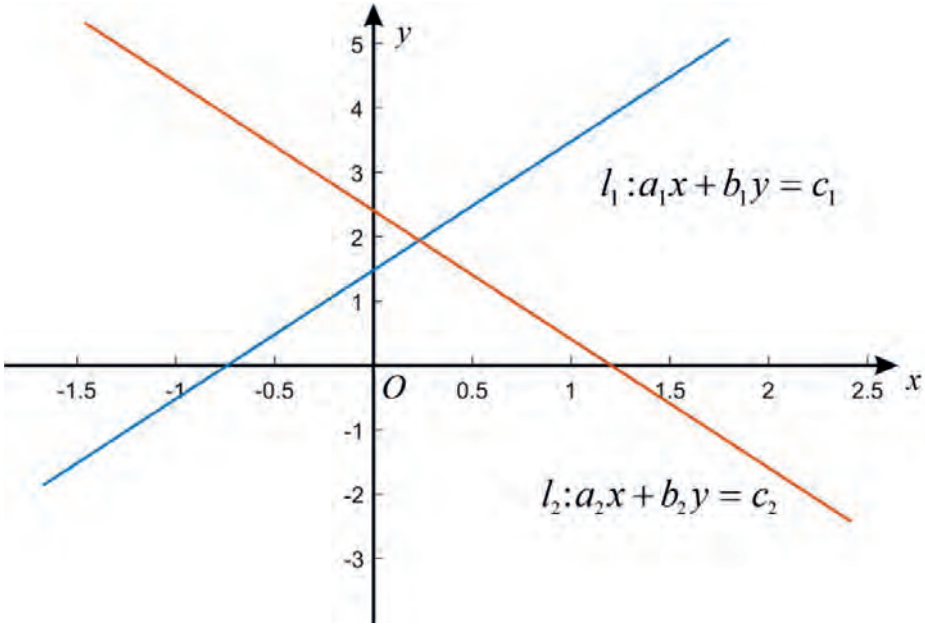


Figure 3.1: Intersecting Lines

b. l_1 and l_2 are two parallel lines and they do not intersect. In this case the given system of linear equations has no solution, see Figure 3.2.

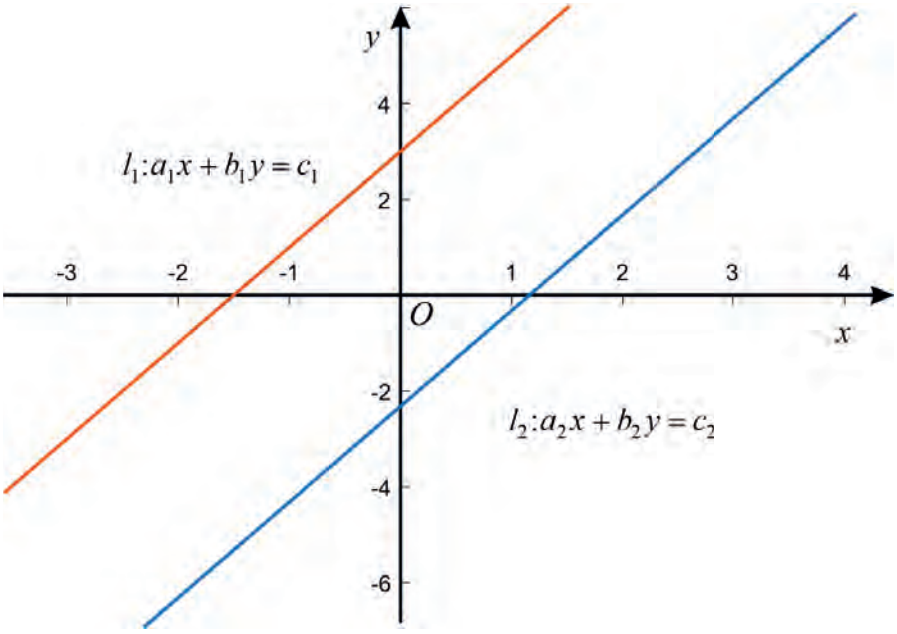


Figure 3.2: Parallel Lines

c. The lines l_1 and l_2 coincide (or identical). In this case, all the ordered pairs of numbers satisfying one of the two equations of the given system on one of the two lines are solutions of the given system and hence the system has infinitely many solutions, see Figure 3.3.

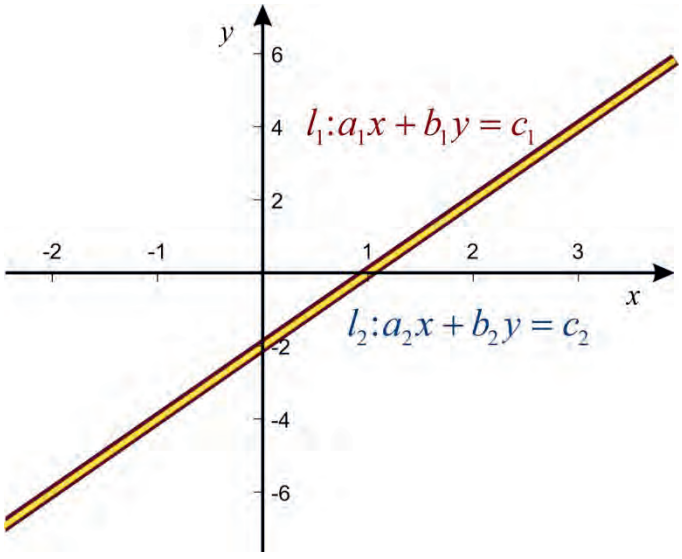


Figure 3.3: Identical Lines

Thus, a given system of linear equations has either **one solution, no solution or infinitely many solutions.**

A consistent system has either only one solution or infinitely many solutions.

Example 8

Determine if each one of the following system of linear equations has only one solution, no solution or infinitely many solutions.

a. $\begin{aligned} x+y&=1 \\ x-y&=3 \end{aligned}$ b. $\begin{aligned} x-y&=0 \\ -2x+2y&=3 \end{aligned}$ c. $\begin{aligned} x+2y&=3 \\ 3x+6y&=9 \end{aligned}$

Solution

a. Adding the two equations gives us $2x = 4 \Longrightarrow x = 2$.

Then, solving for y from the equation $x+y=1 \Rightarrow y=1-2=-1$.

Therefore, $(2,-1)$ is the only solution of the given system.

b. Multiplying the first equation by 2 and adding the two equations gives us $0=3$, which is always false. Therefore, the system has no solution.

c. Multiplying the first equation by 3 and subtracting it from the second equation gives us $0 = 0$. This implies, the two equations are identical and hence the system has infinitely many solutions.

Exercise 3.17

Determine if each one of the following system of linear equations has unique solution, no solution or infinitely many solutions.

a. $\begin{aligned} x-y&=1 \\ x+y&=5 \end{aligned}$ b. $\begin{aligned} x+y&=1 \\ 2x+2y&=4 \end{aligned}$ c. $\begin{aligned} x-2y&=1 \\ -2x+4y&=-2 \end{aligned}$

Homogeneous Systems of Linear Equations

Activity 3.22

Consider the system $\begin{matrix} x+y=0 \\ x-2y=0 \end{matrix}$.

a. Write the given system in matrix form, AX=B.

b. What type of matrix is B in (a)?

The matrix B in your responses of Activity 3.22 is the zero matrix and the system is called a homogenous system.

Definition 3.23

Consider a system of linear equations

$$\begin{matrix} a_{11}x_1+a_{12}x_2+\cdots+a_{1n}x_n=b_1 \\ a_{21}x_1+a_{22}x_2+\cdots+a_{2n}x_n=b_2 \\ \vdots \\ a_{m1}x_1+a_{m2}x_2+\cdots+a_{mn}x_n=b_m \end{matrix}$$

If $b_1=b_2=\cdots=b_m=0$, then the system is called a homogeneous, otherwise, it is called a non-homogeneous system.

Note

A homogenous system is a system of the form $AX=0$.

Example 9

Write the following system in matrix form.

$$\begin{matrix} x+y+z=0 \\ 2y+z=0 \\ x+3z=0 \end{matrix}$$

Solution

The system in matrix form is $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ and the system is homogeneous system.

Remark

Consider a homogeneous system of linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0 \end{aligned}$$

a. The given system has always a solution; namely, $x_1 = x_2 = \cdots = x_n = 0$.

   This solution is called the **trivial** solution of the homogenous system.

b. A solution $(x_1, x_2, \cdots, x_n)^t$ of the given system for which $x_i \neq 0$, for some $i = 1,2, \ldots, n$, is called a **non-trivial** solution of the given homogenous system.

Example 10

Determine the coefficient matrix and the augmented matrix of the following system of linear equations.

$$\begin{aligned} x + 2y + z &= 0 \\ y + z &= 0 \\ x + y &= 0 \end{aligned}$$

Solution

The given system in matrix form is

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Then the coefficient matrix of the system is $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ and the augmented matrix of the system is $\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{array}\right)$

Exercise 3.18

Find the coefficient matrix and the augmented matrix of each of the following system of linear equations.

a. $\begin{aligned} x + y &= 0 \\ 2x - 3y &= 0 \end{aligned}$

b. $\begin{aligned} x + y - z &= 0 \\ 2x + y + z &= 0 \\ x - y - z &= 0 \end{aligned}$

c. $\begin{aligned} 2x + 3y - z &= 0 \\ x - y + z &= 0 \\ x + y - z &= 0 \end{aligned}$

3.6 Solutions of Systems of Linear Equations

In this section, you will learn how to solve systems of linear equations that were introduced in Section 3.5, using the concept of elementary row operations.

Activity 3.23

Solve each of the following systems of Linear Equations.

a. $\begin{aligned} x + y &= 1 \\ y &= 2 \end{aligned}$

b. $\begin{aligned} -x + y &= 3 \\ x + y &= 1 \end{aligned}$

From your solutions in *Activity 3.23*, observe that the two systems have the same solution and such systems are called equivalent systems..

Definition 3.24

Two systems of linear equations over $\mathbb{R}$, the set of real numbers, with the same number of unknowns are said to be equivalent if every solution of one system is a solution of the other system.

Example 1

Solve the following system of linear equations.

$$x + y = 1$$
$$x - y = 3$$

Solution

The augmented matrix of the system is $\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & -1 & 3 \end{array}\right)$.

Then, reduce this matrix into a matrix in Row Echelon Form using appropriate elementary row operations as follows.

$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & -1 & 3 \end{array}\right) \xrightarrow{R_2 \to R_2 - R_1} \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & -2 & 2 \end{array}\right)$ and this last matrix is in Row Echelon Form.

Thus, the systems with augmented matrices $\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & -1 & 3 \end{array}\right)$ and $\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & -2 & 2 \end{array}\right)$ have the same solution set.

Then, solve the system

$$x + y = 1$$
$$-2y = 2$$

From the second equation, we get $y = -1$ and substituting this in the first equation gives us that $x = 1 - (-1) = 2$.

Therefore, the solution set of the given system is $\{(2, -1)\}$.

Note

A system of linear equations that is equivalent to a given system AX = B can be obtained by applying appropriate elementary row operations on the augmented matrix (A|B).

Consider a system AX = B.

(a) If the matrices (A|B) and (C|D)are row equivalent augmented matrices of two systems, then the systems AX = B and CX = D have the same solutions.

(b) The method of solving the system AX = B by reducing (A|B) into Row Echelon Form (REF) is called **Gaussian Elimination Method**..

(c) The method of solving the system AX = B by reducing (A|B) into Reduced Row Echelon Form (RREF) is called **Gauss-Jordan Reduction Method.**

Example 2

Solve the following system of linear equations using Gaussian Elimination Method.

$$x+y+z=1$$
$$x-y-z=5$$
$$x+y-z=3$$

Solution

The augmented matrix of the system is $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 5 \\ 1 & 1 & -1 & 3 \end{array}\right)$.

Reduce this matrix into a matrix in Row Echelon Form using appropriate elementary row operations as follows.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 5 \\ 1 & 1 & -1 & 3 \end{array}\right) \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & -2 & 4 \\ 0 & 0 & -2 & 2 \end{array}\right)$$

and the last matrix is in Row Echelon Form.

Thus, the systems with augmented matrices $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 5 \\ 1 & 1 & -1 & 3 \end{array}\right)$ and $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & -2 & 4 \\ 0 & 0 & -2 & 2 \end{array}\right)$ are equivalent systems and hence they have the same solution set.

Then, the solution set of the system can be obtained using the augment matrix $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & -2 & 4 \\ 0 & 0 & -2 & 2 \end{array}\right)$ and the system with augmented matrix $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & -2 & 4 \\ 0 & 0 & -2 & 2 \end{array}\right)$ is

$$\begin{aligned} x + y + z &= 1 \\ -2y - 2z &= 4 \\ -2z &= 2 \end{aligned}$$

From the last equation, $-2z = 2$, we have $z = -1$.

Consider the second equation,$-2y - 2z = 4$, with $z = -1$ from above and solving for y gives us that $y = -1$.

Use the first equation,$x + y + z = 1$ and $y = -1$ and $z = -1$ from above to solve for x.

Then, solving for x gives you that, $x = 3$.

Therefore, the solution set of the given system is $\{(3, -1, -1)\}$.

Example 3

Solve the following system of linear equations.

$$\begin{aligned} x + 2y - z &= 0 \\ 2x + y - z &= 3 \end{aligned}$$

Solution

The augmented matrix of the system is $\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 2 & 1 & -1 & 3 \end{array}\right)$.

Then reduce the augmented matrix of the system using appropriate elementary row operations as follows.

$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 2 & 1 & -1 & 3 \end{array}\right) \xrightarrow{R_2 \to R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -3 & 1 & 3 \end{array}\right)$ and the last matrix is in REF.

Therefore, the given system is equivalent to the system with augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -3 & 1 & 3 \end{array}\right)$$

That is, the given system is equivalent to the system

$$x + 2y - z = 0$$
$$-3y + z = 3$$

From the second equation, solving for z in terms of y, you get $z = 3y + 3$.

From the first equation, $x = -2y + z = -2y + 3y + 3 = y + 3$.

Therefore, the system has infinitely many solutions and the solution set is given by:

$\{(y+3, y, 3y+3) : y \in \mathbb{R}\}$.

Exercise 3.19

Solve each of the following systems of linear equations using Gaussian Elimination Methods.

a. $\begin{aligned} x + y &= 5 \\ x - 2y &= -4 \end{aligned}$

b. $\begin{aligned} x + y + z &= 3 \\ x - 2y + 3z &= 1 \\ 2x + y - z &= 2 \end{aligned}$

c. $\begin{aligned} x - y - z &= 0 \\ 2x + y + 2z &= 3 \end{aligned}$

3.7 Inverse of a Square Matrix

Activity 3.24

Let $A = \begin{pmatrix} 1 & 3 \\ 1 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & -3 \\ -1 & 1 \end{pmatrix}$. Then find both AB and BA.

In your responses in Activity 3.24, you have observed that $AB = I_2 = BA$. Then matrix B is called an inverse for matrix A.

Definition 3.25

Let A be a square matrix of order n. Then A is said to be invertible (or has an inverse), if there exists a square matrix B of order n such that $AB = I_n = BA$. If such matrix B exists, then it is called an inverse of A.

Let A be an invertible matrix of order n. If B and C are inverses of A, then $BA = I_n = AC$.

Using the associatively property of matrix multiplication, we have

$$B = BI_n = B(AC) = (BA)C = I_nC = C.$$

This implies, $B = C$.

Thus, if A is an invertible matrix, then its inverse is unique.

Notation

If a square matrix A is invertible, then its inverse is unique and it is denoted by A^{-1}.

Example 1

Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$. Find the inverse of A (if it exists).

Solution

If A has an inverse, say $A^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then we have $AA^{-1} = I_2$.

This implies $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ 2c & 2d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Then, by the equality of matrices, $a = 1, b = 0, c = 0$ and $2d = 1 \Rightarrow d = \frac{1}{2}$.

Therefore, A is invertible and $A^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$.

Example 2

Let $A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$. Find its inverse if A is invertible.

Solution

If A has an inverse, say $A^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then we have $AA^{-1} = I_2$.

This implies $\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a+c & b+d \\ 2a+2c & 2b+2d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Thus, we have two systems of linear equations. The first system of linear equations is

$$a + c = 1$$
$$2a + 2c = 0$$

The second system of equations is

$$b + d = 0$$
$$2b + 2d = 1$$

But each one of these two linear systems has no solution.

Therefore, matrix A has no inverse.

Exercise 3.20

Find the inverse of each of the following matrices, if it exists.

a. $\begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$ b. $\begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$ c. $\begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix}$

Properties of Inverses of Matrices

Activity 3.25

Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$. Then find

a. $A^{-1}, (A^{-1})^{-1}$ and B^{-1} b. $(AB)^{-1}$ and $B^{-1}A^{-1}$ c. $(A^t)^{-1}$ and $(A^{-1})^t$

In your responses in Activity 3.25, you have observed that $\left(A^{-1}\right)^{-1} = A$, $(AB)^{-1} = B^{-1}A^{-1}$ and $(A^t)^{-1} = (A^{-1})^t$.

These properties are the following properties of invertible matrices.

Properties of Invertible Matrices

Let A and B be invertible matrices of the same size. Then

a. A^{-1} is invertible and $(A^{-1})^{-1} = A$.

b. A^t is invertible and $(A^t)^{-1} = (A^{-1})^t$.

c. AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

Example 3

Let A and B be two square matrices of order 2 such that $A^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ and $B^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Then, find the inverses of the matrices A^{-1}, AB and A^t.

Solution

a. Given that A is an invertible matrix, A^{-1} is invertible and $\left(A^{-1}\right)^{-1} = A$ and computing $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^{-1}$ we get $A = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$.

b. Since A and B are invertible matrices, AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}.$$

c. Because it is given that A is invertible, A^t is invertible and

$$(A^t)^{-1} = (A^{-1})^t = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^t = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Definition 3.26

An invertible matrix is also called a nonsingular matrix and a matrix that is not invertible is called a singular matrix.

Exercise 3.21

Let $A = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then, determine

a. $(A^{-1})^t$ b. $(A^t)^{-1}$ c. $(AB)^{-1}$ d. $B^{-1}A^{-1}$ e. $(A^{-1})^{-1}$

Invertibility of a Matrix and Elementary Row Operations

Activity 3.26

Consider the matrix $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

a. Find the matrix obtained by applying the elementary row operation $R_1 \to R_1 - 2R_2$ on A.

b. Let B be a matrix obtained by applying the elementary row operation $R_1 \to R_1 - 2R_2$ on I_2.

c. Find the products AB and BA.

From your solution of (c) in Activity 3.26, observe that B is the inverse of A and B is obtained from A by applying the same elementary row operation that transforms A to I_2.

Procedures on How to Determine Invertibility of a Matrix Using Elementary Row Operations

Let A be a square matrix of size n.

Step 1: Augment A with I_n to form the $n \times 2n$ matrix $(A|I_n)$;

Step 2: Apply appropriate elementary row operations and reduce the matrix $(A|I_n)$ to a matrix in Row Echelon Form, say $(B|C)$.

Step 3: If $B = I_n$ in Step 2, then A is invertible and $A^{-1} = C$, otherwise A is not invertible.

Example 4

Using elementary row operations, determine the invertibility of each of the following matrices.

a. $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ b. $B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ c. $C = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$

Solution

a. First augment A with I_2 and obtain $(A|I_2) = \left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right)$.

Then, reduce this matrix to a matrix in Row Echelon Form using appropriate elementary row operations as follows:

$$(A|I_2) = \left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \to R_2 - R_1} \left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \to R_1 - R_2} \left(\begin{array}{cc|cc} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 1 \end{array}\right).$$

Then, the last matrix is in Row Reduced Echelon Form and A is reduced into I_2.

Therefore, A is invertible with its inverse is given by:

$$A^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}.$$

b. Augment B with I_2 and obtain $(B|I_2) = \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right)$.

Then, reduce it to a matrix in Row Echelon Form using appropriate elementary row operations as follows.

$$(B|I_2) = \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \to R_2 - 2R_1} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \to -\frac{1}{2}R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} \end{array}\right)$$

$$\xrightarrow{R_1 \to R_1 - R_2} \left(\begin{array}{cc|cc} 1 & 2 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} \end{array}\right)$$

The last matrix is in Row Reduced Echelon Form, but B is not reduced into I_2.

Hence, B is not invertible.

c. Augment C with I_3 and obtain $(C|I_3) = \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right)$.

$$(C|I_3) = \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \to R_1 - R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & -1 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \to R_1 + R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 1 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \to R_2 - 2R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 1 \\ 0 & 1 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right).$$

The last matrix is in Reduced Row Echelon Form and C is reduced into I_3.

Hence, C is invertible and $C^{-1} = \left(\begin{array}{ccc} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right)$.

Exercise 3.22

Using appropriate elementary row operations determine the invertibility of each of the following matrices and for those that are invertible, find their inverses.

a. $A = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$ b. $B = \begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}$ c. $C = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$

3.8 Applications

In this section, different applications of matrices will be considered.

Example 1

In a triangle, the smallest angle measures 10^o more than half of the largest angle. The middle angle measures 12^o more than the smallest angle. Find the measure of each angle of the triangle.

Solution

Let x be the measure of the smallest angle, y be the measure of the middle angle and z be the measure of the largest angle of the triangle.

Then, from the given information and the sum of the measures of the three angles of a triangle is 180°, you have the following system of linear equations.

$$\begin{aligned} x + y + z &= 180 \\ x &= \frac{1}{2}z + 10 \\ y &= 12 + x. \end{aligned}$$

The system can be written as

$$\begin{aligned} x + y + z &= 180 \\ x - \frac{1}{2}z &= 10 \\ -x + y &= 12. \end{aligned}$$

and the augmented matrix of the system is

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 180 \\ 1 & 0 & -0.5 & 10 \\ -1 & 1 & 0 & 12 \end{array}\right)$$

Apply appropriate elementary row operations to reduce the augmented matrix into REF as follows.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 180 \\ 1 & 0 & -0.5 & 10 \\ -1 & 1 & 0 & 12 \end{array}\right) \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 180 \\ 0 & -1 & -1.5 & -170 \\ 0 & 2 & 1 & 192 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 + 2R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 180 \\ 0 & -1 & -1.5 & -170 \\ 0 & 0 & -2 & -248 \end{array}\right)$$

and this matrix is in Row Echelon Form.

Thus, the given system is equivalent to the system

$$\begin{aligned} x + y + z &= 180 \\ -y - \frac{3}{2}z &= -170 \\ -2z &= -148 \end{aligned}$$

From the last equation, you get $-2z = -148 \Longrightarrow z = 74$.

From the second equation $-y - \frac{3}{2}z = -170$, we have $-y = -170 + \frac{3}{2}(74) = -59$.

This implies $y = 59$.

Then, from the last equation, $x + y + z = 180$, solving for x, gives us

$$x = 180 - (y + z) = 180 - (59 + 74) = 47.$$

Therefore, the three angles of the given triangle are 47^0, $59^0$and 74^0.

Example 2

The perimeter of a given triangle is 30 cm. The shortest side is 4 cm shorter than the longest side of the given triangle. The longest side is 6 cm less than the sum of the other two sides of the given triangle. Find the length of each side of the triangle.

Solution

Let a, b and c be the shortest, the middle and the longest sides of the given triangle respectively. Then, from the given information, you have the following system of linear equations:

$$a + b + c = 30$$
$$c = a + 4$$
$$c + 6 = a + b$$

Then, the given set of linear equations is equivalent to the following system of linear equations:

$$a + b + c = 30$$
$$a - c = -4$$
$$a + b - c = 6$$

and the augmented matrix of the given system is $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 30 \\ 1 & 0 & -1 & -4 \\ 1 & 1 & -1 & 6 \end{array}\right)$.

Use appropriate elementary row operation to reduce the augmented matrix to a matrix in REF as follows:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 30 \\ 1 & 0 & -1 & -4 \\ 1 & 1 & -1 & 6 \end{array}\right) \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 30 \\ 0 & -1 & -2 & -34 \\ 0 & 0 & -2 & -24 \end{array}\right)$$

The last matrix is in REF and the given system has the same solution as the system:

$$a + b + c = 30$$
$$-b - 2c = -34$$
$$-2c = -24$$

From the last equation, you have $-2c = -24 \Longrightarrow c = \frac{-24}{-2} = 12$ and from the second equation,

$b = 34 - 2c = 34 - 2(12) = 34 - 24 = 10.$

Then, from the first equation $a = 30 - (b + c) = 30 - 22 = 8$.

Therefore, the sides of the triangle are 8cm, 10cm and 12cm.

Example 3

If a 30% salt solution is to be mixed with a 20% salt to form a mixture of a 25% salt solution of 500 liters, how much of each is needed?

Solution

Let x represent the amount of 30% salt solution needed and y be the amount of 20% salt solution needed.

The total amount of the mixture must be 500 liters.

Thus, $x + y = 500$

The amount of salt in the end result is 25% of 500 liters, which is $0.25 \times 500 = 125$.

That is, $0.3x + 0.2y = 125$.

Then, we have the following system of linear equations:

$$\begin{aligned} x + y &= 500 \\ 0.3x + 0.2y &= 125 \end{aligned}$$

The augmented matrix of the system is $\left(\begin{array}{cc|c} 1 & 1 & 500 \\ 0.3 & 0.2 & 125 \end{array}\right)$.

$$\left(\begin{array}{cc|c} 1 & 1 & 500 \\ 0.3 & 0.2 & 125 \end{array}\right) \xrightarrow{R_2 \to R_2 - 0.3R_1} \left(\begin{array}{cc|c} 1 & 1 & 500 \\ 0 & -0.1 & -25 \end{array}\right)$$

This implies the given system is equivalent to the system

$$\begin{aligned} x + y &= 500 \\ -0.1y &= -25 \end{aligned}$$

Thus, $-0.1y = -25 \Rightarrow y = \dfrac{-25}{-0.1} = 250$ and $x = 500 - y = 500 - 250 = 250$.

Exercise 3.23

1. In a triangle, the smallest angle measures 10^o less than half of the largest angles. The middle angle measures 20^o more than the smallest angle. Find the measure of each angle of the triangle.
2. The perimeter of a triangle is 55 cm. The measure of the shortest side is 8 cm less than the middle side. The measure of the longest side is 1 cm less than the sum of the other two sides. Find the lengths of the sides.
3. A chemist has 6% salt solution and 12% salt solution. How much of the 6% salt solution and 12% salt solution must be added to get a 300 g of 10% salt solution?

Problem Solving

1. The perimeter of a triangle is 50 m. The longest side of the triangle measures 20 m. more than the shortest side. The middle side is 3 times the measure of the shortest side. Find the lengths of the three sides of the triangle.
2. A train travels 700 kilometers in the same time that a truck travels 500 kilometers. Find the speed of each vehicle if the train's average speed is 10 kilometers per hour faster than the truck's speed.
3. Almaz, Birkti, and Chaltu work in a boutique. One day the three had combined sales of Birr 14800. Almaz sold Birr 1200 more than Birkti. Birkti and Chaltu combined sold Birr 2800 more than Almaz. How much did each person sell on that particular day?

Summary

1. A rectangular array of numbers arranged in rows and columns is called a matrix.
2. A matrix with only one column is called a column matrix. .
3. A matrix with only one row is called a row matrix.
4. A matrix with equal number of rows and columns is called a square matrix.
5. A matrix with all entries 0 is called a zero matrix.
6. A square matrix that has zeros everywhere except possibly along the main diagonal is called diagonal matrix.
7. A diagonal matrix where all elements of the diagonal are equal is called a scalar matrix.
8. The diagonal matrix where all the elements of the diagonal are 1 is called the identity matrix.
9. A square matrix whose elements below the main diagonal are all zero is called an upper triangular matrix.
10. A square matrix whose elements above the main diagonal are all zero is called a lower triangular matrix.
11. Let A $= (a_{ij})_{m\times n}$ and $B = (b_{ij})_{m\times n}$ be two matrices and k be a real number. Then
    a. The sum of A and B is given by, $A + B = (a_{ij} + b_{ij})_{m\times n}$.
    b. The difference A and B is given by $A - B = \left(a_{ij} - b_{ij}\right)_{m\times n}$
    c. The scalar multiple of A by k is defined by $kA = \left(ka_{ij}\right)_{m\times n}$

    12Let $A = \left(a_{ij}\right)_{m\times p}$ and $B = \left(b_{kl}\right)_{p\times n}$ be matrices, where m, n and p are positive integers. The product of A and B, denoted by AB, is the $m \times n$ matrix $AB = \left(c_{ij}\right)_{m\times n}$. where, $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots\ldots + a_{ip}b_{pj}$, for $i = 1, \ldots\ldots m$ and $j = 1, \ldots n$.

12. Given a matrix A, the matrix obtained by interchanging the rows of A to columns and the columns of *A* to rows is called the transpose of A and it is denoted by A^t.
13. The following operations are called Elementary Row operations.
    I. **Swapping:** Interchanging two rows of a matrix.
    II. **Scaling:** Multiplying a row of a matrix by a non-zero constant.
    III. **Pivoting:** Adding a constant multiple of one row of a matrix on another row.
14. A given matrix is said to be in Row Echelon Form, if the following are satisfied:
    a. if there are any rows with no leading entries (rows having zeros entirely) they are at the bottom.
    b. the leading entry (the first non-zero entry) in each row after the first is to the right of the leading entry in the previous row.
15. A is said to be in **Reduced Row Echelon Form (RREF)** if A is in **Row Echelon Form (REF)** and the leading element in each nonzero row is 1 and it is the only nonzero number in its column.

Review Exercise

1. Find the size of each of the following matrices.

a. $A=\begin{pmatrix}1 & 2 & 3\\ -1 & 2 & -3\end{pmatrix}$ b. $B=\begin{pmatrix}-1 & 2 & -3\\ 1 & 2 & 0\\ 0 & 0 & 0\end{pmatrix}$ c. $C=\begin{pmatrix}-1 & 2\\ 0 & 3\\ 3 & -1\end{pmatrix}$

d. $D=\begin{pmatrix}0 & 1 & 0\\ 1 & 2 & 3\\ 3 & -2 & 3\\ 2 & 1 & 0\end{pmatrix}$ e. $E=\begin{pmatrix}0 & 1 & -3 & -1\end{pmatrix}$ f. $F=\begin{pmatrix}1\\ -3\\ 2\end{pmatrix}$

2. Let $A=\begin{pmatrix}1 & 1\\ 2 & 3\end{pmatrix}$, $B=\begin{pmatrix}0 & 1 & 0\\ 3 & 0 & 2\end{pmatrix}$, $C=\begin{pmatrix}1 & 1\\ 3 & 1\end{pmatrix}$ and $D=\begin{pmatrix}1 & 1 & 1\\ 2 & 3 & 2\end{pmatrix}$. Then find each of the following matrices.

a. $A-3C$ b. $AB-CD$ c. $2B+3D$ d. B^tC+D^tA

3. Identify each of the following matrices as symmetric, skew-symmetric, diagonal, scalar, upper triangular or lower triangular matrices.

a. $A=\begin{pmatrix}-1 & 2 & -1\\ 0 & 0 & 0\\ 0 & 0 & 5\end{pmatrix}$ b. $B=\begin{pmatrix}2 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 4\end{pmatrix}$ c. $C=\begin{pmatrix}0 & 2 & -1\\ -2 & 0 & 3\\ 1 & -3 & 0\end{pmatrix}$

d. $D=\begin{pmatrix}1 & 2 & 3\\ 2 & 1 & 1\\ 3 & 1 & 4\end{pmatrix}$ e. $E=\begin{pmatrix}1 & 0 & 0\\ -2 & 0 & 0\\ 4 & 2 & -5\end{pmatrix}$ f. $F=\begin{pmatrix}3 & 0 & 0\\ 0 & 3 & 0\\ 0 & 0 & 3\end{pmatrix}$

4. Find the reduced row echelon form of each of the following matrices.

a. $A=\begin{pmatrix}1 & 2 & 3\\ -2 & 1 & -1\end{pmatrix}$ b. $B=\begin{pmatrix}1 & 2 & 1\\ 2 & 1 & 2\\ 3 & 2 & 3\end{pmatrix}$

5. Solve each of the following systems of linear equations using Gaussian Elimination Method.

a. $\begin{aligned}x-y&=1\\ 2x+y&=2\end{aligned}$ b. $\begin{aligned}x+y+z&=1\\ y+z&=3\\ x-2y+z&=-5\end{aligned}$

6. Determine the inverse of each of the following matrices, if it exists.

a. $A=\begin{pmatrix}1 & 3\\ 1 & 4\end{pmatrix}$ b. $B=\begin{pmatrix}2 & -1\\ -4 & 2\end{pmatrix}$

c. $C=\begin{pmatrix}1 & 2 & 2\\ 1 & -2 & 1\\ 2 & 1 & 2\end{pmatrix}$ d. $D=\begin{pmatrix}1 & 1 & -1\\ 1 & -3 & 3\\ -1 & 2 & -2\end{pmatrix}$

7. A farmer has two types of milk, one that is 20% butterfat and another which is 15% butterfat. How much of each should he use to end up with 50 gallons of 20% butterfat?

UNIT 4

DETERMINANTS AND THEIR PROPERTIES

Unit Outcomes

By the end of this unit, you will be able to:

- Know minor and cofactor of a matrix.
- Evaluate determinant of a matrix.
- Understand properties of a determinant.
- Apply principles of determinants to compute inverse of a matrix.
- Use Cramer's rule to solve system of linear equations.
- Apply determinant concepts to solve real-world situations.

Unit Contents

4.1 Determinants of Matrices of Order 2

4.2 Minors and Cofactors of Elements of Matrices

4.3 Determinants of Matrices of Order 3

4.4 Properties of Determinants

4.5 Inverse of a Square Matrix of Order 2 and 3

4.6 Solutions of Systems of Linear Equations Using Cramer's Rule

4.7 Applications

Summary

Review Exercise

- **Adjoint**
- **cofactor**
- **cramer's rule**
- **determinant**
- **inverse**
- **minor**

Introduction

Every square matrix can be associated with a number called its determinant. Determinants occur in many mathematical topics. The coefficients in system of linear equations are often represented by matrices and the determinant can be used to solve systems of linear equations, if the coefficient matrix is an invertible matrix.

In this unit, determinant of a matrix; basic properties of determinant will be discussed and some applications of determinants in solving systems of linear equations will also be given.

4.1 Determinants of Matrices of Order 2

Activity 4.1

Consider the square matrix $A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of order 2. Suppose $ad-bc\neq 0$. Then

a. show that A is invertible and find A^{-1}.

b. solve the following system of linear equations:

$$ax+by=e$$
$$cx+dy=f$$

From Activity 4.1, you have found that

a. A is invertible and its inverse is given by $A^{-1}=\frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ and

b. The solution set of the given system is $\left\{\left(\frac{ed-fb}{ad-bc},\frac{af-ce}{ad-bc}\right)\right\}$.

In both cases, the number $ad-bc$ is involved and $ad-bc$ is called the determinant of A.

The determinant of a square matrix of order 2 is defined as follows.

Definition 4.1

Let $A=\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ be a square matrix of order 2. The determinant of A, denoted by $\det(A)$ or $|A|$, is the number defined by $\det(A)=a_{11}a_{22}-a_{12}a_{21}$.

Example 1

Find the determinant of $A=\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Solution

Using the definition, $\det(A)=1(4)-2(3)=4-6=-2$

Example 2

Let $A=\begin{pmatrix}1 & x\\ 2 & x+1\end{pmatrix}$. If $\det(A)=0$, then find x.

Solution

If $\det(A)=0$, then $1(x+1)-2x=0\Rightarrow 1-x=0$.

This implies, $x=1$.

Example 3

Find the determinant of $I_2=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$.

Solution

The determinant of the identity matrix for multiplications is given as:

$$\det(I_2)=\begin{vmatrix}1 & 0\\ 0 & 1\end{vmatrix}=1(1)-0(0)=1-0=1.$$

Exercise 4.1

1. Determine the determinant of each of the following matrices.

   a. $A=\begin{pmatrix}1 & 2\\ 1 & 4\end{pmatrix}$ b. $B=\begin{pmatrix}5 & 2\\ 3 & 1\end{pmatrix}$ c. $C=\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}$

2. Determine the determinant of each of the following matrices.

   a. $\begin{pmatrix}1 & 2\\ x & 2x\end{pmatrix}$ b. $\begin{pmatrix}x & 0\\ 0 & x\end{pmatrix}$ c. $\begin{pmatrix}1 & a\\ 0 & 2\end{pmatrix}$

3. Let $A=\begin{pmatrix}3 & x-1\\ 1 & x\end{pmatrix}$. If $\det(A)=0$ then determine the value(s) of x.

4.2 Minors and Cofactors of Elements of Matrices

Minors of Elements of Matrices

Activity 4.2

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. Then find

a. matrix B obtained by crossing-out the 2nd row and the 1st column from A.

b. det(B)

c. matrix C obtained by crossing-out the 1st row and the 2nd column from A.

d. det(C)

From your responses for Activity 4.2, observe that

(i) both B and C are 2×2 matrices;

(ii) the determinant of B is called the minor of a_{21}, the element in the intersection of the second row and the first column of A;

(iii) the determinant of C is called the minor of a_{12}, the element in the intersection of the first row and the second column of A.

Note

If $A = (a)$ is a 1×1 matrix, then the determinant of A is defined by $\det(A) = a$.

Definition 4.2

Let n be a positive integer greater than or equal to 2 and $A = (a_{ij})_{n\times n}$ be a square matrix of order n.

Suppose we know how to define the determinant of any $(n-1)\times(n-1)$ matrix.

Let (i,j), for $i = 1,\ldots,n$ and $j = 1,\ldots,n$, be an ordered pair of positive integers.

Cross out the i^{th} row and j^{th} column of A and obtain an $(n-1)\times(n-1)$ matrix and this matrix is denoted by A_{ij}.

That is,

$$A_{ij} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{pmatrix}_{(n-1)\times(n-1)}$$

Then $\det(A_{ij})$ is called the minor of the entry a_{ij} of A, denoted by M_{ij}.

Example 1

Determine the minors of all the entries of matrix $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 3 & -1 & 5 \end{pmatrix}$.

Solution

i. $A_{11} = \begin{pmatrix} 1 & 0 \\ -1 & 5 \end{pmatrix}$ $A_{12} = \begin{pmatrix} 0 & 0 \\ 3 & 5 \end{pmatrix}$ $A_{13} = \begin{pmatrix} 0 & 1 \\ 3 & -1 \end{pmatrix}$

$A_{21} = \begin{pmatrix} 2 & 1 \\ -1 & 5 \end{pmatrix}$ $A_{22} = \begin{pmatrix} 1 & 1 \\ 3 & 5 \end{pmatrix}$ $A_{23} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$

$A_{31} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ $A_{32} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ $A_{33} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

ii. The minors of A are

$$|A_{11}| = \begin{vmatrix} 1 & 0 \\ -1 & 5 \end{vmatrix} = 5 \qquad |A_{12}| = \begin{vmatrix} 0 & 0 \\ 3 & 5 \end{vmatrix} = 0 \qquad |A_{13}| = \begin{vmatrix} 0 & 1 \\ 3 & -1 \end{vmatrix} = -3,$$

$$|A_{21}| = \begin{vmatrix} 2 & 1 \\ -1 & 5 \end{vmatrix} = 11 \qquad |A_{22}| = \begin{vmatrix} 1 & 1 \\ 3 & 5 \end{vmatrix} = 2 \qquad |A_{23}| = \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix} = -7,$$

$$|A_{31}| = \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} = -1 \qquad |A_{32}| = \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} = 0 \qquad |A_{33}| = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1.$$

Cofactors of Elements of Matrices

Activity 4.3

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. Then compute $(-1)^{i+j}\det(A_{ij})$ for every (i, j) pair of positive integers where $i = 1,2,3$ and $j = 1,2,3$.

From your responses in Activity 4.3, the values $(-1)^{i+j}\det(A_{ij})$ is called the cofactor of the entry a_{ij} for every pair (i, j) of positive integers for $i = 1,2,3$ and $j = 1,2,3$.

Definition 4.3

Let n be a positive integer greater than or equal to 2 and $A = (a_{ij})_{n\times n}$ be a square matrix of order n. Then $C_{ij} = (-1)^{i+j}\det(A_{ij})$ is called the cofactor of a_{ij}, for every pair of positive integers (i, j) for $1 \le i \le n$ and $1 \le j \le n$.

Example 2

Determine the cofactors of all the entries of the matrix $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 3 & -1 & 5 \end{pmatrix}$.

Solution

From Example 1, the minors of all the entries of A are:

$M_{11} = |A_{11}| = 5$ $\quad M_{12} = |A_{12}| = 0$ $\quad M_{13} = |A_{13}| = -3$

$M_{21} = |A_{21}| = 11$ $\quad M_{22} = |A_{22}| = 2$ $\quad M_{23} = |A_{23}| = -7$

$M_{31} = |A_{31}| = -1$ $\quad M_{32} = |A_{32}| = 0$ $\quad M_{33} = |A_{33}| = 1.$

Then the cofactors of all the entries of A are:

$C_{11} = (-1)^{1+1} |A_{11}| = 5$ $\quad C_{12} = (-1)^{1+2} |A_{12}| = 0$ $\quad C_{13} = (-1)^{1+3} |A_{13}| = -3$

$C_{21} = (-1)^{2+1} |A_{21}| = -11$ $\quad C_{22} = (-1)^{2+2} |A_{22}| = 2$ $\quad C_{23} = (-1)^{2+3} |A_{23}| = 7$

$C_{31} = (-1)^{3+1} |A_{31}| = -1$ $\quad C_{32} = (-1)^{3+2} |A_{32}| = 0$ $\quad C_{33} = (-1)^{3+3} |A_{33}| = 1.$

Exercise 4.2

Find the minors and the cofactors of all the entries of the following matrices.

a. $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ $\quad$ b. $B = \begin{pmatrix} 1 & 1 & 3 \\ 0 & 3 & 4 \\ 0 & 0 & 5 \end{pmatrix}$

4.3 Determinants of Matrices of Order 3

Activity 4.4

Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 3 \\ 3 & 4 & 4 \end{pmatrix}$. Then, determine

a. $a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$ $\quad$ b. $a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31}$.

From your responses in Activity 4.3, observe that the answers for both questions are the same number and this number is called the determinant of A.

Definition 4.4

Let $A = (a_{ij})_{3\times3}$ be a square matrix of order 3. Then the determinant of A, det(A) or $|A|$, is the number defined by

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

and this sum is called the cofactor expansion of the determinant along the first row.

It is possible to express determinant of a matrix A as a cofactor expansion along a column of A and gives you the same result, but in this subunit, cofactor expansion of the determinant along a row only is considered.

Example 1

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. Then find the determinant of A using a cofactor expansion along the first row.

Solution

First find the cofactors C_{11}, C_{12} and C_{13} as:

$$C_{11} = (-1)^{1+1}\begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} = 45 - 48 = -3,\ C_{12} = (-1)^{1+2}\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} = -(36 - 42) = 6$$

and $C_{13} = (-1)^{1+3}\begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = 32 - 35 = -3.$

Then the determinant of A is given by:

$$|A| = 1C_{11} + 2C_{12} + 3C_{13} = 1(-3) + 2(6) + 3(-3) = -3 + 12 - 9 = 0.$$

That is, $|A| = 0$.

Example 2

Find the determinant of $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

Solution

Using a cofactor expansion along the first row, the determinant of I_3 is given by:

$$\det(I_3) = 1(-1)^{(1+1)}\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + 0(-1)^{(1+2)}\begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} + 0(-1)^{(1+3)}\begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 1 - 0 + 0 = 1$$

Therefore, the determinant of I_3 is 1.

Exercise 4.3

1. Let $A = \begin{pmatrix} 2 & 7 & 9 \\ 3 & 5 & 6 \\ 7 & 4 & 1 \end{pmatrix}$. Then compute the determinant of A using a cofactor expansion along the first row.

2. Compute the determinant of the matrix $B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

Cofactor Expansion of Determinant along Any Row

Activity 4.5

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 1 \\ 3 & 3 & 2 \end{pmatrix}$. Then evaluate each of the following.

a. $a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$

b. $a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23}$

c. $a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33}$

In your responses in Activity 4.5, you have observed that, the values in (a), (b) and (c) are the same and this number is the determinant of A.

The results in Activity 4.5 tell us that the determinant does not depend on the choice of the row as stated in Theorem 4.1.

Theorem 4.1

Let $A=(a_{ij})_{3\times 3}$ be a square matrix of order 3. Then, the determinant of A, det(A), can be expressed as a cofactor expansion along any row of A.

That is, $\det(A)=a_{i1}C_{i1}+a_{i2}C_{i2}+a_{i3}C_{i3}$ for each $i=1,2,3$.

Example 3

Let $A=\begin{pmatrix}1&2&3\\4&5&6\\7&8&9\end{pmatrix}$. Then, find the determinant of A using cofactor expansion along the three rows of A.

Solution

In Example 1 of these section, you have seen that using cofactor expansion of the determinant along first row, $|A|=a_{11}C_{11}+a_{12}C_{12}+a_{13}C_{13}=0$.

a. The determinant of A using the cofactor expansion along the second row is given by:

$$\det(A)=(-1)^{(2+1)}4\begin{vmatrix}2&3\\8&9\end{vmatrix}+(-1)^{(2+2)}5\begin{vmatrix}1&3\\7&9\end{vmatrix}+(-1)^{(2+3)}6\begin{vmatrix}1&2\\7&8\end{vmatrix}$$
$$=-4(18-24)+5(9-21)-6(8-14)=24-60+36=0.$$

b. The determinant of A using a cofactor expansion along the third row is given by:

$$\det(A) = (-1)^{(3+1)}\,7\begin{vmatrix}2 & 3\\ 5 & 6\end{vmatrix} + (-1)^{(3+2)}\,8\begin{vmatrix}1 & 3\\ 4 & 6\end{vmatrix} + (-1)^{(3+3)}\,9\begin{vmatrix}1 & 2\\ 4 & 5\end{vmatrix}$$
$$= 7(12-15) - 8(6-12) + 9(5-8)$$
$$= 7(-3) - 8(-6) + 9(-3) = -21 + 48 - 27 = 0$$

Thus, the determinant of A using the cofactor expansion along any of its three rows is the same: that is, $|A| = 0$.

Note

Let $A = (a_{ij})_{n\times n}$ be a square matrix of order n.

a. The expression $(-1)^{i+j}$ in the expansion of the determinant determines the algebraic sign in the position (i, j) of a square matrix of order n and these signs form a checkerboard pattern of " + " and " − " that has + in the $(1, 1)$ position. The patterns for $n = 2$ and 3 are given as follows.

$$\underset{n=2}{\begin{pmatrix}+ & -\\ - & +\end{pmatrix}} \text{ and } \underset{n=3}{\begin{pmatrix}+ & - & +\\ - & + & -\\ + & - & +\end{pmatrix}}$$

b. In determining the determinant of a square matrix by cofactor expansion along a row, the best strategy is to use a cofactor expansion along a row with large number of zeros.

Example 4

Compute the determinant of the matrix $A = \begin{pmatrix}1 & 2 & 1\\ 12 & 0 & 0\\ -5 & 11 & 0\end{pmatrix}$.

Solution

As the second row has two zero entries, more zero entries than the other two rows, you can use a cofactor expansion along the second row.

$$\begin{vmatrix} 1 & 2 & 1 \\ 12 & 0 & 0 \\ -5 & 11 & 0 \end{vmatrix} = -12\begin{vmatrix} 2 & 1 \\ 11 & 0 \end{vmatrix} + 0\begin{vmatrix} 1 & 1 \\ -5 & 0 \end{vmatrix} - 0\begin{vmatrix} 1 & 2 \\ -5 & 11 \end{vmatrix}$$

$$= -12\begin{vmatrix} 2 & 1 \\ 11 & 0 \end{vmatrix} = (-12)\times(0-11) = 132$$

Therefore, the determinant of A is 132.

Exercise 4.4

1. Let $A = \begin{pmatrix} 1 & 7 & 2 \\ 0 & 5 & 2 \\ -1 & 4 & 1 \end{pmatrix}$. Then compute the determinant of A using a cofactor expansion along all the three rows.
2. Compute the determinant of each of the following matrices by using cofactor expansion.

   a. $B = \begin{pmatrix} 2 & 1 & 0 \\ 3 & -2 & 2 \\ 1 & 0 & 2 \end{pmatrix}$ b. $C = \begin{pmatrix} 0 & 2 & 1 \\ 4 & 1 & 1 \\ 3 & 2 & 0 \end{pmatrix}$

3. Find the determinant of each of the following matrices.

   a. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$ b. $\begin{pmatrix} 2 & 3 & 4 \\ 0 & 1 & 5 \\ 0 & 0 & -4 \end{pmatrix}$ c. $\begin{pmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \\ 5 & 6 & 3 \end{pmatrix}$

4.4 Properties of Determinants

Determinants of Triangular Matrices

Activity 4.6

Find the determinant of each of the following matrices.

a. $A = \begin{pmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{pmatrix}$ b. $B = \begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}$

From your responses in Activity 4.6, observe that matrices A and B are triangular matrices and

a. the determinant of matrix A is the product of its diagonal entries;
b. the determinant of matrix B is the product of its diagonal entries.

These properties are true in general and are stated as follows.

Theorem 4.2

Let n be a positive integer greater than or equal to 2 and $A = (a_{ij})_{n\times n}$. If A is a triangular matrix, then the determinant of A is the product of its diagonal elements.

That is, if $A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$ or $A = \begin{pmatrix} a_{11} & 0 & \ldots & 0 \\ a_{21} & a_{22} & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{(n-1)1} & a_{(n-2)2} & \ldots & 0 \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{pmatrix}$,

then $|A| = a_{11}a_{22} \ldots a_{nn}$.

Note

1. If A is a diagonal matrix, then A is a triangular matrix and its determinant is the product of its diagonal entries.

   That is, if $A = \begin{pmatrix} a_{11} & 0 & \cdots & 0 & 0 \\ 0 & a_{22} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & 0 & a_{nn} \end{pmatrix}$, then $|A| = a_{11}a_{22} \ldots a_{nn}$

2. For a positive integer n the matrix $I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & 0 & 1 \end{pmatrix}$ is the identity matrix for matrix multiplication and it is a diagonal matrix.

   Thus, $|I_n| = 1 \times 1 \times \ldots \times 1 = 1$.

Example 1

Compute the determinant of $A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$.

Solution

Using the cofactor expansion along the first row:

$$\det(A) = 2\begin{vmatrix} -1 & 4 \\ 0 & 1 \end{vmatrix} - 1\begin{vmatrix} 0 & 4 \\ 0 & 1 \end{vmatrix} + 3\begin{vmatrix} 0 & -1 \\ 0 & 0 \end{vmatrix}$$

$$= 2\times(-1) + (-1)\times 0 + 3\times 0 = -2$$

On the other hand, A is an upper triangular matrix.

Thus, $\det(A) = 2\times(-1)\times 1 = -2$, the product of the diagonal entries of A.

Example 2

Find the determinant of each of the following matrices:

a. $A = \begin{pmatrix} 2 & 2 & 1 \\ 0 & 3 & 5 \\ 0 & 0 & 1 \end{pmatrix}$ b. $B = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{pmatrix}$ c. $C = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

Solution

a. A is an upper triangular matrix and $|A| = 2\times3\times1 = 6$, which is the product of the diagonal entries of A.

b. B is a lower triangular matrix and $|B| = -1 \times 2 \times 1 = -2$, which is the product of the diagonal entries of B.

c. C is a diagonal matrix and hence $|C| = 4\times(-1)\times(-2) = 8$, which is the product of the diagonal entries of C.

Exercise 4.5

Determine the determinant of each of the following matrices:

a. $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & -1 \end{pmatrix}$ b. $B = \begin{pmatrix} 2 & 0 & 0 \\ 2 & -1 & 0 \\ 0 & 1 & -2 \end{pmatrix}$ c. $C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

Determinant and Elementary Row Operations

Let us consider the effects of the three elementary operations of matrices on determinants of matrices. Let us start our discussion by considering the following activity.

Activity 4.7

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then find the determinant of each of the following matrices in terms of the determinant of A.

a. $\begin{pmatrix} c & d \\ a & b \end{pmatrix}$ b. $\begin{pmatrix} 3a & 3b \\ c & d \end{pmatrix}$ c. $\begin{pmatrix} a & b \\ c-2a & d-2b \end{pmatrix}$

From your responses in Activity 4.6, observe that:

a. $\begin{vmatrix} c & d \\ a & b \end{vmatrix} = -|A|$ (Interchanging two rows of a matrix changes the determinant by sign).

b. $\begin{vmatrix} 3a & 3b \\ c & d \end{vmatrix} = 3|A|$ (The determinant of a matrix obtained from A by multiplying a row of A by a scalar is the scalar times the determinant of A).

c. $\begin{vmatrix} a & b \\ c-2a & d-2b \end{vmatrix} = |A|$ (Adding a scalar multiple of a row of A onto another row does not change the determinant).

These three properties are the effects of elementary row operations on the determinant of A and these effects are true in general for any square matrix as stated in Theorem 4.3.

Theorem 4.3

Let A be a square matrix of order n.

a. Interchanging two rows of the matrix changes the sign of the determinant;

That is, if $A \xrightarrow{R_i \leftrightarrow R_j} B$ for $i \neq j$, then $|B| = -|A|$.

b. The determinant of a matrix obtained by multiplying one row of A by a constant gives you the same constant times the determinant of the given matrix.

That is, if, $A \xrightarrow{R_i \to kR_i} B$ and k is any constant, then $|B| = k|A|$.

c. Adding a scalar multiple of one row of A to another row of A does not change the determinant. That is, if $A \xrightarrow{R_i \to R_i + kR_j} B$ for $i \neq j$ and k is any constant, then $|B| = |A|$.

Note

One consequence that follows from property (b) of Theorem 4.3 is if you multiply a square matrix of order n by a scalar k, then the determinant of the new matrix is k^n times the determinant of the given matrix.

That is, if A is a square matrix of order n and k is a scalar, then $|kA| = k^n|A|$.

Example 3

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If $\det(A) = -2$, then find the determinant of each the following matrices:

a. $B = \begin{pmatrix} a & b \\ 2a+c & 2b+d \end{pmatrix}$

b. $C = \begin{pmatrix} 2a & 2b \\ c & d \end{pmatrix}$

c. $D = \begin{pmatrix} 3a & 3b \\ 3c & 3d \end{pmatrix}$

d. $E = \begin{pmatrix} c & d \\ a & d \end{pmatrix}$

Solution

If $\det(A) = -2$, then $|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc = -2$.

a. $|B| = \begin{vmatrix} a & b \\ 2a+c & 2b+d \end{vmatrix} = a(2b+d) - b(2a+c) = ad - bc = -2$. Observe that, matrix B is obtained by replacing the third row of A by 2 times the first row of A plus the second row of A and $|A| = |B| = -2$.

b. $|C| = \begin{vmatrix} 2a & 2b \\ c & d \end{vmatrix} = 2a(d) - 2b(c) = 2(ad - bc) = 2 \times (-2) = -4$. Observe that, matrix C is obtained by scaling the first row of A by -2 and $|C| = 2|A| = 2 \times (-2) = -4$.

c. $|D| = \begin{vmatrix} 3a & 3b \\ 3c & 3d \end{vmatrix} = 3a(3d) - 3b(3c) = 3 \times 3(ad - bc) = 3^2 \times (-2) = 9 \times (-2) = -18$.

Observe that $D = 3A$ and $|D| = 3^2 |A| = 9 \times (-2) = -18$.

d. $|E| = \begin{vmatrix} c & d \\ a & b \end{vmatrix} = cb - da = -(ad - bc) = 2$. Observe that, matrix E is obtained by swapping the first and the second rows of A and $\det(E) = -\det(A) = -(-2) = 2$.

Example 4

Let $A=\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix}$. If $\det(A)=2$, then find the determinant of each of the following matrices.

a. $B=\begin{pmatrix} d & e & f \\ a & b & c \\ g & h & k \end{pmatrix}$

b. $C=\begin{pmatrix} a & b & c \\ d & e & f \\ g+3a & h+3b & k+3c \end{pmatrix}$

c. $D=\begin{pmatrix} a & b & c \\ 5d & 5e & 5f \\ g & h & k \end{pmatrix}$

d. $E=\begin{pmatrix} 4a & 4b & 4c \\ 4d & 4e & 4f \\ 4g & 4h & 4k \end{pmatrix}$

Solution

a. Matrix B is obtained by swapping the first and the second rows of A.
Thus, $\det(B)=-\det(A)=-2$.

b. Matrix C is obtained by replacing the third row of A by 3 times the first row of A plus the third row of A. Thus, $\det(C)=\det(A)=2$.

c. Matrix D is obtained by scaling the second row of A by 5.
Thus, $\det(D)=5\det(A)=5\times 2=10$.

d. A is a 3×3 and $E=4A$.
Thus, $\det(E)=\det(4A)=4^3\det(A)=64\times 2=128$.

Note

Using property (b) in Theorem 4.3 we can introduce many zeros in a row of a square matrix and compute the determinant of the matrix by expanding the determinant along the row with many zeros.

Example 5

Determine the determinant of the following matrix after reducing it to a matrix in REF.

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 3 \end{pmatrix}$$

Solution

First reduce A into an upper triangular matrix using appropriate elementary row operations as follows.

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 3 \end{pmatrix} \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 2 \end{pmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 2R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -2 \end{pmatrix} = B$$

The last matrix, B, is a triangular matrix and hence the determinant of B is the product of its diagonal entries.

That is, $|B| = 1 \times 1 \times (-2) = -2$.

Observe that matrix B is obtained from A by applying three successive elementary row operations. Each of these elementary row operations are of the same type; adding a scalar multiple of one row onto another row.

Thus, all these elementary row operations do not make any change on the determinant of the matrix.

Therefore, the determinants of both A and B are the same and hence

$|A| = |B| = -2$.

Exercise 4.6

1. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If $\det(A) = 2$, then find the determinant of each the following matrices.

   a. $\begin{pmatrix} a & b \\ a+c & b+d \end{pmatrix}$ b. $\begin{pmatrix} a & b \\ 2c & 2d \end{pmatrix}$ c. $\begin{pmatrix} -3a & -3b \\ -3c & -3d \end{pmatrix}$

2. Let $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix}$ and $\det(A) = 3$. Then, compute the determinant of each of the following matrices.

   a. $\begin{pmatrix} a & b & c \\ g & h & k \\ d & e & f \end{pmatrix}$ b. $\begin{pmatrix} 2a & 2b & 2c \\ d & e & f \\ g & h & k \end{pmatrix}$ c. $\begin{pmatrix} a & b & c \\ d & e & f \\ g+3a & h+3b & k+3c \end{pmatrix}$

3. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}$. Determine the determinant of matrix B if:

   a. $A \xrightarrow{R_1 \leftrightarrow R_3} B$ b. $A \xrightarrow{R_2 \to R_2 - 2R_3} B$ c. $A \xrightarrow{R_3 \to \frac{1}{2}R_3} B$

Determinant of Product of Matrices and Determinant of a Transpose

Activity 4.8

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$. Then, compute each of the following.

a. $\det(A)$ and $\det(B)$. b. $\det(AB)$, $\det(A^t)$ and $\det(B^t)$

In your responses in Activity 4.8, observe that;

a. $\det(A) = \det\left(A^t\right)$, that is, the determinant of A and its transpose are equal;

b. $\det(AB) = (\det A)(\det B)$; that is, the determinant of the product of A and B, $\det(AB)$, is equal to the product of the determinant of A and the determinant of B.

In general, we have the following results about the determinant of a transpose and the determinant of product of matrices.

Theorem 4.4

Let A and B be two square matrices of order n. Then

a. $\det(A) = \det(A^t)$ b. $\det(AB) = \det(A)\det(B)$

Note

Using Property (b) of Theorem 4.4 and the associative property of matrix multiplication, if A, B and C area square matrix of order n, , then $|ABC| = |A||B||C|$.

Example 6

Let A and B be square matrices of order 3 with $|A| = 4$ and $|B| = 5$. Then compute each of the following:

a. $\det(AB)$ b. $\det(3A)$ c. $\det(2AB^t)$ d. $\det(A^5)$

Solution

a. $\det(AB) = \det(A)\det(B) = 4 \times 5 = 20.$

b. $\det(3A) = 3^3 \det(A) = 27 \times 4 = 108.$

c. $\det\left(2AB^t\right) = 2^3 \det\left(AB^t\right)$, since AB^t is a square matrix of order 3.

The matrix AB^t is the product of A and B^t.

This implies $\det\left(2AB^t\right) = 2^3 \det(A)\det\left(B^t\right) = 2^3 \det(A)\det(B)$ and

$\det(B) = \det(B^t)$.

Therefore, $\det\left(2AB^t\right) = 2^3 \det(A)\det(B) = 8 \times 4 \times 5 = 160$.

d. $\det\left(A^5\right) = (\det A)^5 = 4^5 = 1024$.

Exercise 4.7

Let A and B be square matrices of order 3 with $|A| = 2$ and $|B| = 3$. Then compute each of the following.

a. $\det(AB)$ b. $\det(2A)$ c. $\det(3AB^t)$ d. $\det(A^4)$

Determinant of Inverse of an Invertible Matrix

Let A be a square matrix of order n. If A is invertible, then $AA^{-1} = I_n$ and

$$\det(I_n) = \det(AA^{-1}) = [\det(A)][\det(A^{-1})] = 1$$

Therefore, $\det(A) \neq 0$, because the product of $\det A$ and $\det(A^{-1})$ is different from zero and hence, $\det(A^{-1}) = \dfrac{1}{\det(A)}$.

Thus, the following theorem is proved.

Theorem 4.5

If A is an invertible matrix of order n, then $\det(A)$ is not zero and the determinant of A^{-1} is given by $\det(A^{-1}) = \dfrac{1}{\det(A)}$.

Example 7

Let A and B be two invertible matrices of order 3. Suppose $\det(A) = 4$ and $\det(B) = 3$. Then find

a. $\det(A^{-1})$ b. $\det(A^{-1}B)$

Solution

a. Since A is invertible and $\det(A) = 4 \neq 0$, then $\det(A^{-1}) = \dfrac{1}{\det(A)} = \dfrac{1}{4}$;

b. $\det\left(A^{-1}B\right) = \det\left(A^{-1}\right)\det\left(B\right) = \left(\dfrac{1}{\det\left(A\right)}\right)\det\left(B\right) = \dfrac{1}{4} \times 3 = \dfrac{3}{4}$

Exercise 4.8

Let A and B be 3×3 matrices with $\det(A) = 2$ and $\det(B) = 5$. Then find

a. $\det(A^{-1})$ b. $\det\left(A^{-1}B\right)$

c. $\det(B^{-1})$ d. $\det(AB^{-1})$

4.5 Inverse of a Square Matrix of Order 2 and 3

There are different applications of determinants. One of these applications that is considered in this section is invertibility of a matrix.

Adjoint of a square Matrix

Activity 4.9

Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 4 & 1 & 4 \end{pmatrix}$. Then find

a. the cofactors of all the entries of A. b. matrix B defined by $B = \left(C_{ij}\right)^{t}_{3\times 3}$.

The matrix B you have obtained in Activity 4.9 is called the adjoint of A and if A is invertible, then matrix B has a role in determining the inverse of A.

Definition 4.5

Let $A = (a_{ij})_{n\times n}$ be a square matrix of order n and $C_{ij} = (-1)^{(i+j)}\det(A_{ij})$ be the cofactor of the element a_{ij} for $i = 1, \ldots, n$ and $j = 1, \ldots, n$. Then, the matrix

$$(C_{ij})^{t}_{n\times n} = \begin{pmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{pmatrix}$$

is called adjoint of A and it is denoted by Adj(A).

Example 1

Determine the adjoint of $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Solution

$A_{11} = (4)$ $A_{12} = (3)$ $A_{21} = (2)$ $A_{22} = (1)$.

Thus, the cofactors of all the elements A are:

$$C_{11} = (-1)^{1+1} \det(A_{11}) = 4 \qquad C_{12} = (-1)^{1+2} \det(A_{12}) = -3$$

$$C_{21} = (-1)^{2+1} \det(A_{21}) = -2 \qquad C_{22} = (-1)^{2+2} \det(A_{22}) = 1.$$

Therefore, the adjoint of A is

$$\text{Adj}(A) = \begin{pmatrix} 4 & -3 \\ -2 & 1 \end{pmatrix}^t = \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$$

Example 2

Determine the adjoint of the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 4 & 3 \\ 1 & 2 & 0 \end{pmatrix}.$$

Solution

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 4 & 3 \\ 2 & 0 \end{vmatrix} = -6 \qquad C_{12} = (-1)^{1+2} \begin{vmatrix} 1 & 3 \\ 1 & 0 \end{vmatrix} = 3 \qquad C_{13} = (-1)^{1+3} \begin{vmatrix} 1 & 4 \\ 1 & 2 \end{vmatrix} = -2$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 2 & 1 \\ 2 & 0 \end{vmatrix} = 2 \qquad C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1 \qquad C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = 0$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix} = 2 \qquad C_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} = -2 \qquad C_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = 2$$

are the cofactors of all the entries of A.

Therefore, the adjoint of A is:

$$\text{Adj(A)} = \begin{pmatrix} C_{11} & C_{12} & C_{31} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix}^t = \begin{pmatrix} -6 & 3 & -2 \\ 2 & -1 & 0 \\ 2 & -2 & 2 \end{pmatrix}^t = \begin{pmatrix} -6 & 2 & 2 \\ 3 & -1 & -2 \\ -2 & 0 & 2 \end{pmatrix}$$

Exercise 4.9

1. Determine the adjoint of $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$.
2. Determine the adjoint of $B = \begin{pmatrix} 1 & 2 & 3 \\ 6 & 1 & 9 \\ 4 & 0 & 5 \end{pmatrix}$.

Adjoint and Determinant

Activity 4.10

Let $A = \begin{pmatrix} 1 & 1 & 2 \\ 3 & 4 & 3 \\ 2 & 2 & 1 \end{pmatrix}$. Then find

a. $\det(A)$ b. Adj(A) c. A(Adj(A)) d. (Adj(A))A

From your responses in Activity 4.10, observe that

$\big(\text{Adj}(\text{A})\big)\text{A} = \det(\text{A})\, I_3 = \text{A}(\text{Adj}(\text{A}))$.

In general, given a square matrix of order n, $\text{A}\big(\text{Adj}(\text{A})\big) = \det(\text{A})\, \text{I}_n = \big(\text{Adj}(\text{A})\big)\text{A}$.

If $\det(\text{A})$ is not equal to zero, then

$$\text{A}\left(\frac{1}{\det(\text{A})}\text{Adj}(\text{A})\right) = \text{I}_n = \left(\frac{1}{\det(\text{A})}\text{Adj}(\text{A})\right)\text{A}$$

This implies, A is invertible and its inverse is given by:

$$\text{A}^{-1} = \frac{1}{\det(\text{A})}\big(\text{Adj}(\text{A})\big)$$

Thus, the following result is proved.

Theorem 4.6

Let A be a square matrix of order n. Then, $A\big(\text{Adj}(A)\big)=\big(\text{Adj}(A)\big)A=\det(A)I_n$ and if the determinant of A is not zero, then A is invertible and its inverse is given by

$$A^{-1}=\frac{1}{|A|}(\text{Adj}(A)).$$

Example 3

Let $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$. Find the inverse of A (if it exists).

Solution

a. From Example 1, $\text{Adj}(A)=\begin{pmatrix}4 & -3\\ -2 & 1\end{pmatrix}^t=\begin{pmatrix}4 & -2\\ -3 & 1\end{pmatrix}$.

b. $\det(A)=4-6=-2\neq 0$.

This implies, A is invertible and its inverse is given by

$$A^{-1}=\frac{1}{\det(A)}\text{Adj}(A)=-\frac{1}{2}\begin{pmatrix}4 & -2\\ -3 & 1\end{pmatrix}=\begin{pmatrix}-2 & 1\\ \frac{3}{2} & -\frac{1}{2}\end{pmatrix}.$$

Example 4

Let $A=\begin{pmatrix}1 & 2 & 1\\ 1 & 4 & 3\\ 1 & 2 & 0\end{pmatrix}$. Find the inverse of A (if it exists).

Solution

From Example 2, $\text{Adj}(A)=\begin{pmatrix}-6 & 2 & 2\\ 3 & -1 & -2\\ -2 & 0 & 2\end{pmatrix}$ and

$$|A|=\begin{vmatrix}4 & 3\\ 2 & 0\end{vmatrix}-2\begin{vmatrix}1 & 3\\ 1 & 0\end{vmatrix}+\begin{vmatrix}1 & 4\\ 1 & 2\end{vmatrix}=-6+6-2=-2\neq 0.$$

Therefore, A is invertible and its inverse is given by

$$A^{-1}=\frac{1}{\det(A)}\text{Adj}(A)=-\frac{1}{2}\begin{pmatrix}-6 & 2 & 2\\ 3 & -1 & -2\\ -2 & 0 & 2\end{pmatrix}=\begin{pmatrix}3 & -1 & -1\\ -\frac{3}{2} & \frac{1}{2} & 1\\ 1 & 0 & -1\end{pmatrix}.$$

Remark

Let A be a square matrix of order n. A is invertible if and only if the determinant of A is not zero. This is equivalent to saying that: A is singular if and only if the determinate of A is zero.

Exercise 4.10

Find the determinant, adjoint and inverse (if it exists) of each one the following matrices

a. $\begin{pmatrix}1 & 1\\ 2 & 3\end{pmatrix}$ b. $\begin{pmatrix}1 & 2 & 3\\ 0 & 1 & 4\\ 0 & 0 & 1\end{pmatrix}$ c. $\begin{pmatrix}1 & 2 & 1\\ 3 & 1 & 4\\ 5 & 1 & 5\end{pmatrix}$

4.6 Solutions of Systems of Linear Equations Using Cramer's Rule

We can use determinants to obtain another method, known as Cramer's Rule, to solve systems of n linear equations in n variables if the coefficient matrix of the system is invertible.

Consider a system of n linear equations $AX = B$, where A is a square matrix of order n. If A is invertible, then the system has a unique solution and this unique solution is given by

$X = A^{-1}B$.

Activity 4.11

Consider a system of two linear equations in two variables given by $\begin{cases} x+y=1 \\ x-y=3 \end{cases}$

a. Solve the given system.

b. Compute $\dfrac{\begin{vmatrix} 1 & 1 \\ 3 & -1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}}$ and $\dfrac{\begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}}$.

From your responses in Activity 4.11, observe that

a. $x = \dfrac{\begin{vmatrix} 1 & 1 \\ 3 & -1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}}$ and $y = \dfrac{\begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}}$;

b. The matrix $\begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix}$ is obtained by replacing the first column of the coefficient matrix of the given system by $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and matrix $\begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$ is obtained by replacing the second column of the coefficient matrix of the given system by $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$

This method is called Cramer's Rule.

Theorem 4.7 (Cramer's Rule for Two Variables)

Given the system $\begin{cases} ax+by=e \\ cx+dy=f \end{cases}$ in two variables x and y, if $ad-bc\neq 0$, then the given system has a unique solution and it is given by $x=\dfrac{\begin{vmatrix} e & b \\ f & d \end{vmatrix}}{ad-bc}$ and $y=\dfrac{\begin{vmatrix} a & e \\ c & f \end{vmatrix}}{ad-bc}$.

Example 1

Use Cramer's rule, if possible, and solve the following system of linear equations:

$$\begin{cases} x+y=1 \\ 2x+y=5 \end{cases}$$

Solution

The given system in matrix form is given by:

$$\begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} 1 \\ 5 \end{pmatrix} \text{ and } \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix}=1-2=-1\neq 0.$$

Thus, the system has a unique solution that is given by:

$$x=\frac{\begin{vmatrix} 1 & 1 \\ 5 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix}}=\frac{1-5}{-1}=\frac{-4}{-1}=4 \text{ and } y=\frac{\begin{vmatrix} 1 & 1 \\ 2 & 5 \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix}}=\frac{5-2}{-1}=\frac{3}{-1}=-3.$$

Therefore, the solution set of the given system of linear equations is $\{(4,-3)\}$.

Exercise 4.11

Use Cramer's rule, if possible, and solve each of the following system of linear equations.

a. $\begin{cases} x+y=2 \\ 2x+3y=4 \end{cases}$

b. $\begin{cases} 2x+y=9 \\ 3x-4y=2 \end{cases}$

c. $\begin{cases} x+3y=8 \\ 2x-3y=4 \end{cases}$

d. $\begin{cases} -x+2y=0 \\ -2x+y=-1 \end{cases}$

Theorem 4.8 (Cramer's Rule for Three Variables)

Given the system

$$\begin{cases} a_{11}x+a_{12}y+a_{13}z=b_1 \\ a_{21}x+a_{22}y+a_{23}z=b_2 \\ a_{31}x+a_{32}y+a_{33}z=b_3 \end{cases}$$

in three variables x, y and z, if $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \neq 0$, then the given system has a unique solution given by $x=\frac{D_x}{D}$, $y=\frac{D_y}{D}$ and $z=\frac{D_z}{D}$, where

$D=\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$, $D_x=\begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}$, $D_y=\begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}$ and

$D_z=\begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$.

Example 2

Using Cramer's Rule (if possible) and solve the following system of linear equations.

$$\begin{cases} 2x+3y-5z=1 \\ x+y-z=2 \\ 2y+z=8 \end{cases}$$

Solution

The given system of linear equations is given in matrix form as:

$$\begin{pmatrix} 2 & 3 & -5 \\ 1 & 1 & -1 \\ 0 & 2 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}=\begin{pmatrix} 1 \\ 2 \\ 8 \end{pmatrix}.$$

Then the determinant of the coefficient matrix is:

$$\mathrm{D}=\begin{vmatrix} 2 & 3 & -5 \\ 1 & 1 & -1 \\ 0 & 2 & 1 \end{vmatrix}=-7\neq 0.$$

Thus, the system has a unique solution and by Cramer's Rule:

$$x=\frac{\begin{vmatrix} 1 & 3 & -5 \\ 2 & 1 & -1 \\ 8 & 2 & 1 \end{vmatrix}}{\begin{vmatrix} 2 & 3 & -5 \\ 1 & 1 & -1 \\ 0 & 2 & 1 \end{vmatrix}}=\frac{-7}{-7}=1, \qquad y=\frac{\begin{vmatrix} 2 & 1 & -5 \\ 1 & 2 & -1 \\ 0 & 8 & 1 \end{vmatrix}}{\begin{vmatrix} 2 & 3 & -5 \\ 1 & 1 & -1 \\ 0 & 2 & 1 \end{vmatrix}}=\frac{-21}{-7}=3 \text{ and}$$

$$z=\frac{\begin{vmatrix} 2 & 3 & 1 \\ 1 & 1 & 2 \\ 0 & 2 & 8 \end{vmatrix}}{\begin{vmatrix} 2 & 3 & -5 \\ 1 & 1 & -1 \\ 0 & 2 & 1 \end{vmatrix}}=\frac{-14}{-7}=2.$$

Therefore, the solution set of the given system of linear equations is $\{(1,3,2)\}$.

Remark

It is worth mentioning that, in finding the solution set of a system of n linear equations in n variables using **Cramer's Rule**, we have to evaluate $n + 1$ determinants. For systems with large number of equations, Gaussian Elimination Method is more efficient.

Exercise 4.12

Use Cramer's rule, if possible, and solve each of the following system of linear equations.

a. $\begin{cases} x+y+z=3 \\ -2x+2y+2z=5 \\ x+y+2z=4 \end{cases}$ b. $\begin{cases} x+y-z=1 \\ 2x+z=4 \\ y-z=5 \end{cases}$ c. $\begin{cases} x+y-z=1 \\ x-y+z=2 \\ 2x+2z=3 \end{cases}$

Systems of Linear Equations with no Solution or Infinitely Many Solutions

Given a system $AX = B$ of n linear equations in n variables, where n is a positive integer, if $|A| = 0$, we cannot apply Cramer's rule to solve the system and further investigation is required.

If $|A| = 0$, then there are two possibilities: the system has no solution or the system has infinitely many solutions.

The cases of no solution or infinitely many solutions of a given system of linear equations are considered in the following two examples.

Example 3

Solve the following system of linear equations (if possible).

$$\begin{cases} -x+2y=3 \\ 3x-6y=0 \end{cases}$$

Solution

The coefficient matrix of the given system is $\begin{pmatrix} -1 & 2 \\ 3 & -6 \end{pmatrix}$ and $\begin{vmatrix} -1 & 2 \\ 3 & -6 \end{vmatrix} = 6 - 6 = 0.$

This implies the system has no unique solution and we cannot use the Cramer's rule. The system has no solution or it has infinitely many solutions and we can use elimination method to determine.

The augmented matrix of the system is $\left(\begin{array}{cc|c} -1 & 2 & 3 \\ 3 & -6 & 0 \end{array}\right)$.

$$\left(\begin{array}{cc|c} -1 & 2 & 3 \\ 3 & -6 & 0 \end{array}\right) \xrightarrow{R_2 \to R_2 + 3R_1} \left(\begin{array}{cc|c} -1 & 2 & 3 \\ 0 & 0 & 9 \end{array}\right)$$

and the last matrix is in REF.

Thus, the given system is equivalent to the system:

$$\begin{cases} -x + 2y = 3 \\ 0 = 9 \end{cases}$$

The last equation, $0 = 9$, is always false.

Therefore, the system has no solution.

Graphing the lines represented by the two equations in Figure 4.1 reveals that the two lines are parallel.

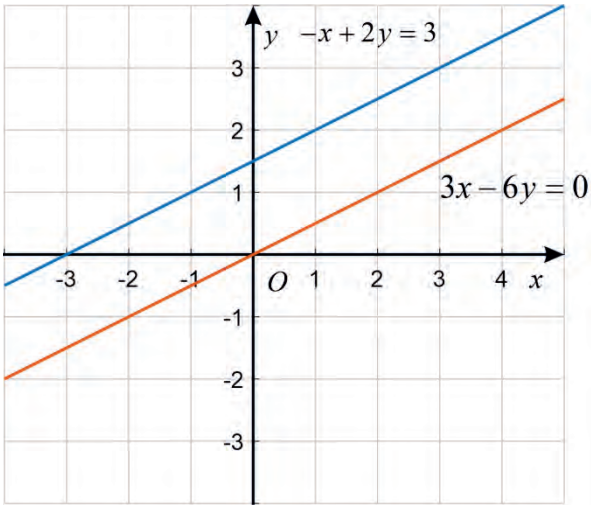


Figure 4.1

Example 4

Solve the following system linear equations.

$$\begin{cases} x-2y+3z=0 \\ 3x+y-2z=0 \\ 2x-4y+6z=0 \end{cases}$$

Solution

Let us first determine the determinant of the coefficient matrix.

The coefficient matrix of the system is $\begin{pmatrix} 1 & -2 & 3 \\ 3 & 1 & -2 \\ 2 & -4 & 6 \end{pmatrix}$ and its determinant is

$$\begin{vmatrix} 1 & -2 & 3 \\ 3 & 1 & -2 \\ 2 & -4 & 6 \end{vmatrix} = 0.$$

This implies the system has no unique solution and we cannot use the Cramer's rule. The system has no solution or it has infinitely many solutions and we can use elimination method to determine.

The augmented matrix of the system is:

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 3 & 1 & -2 & 0 \\ 2 & -4 & 6 & 0 \end{array}\right).$$

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 3 & 1 & -2 & 0 \\ 2 & -4 & 6 & 0 \end{array}\right) \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - 3R_1} \left(\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 0 & 7 & -11 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

and the last matrix is in Row Echelon Form.

Then, the given system of linear equations has the same solution as the system of linear equations:

$$\begin{cases} x-2y+3z=0 \\ 7y-11z=0 \end{cases}.$$

From the last equation, $7y-11z=0 \Rightarrow z=\frac{7}{11}y$.

Substituting this value in the first equation give us

$$x=2y-3z=2y-3\left(\frac{7}{11}y\right)=\frac{1}{11}y,\ y\in\mathbb{R}.$$

Therefore, the given system has infinitely many solutions given by:

$$\left\{\left(\frac{1}{11}y,y,\frac{7}{11}y\right):y\in\mathbb{R}\right\}.$$

Exercise 4.13

Solve each of the following systems of linear equations, if possible.

a. $\begin{cases} x-y=2 \\ -2x+2y=-4 \end{cases}$

b. $\begin{cases} x+y-2z=3 \\ -2x-2y+4z=-6 \\ y-z=5 \end{cases}$

c. $\begin{cases} x+y+z=3 \\ -2x+2y+2z=5 \\ x+y+2z=4 \end{cases}$

d. $\begin{cases} x+y-z=1 \\ x-y+z=2 \\ 2x+2z=3 \end{cases}$

4.7 Applications

Polynomial Interpolation

In studying a set of data that relates two variables x and y, it may be the case that we can use a polynomial to "fit" to the data. If such a polynomial can be established, it can be used to estimate values of x and y which have not been provided.

Example 1

Consider the data points $(0,1),(1,2)$ and $(2,22)$. Find an interpolating polynomial $p(x)$ of degree at most two, and estimate the value of $p(3)$.

Solution

The desired polynomial $p(x)$ is given by $p(x)=a+bx+cx^2$

Then, $p(0)=1=a$, $p(1)=2=a+b+c$ and $p(2)=22=a+2b+4c$.

This gives us a system of linear equations

$$\begin{cases} a=1 \\ a+b+c=2 \\ a+2b+4c=22 \end{cases}$$

and the coefficient matrix of the given system is:

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix} \text{ and } \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{vmatrix} = 1\begin{vmatrix} 1 & 1 \\ 2 & 4 \end{vmatrix} = 1(4-2) = 2 \neq 0.$$

Then, the given system of linear equations has a unique solution.

We can solve the system using Cramer's rule as follows:

$$a=\frac{\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 22 & 2 & 4 \end{vmatrix}}{\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{vmatrix}}=1, \quad b=\frac{\begin{vmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 22 & 4 \end{vmatrix}}{\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{vmatrix}}=-\frac{17}{2}=-8.5, \quad c=\frac{\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 22 \end{vmatrix}}{\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{vmatrix}}=\frac{19}{2}=9.5.$$

Therefore, the required polynomial is:

$p(x)=1-8.5x+9.5x^2$ and $p(3)=1-8.5(3)+9.5(3)^2=61$.

Exercise 4.14

Consider data points $(0,2),(1,5)$ and $(2,25)$. Find an interpolating polynomial $p(x)$ of at most degree two and estimate the values of $p(-1)$ and $p(3)$.

Area of a triangle in the *xy*-plane

Determinant can be used to find the area of a triangle whose three vertices are in the xy −plane.

The area of a triangle with vertices $A(x_1, y_1), B(x_2, y_2)$ and $C(x_3, y_3)$ is in the plane is given by

$$\text{Area}(\Delta ABC) = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \text{ if } \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} > 0 \text{ and}$$

$$\text{Area}(\Delta ABC) = -\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \text{ if } \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} < 0.$$

Example 2

Find the area of the triangle with vertices $A(0,5), B(0,0)$ and $C(5,0)$.

Solution

Using the cofactor expansion along the second row,

$$\begin{vmatrix} 0 & 5 & 1 \\ 0 & 0 & 1 \\ 5 & 0 & 1 \end{vmatrix} = (-1)\begin{vmatrix} 0 & 5 \\ 5 & 0 \end{vmatrix} = 25 > 0.$$

Therefore, Area $(\Delta ABC) = \frac{1}{2}\begin{vmatrix} 0 & 5 & 1 \\ 0 & 0 & 1 \\ 5 & 0 & 1 \end{vmatrix} = \frac{1}{2}(25) = 12.5$ Square units.

Example 3

Find the area of the triangle with vertices $P(1,1), Q(1,3)$ and $R(4,5)$

Solution

Using the cofactor expansion along the first row,

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & 1 \\ 4 & 5 & 1 \end{vmatrix} = 1\begin{vmatrix} 3 & 1 \\ 5 & 1 \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 4 & 1 \end{vmatrix} + 1\begin{vmatrix} 1 & 3 \\ 4 & 5 \end{vmatrix} = -2 + 3 - 7 = -6 < 0.$$

Therefore, Area $(\Delta PQR) = \frac{-1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & 1 \\ 4 & 5 & 1 \end{vmatrix} = -\frac{1}{2}(-6) = 3$ Square units.

Exercise 4.15

Find the area of the triangle with vertices the following three points.

a. $(-2, 0)$, $(0, 2)$ and $(2, 0)$

b. $(1, 0)$, $(2, 1)$ and $(2,3)$

Test for collinear points in the *xy*-plane

Three points in xy −plane are said to be collinear if all the three points lie on the same line. Determinant can be used to determine whether three points in the xy −plane are collinear or not.

Three points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ in the *xy*-plane are collinear if and only if

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

Example 4

Determine if the points $A(1,1)$, $B(2,2)$ and $C(3,3)$ are collinear.

Solution

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = 0 .$$

This implies, the points $A(1,1)$, $B(2,2)$ and $C(3,3)$ are collinear.

Example 5

Determine if the points $A(0,1)$, $B(1,0)$ and $C(0,0)$ in the xy −plane are collinear.

Solution

$$\begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} = -1 \neq 0$$

This implies, the points $A(0,1), B(1,0)$ and $C(0,0)$ are not collinear.

Two point form of equation of a line in the *xy*-plane

We can use determinant to find equation of a line in xy −plane passing through two points. An equation of a line passing through two distinct points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0,$$

where (x, y) is a point on the line.

Example 6

Find an equation of the line passing through the points $P(2,0)$ and $R(0,3)$.

Solution

If (x, y) is a point on the line passing through $P(2,0)$ and $R(0,3)$, then:

$$\begin{vmatrix} x & y & 1 \\ 2 & 0 & 1 \\ 0 & 3 & 1 \end{vmatrix} = 0.$$

This implies $-2\begin{vmatrix} y & 1 \\ 3 & 1 \end{vmatrix} - \begin{vmatrix} x & y \\ 0 & 3 \end{vmatrix} = 0 \Rightarrow -2(y-3)-(3x-0)=0.$

Thus, the general equation of the line passing through the two points $P(2,0)$ and $R(0,3)$ is $3x+2y-6=0$.

Exercise 4.16

1. Determine if the following set of points are collinear.
   a. $(1,1)$, $(3,2)$ and $(4,3)$ b. $(1,0)$, $(2,1)$ and $(3,2)$
2. Find the general form of equation of the line passing through each of the following pairs of points.
   a. $(1,1)$ and $(3,2)$ b. $(1,-1)$ and $(-2,2)$ c. $(-1,1)$ and $(-4,2)$

Problem Solving

1. If the area of a triangle with vertices (–2, 0), (2, 0) and (0, k) is 8 sq. units, then determine the value(s) of k.
2. If the points $(3,k)$, $(k,1)$ and $(6,-1)$ are collinear, then determine the value(s) of k.

Summary

1. If $A=\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, then $\det(A)=a_{11}a_{22}-a_{12}a_{21}$.

2. If $A=(a_{ij})_{n\times n}$, then the matrix obtained by crossing out the i^{th} row and j^{th} column of A is denoted by A_{ij}.

   a. $\det(A_{ij})$ is called the minor of the element a_{ij} and

   b. $C_{ij}=(-1)^{i+j}\det(A_{ij})$ is called the cofactor of a_{ij}.

3. If $A=(a_{ij})_{3\times 3}$, $\det(A)=a_{11}C_{11}+a_{12}C_{12}+a_{13}C_{13}$, expansion of the determinant along the first row.

$$\text{If } A=\begin{pmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{pmatrix} \text{ or } A=\begin{pmatrix} a & b & d \\ 0 & c & e \\ 0 & 0 & f \end{pmatrix}, \text{ then } \det(A)=acf.$$

4. If A and B are two square matrices of order n, then,

   a. $\det(A)=\det(A^t)$

   b. $\det(AB)=\det(A)\det(B)$.

5. If A is an invertible matrix of order n, then $\det(A)$ is not equal to zero and the determinant of A^{-1} is given by

$$\det(A^{-1})=\frac{1}{\det(A)}$$

6. Let $A=(a_{ij})_{n\times n}$ be a square matrix of order n and $C_{ij}=(-1)^{i+j}\det(A_{ij})$ be the cofactor of the entry a_{ij} for $i=1,\ldots,n$ and $j=1,\ldots,n$.

Then matrix $\text{Adj}(A)=(C_{ij})^t_{n\times n}=\begin{pmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{pmatrix}$ is called Adjoint of A.

7. Let A be a square matrix of order n. Then $A\big(\text{Adj}(A)\big)=\big(\text{Adj}(A)\big)A=\det(A)I_n$ and if the determinant of A is not zero, then A is invertible and its inverse is given by

$$A^{-1}=\frac{1}{|A|}(\text{Adj}(A)).$$

Review Exercise

1. Find the minor and the cofactor of each entry of the following matrices.

   a. $A=\begin{pmatrix}1 & 2\\ -3 & 5\end{pmatrix}$ b. $B=\begin{pmatrix}1 & 0 & -1\\ 2 & 3 & 1\\ -1 & 2 & 3\end{pmatrix}$

2. Find the determinant of each of the following matrices.

   a. $A=\begin{pmatrix}1 & 3\\ 2 & 4\end{pmatrix}$ b. $B=\begin{pmatrix}1 & 0 & 0\\ 2 & 3 & 0\\ 3 & -2 & -1\end{pmatrix}$ c. $C=\begin{pmatrix}2 & 1 & 3\\ 1 & -2 & 0\\ -1 & 2 & 4\end{pmatrix}$

   d. $D=\begin{pmatrix}-2 & 3 & -1\\ 0 & 1 & 2\\ 0 & 0 & 3\end{pmatrix}$ e. $E=\begin{pmatrix}-3 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 1\end{pmatrix}$ f. $F=\begin{pmatrix}1 & 2 & 3\\ 0 & 0 & 0\\ -1 & 3 & -2\end{pmatrix}$

3. Let A and B be square matrices of order 3 and $\det(A)=2$ and $\det(B)=-3$. Then find each of the following determinants.

   a. $\det(2AB)$ b. $\det(AB^t)$ c. $\det(A^{-1}B)$

4. Determine the invertiblity of each of the following matrices and find the inverse if it exists.

   a. $A=\begin{pmatrix}1 & 3\\ 0 & 2\end{pmatrix}$ b. $B=\begin{pmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 3 & -2 & -1\end{pmatrix}$ c. $C=\begin{pmatrix}1 & 1 & 1\\ 1 & 2 & 2\\ 1 & 3 & 4\end{pmatrix}$

d. $D=\begin{pmatrix}2 & 3 & -1\\ 0 & 1 & 2\\ 0 & 0 & -1\end{pmatrix}$ e. $E=\begin{pmatrix}1 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 3\end{pmatrix}$ f. $F=\begin{pmatrix}1 & 1 & 1\\ 1 & 2 & 2\\ 1 & 3 & 3\end{pmatrix}$

5. Use Cramer's rule to solve each of the following systems of linear equations:

a. $\begin{cases} x+y=1 \\ 2x+3y=0 \end{cases}$ b. $\begin{cases} x+2y-z=4 \\ 3x-y+z=5 \\ 2x+3y+2z=7 \end{cases}$

6. Find the area of the triangle with vertices the following three set of points.

a. $(-3,0)$, $(0,3)$ and $(1,1)$ b. $(1,-1)$, $(1,2)$ and $(3,2)$

7. Find the general form of equation of the line passing through each of the following pairs of points:

a. $(1,1)$ and $(-1,2)$ b. $(1,2)$ and $(-2,-3)$ c. $(1,0)$ and $(-4,4)$

8. Determine if the following set of points are collinear:

b. $(1,1)$, $(3,2)$ and $(5,-2)$ b. $(1,0)$, $(2,2)$ and $(3,3)$

UNIT 5

VECTORS

Unit Outcomes

By the end of this unit, you will be able to:

- Know basic concepts and procedures about vectors and operation on vectors.
- Know specific facts about vectors.
- Apply principles and theorem about vectors in solving problems involving vectors.

Unit Contents

5.1 Revision on Vectors and Scalars

5.2 Representation of Vectors

5.3 Vector Product

5.4 Application of Scalar and Cross Product

5.5 Application of Vectors

5.6 Applications

Summary

Review Exercise

- coordinate form of a vector
- cross product
- orthogonal vectors
- parallel vectors
- perpendicular vectors
- resolution of vectors
- scalar quantity
- Standard unit vector
- unit vector
- vectors quantity
- zero vector

Introduction

In Grade 9, you have discussed that measurement of any physical quantity can be expressed in terms of a number and a unit. For instance, physical quantities such as length, time, mass, electric current, area, volume, velocity, force, acceleration, work, etc. can be expressed in terms of a number and a unit.

Thus, physical quantities that are used to describe the motion of objects can be divided into two categories. The quantity is either a vector or a scalar. In this unit, you focus on various geometric and algebraic aspects of vector representation and vector algebra.

5.1 Revision on Vectors and Scalars

Activity 5.1

Classify the following physical quantities as scalar and vector quantities:

a. time	b. mass	c. work	d. torque
e. distance	f. displacement	g. length	h. velocity
i. force	j. volume	k. temperature	

From Activity 5.1, you have observed that there are physical quantities that can be expressed by magnitude and direction and there are physical quantities that can be expressed by a certain number associated with a suitable unit only. These physical quantities can be classified as a vector or a scalar physical quantity.

5.1.1 Scalars

Definition 5.1

Physical quantities expressed by a certain number associated with a suitable unit without any mention of direction in space is known as **scalar.** The number describing the quantity of a particular scalar is known as **its magnitude**. For example, time (15 sec), mass (20 kg), length (5 km), area (7 m^2), volume (10 m^3), temperature (37 °C).

5.1.2 Vectors

Definition 5.2

A vector is a physical quantity which can be expressed completely by stating both its magnitude with particular unit and direction. For example, car movement is usually described by giving the speed and the direction, say 60 km/h northeast. The car speed and car direction together form a vector quantity, velocity of the car.

Exercise 5.1

1. Make a list of physical quantities which are not mentioned in the introduction part, distinguish whether they are vectors or scalars.
2. Classify the following physical quantities as scalar and vector quantities.

   a. length b. velocity c. time d. force

5.2 Representation of Vectors

5.2.1 Components of Vectors

Activity 5.2

1. Discuss how a vector quantity can be represented in a coordinate form in a plane by giving examples.
2. If $\mathbf{v} = (1, 1)$ is a vector whose initial point is the origin as shown in Figure 5.1, then find a. the components of **v**. b. the direction of **v**.
3. If $\mathbf{u} = (-1, \sqrt{3})$ is a vector whose initial point is the origin as shown in Figure 5.1, then find a. the components of **v**. b. the direction of **v**.

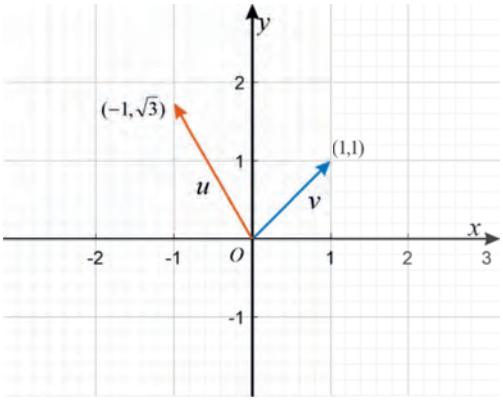


Figure 5.1

From Activity 5.2, you have observed that how vectors can be represented in a coordinate or a column form in a plane.

Definition 5.3

1. If **v** is a vector in the plane whose initial point is the origin, O (0, 0) and whose terminal point is Q (x, y) then the coordinate form of a vector **v** = vector OQ = (x, y) or the column form of a vector **v** = $\begin{pmatrix} x \\ y \end{pmatrix}$ as shown in Figure 5.2. The numbers x and y are called components of a vector **v**, the direction of the vector can be given by the slope m = $\frac{y}{x}$.
2. If **v** is a vector in the plane whose initial point P (x_1 , y_1) and whose terminal point Q(x_2 , y_2) the coordinate form of a vector **v** = vector PQ = ($x_2 - x_1$, $y_2 - y_1$) or the column form of a vector **v** = $\begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$ as shown in Figure 5.3.

   The numbers $x_2 - x_1$ and $y_2 - y_1$ are called components of a vector **v**, the direction of the vector **v** can be given by the inclination of the line with slope $m = \frac{y_2 - y_1}{x_2 - x_1}$.

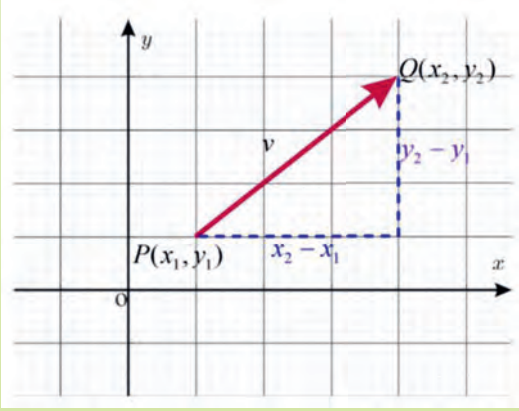


Figure 5.2

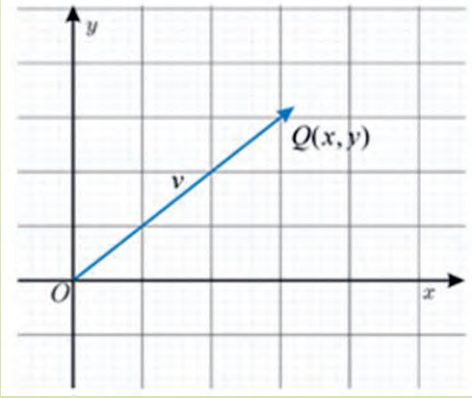


Figure 5.3

Note

1. If both the initial and terminal points lie at the origin, then **v** is the **zero vector** and is given by **v** = (0, 0) or **v** = $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Example 1

Find the coordinate and column forms of the vector **v** that has initial point (0, 0) and terminal point (3, 6).

Solution

Let O = (0, 0) and Q = (3, 6), as shown Figure 5.4.

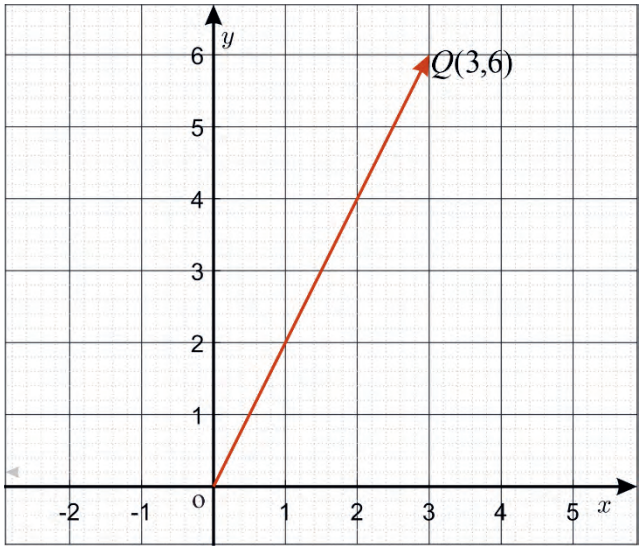


Figure 5.4

Accordingly, coordinate form of vector **v** is **v** = (3, 6) and the column form of vector **v** is $\mathbf{v} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$.

Example 2

Find the coordinate and column forms of the vector **v** that has initial point (1, 2) and terminal point (6, 7).

Solution

Let P = (1, 2) and Q = (6, 7), as shown in Figure 5.4.

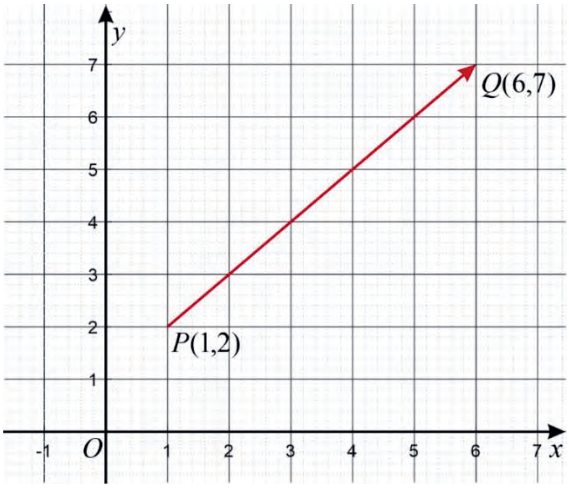


Figure 5.5

Thus, the coordinate form of vector **v** is

v = $\overrightarrow{PQ}$ = ($x_2 - x_1$, $y_2 - y_1$) = (6 – 1, 7 – 2) = (5, 5) and the column form of vector **v** is **v** = $\binom{5}{5}$.

Exercise 5.2

1. Find the coordinate and column forms of the vector **v** that has initial point (0, 0) and terminal point (7, 8).
2. Find the coordinate and column forms of the vector **v** that has initial point (2, 5) and terminal point (5, 10).
3. Find the coordinate and column forms of the vector **v** that has initial point A (a_1, a_2) and terminal point B (b_1, b_2).

5.2.2 Addition and Subtraction of Vectors

a. Addition of Vectors

Activity 5.3

1. What do you know about addition of vectors?
2. Consider vector **u** as a displacement that Kebede walks 5 m due east and then **v** as a displacement that Kebede walks 6 m due north. Find these two displacements as a single displacement.

From Activity 5.3, you can add two vectors geometrically by joining tail to head of the resultant vector **r** = **u** + **v**, which describes addition of vectors and can be found by placing **v** such that its tail is at the same point as the head of **u**. The resultant vector **r**, or addition of vectors **u** + **v**, has its tail at the tail of **u** and its head at the head of **v**, as shown in Figure 5.6, indicating triangular law of vector addition.

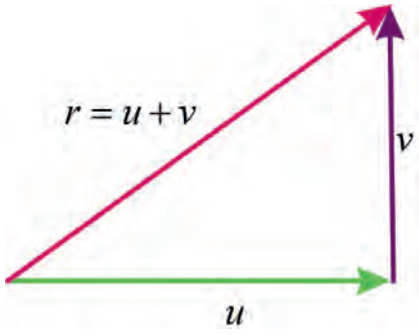


Figure 5.6

Definition 5.4 (Parallelogram method of addition of vectors (head-to-tail method)

If **u** and **v** are two vectors, the sum **u** + **v** is the vector determined by translating vector **v** until its tail coincides with the head of **u**. Then, the directed line segment from the tail of **u** to the head of **v** is the vector **u** + **v**.

Theorem 5.1 (Commutative property of vector addition)

If **u** and **v** are any two vectors, then **u** + **v** = **v** + **u.**

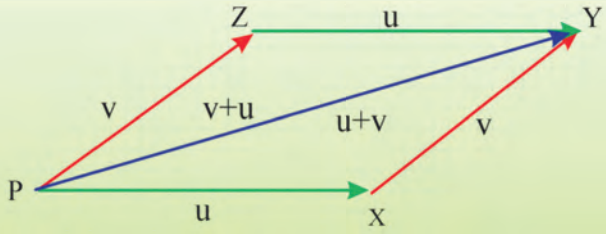


Figure 5.7

Theorem 5.2 (Associative Property of vector Addition)

If **u, v** and **w** are any three vectors, then (**u** + **v**) + **w** = **u** + (**v** + **w**).

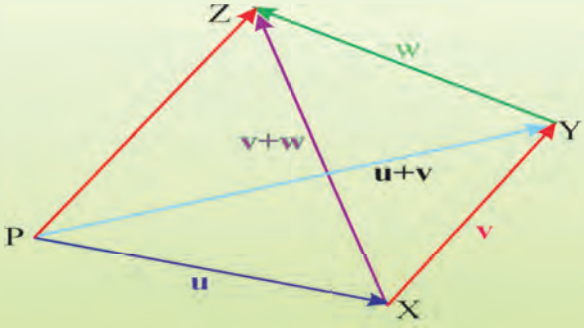


Figure 5.8

Example 1

If **u** = (2, 5), **v** = (7, 3) and **w** = (1, 6) as shown in figure 5.9, then find

a. **u** + **v** b. **v** + **u**

c. (**u** + **v**) + **w** d. **u** + (**v** + **w**)

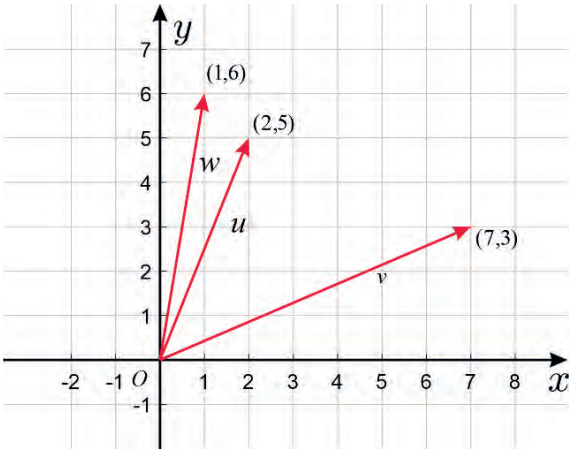


Figure 5.9

Solution

Substitute the given values of **u**, **v** and **w** into the definition of vector addition.

a. $\mathbf{u} + \mathbf{v} = (2, 5) + (7, 3) = (2 + 7, 5 + 3) = (9, 8)$.

b. $\mathbf{v} + \mathbf{u} = (7, 3) + (2, 5) = (7 + 2, 3 + 5) = (9, 8)$.

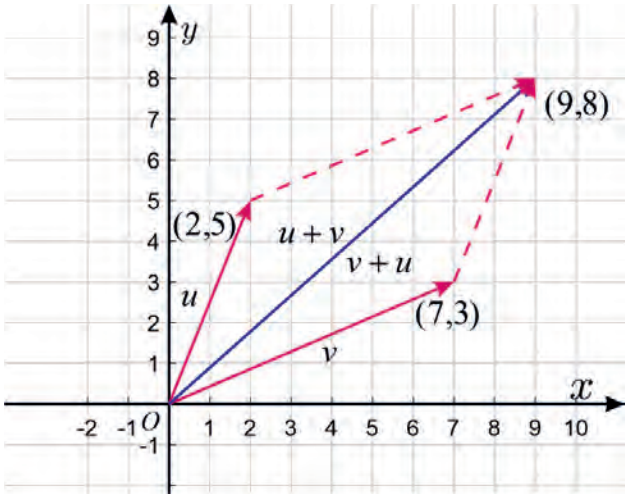


Figure 5.10

c. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = ((2, 5) + (7, 3)) + (1, 6) = (2 + 7, 5 + 3) + (1, 6) = (9, 8) + (1, 6)$
$= (9 + 1, 8 + 6) = (10, 14)$.

d. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (2, 5) + ((7, 3) + (1, 6)) = (2, 5) + (7 + 1, 3 + 6) = (2, 5) + (8, 9)$
$= (2 + 8, 5 + 9) = (10, 14)$.

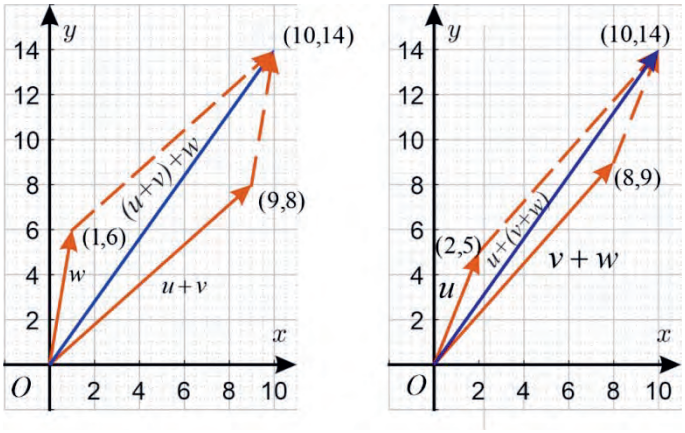


Figure 5.11

Note

1. For any three vectors **u**, **v** and **w**, if **u** = **v** and **v** = **w**, then **u** = **w**.
2. The zero vector **0** has the following property:
   For any vector **u**, **u** + **0** = **0** + **u** = **u**.
3. If **u** = (x_1 , y_1) and **v** = (x_2 , y_2) then **u** + **v** = (x_1 + x_2 , y_1 + y_2).

b. Subtraction of Vectors

Activity 5.4

Suppose that **u** and **v** are two vectors. How can you interpret the subtraction of these vectors in terms of addition?

From activity 5.4, you have observed that **u** – **v** is equal to **u** + (−**v**), that is the vector sum of **u** and −**v**. Now, you can reverse vector **v** and then add using the parallelogram method of addition of vectors.

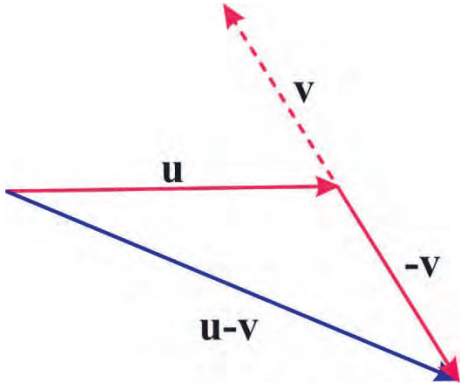


Figure 5.12

This figure shows subtraction of vector **u** – **v** as addition of **u** and (−**v**) using the parallelogram method of addition of vectors.

Moreover, you can also subtract these vectors using the triangular law of vector addition as follows. Draw the position **u** and **v** so that their initial points coincide. Here, the vector from the terminal point of **v** to the terminal point of **u** is then the vector **u** – **v** as shown in Figure 5.13.

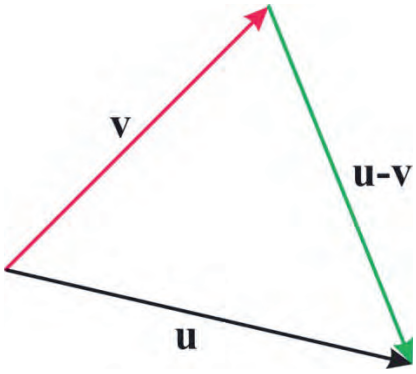


Figure 5.13

Note

1. Subtraction of two vectors **u** and **v** is adding the negative of vector **v** to the vector **u**. That means **u** – **v** = **u** + (–**v**).
2. You can do subtraction of vectors just like how you can do subtraction of scalers. You subtract the corresponding components of vectors while subtracting vectors.
3. If **v** is any non-zero vector and –**v** is the negative of **v**, then **v** + (–**v**) = 0.
4. The subtraction of vectors is not commutative.
5. The subtraction of vectors is not associative.

Example 2

If **u** = (3, 6), **v** = (8, 4) and **w** = (2, 7), then find

a. **u** − **v** b. **v** − **u** c. (**u** − **v**) − **w** d. **u** − (**v** − **w**)

Solution

Substitute the given values of **u**, **v** and **w**.

a. **u** − **v** = **u** + (− **v**) = (3, 6) + (−(8, 4)) = (3, 6) + (−8, −4)

= (3 + (−8), 6 + (−4)) = (−5, 2) .

b. **v** − **u** = (8, 4) − (3, 6) = (8 − 3, 4 − 6) = (5, −2).

c. $(\mathbf{u} - \mathbf{v}) - \mathbf{w} = ((3, 6) - (8, 4)) - (2, 7) = (3 - 8, 6 - 4) - (2, 7)$

$= (-5, 2) - (2, 7) = (-5 - 2, 2 - 7) = (-7, -5).$

d. $\mathbf{u} - (\mathbf{v} - \mathbf{w}) = (3, 6) - ((8, 4) - (2, 7)) = (3, 6) - (8 - 2, 4 - 7)$

$= (3, 6) - (6, -3) = (3 - 6, 6 - (-3)) = (-3, 9).$

Exercise 5.3

1. Find a. $\mathbf{u} + \mathbf{v}$ b. $\mathbf{u} - \mathbf{v}$ c. $(\mathbf{u} + \mathbf{v}) + \mathbf{w}$ d. $(\mathbf{u} - \mathbf{v}) - \mathbf{w}$ if $\mathbf{u} = (3, 4)$, $\mathbf{v} = (5, -1)$ and $\mathbf{w} = (2, -5)$.
2. Write any point A, B, C and F in your notebook and draw diagrams to illustrate the following vector equations:

   a. $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ b. $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CF} = \overrightarrow{AF}$ c. $\overrightarrow{AB} - \overrightarrow{BC} = \overrightarrow{AC}$
3. Graphically add the given vectors **u** and **v** shown in Figure 5.14 by using head to tail rule

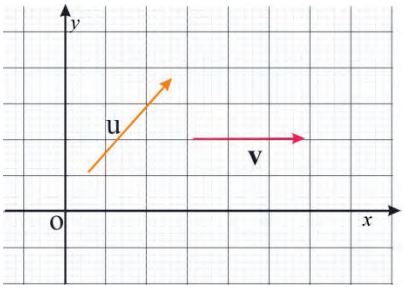


Figure 5.14

5.2.3 Multiplication of Vectors by Scalars

Activity 5.5

Let $\mathbf{u} = (2, 1)$. Then graph the pair of vectors on the same xy−plane.

a. **u** and **w**, where $\mathbf{w} = \mathbf{u} + \mathbf{u} + \mathbf{u}$

b. **u** and $-\mathbf{u}$

c. **u** and $-2\mathbf{w}$, where $\mathbf{w} = \mathbf{u} + \mathbf{u} + \mathbf{u}$

Suppose you have a vector **u**, then if this vector is multiplied by a scalar quantity $k = -1, 2, 3, -3, \frac{-1}{2}$ and so on, you will get the value as shown in Figure 5.15.

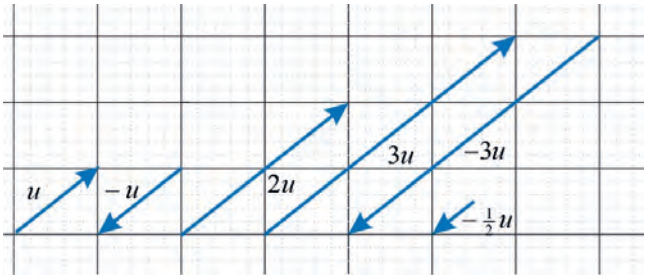


Figure. 5.15

Hence, from the above given set of vectors, you have observed that the direction of vector **u** remains the same when the value of the scalar is positive and the direction becomes exactly opposite when the value of the scalar is negative. In both cases the magnitude keeps changing depending upon the values of the scalar multiple.

Note

The product $k\mathbf{v}$ of a vector **v** and a scalar k is a vector with a magnitude that is $|k|$ times the magnitude of **v**, and with a direction that is the same as the direction of **v** if $k > 0$, and opposite the direction of **v** if $k < 0$. This is called scalar multiplication. If $k = 0$ or $\mathbf{v} = \mathbf{0}$, then $k\mathbf{v} = \mathbf{0}$.

Example 1

Let $\mathbf{u} = (-1, 3)$. Find $7\mathbf{u}$.

Solution

$7\mathbf{u} = 7(-1, 3) = (7(-1), 7(3)) = (-7, 21)$.

Theorem 5.3 (Properties of Scalar Multiplication)

Let **u** and **v** be vectors, let k_1 and k_2 be scalars. Then the following properties are true.

1	The magnitude of the vector which is multiplied by a scalar is equal to the absolute value of the scalar times the magnitude of the vector.	$\lvert k_1\mathbf{u}\rvert = \lvert k_1\rvert\lvert\mathbf{u}\rvert$
2	Associative Property	$k_1(k_2\,\mathbf{u}) = (k_1k_2)\,\mathbf{u}$
3	Distributive Property	i. $(k_1 + k_2)\,\mathbf{u} = k_1\mathbf{u} + k_2\mathbf{u}$ ii. $k_1(\mathbf{u} + \mathbf{v}) = k_1\mathbf{u} + k_1\mathbf{v}$
4	Identity Property	$1\cdot\mathbf{u} = \mathbf{u}$
5	Multiplicative Property of 0	$0\cdot\mathbf{u} = 0$

Example 2

If $\mathbf{u} = (5, -9)$ and $\mathbf{v} = (3, 5)$, then find

a. $2(3\mathbf{u})$ **b.** $(3 + 2)\,\mathbf{u}$ **c.** $3(\mathbf{u} + \mathbf{v})$ **d.** $1\cdot\mathbf{v}$ **e.** $0\cdot\mathbf{v}$

Solution

a. $2(3\mathbf{u}) = 2(3(5, -9)) = 2((3(5), 3(-9))) = 2(15, -27) = (30, -54)$

b. $(3 + 2)\,\mathbf{u} = (3 + 2)\,(5, -9) = 5(5, -9) = (25, -45)$

c. $3(\mathbf{u} + \mathbf{v}) = 3((5, -9) + (3, 5))$

$= 3(5, -9) + 3\,(3, 5) = (15, -27) + (9 + 15)$

$= (24, -12)$

d. $1\cdot\mathbf{v} = 1\cdot(3, 5) = (3, 5)$

e. $0\cdot\mathbf{v} = 0\cdot(3, 5) = (0, 0)$

Exercise 5.4

1. Let **u** = (11, 16) and **v** = (–8, 12). Find

   a. 5**u** b. 3(4**u**) c. (5 + 3) **u** d. 4(**u** + **v**) e. 1 · **v** f. 0 · **v**

2. In the triangle ABC, $\overrightarrow{AB}$ represents **u** and $\overrightarrow{BC}$ represents **v**. If D is the midpoint of AB, then express $\overrightarrow{AC}$, $\overrightarrow{CA}$ and $\overrightarrow{DC}$ in terms of **u** and **v**.

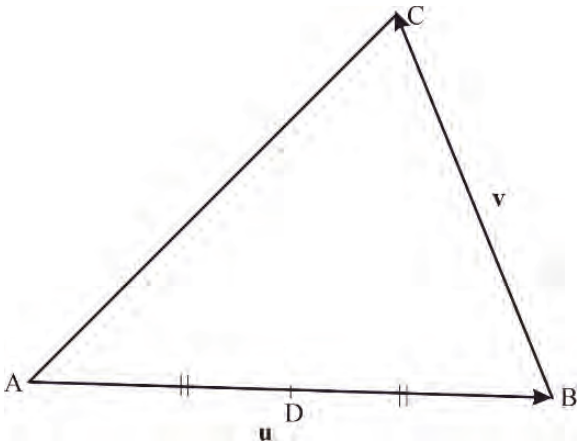


Figure 5. 16

3. ABCD is a parallelogram. $\overrightarrow{AB}$ represents **u** and $\overrightarrow{BC}$ represents **v**. If M is the midpoint of AC, and N is the midpoint of BD, find $\overrightarrow{BM}$ and $\overrightarrow{AN}$ in terms of **u** and **v**, afterwards show that M and N are coincident.

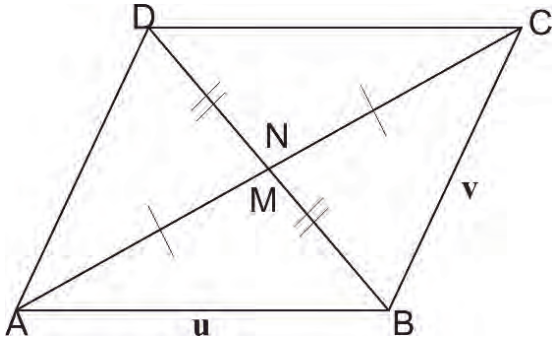


Figure 5. 17

5.2.4 Unit Vectors

Activity 5.6

For vectors in the plane, using standard unit vector " **i** " and "**j**" as shown in Figure 5.18, express a, b, c, d and e in terms of **i** and **j**.

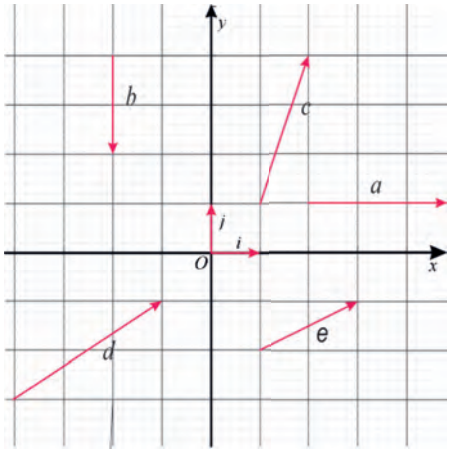


Figure 5.18

From Activity 5.6, you should have observed that, for example, $\mathbf{c} = \mathbf{i} + 3\mathbf{j}$ and $\mathbf{e} = 2\mathbf{i} + \mathbf{j}$. Then, the vector $\mathbf{c} + \mathbf{e}$ is simply $3\mathbf{i} + 4\mathbf{j}$. Adding vectors in terms of **i** and **j** is just a matter of adding components.

Consider a vector **u** whose initial point is the origin O (0, 0) and whose terminal point is the point P (a, b) as shown in Figure 5.19.

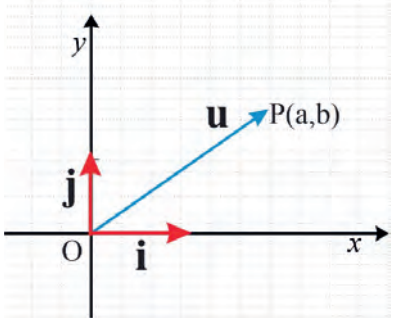


Figure 5.19

Hence, if a vector $\mathbf{u} = (a, b)$, then $\mathbf{u} = (a, b) = (a, 0) + (0, b) = a(1, 0) + b(0, 1)$

$= a\mathbf{i} + b\mathbf{j}$ where the unit vectors are $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$.

Example 1

Express the following vectors in terms of the standard unit vectors **i** and **j**:

a. $(5, -6)$ b. $(-4, 8)$ c. $(-7, 0)$ d. $(0, 3)$

Solution

a. $(5, -6) = (5, 0) + (0, -6) = 5\,(1, 0) + (-6)\,(0, 1) = 5\mathbf{i} + (-6)\,\mathbf{j} = 5\mathbf{i} - 6\mathbf{j}$.

b. $(-4, 8) = (-4, 0) + (0, 8) = -4\,(1, 0) + 8\,(0, 1) = -4\mathbf{i} + 8\mathbf{j}$.

c. $(-7, 0) = (-7, 0) + (0, 0) = -7\,(1, 0) + 0\,(0, 1) = -7\mathbf{i} + 0\mathbf{j} = -7\mathbf{i}$.

d. $(0, 3) = (0, 0) + (0, 3) = 0\,(1, 0) + 3(0, 1) = 0\mathbf{i} + 3\mathbf{j} = 3\mathbf{j}$.

Example 2

Express each of the following vectors as a vector in a coordinate:

a. $\mathbf{i} + 4\mathbf{j}$ b. $3\mathbf{i} - 3\mathbf{j}$ c. $-7\mathbf{i} + 8\mathbf{j}$

Solution

a. $\mathbf{i} + 4\mathbf{j} = (1, 0) + 4(0, 1) = (1, 0) + (0, 4) = (1 + 0, 0 + 4) = (1, 4)$.

b. $3\mathbf{i} - 3\,\mathbf{j} = 3(1, 0) + -\,3(0, 1) = (3, 0) + (0, -3) = (3 + 0, 0 + (-3)) = (3, -3)$.

c. $-7\mathbf{i} + 8\mathbf{j} = -7(1, 0) + 8(0, 1) = (-7, 0) + (0, 8) = (-7 + 0, 0 + 8) = (-7, 8)$.

Exercise 5.5

1. Express the vector $(3, 4)$ in terms of the standard unit vectors **i** and **j**.
2. Express $\overrightarrow{PQ}$ in terms of unit vectors **i** and **j**, when the end points are P $(6, 2)$ and Q $(1, 3)$.
3. Let **u** be the vector with initial point $(-4, 6)$ and terminal point $(-10, 5)$. Write **u** in terms of the standard unit vectors **i** and **j**. Also, represent it graphically.
4. Write the following vectors as a vector in the coordinate:

a. $\mathbf{i} + \mathbf{j}$ b. $-2\mathbf{i} + \frac{1}{5}\mathbf{j}$ c. $-3\mathbf{i} + (-5\mathbf{j})$ d. $\frac{3}{2}\mathbf{i} - 6\mathbf{j}$

5.2.5 Norm of Vectors

If $\overrightarrow{AB}$ is a vector with initial point (x_1, y_1) and terminal point (x_2, y_2) as shown in Figure 5.20, then its position vector **u** is determined as

$\mathbf{u} = (x_2 - x_1, y_2 - y_1) = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j}$

Thus, $(x_2 - x_1)$ and $(y_2 - y_1)$ are the coordinates of **u** with respect to the base (**i**, **j**).

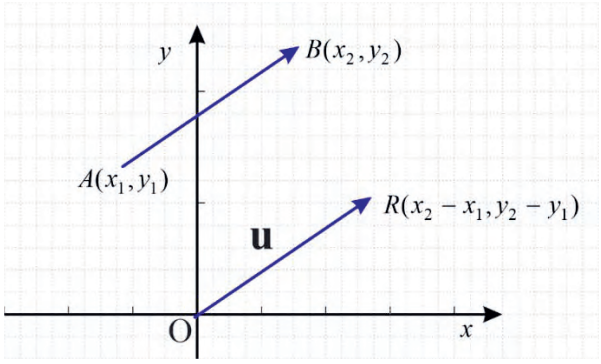


Figure 5.20

The norm or the magnitude of a vector $\mathbf{u} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j}$ is given by

$|\mathbf{u}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Example 1

Find the norms of the following vectors:

a. $(3, -3)$ b. $(4, 6)$ c. $(-4, 0)$ d. $(0, 6)$

Solution

a. The norm or magnitude of vector $\mathbf{u} = (3, -3)$ is

$|\mathbf{u}| = \sqrt{3^2 + (-3)^2} = \sqrt{9 + 9} = 3\sqrt{2}$.

b. The norm or magnitude of vector $\mathbf{u} = (4, 6)$ is

$|\mathbf{u}| = \sqrt{4^2 + 6^2} = \sqrt{16 + 36} = 2\sqrt{13}$.

c. The norm or magnitude of vector $\mathbf{u} = (-4, 0)$ is

$|\mathbf{u}| = \sqrt{(-4)^2 + 0^2} = \sqrt{16 + 0} = 4$.

d. The norm or magnitude of vector $\mathbf{u} = (0, 6)$ is $|\mathbf{u}| = \sqrt{0^2 + (6)^2} = \sqrt{0 + 36} = 6$.

Definition 5.5

Any vector that has magnitude which equals to 1 is called unit vectors and denoted by $\hat{\mathbf{v}}$ (read as **v** cap). Unit vectors are generally used to denote the direction of a vector.

- If **u** is any non -zero vector then the unit vector in the direction of **u** is given as $\hat{\mathbf{u}} = \mathbf{u} \times \frac{1}{|\mathbf{u}|}$.
- For vectors in two dimensions, in general $\mathbf{u} = x\mathbf{i} + y\mathbf{j}$ where x and y are scalars and its magnitude is given by the length or norms of vector OA. Where OA $= \sqrt{x^2 + y^2}$

  $\Rightarrow |\mathbf{u}| = \sqrt{x^2 + y^2}$

  In this case, we say that **u** is expressed as a linear combination of **i** and **j**.

Example 2

If $\mathbf{u} = (1, 2)$ and $\mathbf{v} = (3, 4)$, then find:

a. the unit vector in the direction of **v**.

b. the unit vector in the direction of $\mathbf{u} + \mathbf{v}$.

c. the unit vector in the direction of $\mathbf{u} - \mathbf{v}$.

Solution

a. $|\mathbf{v}| = \sqrt{x^2 + y^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

Substitute the given values of **v** and $|\mathbf{v}|$ into $\hat{\mathbf{v}} = \mathbf{v} \times \frac{1}{|\mathbf{v}|}$

Hence, $\hat{\mathbf{v}} = \mathbf{v} \times \frac{1}{|\mathbf{v}|}$

$\hat{\mathbf{v}} = (3, 4) \times \frac{1}{5} = (\frac{3}{5}, \frac{4}{5})$.

Thus, the unit vector in the direction of **v** is $(\frac{3}{5}, \frac{4}{5}) = \frac{3}{5}\boldsymbol{i} + \frac{4}{5}\boldsymbol{j}$.

b. $\mathbf{u} + \mathbf{v} = (1, 2) + (3,4) = (4,6)$

The norm of $|\mathbf{u} + \mathbf{v}| = \sqrt{x^2 + y^2} = \sqrt{4^2 + 6^2} = \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$

Substitute the given values of $\mathbf{u}+\mathbf{v}$ and $|\mathbf{u}+\mathbf{v}|$ into $\widehat{(\mathbf{u}+\mathbf{v})} = (\mathbf{u}+\mathbf{v}) \times \frac{1}{|\mathbf{u}+\mathbf{v}|}$

Hence, $\widehat{(\mathbf{u}+\mathbf{v})} = (\mathbf{u}+\mathbf{v}) \times \frac{1}{|\mathbf{u}+\mathbf{v}|}$

$$= (4, 6) \times \frac{1}{2\sqrt{13}} = \left(\frac{4}{2\sqrt{13}}, \frac{6}{2\sqrt{13}}\right) = \left(\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}}\right) = \left(\frac{2\sqrt{13}}{13}, \frac{3\sqrt{13}}{13}\right)$$

Thus, the unit vector in the direction of $\mathbf{u}+\mathbf{v}$ is $\left(\frac{2\sqrt{13}}{13}, \frac{3\sqrt{13}}{13}\right) = \frac{2\sqrt{13}}{13}\mathbf{i} + \frac{3\sqrt{13}}{13}\mathbf{j}$.

c. $\mathbf{u}-\mathbf{v} = (1, 2) - (3, 4) = (-2, -2)$.

The norm of $|\mathbf{u}-\mathbf{v}| = \sqrt{x^2+y^2} = \sqrt{(-2)^2+(-2)^2} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$.

Substitute the given values of $\mathbf{u}-\mathbf{v}$ and $|\mathbf{u}-\mathbf{v}|$ into $\widehat{(\mathbf{u}-\mathbf{v})} = (\mathbf{u}-\mathbf{v}) \times \frac{1}{|\mathbf{u}-\mathbf{v}|}$.

Hence, $\widehat{(\mathbf{u}-\mathbf{v})} = (\mathbf{u}-\mathbf{v}) \times \frac{1}{|\mathbf{u}-\mathbf{v}|} = (-2, -2) \times \frac{1}{2\sqrt{2}} = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$.

Thus, the unit vector in the direction of $\mathbf{u}-\mathbf{v}$ is $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) = -\frac{\sqrt{2}}{2}\mathbf{i} - \frac{\sqrt{2}}{2}\mathbf{j}$.

Exercise 5.6

1. Find the norm (or magnitude) of each of the following vectors.

   a. $\mathbf{u} = (1, 1)$ b. $\mathbf{v} = (-2, 1)$ c. $\mathbf{u} = (5, 0)$

   d. $\mathbf{v} = \left(\frac{1}{4}, \frac{3}{2}\right)$ e. $\mathbf{u} = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{5}\right)$

2. If $\mathbf{u} = 3\mathbf{i} + 5\mathbf{j}$ and $\mathbf{v} = 7\mathbf{i} - \mathbf{j}$, then find

   a. $|\mathbf{u}+\mathbf{v}|$ b. $|\mathbf{u}-\mathbf{v}|$ c. $r(\mathbf{u}+2\mathbf{v}), r \in \mathbb{R}$ d. $|\frac{1}{4}\mathbf{u}-\mathbf{v}|$

3. Find a unit vector in the direction of the vector (3, 6).

4. Find a unit vector in the direction opposite to the vector (1, 3).

5. Find two, unit vectors, one in the same direction and the other opposite to the vector $\mathbf{u} = (a, b) \neq 0$.

Definition 5.6

Two vectors **u** and **v** are parallel **if u** = k**v** for some non-zero k. furthermore, if $k > 0$ they are actually in the same direction, if $k < 0$, they are in opposite direction.

Note

1. Two vectors $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j}$ are equal if and only if their components are equal. i.e., $u_1 = v_1$ and $u_2 = v_2$.
2. Two or more vectors are said to be equal if they have the same magnitude and the same direction.

Example 3

Determine which of the vectors are equal to each other as shown in Figure 5.21.

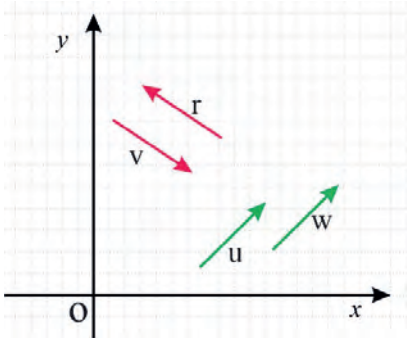


Figure 5.21

Solution

You will compare the given vectors to determine their magnitudes and directions.

First, you determine the magnitudes of the vectors.

It is obvious that $|\mathbf{u}| = |\mathbf{w}|$ and $|\mathbf{v}| = |\mathbf{r}|$. The magnitudes of the vectors **u** and **w** as well as the magnitude of vectors **v** and **r** are the same.

To compare the directions, it can be observed that the vectors **u** and **w** are parallel to each other with arrows on the same direction whereas the vectors **v** and **r** are parallel to each other with arrows on the opposite direction.

Thus, we can conclude that:

1. If two vectors have the same magnitude and direction, they are equal.
2. If two vectors have the same magnitude but opposite directions, one is inverse vector to the other.

Example 4

If $\mathbf{u} = 2\mathbf{i} + \mathbf{j}$ and $\mathbf{v} = \mathbf{i} - 2\mathbf{j}$, which of the vectors below are parallel to **u** or **v**?

1. $a = -2\mathbf{i} - \mathbf{j}$
2. $b = 5\mathbf{i} - 10\mathbf{j}$
3. $c = 4\mathbf{i} - 8\mathbf{j}$
4. $d = 6\mathbf{i} + 3\mathbf{j}$
5. $e = -2\mathbf{i} + 4\mathbf{j}$
6. $f = -4\mathbf{i} - 2\mathbf{j}$

Solution

The given vectors are expressed in terms of the unit vectors i and j. To determine whether or not they are parallel, you can check if their respective components can be expressed as scalar multiples of each other or not. Therefore, vectors **a, d** and **f** are parallel to vector **u** whereas vectors **b, c** and **e** are parallel to vector **v.**

Exercise 5.7

1. Determine whether the following two vectors are parallel or not. Then determine the magnitude of the two vectors.

a. $\mathbf{u} = (3\mathbf{i} + 4\mathbf{j})$ and $\mathbf{v} = (9\mathbf{i} + 12\mathbf{j})$

b. $\mathbf{w} = (6\mathbf{i} + 8\mathbf{j})$ and $\mathbf{u} = -3\mathbf{i} - 4\mathbf{j}$

c. $\mathbf{v} = (\frac{1}{4}, \frac{3}{2})$ and $\mathbf{u} = (\frac{1}{2}, \frac{3}{4})$

2. The column vectors **u** and **v** are defined by $\mathbf{u} = \begin{pmatrix} 8-x \\ 6-y \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} x-4 \\ y+2 \end{pmatrix}$.

Given that $\mathbf{u} = \mathbf{v}$

a. Find the values of x and y.

b. Find the values of $|\mathbf{u}|$ and $|\mathbf{v}|$.

5.3 Vector Product

The products of vectors are discussed in this sub-unit as follows:

5.3.1 Scalar (inner or dot) Product

Activity 5.7

1. Find the angle between the following pairs of vectors.

   a. $4\mathbf{i}$ and $2\mathbf{j}$ b. $\mathbf{i}$ and $3\mathbf{i}$ c. $2\mathbf{i}$ and $-\mathbf{i}$ d. $\mathbf{i}+\mathbf{j}$ and $3\mathbf{j}$ e. $\mathbf{i}$ and $\mathbf{i}+\sqrt{3}\mathbf{j}$

2. Consider two vectors such as $\mathbf{u}$ and $\mathbf{v}$ both of which have the same initial point. The angle formed by these two vectors is θ as shown in figure 5.22. Discuss how you can express θ in terms of $|\mathbf{u}|$ and $|\mathbf{v}|$.

In previous section, you have seen three vector operations: vector addition, subtraction and multiplication by a scalar, each of which provides another **vector.** In this section, you will study another vector operation like the inner or dot product. This product provides a **scalar,** rather than **a vector.**

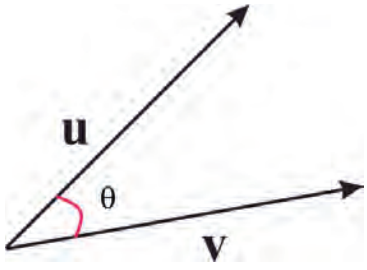


Figure 5.22

As shown in Figure. 5.22, the tails of the two vectors coincide and the angle between the vectors has been labelled θ. It is expressed θ in terms of $|\mathbf{u}|$ and $|\mathbf{v}|$ as $\mathbf{u}\cdot\mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta$ which defines the scalar product of two vectors such as $\mathbf{u}$ and $\mathbf{v}$. It is very important to use the dot in the formula. The dot is the symbol for the **scalar product**, and is the reason why the scalar product is also known as **the dot product.**

The angle θ is always chosen to lie between 0 and π, and the tails of the two vectors must coincide.

Theorem 5.4

If **u** and **v** are vectors and θ is the angle between **u** and **v**, then the dot product of **u** and **v**, denoted by **u** · **v** is given by **u** · **v** = $|\mathbf{u}||\mathbf{v}|\cos\theta$

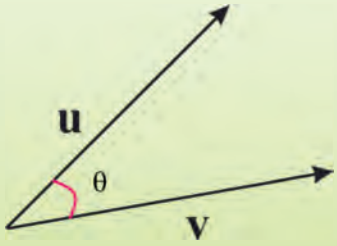


Figure. 5.23

Example 1

Suppose the vector **u** has norm 8 and the vector **v** has norm 7. Suppose also that the angle, θ, between these vectors is 30° as shown in Figure 5.24. Calculate **u** · **v**.

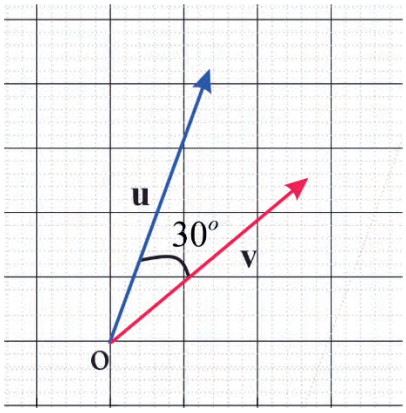


Figure 5.24

Solution

u · **v** = $|\mathbf{u}||\mathbf{v}|\cos\theta$

$= 8\times7\cos 30$

$= 56\left(\frac{\sqrt{3}}{2}\right)$

$= 28\sqrt{3}$.

Example 2

Find the dot product of the vectors **u** and **v** when

a. **u** = (0, 2) and **v** = (0, 4)

b. **u** = (3, 0) and **v** = (2, $2\sqrt{3}$)

Solution

From the Theorem 5.4 you have

a. $|\mathbf{u}| = 2$, $|\mathbf{v}| = 4$ and $\theta = 0^\circ \Rightarrow \mathbf{u} \cdot \mathbf{v} = 2 \times 4 \times \cos 0^\circ = 8 \times 1 = 8$.

b. Let us show these vectors graphically as shown in Figure 5.25.

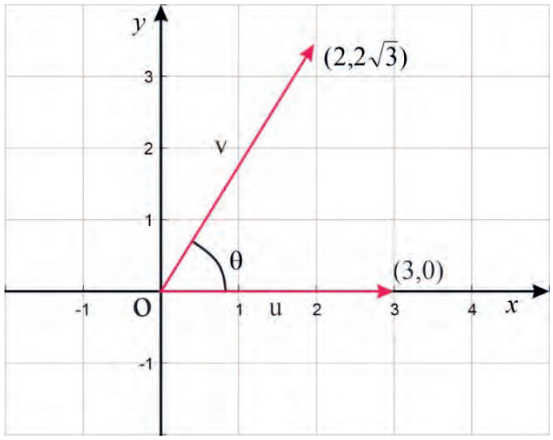


Figure 5.25

$|\mathbf{u}| = 3$, $|\mathbf{v}| = \sqrt{(2\sqrt{3})^2 + 2^2} = 4$ and $\cos\theta = \frac{2}{4} = \frac{1}{2}$ why?

Then, $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos\theta$

$= 3 \times 4 \times \frac{1}{2} = 6$

Before discussing the next properties of dot product, recall the relation between degree and radian that you have learnt in grade 10 such as, $45^\circ = \frac{\pi}{4}$, $90^\circ = \frac{\pi}{2}$, $180^\circ = \pi$, $270^\circ = \frac{3\pi}{2}$ and $360^\circ = 2\pi$.

Note

1. $\mathbf{i} \cdot \mathbf{j} = 0$, since $\cos\left(\frac{\pi}{2}\right) = 0$.
2. $\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = 1$, since $\cos(0) = 1$.
3. If either $\mathbf{u}$ or $\mathbf{v}$ is 0 then $\mathbf{u} \cdot \mathbf{v} = 0$.
4. $\mathbf{u} \cdot \mathbf{u} > 0$ if $\mathbf{u} \neq 0$ and $\mathbf{u} \cdot \mathbf{u} = 0$ if $\mathbf{u} = 0$.
5. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ …. dot product of vector is commutative.
6. If the vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel and pointing to the same direction, then $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|$.
7. If the vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel and pointing to the opposite direction, then $\mathbf{u} \cdot \mathbf{v} = -|\mathbf{u}||\mathbf{v}|$.
8. In particular, for any vector $\mathbf{u}$, you have $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$, you write $\mathbf{u}^2$ to mean $|\mathbf{u}|^2$
9. If the vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular or orthogonal, then $\mathbf{u} \cdot \mathbf{v} = 0$, since $\cos\left(\frac{\pi}{2}\right) = 0$.
10. If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors, then the cosine of the angle between $\mathbf{u}$ and $\mathbf{v}$ is given by $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

Exercise 5.8

1. Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors with $|\mathbf{u}| = 4$ and $|\mathbf{v}| = 2$ and $\theta = 45°$.
   Find their dot product.
2. Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors with $|\mathbf{u}| = 7$ and $|\mathbf{v}| = 3$ and $\theta = \frac{\pi}{2}$.
   Find their dot product.
3. Find the dot product of the vectors $\mathbf{u}$ and $\mathbf{v}$ when
   a. $\mathbf{u} = (2, 0)$ and $\mathbf{v} = (4, 0)$.
   b. $\mathbf{u} = (0, 3)$ and $\mathbf{v} = (2\sqrt{3}, 2)$.

Theorem 5.5 (Properties of the dot product)

If **u, v** and **w** are vectors and k is a scalar then

1. $k\,(\mathbf{u} \cdot \mathbf{v}) = (k\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (k\mathbf{v})$
2. Distributive property
   i) $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
   ii) $(\mathbf{v} + \mathbf{w}) \cdot \mathbf{u} = \mathbf{v} \cdot \mathbf{u} + \mathbf{w} \cdot \mathbf{u}$

Corollary 5.1

If $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ are vectors, then $\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2$.

Proof

$$\mathbf{u} \cdot \mathbf{v} = (u_1\mathbf{i} + u_2\mathbf{j}) \cdot (v_1\mathbf{i} + v_2\mathbf{j})$$

$$= u_1\mathbf{i} \cdot (v_1\mathbf{i} + v_2\mathbf{j}) + u_2\mathbf{j} \cdot (v_1\mathbf{i} + v_2\mathbf{j})$$

$$= u_1\mathbf{i} \cdot v_1\mathbf{i} + u_1\mathbf{i} \cdot v_2\mathbf{j} + u_2\mathbf{j} \cdot v_1\mathbf{i} + u_2\mathbf{j} \cdot v_2\,\mathbf{j}$$

$$= u_1v_1\mathbf{i} \cdot \mathbf{i} + u_1v_2\mathbf{i} \cdot \mathbf{j} + u_2v_1\mathbf{j} \cdot \mathbf{i} + u_2v_2\mathbf{j} \cdot \mathbf{j}$$

$$= u_1v_1 + u_2v_2.$$

Example 3

Find the dot product of the vectors $\mathbf{u} = 5\mathbf{i} + 3\mathbf{j}$ and $\mathbf{v} = 7\mathbf{i} - 2\mathbf{j}$.

Solution

$\mathbf{u} \cdot \mathbf{v} = (5\mathbf{i} + 3\mathbf{j}) \cdot (7\mathbf{i} - 2\mathbf{j}) = 5 \times 7 + 3 \times (-2) = 29.$

Example 4

Find the cosine of the angle formed by two vectors $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{v} = -2\mathbf{i} + 3\mathbf{j}$.